

Name	Value	Data Type	Scope
abort	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>abort - MainProgram/selfdestruct - *3(OTE), 0(XIO)</i>			
abortable	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>abortable - MainProgram/selfdestruct - *2(OTE), 2(XIC), 3(XIC)</i>			
acel	300	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>acel - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>acel - MainProgram/ThugShake - *B2, #3(Action,Action_027 (Body)), *B4, #2(Action,Action_030 (Body))</i>			
Action_019		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_019 - MainProgram/CONTROL - *C9(Action,Action_019)</i>			
Action_021		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_021 - MainProgram/CONTROL - *B11(Action,Action_021)</i>			
Action_022		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_022 - MainProgram/CONTROL - *B11(Action,Action_022)</i>			
Action_023		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_023 - MainProgram/CONTROL - *B12(Action,Action_023)</i>			
Action_024		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_024 - MainProgram/CONTROL - *B1(Action,Action_024)</i>			
Action_025		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_025 - MainProgram/CONTROL - *B21(Action,Action_025)</i>			
Action_026		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_026 - MainProgram/CONTROL - *C30(Action,Action_026)</i>			
Action_027		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_027 - MainProgram/ThugShake - *B2(Action,Action_027)</i>			
Action_028		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_028 - MainProgram/ThugShake - *B1(Action,Action_028)</i>			
Action_029		SFC_ACTION	MainProgram
Constant	No		

Action_029 (Continued)				
External Access:	Read/Write			
<i>Action_029 - MainProgram/ThugShake - *B3(Action,Action_029)</i>				
Action_030			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_030 - MainProgram/ThugShake - *B4(Action,Action_030)</i>				
Action_031			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_031 - MainProgram/SetWindingDia - *B2(Action,Action_031)</i>				
Action_032			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_032 - MainProgram/SetWindingDia - *B3(Action,Action_032)</i>				
Action_034			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_034 - MainProgram/CONTROL - *B14(Action,Action_034)</i>				
Action_039			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_039 - MainProgram/CONTROL - *B23(Action,Action_039)</i>				
Action_040			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_040 - MainProgram/CONTROL - *B27(Action,Action_040)</i>				
Action_041			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_041 - MainProgram/CONTROL - *B38(Action,Action_041)</i>				
Action_042			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_042 - MainProgram/CONTROL - *B38(Action,Action_042)</i>				
Action_043			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_043 - MainProgram/CONTROL - *B15(Action,Action_043)</i>				
Action_044			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_044 - MainProgram/SetWindingDia - *B3(Action,Action_044)</i>				
Action_045			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_045 - MainProgram/SetWindingDia - *B1(Action,Action_045)</i>				
Action_046			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_046 - MainProgram/SetWindingDia - *B4(Action,Action_046)</i>				

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Capstone_Test (Controller)			8/12/2025 1:37:10 AM
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Action_047			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_047 - MainProgram/SetWindingDia - *B6(Action,Action_047)</i>			
Action_048			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_048 - MainProgram/SetWindingDia - *B7(Action,Action_048)</i>			
Action_049			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_049 - MainProgram/CONTROL - *B13(Action,Action_049)</i>			
Action_051			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_051 - MainProgram/CONTROL - *B18(Action,Action_051)</i>			
Action_052			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_052 - MainProgram/CONTROL - *B19(Action,Action_052)</i>			
Action_053			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_053 - MainProgram/CONTROL - *B20(Action,Action_053)</i>			
Action_054			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_054 - MainProgram/CONTROL - *B6(Action,Action_054)</i>			
Action_056			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_056 - MainProgram/SetWindingDia - *B5(Action,Action_056)</i>			
Action_057			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_057 - MainProgram/CONTROL - *B16(Action,Action_057)</i>			
Action_058			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_058 - MainProgram/CONTROL - *B17(Action,Action_058)</i>			
Action_059			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_059 - MainProgram/CONTROL - *B17(Action,Action_059)</i>			
Action_060			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_060 - MainProgram/CONTROL - *B36(Action,Action_060)</i>			
Action_061			SFC_ACTION
Constant	No		MainProgram
External Access:	Read/Write		
<i>Action_061 - MainProgram/CONTROL - *B37(Action,Action_061)</i>			
			Logix Designer

Action_062		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_062 - MainProgram/CONTROL - *C31(Action,Action_062)</i>			
Action_063		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_063 - MainProgram/CONTROL - *B39(Action,Action_063)</i>			
Action_064		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_064 - MainProgram/CONTROL - *B6(Action,Action_064)</i>			
Action_065		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_065 - MainProgram/CONTROL - *B7(Action,Action_065)</i>			
Action_066		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_066 - MainProgram/CONTROL - *B4(Action,Action_066)</i>			
Action_067		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_067 - MainProgram/CONTROL - *B35(Action,Action_067)</i>			
Action_068		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_068 - MainProgram/CONTROL - *C32(Action,Action_068)</i>			
Action_069		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_069 - MainProgram/CONTROL - *C32(Action,Action_069)</i>			
Action_070		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_070 - MainProgram/CONTROL - *B35(Action,Action_070)</i>			
AdhesiveHeaterDutyCycle		REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AdhesiveHeaterDutyCycle - MainProgram/HeatExtruder - 4(MUL)</i>			
AdhesiveHeaterOnTime		DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AdhesiveHeaterOnTime - MainProgram/HeatExtruder - *4(MUL), 5(LEQ)</i>			
AdhesiveTimer		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AdhesiveTimer - MainProgram/HeatExtruder - *4(TON)</i>			
AdhesiveTimer.PRE		DINT	
<i>AdhesiveTimer.PRE - MainProgram/HeatExtruder - 4(MUL)</i>			
AdhesiveTimer.ACC		DINT	
<i>AdhesiveTimer.ACC - MainProgram/HeatExtruder - 5(LEQ)</i>			
AdhesiveTimer.DN		BOOL	

AdhesiveTimer (Continued) <i>AdhesiveTimer.DN - MainProgram/HeatExtruder - 4(XIO)</i>		
Alarm_DOOR External Access: Read/Write <i>Alarm_DOOR - MainProgram/Alarms - *4(ALMD)</i>	ALARM_DIGITAL	MainProgram
Alarm_ESTOP External Access: Read/Write <i>Alarm_ESTOP - MainProgram/Alarms - *1(ALMD)</i>	ALARM_DIGITAL	MainProgram
Alarm_Manual External Access: Read/Write <i>Alarm_Manual - MainProgram/Alarms - *5(ALMD)</i>	ALARM_DIGITAL	MainProgram
Alarm_MIRCOSCAN External Access: Read/Write <i>Alarm_MIRCOSCAN - MainProgram/Alarms - *3(ALMD)</i>	ALARM_DIGITAL	MainProgram
Alarm_NANOSCAN External Access: Read/Write <i>Alarm_NANOSCAN - MainProgram/Alarms - *2(ALMD)</i>	ALARM_DIGITAL	MainProgram
Alarm_RobotFault External Access: Read/Write <i>Alarm_RobotFault - MainProgram/Alarms - *10(ALMD)</i>	ALARM_DIGITAL	MainProgram
Alarm_Safety External Access: Read/Write <i>Alarm_Safety - MainProgram/Alarms - *0(ALMD)</i>	ALARM_DIGITAL	MainProgram
Alarm_Servo External Access: Read/Write <i>Alarm_Servo - MainProgram/Alarms - *6(ALMD)</i>	ALARM_DIGITAL	MainProgram
Alarm_Spool1 External Access: Read/Write <i>Alarm_Spool1 - MainProgram/Alarms - *7(ALMD)</i>	ALARM_DIGITAL	MainProgram
Alarm_Spool2 External Access: Read/Write <i>Alarm_Spool2 - MainProgram/Alarms - *8(ALMD)</i>	ALARM_DIGITAL	MainProgram
Alarm_Spool3 External Access: Read/Write <i>Alarm_Spool3 - MainProgram/Alarms - *9(ALMD)</i>	ALARM_DIGITAL	MainProgram
AOI_Motor Constant No External Access: Read/Write	AOI_Motor	Capstone_Test
AOI_Motor.DDO <i>AOI_Motor.DDO - MainProgram/MoveWinding - *8(OM_AZ_DDO)</i>	OM_AZ_DDO	
AOI_Motor.DDO.EnableIn 1 Enable Input - System Defined Parameter	BOOL	
AOI_Motor.DDO.EnableOut 1 Enable Output - System Defined Parameter	BOOL	
AOI_Motor.DDO.Set_Exec 0 Execute AOI <i>AOI_Motor.DDO.Set_Exec - MainProgram/MoveWinding - *4(OTL), *5(OTU), *6(OTU)</i>	BOOL	
AOI_Motor.DDO.Set_OpeType 1 Direct data operation operation type	INT	
AOI_Motor.DDO.Set_Position 284774 Direct data operation position <i>AOI_Motor.DDO.Set_Position - MainProgram/MoveWinding - *3(MUL), 4(NEQ)</i>	DINT	

AOI_Motor (Continued)		
AOI_Motor.DDO.Set_Velocity	750	DINT
Direct data operation operating speed		
<i>AOI_Motor.DDO.Set_Velocity - MainProgram/MoveWinding - *3(MOV)</i>		
AOI_Motor.DDO.Set_Acceleration	300	DINT
Direct data operation starting/changing rate		
<i>AOI_Motor.DDO.Set_Acceleration - MainProgram/MoveWinding - *3(MOV)</i>		
AOI_Motor.DDO.Set_Deceleration	750	DINT
Direct data operation stopping deceleration		
<i>AOI_Motor.DDO.Set_Deceleration - MainProgram/MoveWinding - *3(MOV)</i>		
AOI_Motor.DDO.Set_Current	1000	INT
Direct data operation operating current		
AOI_Motor.DDO.Set_Send	0	INT
Direct data operation forwarding destination		
AOI_Motor.DDO.Sts_Busy	0	BOOL
Executing status in AOI		
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/MoveWinding - 7(XIC)</i>		
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/SetWindingDia - B4, #1(Tran,Tran_041 (Condition))</i>		
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/ThugShake - B2, #1(Tran,Tran_027 (Condition)), C4, #1(Tran,Tran_029 (Condition))</i>		
AOI_Motor.DDO.Sts_Err	0	BOOL
Error status in AOI		
<i>AOI_Motor.DDO.Sts_Err - MainProgram/MoveWinding - 4(XIO), 6(XIC)</i>		
AOI_Motor.DDO.Sts_Done	0	BOOL
Complete status in AOI		
<i>AOI_Motor.DDO.Sts_Done - MainProgram/MoveWinding - 5(XIC)</i>		
AOI_Motor.DDO.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.DDO.Read_Command_position	284774	DINT
Command position monitor[step]		
AOI_Motor.DDO.Read_Feedback_position	284774	DINT
Feedback position monitor[step]		
AOI_Motor.DDO.Read_Torque_monitor	-2	INT
Torque monitor as a percentage of the maximum holding torque(1=0.1%)		
AOI_Motor.DDO.Read_Cumulative_load_monitor	2083663	DINT
Cumulative load monitor		
AOI_Motor.Move.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Move.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Move.Set_Exe	0	BOOL
Execute AOI		
AOI_Motor.Move.Set_OpeDataNo	0	INT
Selected Operation data No.		
AOI_Motor.Move.Sts_Busy	0	BOOL
Executing status in AOI		
AOI_Motor.Move.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.Move.Sts_Done	1	BOOL
Complete status in AOI		
AOI_Motor.Move.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.Move.Read_Command_position	-2692	DINT
Command position monitor[step]		
AOI_Motor.Move.Read_Feedback_position	-2692	DINT
Feedback position monitor[step]		
AOI_Motor.Move.Read_Torque_monitor		

AOI_Motor (Continued)		
11	INT	
Torque monitor as a percentage of the maximum holding torque(1=0.1%)		
AOI_Motor.Move.Read_Cumulative_load_monitor		
677105	DINT	
Cumulative load monitor		
AOI_Motor.Monitor	OM_AZ_MONITOR	
<i>AOI_Motor.Monitor - MainProgram/MoveWinding - *1(OM_AZ_MONITOR)</i>		
AOI_Motor.Monitor.EnableIn	1	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Monitor.EnableOut	1	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Monitor.Read_Present_alarm		
16#0000	INT	
Alarm code monitor[Hex]		
AOI_Motor.Monitor.Read_Feedback_position		
284774	DINT	
Feedback position monitor[step]		
AOI_Motor.Monitor.Read_Feedback_speed		
0	DINT	
Feedback speed monitor[Hz]		
AOI_Motor.Monitor.Read_Command_position		
284774	DINT	
Command position monitor[step]		
<i>CurrentWindingPos - MainProgram/MoveWinding - 2(DIV), 4(NEQ)</i>		
AOI_Motor.Monitor.Read_Torque_monitor		
-2	INT	
Torque monitor as a percentage of the maximum holding torque[1=0.1%]		
AOI_Motor.Monitor.Read_CST_operating_current		
500	INT	
Operation current monitor in alpha control mode(CST) (1=0.1%)		
AOI_Motor.Monitor.Read_Information		
16#0000_0000	DINT	
Information monitor before an alarm is generated.		
AOI_Motor.Monitor.Read_Driver_temperature		
430	DINT	
Driver temperature monitor(1=0.1deg)		
AOI_Motor.Monitor.Read_Motor_temperature		
459	DINT	
Motor temperature monitor(1=0.1deg)		
AOI_Motor.Monitor.Read_Cumulative_load_monitor		
2083663	DINT	
Cumulative load monitor		
AOI_Motor.Monitor.Read_Tripmeter		
317	DINT	
Travel distance monitor[x1000rev]		
AOI_Motor.ReadPosition.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.ReadPosition.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.ReadPosition.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.ReadPosition.ReadPRMID		
16#0c01	INT	
Read parameter ID		
AOI_Motor.ReadPosition.ReadData		
5000	DINT	
Read data		
AOI_Motor.ReadPosition.Sts_Busy	1	BOOL
Executing status in AOI		
AOI_Motor.ReadPosition.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.ReadPosition.Sts_Done		

AOI_Motor (Continued)		
Complete status in AOI	0	BOOL
AOI_Motor.WritePosition.EnableIn		
Enable Input - System Defined Parameter	0	BOOL
AOI_Motor.WritePosition.EnableOut		
Enable Output - System Defined Parameter	0	BOOL
AOI_Motor.WritePosition.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.WritePosition.WritePRMID		
Write parameter ID	16#0c01	INT
AOI_Motor.WritePosition.WriteData		
Write data	4000	DINT
AOI_Motor.WritePosition.Sts_Busy		
Executing status in AOI	0	BOOL
AOI_Motor.WritePosition.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.WritePosition.Sts_Done		
Complete status in AOI	1	BOOL
AOI_Motor.Home.EnableIn	1	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Home.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Home.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.Home.Sts_Busy	0	BOOL
Executing status in AOI		
AOI_Motor.Home.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.Home.Sts_Done	0	BOOL
Complete status in AOI		
AOI_Motor.Home.Read_Present_alarm		
Alarm code monitor[Hex]	16#0000	INT
AOI_Motor.Home.Read_Feedback_position		
Feedback position monitor[step]	0	DINT
AOI_Motor.WriteSpeed.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.WriteSpeed.EnableOut		
Enable Output - System Defined Parameter	0	BOOL
AOI_Motor.WriteSpeed.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.WriteSpeed.WritePRMID		
Write parameter ID	16#0c02	INT
AOI_Motor.WriteSpeed.WriteData	1000	DINT
Write data		
AOI_Motor.WriteSpeed.Sts_Busy	0	BOOL
Executing status in AOI		
AOI_Motor.WriteSpeed.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.WriteSpeed.Sts_Done	1	BOOL
Complete status in AOI		
AOI_Motor.ReadSpeed.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.ReadSpeed.EnableOut	0	BOOL

AOI_Motor (Continued)			
Enable Output - System Defined Parameter			
AOI_Motor.ReadSpeed.Set_Exec	0	BOOL	
Execute AOI			
AOI_Motor.ReadSpeed.ReadPRMID	16#0c02	INT	
Read parameter ID			
AOI_Motor.ReadSpeed.ReadData	1000	DINT	
Read data			
AOI_Motor.ReadSpeed.Sts_Busy	0	BOOL	
Executing status in AOI			
AOI_Motor.ReadSpeed.Sts_Err	0	BOOL	
Error status in AOI			
AOI_Motor.ReadSpeed.Sts_Done	1	BOOL	
Complete status in AOI			
AutoONS	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
<i>AutoONS - MainProgram/RobotAutoControl - *4(ONS)</i>			
 CameraFlash	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
<i>CameraFlash - MainProgram/FillValues - #47</i>			
<i>CameraFlash - MainProgram/MainRoutine - *9(OTE)</i>			
cBypass		COUNTER	SafetyProgram
Constant No			
External Access: Read/Write			
<i>cBypass - SafetyProgram  /MainRoutine - *10(CTU), *11(RES)</i>			
cBypass.ACC	1	DINT	
<i>cBypass.ACC - SafetyProgram  /MainRoutine - 11(EQU)</i>			
 ClampMotorFwd	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
<i>ClampMotorFwd - MainProgram/CONTROL - *B1, #14(Action,Action_024 (Body)), *B16, #1(Action,Action_057 (Body)), *B17, #1(Action,Action_058 (Body)), *B19, #1(Action,Action_052 (Body)), *B20, #1(Action,Action_053 (Body)), *B23, #34(Action,Action_039 (Body))</i>			
<i>ClampMotorFwd - MainProgram/FillValues - #50</i>			
<i>ClampMotorFwd - MainProgram/pause - #24, *#46, *#70</i>			
<i>ClampMotorFwd - MainProgram/StopAll - *#16</i>			
 ClampMotorRev	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
<i>ClampMotorRev - MainProgram/CONTROL - *B1, #15(Action,Action_024 (Body)), *B17, #1(Action,Action_059 (Body)), *B18, #1(Action,Action_051 (Body)), *B35, #2(Action,Action_067 (Body)), *B36, #1(Action,Action_060 (Body))</i>			
<i>ClampMotorRev - MainProgram/FillValues - #51</i>			
<i>ClampMotorRev - MainProgram/pause - #25, *#47, *#71</i>			
<i>ClampMotorRev - MainProgram/StopAll - *#17</i>			
 Coil		Coil	Capstone_Test
Constant No			
External Access: Read/Write			
Coil.Inductance	8.92412616e-004	REAL	
mH			
Coil.Mass	2.11115021e-005	REAL	
Kg			
Coil.Diameter	0.18	REAL	
m			
<i>Coil.Diameter - MainProgram/CONTROL - B13, #1(Action,Action_049 (Body))</i>			
Coil.Turns_X	2	INT	
<i>Coil.Turns_X - MainProgram/CONTROL - A34, #1(Tran,Tran_036 (Condition))</i>			

Coil (Continued)			
Coil.Turns_Z	2	INT	
<i>Coil.Turns_Z - MainProgram/CONTROL - A34, #1(Tran,Tran_036 (Condition)), B23, #16(Action,Action_039 (Body)), C30, #1(Tran,Tran_064 (Condition))</i>			
Coil.Current	0.15	REAL	
<i>A</i>			
Coil.Length	0.003	REAL	
<i>m</i>			
Coil.Gauge	16	DINT	
<i>Coil.Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)</i>			
CoilIndex_X	1	DINT	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>CoilIndex_X - MainProgram/CONTROL - *B1, #16(Action,Action_024 (Body)), *B23, #25(Action,Action_039 (Body)), A34, #1(Tran,Tran_036 (Condition)), B23, #17(Action,Action_039 (Body)), B23, #25(Action,Action_039 (Body)), B23, #29(Action,Action_039 (Body)), B23, #8(Action,Action_039 (Body))</i>			
<i>CoilIndex_X - MainProgram/StopAll - *#18</i>			
CoilIndex_Z	1	DINT	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>CoilIndex_Z - MainProgram/CONTROL - *B1, #17(Action,Action_024 (Body)), *B23, #21(Action,Action_039 (Body)), *B23, #24(Action,Action_039 (Body)), A34, #1(Tran,Tran_036 (Condition)), B23, #16(Action,Action_039 (Body)), B23, #17(Action,Action_039 (Body)), B23, #21(Action,Action_039 (Body)), B23, #33(Action,Action_039 (Body)), C30, #1(Tran,Tran_064 (Condition))</i>			
<i>CoilIndex_Z - MainProgram/StopAll - *#19</i>			
CoilPickupDeviation	10	DINT	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>CoilPickupDeviation - MainProgram/CONTROL - B36, #3(Action,Action_060 (Body)), B39, #1(Action,Action_063 (Body))</i>			
countdown		TIMER	Capstone_Test
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>countdown - MainProgram/selfdestruct - *1(TON)</i>			
countdown.TT	0	BOOL	
<i>countdown.TT - MainProgram/selfdestruct - 2(XIC)</i>			
CurrentTime	0	DINT	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>CurrentTime - MainProgram/FillValues - #63, #64, *#65</i>			
CurrentWindingPos	284774	DINT	MainProgram
<i>Command position monitor[step]</i>			
<i>AliasFor: AOI_Motor.Monitor.Read_Command_position(C)</i>			
<i>Base Tag: AOI_Motor.Monitor.Read_Command_position(C)</i>			
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>CurrentWindingPos - MainProgram/MoveWinding - 2(DIV), 4(NEQ)</i>			
CycleTime	0	DINT	Capstone_Test
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>CycleTime - MainProgram/CONTROL - A30, #1(Tran,Tran_065 (Condition)), F34, #1(Tran,Tran_040 (Condition))</i>			
<i>CycleTime - MainProgram/FillValues - #63, *#63</i>			
decel	750	DINT	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>decel - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>decel - MainProgram/ThugShake - *B2, #4(Action,Action_027 (Body)), *B4, #3(Action,Action_030 (Body))</i>			

- DoorInterlock1	0	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt02.Data(C)		
Base Tag:	Local:1:I.Pt02.Data(C)		
Constant	No		
External Access:	Read/Write		
<i>DoorInterlock1 - SafetyProgram ▢ /MainRoutine - 2(XIO)</i>			
- DoorInterlock2	0	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt03.Data(C)		
Base Tag:	Local:1:I.Pt03.Data(C)		
Constant	No		
External Access:	Read/Write		
<i>DoorInterlock2 - SafetyProgram ▢ /MainRoutine - 2(XIO)</i>			
- DoorInterlockTripped	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
<i>DoorInterlockTripped - SafetyProgram ▢ /MainRoutine - *2(OTE), 6(XIC), 7(XIC)</i>			
<i>\SafetyProgram.DoorInterlockTripped - MainProgram/Alarms - 4(XIC)</i>			
DoTheThugShake	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>DoTheThugShake - MainProgram/CONTROL - *B15, #1(Action,Action_043 (Body)), A15, #1(Tran,Tran_039 (Condition))</i>			
<i>DoTheThugShake - MainProgram/SetWindingDia - *B3, #1(Action,Action_044 (Body)), *B5, #1(Action,Action_056 (Body)), B3, #1(Tran,Tran_044 (Condition)), B5, #1(Tran,Tran_057 (Condition))</i>			
<i>DoTheThugShake - MainProgram/ThugShake - *B4, #1(Action,Action_030 (Body)), B1, #1(Tran,Tran_026 (Condition))</i>			
ENABLE	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ENABLE - MainProgram/CONTROL - A8, #1(Tran,Tran_019 (Condition))</i>			
<i>ENABLE - MainProgram/MainRoutine - 0(XIO)</i>			
<i>ENABLE - MainProgram/WindInductor - *4(OTE), 2(XIO), 3(XIO)</i>			
- EStop1	1	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt00.Data(C)		
Base Tag:	Local:1:I.Pt00.Data(C)		
Constant	No		
External Access:	Read/Write		
<i>EStop1 - SafetyProgram ▢ /MainRoutine - 1(XIO)</i>			
- EStop2	1	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt01.Data(C)		
Base Tag:	Local:1:I.Pt01.Data(C)		
Constant	No		
External Access:	Read/Write		
<i>EStop2 - SafetyProgram ▢ /MainRoutine - 1(XIO)</i>			
- EStopTripped	0	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
<i>EStopTripped - SafetyProgram ▢ /MainRoutine - *1(OTE), 6(XIC), 7(XIC)</i>			
<i>\SafetyProgram.EStopTripped - MainProgram/Alarms - 1(XIC)</i>			
ExtendCutting	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ExtendCutting - MainProgram/CONTROL - *B35, #1(Action,Action_067 (Body)), *B35, #1(Action,Action_070 (Body))</i>			
<i>ExtendCutting - MainProgram/FillValues - #57</i>			
ExtruderMotor	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		

ExtruderMotor (Continued)

*ExtruderMotor - MainProgram/CONTROL - *B23, #30(Action,Action_039 (Body)), *B35, #2(Action,Action_070 (Body)), *C30, #1(Action,Action_026 (Body))*
ExtruderMotor - MainProgram/FillValues - #58

ExtruderOn 0 BOOL MainProgram
 Constant No
 External Access: Read/Write
*ExtruderOn - MainProgram/CONTROL - *B20, #5(Action,Action_053 (Body))*
*ExtruderOn - MainProgram/HeatExtruder - *0(OTE), 0(XIC), 1(XIC), 2(XIC), 3(XIC), 4(XIC), 5(XIC)*

Fanuc_Robot:I AB:ETHERNET_MODULE_INT_8Bytes:I:0
 Capstone_Test
 Constant No
 External Access: Read/Write
Fanuc_Robot:I.Data[0] 0 INT
 In From Robot
Fanuc_Robot:I.Data[0].0 0 BOOL
 In From Robot
RobotAtHome1 - MainProgram/CONTROL - A39, #1(Tran,Tran_037 (Condition))
Fanuc_Robot:I.Data[0].1 0 BOOL
 In From Robot
RobotAtCutting2 - MainProgram/CONTROL - A11, #1(Tran,Tran_021 (Condition))
Fanuc_Robot:I.Data[0].2 0 BOOL
 In From Robot
RobotWireInserted3 - MainProgram/CONTROL - A19, #1(Tran,Tran_051 (Condition))
Fanuc_Robot:I.Data[1] 168 INT
 Robot Status
Fanuc_Robot:I.Data[1].0 0 BOOL
 Cmd Enabled
Fanuc_Robot:I.Data[1].1 0 BOOL
 System Ready
Fanuc_Robot:I.Data[1].1 - MainProgram/Alarms - 10(ALMD)
Fanuc_Robot:I.Data[1].1 - MainProgram/RobotAutoControl - 5(XIC)
Fanuc_Robot:I.Data[1].2 0 BOOL
 PRG Running
Fanuc_Robot:I.Data[1].2 - MainProgram/RobotAutoControl - 6(XIC), 7(XIO)
Fanuc_Robot:I.Data[1].5 1 BOOL
 Fault
Fanuc_Robot:I.Data[1].5 - MainProgram/Alarms - 10(XIC)
Fanuc_Robot:I.Data[1].5 - MainProgram/RobotAutoControl - 7(XIO)

Fanuc_Robot:O AB:ETHERNET_MODULE_INT_8Bytes:O:0
 Capstone_Test
 Constant No
 External Access: Read/Write
Fanuc_Robot:O.Data[0] 26 INT
 Out to Robot
Fanuc_Robot:O.Data[0].0 0 BOOL
 Out to Robot
*RobotToCutting - MainProgram/CONTROL - *B1, #9(Action,Action_024 (Body)), *B11, #2(Action,Action_021 (Body)), *B11, #2(Action,Action_022 (Body))*
*RobotToCutting - MainProgram/pause - #20, *#42, *#66*
*RobotToCutting - MainProgram/StopAll - *#12*
Fanuc_Robot:O.Data[0].1 1 BOOL
 Out to Robot
*RobotLoaded2 - MainProgram/CONTROL - *B1, #10(Action,Action_024 (Body)), *B12, #4(Action,Action_023 (Body)), *B18, #2(Action,Action_051 (Body))*
*RobotLoaded2 - MainProgram/pause - #21, *#43, *#67*
*RobotLoaded2 - MainProgram/StopAll - *#13*
Fanuc_Robot:O.Data[0].2 0 BOOL
 Out to Robot
*RobotMoveNextLayer3 - MainProgram/CONTROL - *B1, #11(Action,Action_024 (Body)), *B12, #3(Action,Action_023 (Body)), *B23, #18(Action,Action_039 (Body)), *B27, #1(Action,Action_040 (Body))*

Fanuc_Robot:O (Continued)			
RobotMoveNextLayer3 - MainProgram/pause - #22, *#44, *#68			
RobotMoveNextLayer3 - MainProgram/StopAll - *#14			
Fanuc_Robot:O.Data[0].3	1	BOOL	
Out to Robot			
RobotDirection4 - MainProgram/CONTROL - *B1, #13(Action,Action_024 (Body)), *B12, #2(Action,Action_023 (Body)), *B23, #10(Action,Action_039 (Body)), *B23, #12(Action,Action_039 (Body))			
RobotDirection4 - MainProgram/pause - #23, *#45, *#69			
RobotDirection4 - MainProgram/StopAll - *#15			
Fanuc_Robot:O.Data[0].4	1	BOOL	
Out to Robot			
RobotDoneGetThisShitOutOfHere4 - MainProgram/CONTROL - *B1, #12(Action,Action_024 (Body)), *B38, #1(Action,Action_041 (Body)), *B38, #1(Action,Action_042 (Body))			
Fanuc_Robot:O.Data[0].5	0	BOOL	
Out to Robot			
RobotMoveBack6 - MainProgram/CONTROL - *B20, #2(Action,Action_053 (Body)), *B21, #1(Action,Action_025 (Body))			
Fanuc_Robot:O.Data[1]	143	INT	
AutoMode			
Fanuc_Robot:O.Data[1].0	1	BOOL	
IMSTP			
Fanuc_Robot:O.Data[1].0 - MainProgram/RobotAutoControl - *3(OTE)			
Fanuc_Robot:O.Data[1].1	1	BOOL	
Hold			
Fanuc_Robot:O.Data[1].1 - MainProgram/RobotAutoControl - *3(OTE)			
Fanuc_Robot:O.Data[1].2	1	BOOL	
SFSPD			
Fanuc_Robot:O.Data[1].2 - MainProgram/RobotAutoControl - *3(OTE)			
Fanuc_Robot:O.Data[1].4	0	BOOL	
Fault Reset			
Fanuc_Robot:O.Data[1].4 - MainProgram/RobotAutoControl - *11(OTE)			
Fanuc_Robot:O.Data[1].5	0	BOOL	
Start			
Fanuc_Robot:O.Data[1].5 - MainProgram/RobotAutoControl - *12(OTE)			
Fanuc_Robot:O.Data[1].7	1	BOOL	
Enable			
Fanuc_Robot:O.Data[1].7 - MainProgram/RobotAutoControl - *3(OTE)			
GoodToRun	0	BOOL	MainProgram
Constant			No
External Access:			Read/Write
GoodToRun - MainProgram/MainRoutine - *2(OTE), 2(XIC), 4(XIC)			
GoodToRun - MainProgram/pause - #2, #5			
GoodToRun - MainProgram/WindInductor - 0(XIC)			
GoodToRunFalling		FBD_ONESHOT	MainProgram
Constant			No
External Access:			Read/Write
GoodToRunFalling - MainProgram/pause - *#3			
GoodToRunFalling.InputBit	1	BOOL	
GoodToRunFalling.InputBit - MainProgram/pause - *#2			
GoodToRunFalling.OutputBit	0	BOOL	
GoodToRunFalling.OutputBit - MainProgram/pause - #8			
GoodToRunRising		FBD_ONESHOT	MainProgram
Constant			No
External Access:			Read/Write
GoodToRunRising - MainProgram/pause - *#6			
GoodToRunRising.InputBit	1	BOOL	
GoodToRunRising.InputBit - MainProgram/pause - *#5			
GoodToRunRising.OutputBit	0	BOOL	
GoodToRunRising.OutputBit - MainProgram/pause - #54			
Heater	0	BOOL	Capstone_Test
Constant			No

Heater (Continued)			
External Access:	Read/Write		
<i>Heater - MainProgram/FillValues - #59</i>			
<i>Heater - MainProgram/HeatExtruder - *1(OTE), *5(OTE)</i>			
HeaterAtTemp	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>HeaterAtTemp - MainProgram/CONTROL - A20, #1(Tran,Tran_053 (Condition))</i>			
<i>HeaterAtTemp - MainProgram/HeatExtruder - *2(OTE), 2(XIC)</i>			
HeaterAtTempTimer		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>HeaterAtTempTimer - MainProgram/HeatExtruder - *3(TON)</i>			
HeaterAtTempTimer.DN	0	BOOL	
<i>HeaterAtTempTimer.DN - MainProgram/HeatExtruder - 2(XIC)</i>			
HeightMotorDown	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>HeightMotorDown - MainProgram/CONTROL - *B1, #19(Action,Action_024 (Body)), *B6, #1(Action,Action_064 (Body)), *B7, #1(Action,Action_065 (Body)), *C30, #2(Action,Action_026 (Body)), *C31, #1(Action,Action_062 (Body))</i>			
<i>HeightMotorDown - MainProgram/FillValues - #53</i>			
<i>HeightMotorDown - MainProgram/pause - #27, *#49, *#73</i>			
<i>HeightMotorDown - MainProgram/StopAll - *#21</i>			
HeightMotorUp	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>HeightMotorUp - MainProgram/CONTROL - *B1, #20(Action,Action_024 (Body)), *B35, #3(Action,Action_067 (Body)), *B36, #2(Action,Action_060 (Body)), *B4, #1(Action,Action_066 (Body)), *B6, #1(Action,Action_054 (Body)), *C32, #1(Action,Action_068 (Body)), *C32, #1(Action,Action_069 (Body))</i>			
<i>HeightMotorUp - MainProgram/FillValues - #54</i>			
<i>HeightMotorUp - MainProgram/pause - #28, *#50, *#74</i>			
<i>HeightMotorUp - MainProgram/StopAll - *#22</i>			
HMI		HMIControl	Capstone_Test
Constant	No		
External Access:	Read/Write		
HMI.Manual.Spool.FWD1	0	BOOL	
<i>HMI.Manual.Spool.FWD1 - MainProgram/FillValues - #40</i>			
HMI.Manual.Spool.FWD2	0	BOOL	
<i>HMI.Manual.Spool.FWD2 - MainProgram/FillValues - #42</i>			
HMI.Manual.Spool.FWD3	0	BOOL	
<i>HMI.Manual.Spool.FWD3 - MainProgram/FillValues - #44</i>			
HMI.Manual.Spool.REV1	0	BOOL	
<i>HMI.Manual.Spool.REV1 - MainProgram/FillValues - #41</i>			
HMI.Manual.Spool.REV2	0	BOOL	
<i>HMI.Manual.Spool.REV2 - MainProgram/FillValues - #43</i>			
HMI.Manual.Spool.REV3	0	BOOL	
<i>HMI.Manual.Spool.REV3 - MainProgram/FillValues - #45</i>			
HMI.Manual.Cut	0	BOOL	
<i>HMI.Manual.Cut - MainProgram/FillValues - #57</i>			
HMI.Manual.CameraLight	0	BOOL	
<i>HMI.Manual.CameraLight - MainProgram/FillValues - #47</i>			
HMI.Manual.ExtrudeHeater	0	BOOL	
<i>HMI.Manual.ExtrudeHeater - MainProgram/FillValues - #59</i>			
HMI.Manual.ExtrudeMotor	0	BOOL	
<i>HMI.Manual.ExtrudeMotor - MainProgram/FillValues - #58</i>			
HMI.Manual.WindingSolenoid	0	BOOL	
<i>HMI.Manual.WindingSolenoid - MainProgram/FillValues - #60</i>			
HMI.Manual.ClampMotorFWD	0	BOOL	
<i>HMI.Manual.ClampMotorFWD - MainProgram/FillValues - #50</i>			

HMI (Continued)			
HMI.Manual.ClampMotorREV	0	BOOL	
<i>HMI.Manual.ClampMotorREV - MainProgram/FillValues - #51</i>			
HMI.Manual.HeightMotorUp	0	BOOL	
<i>HMI.Manual.HeightMotorUp - MainProgram/FillValues - #54</i>			
HMI.Manual.HeightMotorDown	0	BOOL	
<i>HMI.Manual.HeightMotorDown - MainProgram/FillValues - #53</i>			
HMI.StartPB	0	BOOL	
<i>HMI.StartPB - MainProgram/MainRoutine - 2(XIC)</i>			
HMI.ResetPB	0	BOOL	
<i>HMI.ResetPB - MainProgram/MainRoutine - 0(XIC)</i>			
HMI.StopPB	0	BOOL	
<i>HMI.StopPB - MainProgram/MainRoutine - 2(XIO)</i>			
<i>HMI.StopPB - MainProgram/pause - #8</i>			
HMI.RobotFaultReset	0	BOOL	
<i>HMI.RobotFaultReset - MainProgram/RobotAutoControl - 8(XIC)</i>			
InductorSent	0	BOOL	MainProgram
PC sets this true when the inductor data has been sent			
Constant No			
External Access: Read/Write			
<i>InductorSent - MainProgram/CONTROL - *B1, #18(Action,Action_024 (Body))</i>			
<i>InductorSent - MainProgram/pause - #26, *#48, *#72</i>			
<i>InductorSent - MainProgram/StopAll - *#20</i>			
<i>InductorSent - MainProgram/WindInductor - 1(XIC)</i>			
InitialLayers	20	DINT	MainProgram
Constant No			
External Access: Read/Write			
<i>InitialLayers - MainProgram/CONTROL - *B12, #6(Action,Action_023 (Body))</i>			
<i>InitialLayers - MainProgram/FillValues - #69, #69</i>			
Input_RemoteIO:I		AB:1732_DI16:I:0	Capstone_Test
Constant No			
External Access: Read/Write			
Input_RemoteIO:I.Data.0	0	BOOL	
StartPB			
<i>PB_Start - MainProgram/MainRoutine - 2(XIC)</i>			
Input_RemoteIO:I.Data.2	1	BOOL	
<i>Input_RemoteIO:I.Data.2 - MainProgram/FillValues - #10</i>			
Input_RemoteIO:I.Data.3	1	BOOL	
<i>Input_RemoteIO:I.Data.3 - MainProgram/FillValues - #11</i>			
Input_RemoteIO:I.Data.5	0	BOOL	
<i>Input_RemoteIO:I.Data.5 - MainProgram/FillValues - #12</i>			
Input_RemoteIO:I.Data.6	0	BOOL	
<i>WireAtEndOfLine - MainProgram/CONTROL - A12, #1(Tran,Tran_022 (Condition))</i>			
Input_RemoteIO:I.Data.9	1	BOOL	
<i>WireAtCutting - MainProgram/CONTROL - A9, #1(Tran,Tran_020 (Condition)), C10, #1(Tran,Tran_063 (Condition)), C9, #1(Tran,Tran_061 (Condition))</i>			
Layers	20	DINT	MainProgram
Constant No			
External Access: Read/Write			
<i>Layers - MainProgram/CONTROL - *B12, #5(Action,Action_023 (Body)), B12, #6(Action,Action_023 (Body))</i>			
<i>Layers - MainProgram/FillValues - #69</i>			
LoadCommand	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
<i>LoadCommand - MainProgram/WindInductor - *3(OTE)</i>			
Local:1:I		AB:5000_SDI8:I:0	Capstone_Test
Constant No			
External Access: Read/Write			

Local:1:I (Continued)		
Local:1:I.Pt00.Data	1	BOOL
<i>EStop1 - SafetyProgram /MainRoutine - 1(XIO)</i>		
Local:1:I.Pt01.Data	1	BOOL
<i>EStop2 - SafetyProgram /MainRoutine - 1(XIO)</i>		
Local:1:I.Pt02.Data	0	BOOL
<i>DoorInterlock1 - SafetyProgram /MainRoutine - 2(XIO)</i>		
Local:1:I.Pt03.Data	0	BOOL
<i>DoorInterlock2 - SafetyProgram /MainRoutine - 2(XIO)</i>		
Local:1:I.Pt04.Data	0	BOOL
<i>nanoScanCh1 - SafetyProgram /MainRoutine - 3(XIO)</i>		
Local:1:I.Pt05.Data	0	BOOL
<i>nanoScanCh2 - SafetyProgram /MainRoutine - 3(XIO)</i>		
Local:1:I.Pt06.Data	0	BOOL
<i>PB_Reset - MainProgram/Alarms - 0(ALMD), 1(ALMD), 2(ALMD), 3(ALMD), 4(ALMD)</i>		
<i>PB_Reset - MainProgram/RobotAutoControl - 0(XIC)</i>		
<i>PB_Reset - SafetyProgram /MainRoutine - 10(XIC), 6(XIO), 8(XIC)</i>		
Local:1:I.Pt07.Data	1	BOOL
<i>NanoScanWarning - SafetyProgram /MainRoutine - 5(XIC)</i>		
Local:2:O		
Constant	No	AB:5000_DO8:O:0
External Access:	Read/Write	Capstone_Test
Local:2:O.Pt00		CHANNEL_DO:O:0
Robot Safety Relay K1		
Local:2:O.Pt00.Data	1	BOOL
Robot Safety Relay K1		
<i>Local:2:O.Pt00.Data - SafetyProgram /MainRoutine - *9(OTE), 8(XIC)</i>		
Local:2:O.Pt01		CHANNEL_DO:O:0
Robot Safety Relay K2		
Local:2:O.Pt01.Data	1	BOOL
Robot Safety Relay K2		
<i>Local:2:O.Pt01.Data - SafetyProgram /MainRoutine - *9(OTE), 8(XIC)</i>		
Local:2:O.Pt02.Data	0	BOOL
Robot Reset Relay		
<i>Local:2:O.Pt02.Data - SafetyProgram /MainRoutine - *8(OTE)</i>		
Local:4:O		
Constant	No	AB:5000_DO16:O:0
External Access:	Read/Write	Capstone_Test
Local:4:O.Pt00.Data	0	BOOL
Top Feed Motor Fwd		
<i>Local:4:O.Pt00.Data - MainProgram/FillValues - *#40</i>		
Local:4:O.Pt01.Data	0	BOOL
Top Feed Motor Rev		
<i>Local:4:O.Pt01.Data - MainProgram/FillValues - *#41</i>		
Local:4:O.Pt02.Data	0	BOOL
Mid Feed Motor Fwd		
<i>Local:4:O.Pt02.Data - MainProgram/FillValues - *#42</i>		
Local:4:O.Pt03.Data	0	BOOL
Mid Feed Motor Rev		
<i>Local:4:O.Pt03.Data - MainProgram/FillValues - *#43</i>		
Local:4:O.Pt04.Data	0	BOOL
Bottom Feed Motor Fwd		
<i>Local:4:O.Pt04.Data - MainProgram/FillValues - *#44</i>		
Local:4:O.Pt05.Data	0	BOOL
Bottom Feed Motor Rev		
<i>Local:4:O.Pt05.Data - MainProgram/FillValues - *#45</i>		
Local:4:O.Pt06		CHANNEL_DO:O:0
Camera Flash		
Local:4:O.Pt06.Data	0	BOOL
Reset Robot Safety Relay		
<i>Local:4:O.Pt06.Data - MainProgram/FillValues - *#47</i>		
Local:4:O.Pt07.Data	1	BOOL

Local:4:O (Continued)		
<i>StackLightGreen - MainProgram/MainRoutine - *5(OTE)</i>		
<i>StackLightGreen - MainProgram/pause - #10, *#32, *#56</i>		
<i>StackLightGreen - MainProgram/StopAll - *#2</i>		
Local:4:O.Pt08.Data	1	BOOL
<i>StackLightYellow - MainProgram/MainRoutine - *6(OTE)</i>		
<i>StackLightYellow - MainProgram/pause - #11, *#33, *#57</i>		
<i>StackLightYellow - MainProgram/StopAll - *#3</i>		
Local:4:O.Pt09.Data	0	BOOL
<i>StackLightRed - MainProgram/MainRoutine - *7(OTE)</i>		
<i>StackLightRed - MainProgram/pause - #9, *#31, *#55</i>		
<i>StackLightRed - MainProgram/StopAll - *#1</i>		
Local:4:O.Pt10.Data	0	BOOL
<i>Local:4:O.Pt10.Data - MainProgram/FillValues - *#53</i>		
Local:4:O.Pt11.Data	0	BOOL
<i>Local:4:O.Pt11.Data - MainProgram/FillValues - *#54</i>		
Local:4:O.Pt12.Data	0	BOOL
<i>Local:4:O.Pt12.Data - MainProgram/FillValues - *#50</i>		
Local:4:O.Pt13.Data	0	BOOL
<i>Local:4:O.Pt13.Data - MainProgram/FillValues - *#51</i>		
Local:4:O.Pt14.Data	0	BOOL
<i>ResetLight - MainProgram/MainRoutine - *7(OTE)</i>		
Local:5:I		
Constant	No	AB:5000_AI8:I:0
External Access:	Read/Write	Capstone_Test
Local:5:I.Ch00.Data	164.73665	REAL
<i>Local:5:I.Ch00.Data - MainProgram/Alarms - 7(LEQ)</i>		
<i>Local:5:I.Ch00.Data - MainProgram/FillValues - #19, #20</i>		
Local:5:I.Ch01.Data	625.2037	REAL
<i>Local:5:I.Ch01.Data - MainProgram/Alarms - 8(LEQ)</i>		
Local:5:I.Ch02.Data	600.10535	REAL
<i>Local:5:I.Ch02.Data - MainProgram/Alarms - 9(LEQ)</i>		
<i>Local:5:I.Ch02.Data - MainProgram/FillValues - #31, #32</i>		
Local:5:I.Ch03.Data	-2.61334376e-003	REAL
<i>Local:5:I.Ch03.Data - MainProgram/FillValues - #16</i>		
MANLY		
Constant	No	0
External Access:	Read/Write	Capstone_Test
<i>MANLY - MainProgram/Alarms - 5(XIC)</i>		
<i>MANLY - MainProgram/MainRoutine - 0(XIO), 4(XIO)</i>		
microScan:SI		
Constant	No	_0328:MICS3_CBAZXXIZ1_F211ADCF:SI:0
External Access:	Read/Write	Capstone_Test
microScan:SI.Data[0]	1	SINT
Mode		
microScan:SI.Data[1]	1	SINT
Safe CutOff Path(1-8)		
microScan:SI.Data[2]	1	SINT
NonSafe CutOff Path(1-8)		
<i>microScan:SI.Data[2] - SafetyProgram /MainRoutine - 4(LES), 5(EQU)</i>		
microScan:SI.Data[3]	0	SINT
Reserved		
microScan:SI.Data[4]	1	SINT
Current Monitoring Case No.		
microScan:SI.Data[5]	0	SINT
Reset Required CutOff Path(1-8)		
microScan:SI.Data[6]	0	SINT
Reserved		
microScan:SI.Data[7]	0	SINT
Errors		

 microScan:SO			_0328:MICS3_CBAZXXIZ1_DD7B81FB:SO:0	Capstone_Test
Constant	No			
External Access:	Read/Write			
microScan:SO.Data[0]	3	SINT		
Mode				
<i>microScan:SO.Data[0] - SafetyProgram  /MainRoutine - *0(MOV)</i>				
microScan:SO.Data[0].0	1	BOOL		
Trigger Run-Mode				
microScan:SO.Data[1]	0	SINT		
Reserved				
microScan:SO.Data[2]	0	SINT		
Reserved				
microScan:SO.Data[3]	1	SINT		
Control Inputs				
<i>microScan:SO.Data[3] - SafetyProgram  /MainRoutine - *0(MOV)</i>				
microScan:SO.Data[3].0	1	BOOL		
Control input 1 (A1)				
microScan:SO.Data[4]	0	SINT		
Reserved				
microScan:SO.Data[5]	0	SINT		
Trigger Reset CutOff Path (1-8)				
microScan:SO.Data[6]	0	SINT		
Reserved				
microScan:SO.Data[7]	0	SINT		
Trigger Device Reboot				
MicroScanTripped	0	BOOL		SafetyProgram
Constant	No			
External Access:	Read/Write			
<i>MicroScanTripped - SafetyProgram  /MainRoutine - *4(OTE), 6(XIC), 7(XIC)</i>				
<i>\SafetyProgram.MicroScanTripped - MainProgram/Alarms - 3(XIC)</i>				
nanoScanCh1	0	BOOL		SafetyProgram
AliasFor:	Local:1:I.Pt04.Data(C)			
Base Tag:	Local:1:I.Pt04.Data(C)			
Constant	No			
External Access:	Read/Write			
<i>nanoScanCh1 - SafetyProgram  /MainRoutine - 3(XIO)</i>				
nanoScanCh2	0	BOOL		SafetyProgram
AliasFor:	Local:1:I.Pt05.Data(C)			
Base Tag:	Local:1:I.Pt05.Data(C)			
Constant	No			
External Access:	Read/Write			
<i>nanoScanCh2 - SafetyProgram  /MainRoutine - 3(XIO)</i>				
NanoScanTripped	1	BOOL		SafetyProgram
Constant	No			
External Access:	Read/Write			
<i>NanoScanTripped - SafetyProgram  /MainRoutine - *3(OTE), 6(XIC), 7(XIC)</i>				
<i>\SafetyProgram.NanoScanTripped - MainProgram/Alarms - 2(XIC)</i>				
NanoScanWarning	1	BOOL		SafetyProgram
AliasFor:	Local:1:I.Pt07.Data(C)			
Base Tag:	Local:1:I.Pt07.Data(C)			
Constant	No			
External Access:	Read/Write			
<i>NanoScanWarning - SafetyProgram  /MainRoutine - 5(XIC)</i>				
 NewLoadedButtonHMI	0	BOOL		Capstone_Test
Constant	No			
External Access:	Read/Write			
<i>NewLoadedButtonHMI - MainProgram/WindInductor - 4(XIC)</i>				

OldTime	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>OldTime - MainProgram/FillValues - #63, *#64</i>			
Output_RemoteIO:O		AB:1732_DO16:O:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Output_RemoteIO:O.Data.0	0	BOOL	
<i>Output_RemoteIO:O.Data.0 - MainProgram/FillValues - *#57</i>			
Output_RemoteIO:O.Data.2	0	BOOL	
<i>Output_RemoteIO:O.Data.2 - MainProgram/FillValues - *#58</i>			
Output_RemoteIO:O.Data.4	0	BOOL	
<i>Output_RemoteIO:O.Data.4 - MainProgram/FillValues - *#59</i>			
Output_RemoteIO:O.Data.6	0	BOOL	
<i>Output_RemoteIO:O.Data.6 - MainProgram/FillValues - *#60</i>			
PB_Reset	0	BOOL	Capstone_Test
AliasFor:	Local:1:I.Pt06.Data		
Base Tag:	Local:1:I.Pt06.Data		
Constant	No		
External Access:	Read/Write		
<i>PB_Reset - MainProgram/Alarms - 0(ALMD), 1(ALMD), 2(ALMD), 3(ALMD), 4(ALMD)</i>			
<i>PB_Reset - MainProgram/RobotAutoControl - 0(XIC)</i>			
<i>PB_Reset - SafetyProgram /MainRoutine - 10(XIC), 6(XIO), 8(XIC)</i>			
PB_Start	0	BOOL	Capstone_Test
StartPB			
AliasFor:	Input_RemoteIO:I.Data.0		
Base Tag:	Input_RemoteIO:I.Data.0		
Constant	No		
External Access:	Read/Write		
<i>PB_Start - MainProgram/MainRoutine - 2(XIC)</i>			
prog	0.0	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>prog - MainProgram/FillValues - #70, *#69</i>			
proggy	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>proggy - MainProgram/FillValues - #71, *#70</i>			
Progress	0	DINT	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>Progress - MainProgram/FillValues - *#71</i>			
Ready	0	BOOL	MainProgram
Bit for PC "I am ready for the next inductor"			
Constant	No		
External Access:	Read/Write		
<i>Ready - MainProgram/WindInductor - *0(OTE), 0(XIC), 1(XIC)</i>			
RequiredTemp	22.5	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>RequiredTemp - MainProgram/HeatExtruder - 1(LEQ), 2(GEQ)</i>			
ResetLight	0	BOOL	Capstone_Test
AliasFor:	Local:4:O.Pt14.Data		
Base Tag:	Local:4:O.Pt14.Data		
Constant	No		

ResetLight (Continued)			
External Access:	Read/Write		
<i>ResetLight - MainProgram/MainRoutine - *7(OTE)</i>			
 RobotAtCutting2	0	BOOL	Capstone_Test
In From Robot			
AliasFor:	Fanuc_Robot:I.Data[0].1		
Base Tag:	Fanuc_Robot:I.Data[0].1		
Constant	No		
External Access:	Read/Write		
<i>RobotAtCutting2 - MainProgram/CONTROL - A11, #1(Tran,Tran_021 (Condition))</i>			
 RobotAtHome1	0	BOOL	Capstone_Test
In From Robot			
AliasFor:	Fanuc_Robot:I.Data[0].0		
Base Tag:	Fanuc_Robot:I.Data[0].0		
Constant	No		
External Access:	Read/Write		
<i>RobotAtHome1 - MainProgram/CONTROL - A39, #1(Tran,Tran_037 (Condition))</i>			
RobotAutoAutoTimer		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>RobotAutoAutoTimer - MainProgram/RobotAutoControl - *0(TON)</i>			
RobotAutoAutoTimer.DN	1	BOOL	
<i>RobotAutoAutoTimer.DN - MainProgram/RobotAutoControl - 2(XIC)</i>			
RobotAutoStep	2	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
RobotAutoStep.0	0	BOOL	
Fault Reset			
<i>RobotAutoStep.0 - MainProgram/RobotAutoControl - *4(OTE), 4(XIC), 5(XIC), 8(XIC)</i>			
RobotAutoStep.1	1	BOOL	
Start Program			
<i>RobotAutoStep.1 - MainProgram/RobotAutoControl - *5(OTE), 4(XIO), 5(XIC), 6(XIC), 9(XIC)</i>			
RobotAutoStep.2	0	BOOL	
Robot Program Running			
<i>RobotAutoStep.2 - MainProgram/RobotAutoControl - *6(OTE), 5(XIO), 6(XIC), 7(XIC)</i>			
RobotAutoStep.3	0	BOOL	
Robot Program Done			
<i>RobotAutoStep.3 - MainProgram/RobotAutoControl - *7(OTE), 10(XIC), 6(XIO), 7(XIC)</i>			
 RobotDirection4	1	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].3		
Base Tag:	Fanuc_Robot:O.Data[0].3		
Constant	No		
External Access:	Read/Write		
<i>RobotDirection4 - MainProgram/CONTROL - *B1, #13(Action,Action_024 (Body)), *B12, #2(Action,Action_023 (Body)), *B23, #10(Action,Action_039 (Body)), *B23, #12(Action,Action_039 (Body))</i>			
<i>RobotDirection4 - MainProgram/pause - #23, *#45, *#69</i>			
<i>RobotDirection4 - MainProgram/StopAll - *#15</i>			
 RobotDoneGetThisShitOutOfHere4	1	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].4		
Base Tag:	Fanuc_Robot:O.Data[0].4		
Constant	No		
External Access:	Read/Write		
<i>RobotDoneGetThisShitOutOfHere4 - MainProgram/CONTROL - *B1, #12(Action,Action_024 (Body)), *B38, #1(Action,Action_041 (Body)), *B38, #1(Action,Action_042 (Body))</i>			

RobotEnable	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>RobotEnable - MainProgram/FillValues - *#3</i>			
<i>RobotEnable - MainProgram/pause - #29, *#51, *#75</i>			
<i>RobotEnable - MainProgram/RobotAutoControl - *0(OTL), *1(OTU), 0(XIC), 3(XIC)</i>			
RobotLoaded2	1	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].1		
Base Tag:	Fanuc_Robot:O.Data[0].1		
Constant	No		
External Access:	Read/Write		
<i>RobotLoaded2 - MainProgram/CONTROL - *B1, #10(Action,Action_024 (Body)), *B12, #4(Action,Action_023 (Body)), *B18, #2(Action,Action_051 (Body))</i>			
<i>RobotLoaded2 - MainProgram/pause - #21, *#43, *#67</i>			
<i>RobotLoaded2 - MainProgram/StopAll - *#13</i>			
RobotMoveBack6	0	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].5		
Base Tag:	Fanuc_Robot:O.Data[0].5		
Constant	No		
External Access:	Read/Write		
<i>RobotMoveBack6 - MainProgram/CONTROL - *B20, #2(Action,Action_053 (Body)), *B21, #1(Action,Action_025 (Body))</i>			
RobotMoveNextLayer3	0	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].2		
Base Tag:	Fanuc_Robot:O.Data[0].2		
Constant	No		
External Access:	Read/Write		
<i>RobotMoveNextLayer3 - MainProgram/CONTROL - *B1, #11(Action,Action_024 (Body)), *B12, #3(Action,Action_023 (Body)), *B23, #18(Action,Action_039 (Body)), *B27, #1(Action,Action_040 (Body))</i>			
<i>RobotMoveNextLayer3 - MainProgram/pause - #22, *#44, *#68</i>			
<i>RobotMoveNextLayer3 - MainProgram/StopAll - *#14</i>			
RobotRelayReset	0	BOOL	Capstone_Test
Reset Robot Safety Relay			
AliasFor:	Local:4:O.Pt06.Data		
Base Tag:	Local:4:O.Pt06.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt06.Data - MainProgram/FillValues - *#47</i>			
RobotStartAuto	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>RobotStartAuto - MainProgram/RobotAutoControl - *2(OTE), 4(XIC)</i>			
RobotToCutting	0	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].0		
Base Tag:	Fanuc_Robot:O.Data[0].0		
Constant	No		
External Access:	Read/Write		
<i>RobotToCutting - MainProgram/CONTROL - *B1, #9(Action,Action_024 (Body)), *B11, #2(Action,Action_021 (Body)), *B11, #2(Action,Action_022 (Body))</i>			
<i>RobotToCutting - MainProgram/pause - #20, *#42, *#66</i>			
<i>RobotToCutting - MainProgram/StopAll - *#12</i>			
RobotWireInserted3	0	BOOL	Capstone_Test
also means wire in			
AliasFor:	Fanuc_Robot:I.Data[0].2		

RobotWireInserted3 (Continued)			
Base Tag:	Fanuc_Robot:I.Data[0].2		
Constant	No		
External Access:	Read/Write		
RobotWireInserted3 - MainProgram/CONTROL - A19, #1(Tran,Tran_051 (Condition))			
 SafetyBit	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
SafetyBit - MainProgram/Alarms - 0(XIC), 1(XIC), 2(XIC), 3(XIC), 4(XIC)			
SafetyBit - MainProgram/MainRoutine - 10(XIC), 4(XIO), 5(XIO), 6(XIO), 7(XIC)			
SafetyBit - MainProgram/RobotAutoControl - 1(XIC), 4(XIO), 5(XIO), 6(XIO)			
SafetyBit - SafetyProgram  /MainRoutine - *7(OTE), 7(XIC), 9(XIO)			
- SafetyBypass	0	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
SafetyBypass - SafetyProgram  /MainRoutine - *10(OTL), *11(OTU), 3(XIO), 4(XIO)			
- SafetyOverride	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
SafetyOverride - SafetyProgram  /MainRoutine - 6(XIO), 6(XIO), 6(XIO), 7(XIO), 7(XIO), 7(XIO)			
- SafetyResetPBBit	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
SafetyResetPBBit - SafetyProgram  /MainRoutine - *6(OTE), 7(XIC)			
 selfdestruct	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
selfdestruct - MainProgram/MainRoutine - 0(XIC)			
selfdestruct - MainProgram/selfdestruct - 0(XIC), 2(XIO), 3(XIC)			
SettingDia	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
SettingDia - MainProgram/CONTROL - A14, #1(Tran,Tran_049 (Condition))			
SettingDia - MainProgram/SetWindingDia - *B1, #2(Action,Action_045 (Body)), *B2, #1(Action,Action_031 (Body)), *B7, #2(Action,Action_048 (Body))			
SolenoidAlarm		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
SolenoidAlarm - MainProgram/MainRoutine - *8(TON)			
SolenoidAlarm.DN	0	BOOL	
SolenoidAlarm.DN - MainProgram/MainRoutine - 9(XIC)			
speed	750	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
speed - MainProgram/MainRoutine - 1(JSR), 4(JSR)			
 Spool0F	0	BOOL	Capstone_Test
Top Feed Motor Fwd			
AliasFor:	Local:4:O.Pt00.Data		
Base Tag:	Local:4:O.Pt00.Data		
Constant	No		
External Access:	Read/Write		
Local:4:O.Pt00.Data - MainProgram/FillValues - *#40			
 Spool0R	0	BOOL	Capstone_Test
Top Feed Motor Rev			

Spool0R (Continued)			
AliasFor:	Local:4:O.Pt01.Data		
Base Tag:	Local:4:O.Pt01.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt01.Data - MainProgram/FillValues - *#41</i>			
 Spool1F	0	BOOL	Capstone_Test
Mid Feed Motor Fwd			
AliasFor:	Local:4:O.Pt02.Data		
Base Tag:	Local:4:O.Pt02.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt02.Data - MainProgram/FillValues - *#42</i>			
 Spool1R	0	BOOL	Capstone_Test
Mid Feed Motor Fwd			
AliasFor:	Local:4:O.Pt02.Data		
Base Tag:	Local:4:O.Pt02.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt02.Data - MainProgram/FillValues - *#42</i>			
 Spool2F	0	BOOL	Capstone_Test
Bottom Feed Motor Fwd			
AliasFor:	Local:4:O.Pt04.Data		
Base Tag:	Local:4:O.Pt04.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt04.Data - MainProgram/FillValues - *#44</i>			
 Spool2R	0	BOOL	Capstone_Test
Bottom Feed Motor Rev			
AliasFor:	Local:4:O.Pt05.Data		
Base Tag:	Local:4:O.Pt05.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt05.Data - MainProgram/FillValues - *#45</i>			
 SpoolArray		Spool[3]	Capstone_Test
Constant	No		
External Access:	Read/Write		
SpoolArray[0].REV	0	BOOL	
out			
<i>SpoolArray[0].REV - MainProgram/CONTROL - *B1, #3(Action,Action_024 (Body))</i>			
<i>SpoolArray[0].REV - MainProgram/FillValues - #41</i>			
<i>SpoolArray[0].REV - MainProgram/pause - #14, *#36, *#60</i>			
<i>SpoolArray[0].REV - MainProgram/StopAll - *#6</i>			
SpoolArray[0].FWD	0	BOOL	
out			
<i>SpoolArray[0].FWD - MainProgram/CONTROL - *B1, #2(Action,Action_024 (Body))</i>			
<i>SpoolArray[0].FWD - MainProgram/FillValues - #40</i>			
<i>SpoolArray[0].FWD - MainProgram/pause - #13, *#35, *#59</i>			
<i>SpoolArray[0].FWD - MainProgram/StopAll - *#5</i>			
SpoolArray[0].Mass	164.73665	REAL	
current mass reading			
<i>SpoolArray[0].Mass - MainProgram/FillValues - *#20, *#22</i>			
SpoolArray[0].Length	0.0	REAL	
approximated legth based on Mass			
SpoolArray[0].Loaded	1	BOOL	
if is in loaded state			
<i>SpoolArray[0].Loaded - MainProgram/FillValues - *#10</i>			
SpoolArray[0].Gauge	-1	DINT	
Wire Gauge			

SpoolArray (Continued)			
SpoolArray[1].REV	0	BOOL	
out			
<i>SpoolArray[1].REV - MainProgram/CONTROL - *B1, #5(Action,Action_024 (Body))</i>			
<i>SpoolArray[1].REV - MainProgram/FillValues - #43</i>			
<i>SpoolArray[1].REV - MainProgram/pause - #16, *#38, *#62</i>			
<i>SpoolArray[1].REV - MainProgram/StopAll - *#8</i>			
SpoolArray[1].FWD	0	BOOL	
out			
<i>SpoolArray[1].FWD - MainProgram/CONTROL - *B1, #4(Action,Action_024 (Body))</i>			
<i>SpoolArray[1].FWD - MainProgram/FillValues - #42</i>			
<i>SpoolArray[1].FWD - MainProgram/pause - #15, *#37, *#61</i>			
<i>SpoolArray[1].FWD - MainProgram/StopAll - *#7</i>			
SpoolArray[1].Mass	163.34396	REAL	
current mass reading			
SpoolArray[1].Length	0.0	REAL	
approximated legth based on Mass			
SpoolArray[1].Loaded	1	BOOL	
if is in loaded state			
<i>SpoolArray[1].Loaded - MainProgram/FillValues - *#11</i>			
SpoolArray[1].Gauge	-1	DINT	
Wire Gauge			
SpoolArray[2].REV	0	BOOL	
out			
<i>SpoolArray[2].REV - MainProgram/CONTROL - *B1, #7(Action,Action_024 (Body))</i>			
<i>SpoolArray[2].REV - MainProgram/FillValues - #45</i>			
<i>SpoolArray[2].REV - MainProgram/pause - #18, *#40, *#64</i>			
<i>SpoolArray[2].REV - MainProgram/StopAll - *#10</i>			
SpoolArray[2].FWD	0	BOOL	
out			
<i>SpoolArray[2].FWD - MainProgram/CONTROL - *B1, #6(Action,Action_024 (Body)), *B11, #1(Action,Action_021 (Body)), *B11, #1(Action,Action_022 (Body)), *B12, #1(Action,Action_023 (Body)), *B23, #3(Action,Action_039 (Body)), *B37, #1(Action,Action_061 (Body)), *C9, #1(Action,Action_019 (Body))</i>			
<i>SpoolArray[2].FWD - MainProgram/FillValues - #44</i>			
<i>SpoolArray[2].FWD - MainProgram/pause - #17, *#39, *#63</i>			
<i>SpoolArray[2].FWD - MainProgram/StopAll - *#9</i>			
SpoolArray[2].Mass	600.10535	REAL	
current mass reading			
<i>SpoolArray[2].Mass - MainProgram/FillValues - *#32, *#34</i>			
SpoolArray[2].Length	0.0	REAL	
approximated legth based on Mass			
SpoolArray[2].Loaded	0	BOOL	
if is in loaded state			
<i>SpoolArray[2].Loaded - MainProgram/FillValues - *#12</i>			
SpoolArray[2].Gauge	16	DINT	
Wire Gauge			
<i>SpoolArray[2].Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)</i>			
StackLightGreen	1	BOOL	Capstone_Test
AliasFor:	Local:4:O.Pt07.Data		
Base Tag:	Local:4:O.Pt07.Data		
Constant	No		
External Access:	Read/Write		
<i>StackLightGreen - MainProgram/MainRoutine - *5(OTE)</i>			
<i>StackLightGreen - MainProgram/pause - #10, *#32, *#56</i>			
<i>StackLightGreen - MainProgram/StopAll - *#2</i>			
StackLightRed	0	BOOL	Capstone_Test
AliasFor:	Local:4:O.Pt09.Data		
Base Tag:	Local:4:O.Pt09.Data		
Constant	No		
External Access:	Read/Write		
<i>StackLightRed - MainProgram/MainRoutine - *7(OTE)</i>			
<i>StackLightRed - MainProgram/pause - #9, *#31, *#55</i>			

StackLightRed (Continued)			
<i>StackLightRed - MainProgram/StopAll - *#1</i>			
StackLightYellow	1	BOOL	Capstone_Test
AliasFor:	Local:4:O.Pt08.Data		
Base Tag:	Local:4:O.Pt08.Data		
Constant	No		
External Access:	Read/Write		
<i>StackLightYellow - MainProgram/MainRoutine - *6(OTE)</i>			
<i>StackLightYellow - MainProgram/pause - #11, *#33, *#57</i>			
<i>StackLightYellow - MainProgram/StopAll - *#3</i>			
StartHeating	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>StartHeating - MainProgram/HeatExtruder - 0(XIC)</i>			
Step_021		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_021 - MainProgram/CONTROL - *A1(Step,Step_021)</i>			
<i>Step_021 - MainProgram/MainRoutine - 0(SFR)</i>			
Step_021.T	201	DINT	
<i>Step_021.T - MainProgram/CONTROL - A4, #1(Tran,Tran_073 (Condition))</i>			
Step_022		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
Step_022.T	0	DINT	
<i>Step_022.T - MainProgram/WindInductor - 4(LES)</i>			
Step_023		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_023 - MainProgram/CONTROL - *A11(Step,Step_023)</i>			
Step_024		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_024 - MainProgram/CONTROL - *A11(Step,Step_024)</i>			
Step_025		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_025 - MainProgram/CONTROL - *A12(Step,Step_025)</i>			
Step_028		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_028 - MainProgram/CONTROL - *A21(Step,Step_028)</i>			
Step_028.T	1501	DINT	
<i>Step_028.T - MainProgram/CONTROL - A21, #1(Tran,Tran_042 (Condition))</i>			
Step_029		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_029 - MainProgram/CONTROL - *A28(Step,Step_029)</i>			
Step_029.T	4017	DINT	
<i>Step_029.T - MainProgram/CONTROL - A28, #1(Tran,Tran_038 (Condition))</i>			
Step_030		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_030 - MainProgram/ThugShake - *B1(Step,Step_030)</i>			

Step_031 Constant No External Access: Read/Write <i>Step_031 - MainProgram/ThugShake - *B2(Step,Step_031)</i>	SFC_STEP	MainProgram
Step_031.T <i>Step_031.T - MainProgram/ThugShake - B2, #1(Tran,Tran_027 (Condition))</i>	140 DINT	
Step_032 Constant No External Access: Read/Write <i>Step_032 - MainProgram/ThugShake - *B3(Step,Step_032)</i>	SFC_STEP	MainProgram
Step_032.T <i>Step_032.T - MainProgram/ThugShake - C4, #1(Tran,Tran_029 (Condition))</i>	0 DINT	
Step_033 Constant No External Access: Read/Write <i>Step_033 - MainProgram/ThugShake - *B4(Step,Step_033)</i>	SFC_STEP	MainProgram
Step_034 Constant No External Access: Read/Write <i>Step_034 - MainProgram/SetWindingDia - *B1(Step,Step_034)</i>	SFC_STEP	MainProgram
Step_037 Constant No External Access: Read/Write <i>Step_037 - MainProgram/CONTROL - *A23(Step,Step_037)</i>	SFC_STEP	MainProgram
Step_037.T <i>Step_037.T - MainProgram/CONTROL - A28, #1(Tran,Tran_025 (Condition))</i>	11 DINT	
Step_038 Constant No External Access: Read/Write <i>Step_038 - MainProgram/CONTROL - *A38(Step,Step_038)</i>	SFC_STEP	MainProgram
Step_040 Constant No External Access: Read/Write <i>Step_040 - MainProgram/CONTROL - *C30(Step,Step_040)</i>	SFC_STEP	MainProgram
Step_040.T <i>Step_040.T - MainProgram/CONTROL - C31, #1(Tran,Tran_066 (Condition))</i>	901 DINT	
Step_041 Constant No External Access: Read/Write <i>Step_041 - MainProgram/CONTROL - *A14(Step,Step_041)</i>	SFC_STEP	MainProgram
Step_041.T <i>Step_041.T - MainProgram/CONTROL - A14, #1(Tran,Tran_045 (Condition))</i>	101 DINT	
Step_042 Constant No External Access: Read/Write <i>Step_042 - MainProgram/SetWindingDia - *B2(Step,Step_042)</i>	SFC_STEP	MainProgram
Step_042.T <i>Step_042.T - MainProgram/SetWindingDia - B2, #1(Tran,Tran_035 (Condition))</i>	501 DINT	
Step_043 Constant No External Access: Read/Write <i>Step_043 - MainProgram/SetWindingDia - *B3(Step,Step_043)</i>	SFC_STEP	MainProgram
Step_043.T <i>Step_043.T - MainProgram/SetWindingDia - B4, #1(Tran,Tran_041 (Condition))</i>	4120 DINT	

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Step_045		SFC_STEP	MainProgram
Constant		No	
External Access:		Read/Write	
Step_045 - MainProgram/CONTROL - *A15(Step,Step_045)			
Step_046		SFC_STEP	MainProgram
Constant		No	
External Access:		Read/Write	
Step_046 - MainProgram/SetWindingDia - *B3(Step,Step_046)			
Step_047		SFC_STEP	MainProgram
Constant		No	
External Access:		Read/Write	
Step_047 - MainProgram/SetWindingDia - *B4(Step,Step_047)			
Step_047.T		101	DINT
Step_047.T - MainProgram/SetWindingDia - B5, #1(Tran,Tran_046 (Condition))			
Step_048		SFC_STEP	MainProgram
Constant		No	
External Access:		Read/Write	
Step_048 - MainProgram/SetWindingDia - *B6(Step,Step_048)			
Step_048.T		629	DINT
Step_048.T - MainProgram/SetWindingDia - B6, #1(Tran,Tran_047 (Condition))			
Step_049		SFC_STEP	MainProgram
Constant		No	
External Access:		Read/Write	
Step_049 - MainProgram/SetWindingDia - *B7(Step,Step_049)			
Step_050		SFC_STEP	MainProgram
Constant		No	
External Access:		Read/Write	
Step_050 - MainProgram/CONTROL - *A13(Step,Step_050)			
Step_050.T		101	DINT
Step_050.T - MainProgram/CONTROL - A14, #1(Tran,Tran_049 (Condition))			
Step_052		SFC_STEP	MainProgram
Constant		No	
External Access:		Read/Write	
Step_052 - MainProgram/CONTROL - *A18(Step,Step_052)			
Step_053		SFC_STEP	MainProgram
Constant		No	
External Access:		Read/Write	
Step_053 - MainProgram/CONTROL - *A19(Step,Step_053)			
Step_053.T		1001	DINT
Step_053.T - MainProgram/CONTROL - A19, #1(Tran,Tran_052 (Condition))			
Step_054		SFC_STEP	MainProgram
Constant		No	
External Access:		Read/Write	
Step_054 - MainProgram/CONTROL - *A20(Step,Step_054)			
Step_054.T		180000	DINT
Step_054.T - MainProgram/CONTROL - A20, #1(Tran,Tran_053 (Condition))			
Step_056		SFC_STEP	MainProgram
Constant		No	
External Access:		Read/Write	
Step_056 - MainProgram/CONTROL - *A37(Step,Step_056)			
Step_056.T		1500	DINT
Step_056.T - MainProgram/CONTROL - A38, #1(Tran,Tran_055 (Condition))			
Step_057		SFC_STEP	MainProgram
Constant		No	

Step_057 (Continued)			
External Access:	Read/Write		
<i>Step_057 - MainProgram/SetWindingDia - *B5(Step,Step_057)</i>			
Step_059		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_059 - MainProgram/CONTROL - *A16(Step,Step_059)</i>			
Step_059.T	2001	DINT	
<i>Step_059.T - MainProgram/CONTROL - A16, #1(Tran,Tran_058 (Condition))</i>			
Step_060		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_060 - MainProgram/CONTROL - *A17(Step,Step_060)</i>			
Step_060.T	101	DINT	
<i>Step_060.T - MainProgram/CONTROL - A17, #1(Tran,Tran_059 (Condition))</i>			
Step_061		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_061 - MainProgram/CONTROL - *A17(Step,Step_061)</i>			
Step_061.T	901	DINT	
<i>Step_061.T - MainProgram/CONTROL - A18, #1(Tran,Tran_060 (Condition))</i>			
Step_062		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_062 - MainProgram/CONTROL - *A35(Step,Step_062)</i>			
Step_062.T	101	DINT	
<i>Step_062.T - MainProgram/CONTROL - A35, #1(Tran,Tran_062 (Condition))</i>			
Step_063		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_063 - MainProgram/CONTROL - *A8(Step,Step_063)</i>			
Step_064		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_064 - MainProgram/CONTROL - *C9(Step,Step_064)</i>			
Step_065		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_065 - MainProgram/CONTROL - *A30(Step,Step_065)</i>			
Step_066		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_066 - MainProgram/CONTROL - *C31(Step,Step_066)</i>			
Step_066.T	101	DINT	
<i>Step_066.T - MainProgram/CONTROL - C31, #1(Tran,Tran_067 (Condition))</i>			
Step_067		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_067 - MainProgram/CONTROL - *A29(Step,Step_067)</i>			
Step_067.T	1	DINT	
<i>Step_067.T - MainProgram/CONTROL - A30, #1(Tran,Tran_065 (Condition))</i>			
Step_068		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		

Step_068 (Continued)			
<i>Step_068 - MainProgram/CONTROL - *A33(Step,Step_068)</i>			
Step_068.T	0	DINT	
<i>Step_068.T - MainProgram/CONTROL - F34, #1(Tran,Tran_040 (Condition))</i>			
Step_069		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_069 - MainProgram/CONTROL - *A6(Step,Step_069)</i>			
Step_069.T	101	DINT	
<i>Step_069.T - MainProgram/CONTROL - A6, #1(Tran,Tran_072 (Condition))</i>			
Step_070		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_070 - MainProgram/CONTROL - *A36(Step,Step_070)</i>			
Step_071		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_071 - MainProgram/CONTROL - *A39(Step,Step_071)</i>			
Step_072		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_072 - MainProgram/CONTROL - *A6(Step,Step_072)</i>			
Step_072.T	251	DINT	
<i>Step_072.T - MainProgram/CONTROL - A7, #1(Tran,Tran_071 (Condition))</i>			
Step_073		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_073 - MainProgram/CONTROL - *A7(Step,Step_073)</i>			
Step_074		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_074 - MainProgram/CONTROL - *A4(Step,Step_074)</i>			
Step_074.T	1201	DINT	
<i>Step_074.T - MainProgram/CONTROL - A5, #1(Tran,Tran_068 (Condition))</i>			
Step_075		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_075 - MainProgram/CONTROL - *A35(Step,Step_075)</i>			
Step_075.T	701	DINT	
<i>Step_075.T - MainProgram/CONTROL - A36, #1(Tran,Tran_074 (Condition))</i>			
Step_076		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_076 - MainProgram/CONTROL - *C32(Step,Step_076)</i>			
Step_076.T	501	DINT	
<i>Step_076.T - MainProgram/CONTROL - C32, #1(Tran,Tran_075 (Condition))</i>			
Step_077		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_077 - MainProgram/CONTROL - *C32(Step,Step_077)</i>			
Step_077.T	101	DINT	
<i>Step_077.T - MainProgram/CONTROL - C33, #1(Tran,Tran_076 (Condition))</i>			
STEP_CoilKnown	0	BOOL	MainProgram
Constant	No		

STEP_CoilKnown (Continued)			
External Access:	Read/Write		
<i>STEP_CoilKnown - MainProgram/WindInductor - *1(OTE), 0(XIO), 1(XIC), 2(XIC), 3(XIC)</i>			
Step_GettingSpoolCurrent	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_GettingSpoolCurrent - MainProgram/WindInductor - *3(OTE), 0(XIO), 1(XIO), 3(XIC), 4(XIC)</i>			
STEP_OperatorSwapsSpool	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>STEP_OperatorSwapsSpool - MainProgram/WindInductor - *2(OTE), 0(XIO), 1(XIO), 2(XIC), 4(XIC)</i>			
STEP_WireLoaded	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>STEP_WireLoaded - MainProgram/WindInductor - *4(OTE), 2(XIO), 3(XIO), 4(XIC)</i>			
StopHeating	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>StopHeating - MainProgram/CONTROL - *B20, #4(Action,Action_053 (Body)), *B35, #3(Action,Action_070 (Body))</i>			
<i>StopHeating - MainProgram/HeatExtruder - 0(XIO), 2(XIO), 3(XIO)</i>			
SystemState	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>SystemState - MainProgram/MainRoutine - *1(MOV)</i>			
SystemState.0	0	BOOL	
<i>SystemState.0 - MainProgram/pause - #55, *#9</i>			
SystemState.1	0	BOOL	
<i>SystemState.1 - MainProgram/pause - #56, *#10</i>			
SystemState.2	0	BOOL	
<i>SystemState.2 - MainProgram/pause - #57, *#11</i>			
SystemState.3	0	BOOL	
<i>SystemState.3 - MainProgram/pause - #58, *#12</i>			
SystemState.4	0	BOOL	
<i>SystemState.4 - MainProgram/pause - #59, *#13</i>			
SystemState.5	0	BOOL	
<i>SystemState.5 - MainProgram/pause - #60, *#14</i>			
SystemState.6	0	BOOL	
<i>SystemState.6 - MainProgram/pause - #61, *#15</i>			
SystemState.7	0	BOOL	
<i>SystemState.7 - MainProgram/pause - #62, *#16</i>			
SystemState.8	0	BOOL	
<i>SystemState.8 - MainProgram/pause - #63, *#17</i>			
SystemState.9	0	BOOL	
<i>SystemState.9 - MainProgram/pause - #64, *#18</i>			
SystemState.10	0	BOOL	
<i>SystemState.10 - MainProgram/pause - #65, *#19</i>			
SystemState.11	0	BOOL	
<i>SystemState.11 - MainProgram/pause - #66, *#20</i>			
SystemState.12	0	BOOL	
<i>SystemState.12 - MainProgram/pause - #67, *#21</i>			
SystemState.13	0	BOOL	
<i>SystemState.13 - MainProgram/pause - #68, *#22</i>			
SystemState.14	0	BOOL	
<i>SystemState.14 - MainProgram/pause - #69, *#23</i>			
SystemState.15	0	BOOL	
<i>SystemState.15 - MainProgram/pause - #70, *#24</i>			
SystemState.16	0	BOOL	
<i>SystemState.16 - MainProgram/pause - #71, *#25</i>			
SystemState.17	0	BOOL	

SystemState (Continued)			
<i>SystemState.17 - MainProgram/pause - #72, *#26</i>			
SystemState.18	0	BOOL	
<i>SystemState.18 - MainProgram/pause - #73, *#27</i>			
SystemState.19	0	BOOL	
<i>SystemState.19 - MainProgram/pause - #74, *#28</i>			
SystemState.20	0	BOOL	
<i>SystemState.20 - MainProgram/pause - #75, *#29</i>			
- tBypass			
Constant	No	TIMER	SafetyProgram
External Access: Read/Write			
<i>tBypass - SafetyProgram /MainRoutine - *10(TON)</i>			
tBypass.DN	0	BOOL	
<i>tBypass.DN - SafetyProgram /MainRoutine - 10(XIC)</i>			
TempDint			
Constant	100	DINT	MainProgram
Constant No			
External Access: Read/Write			
<i>TempDint - MainProgram/CONTROL - *B1, #22(Action,Action_024 (Body)), *B37, #3(Action,Action_061 (Body)), B1, #23(Action,Action_024 (Body)), B37, #4(Action,Action_061 (Body))</i>			
<i>TempDint - MainProgram/SetWindingDia - *B6, #1(Action,Action_047 (Body)), B6, #2(Action,Action_047 (Body))</i>			
Temperature			
Constant	-2.61334376e-003	REAL	Capstone_Test
Constant No			
External Access: Read/Write			
<i>Temperature - MainProgram/FillValues - *#16</i>			
<i>Temperature - MainProgram/HeatExtruder - 1(GEQ), 1(LEQ), 2(GEQ), 2(GEQ), 2(LES), 3(LES), 4(LES), 5(LES)</i>			
- test			
Constant	0	BOOL	SafetyProgram
Constant No			
External Access: Read/Write			
<i>test - SafetyProgram /MainRoutine - 10(XIC), 11(XIC)</i>			
ThugShakeIndex			
Constant	0	DINT	MainProgram
Constant No			
External Access: Read/Write			
<i>ThugShakeIndex - MainProgram/ThugShake - *B1, #1(Action,Action_028 (Body)), *B2, #1(Action,Action_027 (Body)), B2, #1(Action,Action_027 (Body)), B4, #1(Tran,Tran_028 (Condition)), C4, #1(Tran,Tran_029 (Condition))</i>			
timeout			
Constant	No	TIMER	MainProgram
Constant No			
External Access: Read/Write			
<i>timeout - MainProgram/selfdestruct - *0(TON)</i>			
timeout.DN	0	BOOL	
<i>timeout.DN - MainProgram/selfdestruct - 0(XIC), 1(XIC)</i>			
Tran_019			
Constant	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
<i>Tran_019 - MainProgram/CONTROL - *A8(Tran,Tran_019), *A8, #1(Tran,Tran_019 (Condition))</i>			
Tran_020			
Constant	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
<i>Tran_020 - MainProgram/CONTROL - *A9(Tran,Tran_020), *A9, #1(Tran,Tran_020 (Condition))</i>			
Tran_021			
Constant	1	BOOL	MainProgram
Constant No			
External Access: Read/Write			
<i>Tran_021 - MainProgram/CONTROL - *A11(Tran,Tran_021), *A11, #1(Tran,Tran_021 (Condition))</i>			
Tran_022			
Constant	1	BOOL	MainProgram
Constant No			

Tran_022 (Continued)			
External Access:	Read/Write		
<i>Tran_022 - MainProgram/CONTROL - *A12(Tran,Tran_022), *A12, #1(Tran,Tran_022 (Condition))</i>			
Tran_025	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_025 - MainProgram/CONTROL - *A28(Tran,Tran_025), *A28, #1(Tran,Tran_025 (Condition))</i>			
Tran_026	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_026 - MainProgram/ThugShake - *B1(Tran,Tran_026), *B1, #1(Tran,Tran_026 (Condition))</i>			
Tran_027	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_027 - MainProgram/ThugShake - *B2(Tran,Tran_027), *B2, #1(Tran,Tran_027 (Condition))</i>			
Tran_028	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_028 - MainProgram/ThugShake - *B4(Tran,Tran_028), *B4, #1(Tran,Tran_028 (Condition))</i>			
Tran_029	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_029 - MainProgram/ThugShake - *C4(Tran,Tran_029), *C4, #1(Tran,Tran_029 (Condition))</i>			
Tran_030	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_030 - MainProgram/ThugShake - *B4(Tran,Tran_030), *B4, #1(Tran,Tran_030 (Condition))</i>			
Tran_031	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_031 - MainProgram/SetWindingDia - *B1(Tran,Tran_031), *B1, #1(Tran,Tran_031 (Condition))</i>			
Tran_034	1	BOOL	MainProgram
Ready to wind			
Constant	No		
External Access:	Read/Write		
<i>Tran_034 - MainProgram/CONTROL - *A13(Tran,Tran_034), *A13, #1(Tran,Tran_034 (Condition))</i>			
Tran_035	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_035 - MainProgram/SetWindingDia - *B2(Tran,Tran_035), *B2, #1(Tran,Tran_035 (Condition))</i>			
Tran_036	1	BOOL	MainProgram
Done Winding			
Constant	No		
External Access:	Read/Write		
<i>Tran_036 - MainProgram/CONTROL - *A34(Tran,Tran_036), *A34, #1(Tran,Tran_036 (Condition))</i>			
Tran_037	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_037 - MainProgram/CONTROL - *A39(Tran,Tran_037), *A39, #1(Tran,Tran_037 (Condition))</i>			
Tran_038	1	BOOL	MainProgram
robot takes ~ 500ms and wait until winding is done moving			
Constant	No		

Tran_038 (Continued)			
External Access:	Read/Write		
<i>Tran_038 - MainProgram/CONTROL - *A28(Tran,Tran_038), *A28, #1(Tran,Tran_038 (Condition))</i>			
Tran_039	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_039 - MainProgram/CONTROL - *A15(Tran,Tran_039), *A15, #1(Tran,Tran_039 (Condition))</i>			
Tran_040	1	BOOL	MainProgram
Still winding, allow time for done winding to be true before going here			
Constant	No		
External Access:	Read/Write		
<i>Tran_040 - MainProgram/CONTROL - *F34(Tran,Tran_040), *F34, #1(Tran,Tran_040 (Condition))</i>			
Tran_041	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_041 - MainProgram/SetWindingDia - *B4(Tran,Tran_041), *B4, #1(Tran,Tran_041 (Condition))</i>			
Tran_042	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_042 - MainProgram/CONTROL - *A21(Tran,Tran_042), *A21, #1(Tran,Tran_042 (Condition))</i>			
Tran_044	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_044 - MainProgram/SetWindingDia - *B3(Tran,Tran_044), *B3, #1(Tran,Tran_044 (Condition))</i>			
Tran_045	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_045 - MainProgram/CONTROL - *A14(Tran,Tran_045), *A14, #1(Tran,Tran_045 (Condition))</i>			
Tran_046	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_046 - MainProgram/SetWindingDia - *B5(Tran,Tran_046), *B5, #1(Tran,Tran_046 (Condition))</i>			
Tran_047	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_047 - MainProgram/SetWindingDia - *B6(Tran,Tran_047), *B6, #1(Tran,Tran_047 (Condition))</i>			
Tran_048	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_048 - MainProgram/SetWindingDia - *B7(Tran,Tran_048), *B7, #1(Tran,Tran_048 (Condition))</i>			
Tran_049	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_049 - MainProgram/CONTROL - *A14(Tran,Tran_049), *A14, #1(Tran,Tran_049 (Condition))</i>			
Tran_051	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_051 - MainProgram/CONTROL - *A19(Tran,Tran_051), *A19, #1(Tran,Tran_051 (Condition))</i>			
Tran_052	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_052 - MainProgram/CONTROL - *A19(Tran,Tran_052), *A19, #1(Tran,Tran_052 (Condition))</i>			

Tran_053	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_053 - MainProgram/CONTROL - *A20(Tran,Tran_053), *A20, #1(Tran,Tran_053 (Condition))</i>			
Tran_054	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_054 - MainProgram/CONTROL - *A30(Tran,Tran_054), *A30, #1(Tran,Tran_054 (Condition))</i>			
Tran_055	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_055 - MainProgram/CONTROL - *A38(Tran,Tran_055), *A38, #1(Tran,Tran_055 (Condition))</i>			
Tran_057	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_057 - MainProgram/SetWindingDia - *B5(Tran,Tran_057), *B5, #1(Tran,Tran_057 (Condition))</i>			
Tran_058	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_058 - MainProgram/CONTROL - *A16(Tran,Tran_058), *A16, #1(Tran,Tran_058 (Condition))</i>			
Tran_059	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_059 - MainProgram/CONTROL - *A17(Tran,Tran_059), *A17, #1(Tran,Tran_059 (Condition))</i>			
Tran_060	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_060 - MainProgram/CONTROL - *A18(Tran,Tran_060), *A18, #1(Tran,Tran_060 (Condition))</i>			
Tran_061	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_061 - MainProgram/CONTROL - *C9(Tran,Tran_061), *C9, #1(Tran,Tran_061 (Condition))</i>			
Tran_062	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_062 - MainProgram/CONTROL - *A35(Tran,Tran_062), *A35, #1(Tran,Tran_062 (Condition))</i>			
Tran_063	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_063 - MainProgram/CONTROL - *C10(Tran,Tran_063), *C10, #1(Tran,Tran_063 (Condition))</i>			
Tran_064	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_064 - MainProgram/CONTROL - *C30(Tran,Tran_064), *C30, #1(Tran,Tran_064 (Condition))</i>			
Tran_065	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_065 - MainProgram/CONTROL - *A30(Tran,Tran_065), *A30, #1(Tran,Tran_065 (Condition))</i>			
Tran_066	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_066 - MainProgram/CONTROL - *C31(Tran,Tran_066), *C31, #1(Tran,Tran_066 (Condition))</i>			

Tran_067	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_067 - MainProgram/CONTROL - *C31(Tran,Tran_067), *C31, #1(Tran,Tran_067 (Condition))</i>			
Tran_068	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_068 - MainProgram/CONTROL - *A5(Tran,Tran_068), *A5, #1(Tran,Tran_068 (Condition))</i>			
Tran_069	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_069 - MainProgram/CONTROL - *A37(Tran,Tran_069), *A37, #1(Tran,Tran_069 (Condition))</i>			
Tran_070	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_070 - MainProgram/CONTROL - *A39(Tran,Tran_070), *A39, #1(Tran,Tran_070 (Condition))</i>			
Tran_071	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_071 - MainProgram/CONTROL - *A7(Tran,Tran_071), *A7, #1(Tran,Tran_071 (Condition))</i>			
Tran_072	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_072 - MainProgram/CONTROL - *A6(Tran,Tran_072), *A6, #1(Tran,Tran_072 (Condition))</i>			
Tran_073	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_073 - MainProgram/CONTROL - *A4(Tran,Tran_073), *A4, #1(Tran,Tran_073 (Condition))</i>			
Tran_074	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_074 - MainProgram/CONTROL - *A36(Tran,Tran_074), *A36, #1(Tran,Tran_074 (Condition))</i>			
Tran_075	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_075 - MainProgram/CONTROL - *C32(Tran,Tran_075), *C32, #1(Tran,Tran_075 (Condition))</i>			
Tran_076	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_076 - MainProgram/CONTROL - *C33(Tran,Tran_076), *C33, #1(Tran,Tran_076 (Condition))</i>			
tRobotFaultReset		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>tRobotFaultReset - MainProgram/RobotAutoControl - *8(TON)</i>			
tRobotFaultReset.TT	0	BOOL	
<i>tRobotFaultReset.TT - MainProgram/RobotAutoControl - 11(XIC)</i>			
tRobotProgDone		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>tRobotProgDone - MainProgram/RobotAutoControl - *10(TON)</i>			
tRobotProgDone.DN	0	BOOL	
<i>tRobotProgDone.DN - MainProgram/RobotAutoControl - 3(XIO)</i>			

Capstone_Test - Safety Project Tag Listing		Page 36	
Capstone_Test (Controller)		8/12/2025 1:37:18 AM	
		D:\CapstoneFinal.ACD	
- tRobotRelay		TIMER	SafetyProgram
Constant	No		
External Access:	Read/Write		
<i>tRobotRelay - SafetyProgram /MainRoutine - *9(TON)</i>			
tRobotRelay.DN	1	BOOL	
<i>tRobotRelay.DN - SafetyProgram /MainRoutine - 8(XIC)</i>			
tRobotStart		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>tRobotStart - MainProgram/RobotAutoControl - *9(TON)</i>			
tRobotStart.TT	0	BOOL	
<i>tRobotStart.TT - MainProgram/RobotAutoControl - 12(XIC)</i>			
TurnsToStepsConstant		REAL	MainProgram
Constant	2857.25		
External Access:	No		
<i>TurnsToStepsConstant - MainProgram/MoveWinding - 2(DIV), 3(MUL)</i>			
WarningBit		BOOL	Capstone_Test
Constant	1		
External Access:	No		
<i>WarningBit - MainProgram/MainRoutine - 6(XIC)</i>			
<i>WarningBit - SafetyProgram /MainRoutine - *5(OTE)</i>			
Winding		ServoMotor	MainProgram
Constant	No		
External Access:	Read/Write		
Winding.POS	99.667	REAL	
request angle relative to current pos			
<i>Winding.POS - MainProgram/MoveWinding - *0(SBR), 3(MUL)</i>			
Winding.SPEED	750	DINT	
deg/s			
<i>Winding.SPEED - MainProgram/MoveWinding - *0(SBR), 3(MOV)</i>			
Winding.ACCEL	300	DINT	
deg/s/s			
<i>Winding.ACCEL - MainProgram/MoveWinding - *0(SBR), 3(MOV)</i>			
Winding.DECCEL	750	DINT	
deg/s/s			
<i>Winding.DECCEL - MainProgram/MoveWinding - *0(SBR), 3(MOV)</i>			
Winding_Servo:I		_00BB:AZD_AEP_A3E1851B:I:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Winding_Servo:I.Data		INT	
<i>Winding_Servo:I.Data - MainProgram/MoveWinding - *1(OM_AZ_MONITOR), *8(OM_AZ_DDO)</i>			
Winding_Servo:I.Data[0]	16416	INT	
Remote I/O (R-OUT)			
Winding_Servo:I.Data[0].0	0	BOOL	
R-OUT0			
Winding_Servo:I.Data[1]	0	INT	
Operation data number selection_R			
Winding_Servo:I.Data[2]	612	INT	
Fixed I/O (OUT)			
Winding_Servo:I.Data[2].0	0	BOOL	
SEQ-BSY			
Winding_Servo:I.Data[2].7	0	BOOL	
ALM-A			
<i>Winding_Servo:I.Data[2].7 - MainProgram/Alarms - 6(XIC)</i>			
Winding_Servo:I.Data[3]	0	INT	
Present Alarm			
Winding_Servo:I.Data[4]	22630	INT	
Feedback position (lower)			
Winding_Servo:I.Data[5]	4	INT	
Logix Designer			

Winding_Servo:I (Continued)		
Feedback position (upper)		
Winding_Servo:I.Data[6]	0	INT
Feedback speed [Hz] (lower)		
Winding_Servo:I.Data[7]	0	INT
Feedback speed [Hz] (upper)		
Winding_Servo:I.Data[8]	22630	INT
Command position (lower)		
Winding_Servo:I.Data[9]	4	INT
Command position (upper)		
Winding_Servo:I.Data[10]	-6	INT
Torque monitor		
Winding_Servo:I.Data[11]	500	INT
CST operating current		
Winding_Servo:I.Data[12]	0	INT
Information (lower)		
Winding_Servo:I.Data[13]	0	INT
Information (upper)		
Winding_Servo:I.Data[14]	0	INT
Reserved		
Winding_Servo:I.Data[15]	3074	INT
Read parameter ID_R		
Winding_Servo:I.Data[16]	0	INT
Read/write status		
Winding_Servo:I.Data[17]	0	INT
Write parameter ID_R		
Winding_Servo:I.Data[18]	-21072	INT
Read data (lower)		
Winding_Servo:I.Data[19]	1	INT
Read data (upper)		
Winding_Servo:I.Data[20]	430	INT
Assignable monitor 0 (lower)		
Winding_Servo:I.Data[21]	0	INT
Assignable monitor 0 (upper)		
Winding_Servo:I.Data[22]	459	INT
Assignable monitor 1 (lower)		
Winding_Servo:I.Data[23]	0	INT
Assignable monitor 1 (upper)		
Winding_Servo:I.Data[24]	-13489	INT
Assignable monitor 2 (lower)		
Winding_Servo:I.Data[25]	31	INT
Assignable monitor 2 (upper)		
Winding_Servo:I.Data[26]	317	INT
Assignable monitor 3 (lower)		
Winding_Servo:I.Data[27]	0	INT
Assignable monitor 3 (upper)		
Winding_Servo:O		
Constant	No	
External Access:	Read/Write	
<i>Winding_Servo:O - MainProgram/MoveWinding - *8(OM_AZ_DDO)</i>		
Winding_Servo:O.Data[0]	0	INT
Remote I/O (R-IN)		
Winding_Servo:O.Data[0].0	0	BOOL
R-IN0		
Winding_Servo:O.Data[1]	0	INT
Operation data number selection		
Winding_Servo:O.Data[2]	512	INT
Fixed I/O (IN)		
Winding_Servo:O.Data[2].0	0	BOOL
FW-JOG		
Winding_Servo:O.Data[2].5	0	BOOL
STOP		
<i>Winding_Servo:O.Data[2].5 - MainProgram/MainRoutine - *7(OTE)</i>		
		_00BB:AZD_AEP_60D82B42:O:0
		Capstone_Test

Winding_Servo:O (Continued)			
Winding_Servo:O.Data[2].6	0	BOOL	
FREE			
<i>Winding_Servo:O.Data[2].6 - MainProgram/FillValues - *#7</i>			
Winding_Servo:O.Data[2].7	0	BOOL	
ALM-RST			
<i>Winding_Servo:O.Data[2].7 - MainProgram/Alarms - 6(ALMD)</i>			
<i>Winding_Servo:O.Data[2].7 - MainProgram/MoveWinding - *6(OTE)</i>			
Winding_Servo:O.Data[3]	1	INT	
Direct data operation type			
Winding_Servo:O.Data[4]	22630	INT	
Direct data operation position (lower)			
Winding_Servo:O.Data[5]	4	INT	
Direct data operation position (upper)			
Winding_Servo:O.Data[6]	750	INT	
Direct data operation operating speed (lower)			
Winding_Servo:O.Data[7]	0	INT	
Direct data operation operating speed (upper)			
Winding_Servo:O.Data[8]	300	INT	
Direct data operation starting/changing rate (lower)			
Winding_Servo:O.Data[9]	0	INT	
Direct data operation starting/changing rate (upper)			
Winding_Servo:O.Data[10]	750	INT	
Direct data operation stopping deceleration (lower)			
Winding_Servo:O.Data[11]	0	INT	
Direct data operation stopping deceleration (upper)			
Winding_Servo:O.Data[12]	1000	INT	
Direct data operation operating current			
Winding_Servo:O.Data[13]	0	INT	
Direct data operation forwarding destination			
Winding_Servo:O.Data[14]	0	INT	
Reserved			
Winding_Servo:O.Data[15]	3074	INT	
Read Parameter ID			
Winding_Servo:O.Data[16]	0	INT	
Write Request			
Winding_Servo:O.Data[17]	3074	INT	
Write Parameter ID			
Winding_Servo:O.Data[18]	1000	INT	
Write data (lower)			
Winding_Servo:O.Data[19]	0	INT	
Write data (upper)			
WindingCurrentDia	170	DINT	MainProgram
Constant No			
External Access: Read/Write			
<i>WindingCurrentDia - MainProgram/CONTROL - *B13, #1(Action,Action_049 (Body)), *B36, #3(Action,Action_060 (Body)), *B39, #1(Action,Action_063 (Body)), A37, #1(Tran,Tran_069 (Condition)), B36, #3(Action,Action_060 (Body)), B39, #1(Action,Action_063 (Body))</i>			
<i>WindingCurrentDia - MainProgram/SetWindingDia - B1, #1(Tran,Tran_031 (Condition)), B3, #1(Action,Action_032 (Body)), B7, #1(Action,Action_048 (Body))</i>			
WindingFree	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
<i>WindingFree - MainProgram/FillValues - #7</i>			
WindingMoving	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
<i>WindingMoving - MainProgram/CONTROL - A28, #1(Tran,Tran_038 (Condition)), A34, #1(Tran,Tran_036 (Condition)), A38, #1(Tran,Tran_055 (Condition)), A39, #1(Tran,Tran_070 (Condition)), A4, #1(Tran,Tran_073 (Condition))</i>			
<i>WindingMoving - MainProgram/MoveWinding - *7(OTE)</i>			
<i>WindingMoving - MainProgram/SetWindingDia - B6, #1(Tran,Tran_047 (Condition))</i>			

WindingOldDia	170	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingOldDia - MainProgram/CONTROL - A37, #1(Tran,Tran_069 (Condition))</i>			
<i>WindingOldDia - MainProgram/SetWindingDia - *B7, #1(Action,Action_048 (Body)), B1, #1(Tran,Tran_031 (Condition)), B3, #1(Action,Action_032 (Body))</i>			
WindingSolenoid	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>WindingSolenoid - MainProgram/CONTROL - *B14, #1(Action,Action_034 (Body))</i>			
<i>WindingSolenoid - MainProgram/FillValues - #60</i>			
<i>WindingSolenoid - MainProgram/MainRoutine - 8(XIC)</i>			
<i>WindingSolenoid - MainProgram/pause - #12, *#34, *#58</i>			
<i>WindingSolenoid - MainProgram/SetWindingDia - *B1, #1(Action,Action_045 (Body)), *B2, #2(Action,Action_031 (Body)), *B4, #1(Action,Action_046 (Body))</i>			
<i>WindingSolenoid - MainProgram/StopAll - *#4</i>			
WindingTurns	99.667	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingTurns - MainProgram/CONTROL - *B1, #23(Action,Action_024 (Body)), *B23, #6(Action,Action_039 (Body)), *B37, #4(Action,Action_061 (Body)), *B37, #5(Action,Action_061 (Body)), B1, #22(Action,Action_024 (Body)), B23, #6(Action,Action_039 (Body)), B37, #3(Action,Action_061 (Body)), B37, #5(Action,Action_061 (Body))</i>			
<i>WindingTurns - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>WindingTurns - MainProgram/MoveWinding - *2(DIV)</i>			
<i>WindingTurns - MainProgram/SetWindingDia - *B3, #1(Action,Action_032 (Body)), *B6, #2(Action,Action_047 (Body)), B3, #1(Action,Action_032 (Body)), B6, #1(Action,Action_047 (Body))</i>			
<i>WindingTurns - MainProgram/ThugShake - *B2, #2(Action,Action_027 (Body)), *B3, #1(Action,Action_029 (Body)), B2, #2(Action,Action_027 (Body)), B3, #1(Action,Action_029 (Body))</i>			
WireAtCutting	1	BOOL	Capstone_Test
AliasFor:	Input_RemoteIO:I.Data.9		
Base Tag:	Input_RemoteIO:I.Data.9		
Constant	No		
External Access:	Read/Write		
<i>WireAtCutting - MainProgram/CONTROL - A9, #1(Tran,Tran_020 (Condition)), C10, #1(Tran,Tran_063 (Condition)), C9, #1(Tran,Tran_061 (Condition))</i>			
WireAtEndOfLine	0	BOOL	Capstone_Test
AliasFor:	Input_RemoteIO:I.Data.6		
Base Tag:	Input_RemoteIO:I.Data.6		
Constant	No		
External Access:	Read/Write		
<i>WireAtEndOfLine - MainProgram/CONTROL - A12, #1(Tran,Tran_022 (Condition))</i>			
WireInRobot	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WireInRobot - MainProgram/CONTROL - *B1, #8(Action,Action_024 (Body))</i>			
<i>WireInRobot - MainProgram/pause - #19, *#41, *#65</i>			
<i>WireInRobot - MainProgram/StopAll - *#11</i>			
wireLoadDist	3000	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>wireLoadDist - MainProgram/FillValues - *#67</i>			

- Controller Capstone_Test
- Controller Fault Handler
- Power-Up Handler

Tasks

- MainTask
 - MainProgram
 - MainRoutine
 - Alarms
 - CONTROL
 - FillValues

Progrsm that fills various values throughout the program
 - HeatExtruder
 - MoveWinding

INPUTS: pos speed accel deccel
 - pause
 - RobotAutoControl
 - selfdestruct
 - SetWindingDia

sets the diameter for winding mechanism
 - StopAll
 - ThugShake

Shakes winding to hopefully get the solenoid to seat into a hole
 - WindInductor
 - SafetyTask
 - SafetyProgram
 - MainRoutine

Unscheduled

Motion Groups

Ungrouped Axes

Add-On Instructions

- CurrentToSpoolIndex

Intakes a current and ouputs a spool index
- Logic
- OM_AZ_DDO
 - Logic
- OM_AZ_MONITOR
 - Logic
- OM_AZ_MOVE
 - Logic
- OM_AZ_READ
 - Logic
- OM_AZ_WRITE
 - Logic
- OM_AZ_ZHOME
 - Logic

Data Types

- User-Defined
 - AOI_Motor
 - Coil
 - DCMotorDirectionControl
 - HMIControl
 - HMIManual
 - MicroScan
 - MicroScanIn
 - MicroScanOut
 - ServoMotor
 - Spool
- Strings

**Add-On-Defined****CurrentToSpoolIndex**

Intakes a current and ouputs a spool index

OM_AZ_DDO**OM_AZ_MONITOR****OM_AZ_MOVE****OM_AZ_READ****OM_AZ_WRITE****OM_AZ_ZHOME****Module-Defined****AB:1732_D2:I:0****AB:1732_DI16:C:0****AB:1732_DI16:I:0****AB:1732_DO16:C:0****AB:1732_DO16:O:0****AB:5000_AI8:C:0****AB:5000_AI8:I:0****AB:5000_AI8:O:0****AB:5000_AI_Channel:C:0****AB:5000_DI16:C:0****AB:5000_DI16:I:0****AB:5000_DI_Channel:C:0****AB:5000_DO16:C:0****AB:5000_DO16:I:0****AB:5000_DO16:O:0****AB:5000_DO8:O:0****AB:5000_DO_Channel:C:0****AB:5000_HART_Command_Control_Struct:I:0****AB:5000_HART_Command_Control_Struct:O:0****AB:5000_HART_Static_Struct:I:0****AB:5000_SafetyReadback_Channel:I:0****AB:5000_SafetyReadback_Channel:I:1****AB:5000_SDI8:I:0****AB:5000_SDO8:I:1****AB:5000_String16_Struct:I:0****AB:5000_String32_Struct:I:0****AB:ETHERNET_MODULE:C:0****AB:ETHERNET_MODULE_DINT_16Bytes:I:0****AB:ETHERNET_MODULE_DINT_16Bytes:O:0****AB:ETHERNET_MODULE_DINT_496Bytes:O:0****AB:ETHERNET_MODULE_DINT_500Bytes:I:0****AB:ETHERNET_MODULE_INT_8Bytes:I:0****AB:ETHERNET_MODULE_INT_8Bytes:O:0****AB:ETHERNET_MODULE_SINT_124Bytes:O:0****AB:ETHERNET_MODULE_SINT_125Bytes:I:0****AB:ETHERNET_MODULE_SINT_16Bytes:O:0****AB:ETHERNET_MODULE_SINT_8Bytes:I:0****CHANNEL_AI:I:0****CHANNEL_AI:O:0****CHANNEL_AI_CAL:O:0****CHANNEL_AI_CJ_DIAG:I:0****CHANNEL_AI_DIAG:I:0****CHANNEL_AI_DIAG_CAL:I:0****CHANNEL_AI_HART:I:0****CHANNEL_AO:O:0****CHANNEL_AO_CAL:O:0****CHANNEL_AO_DIAG:I:0****CHANNEL_AO_DIAG_CAL:I:0**

101
010

CHANNEL_DI:I:0

101
010

CHANNEL_DI_COUNTER:I:0

101
010

CHANNEL_DI_COUNTER:O:0

101
010

CHANNEL_DI_TIMESTAMP:I:0

101
010

CHANNEL_DI_TIMESTAMP:O:0

101
010

CHANNEL_DO:I:0

101
010

CHANNEL_DO:O:0

101
010

CHANNEL_DO_DIAG:I:0

101
010

CHANNEL_DO_OVERRIDE:O:0

101
010

CHANNEL_DO_SCHEDULED:O:0

101
010

CHANNEL_HSC:I:0

101
010

CHANNEL_HSC:O:0

101
010

CHANNEL_SDI:I:0

101
010

FR:Standard_RobotPlus_8Bytes:I1:0

101
010

FR:Standard_RobotPlus_8Bytes:I5:0

101
010

FR:Standard_RobotPlus_8Bytes:O1:0

101
010

FR:Standard_RobotPlus_8Bytes:O5:0

101
010

PAX_HART_DEVICE:I:0

101
010

_00BB:AZD_AEP_1000673B:I:0

101
010

_00BB:AZD_AEP_41B5110B:O:0

101
010

_00BB:AZD_AEP_60D82B42:O:0

101
010

_00BB:AZD_AEP_A3E1851B:I:0

101
010

_0328:MICS3_CBAZXXIZ1_7FDE013E:I:0

101
010

_0328:MICS3_CBAZXXIZ1_DD7B81FB:SO:0

101
010

_0328:MICS3_CBAZXXIZ1_F211ADCF:SI:0

Trends

I/O Configuration

5069 Backplane

- 

[0] 5069-L330ERMS3 Capstone_Test
- 

[1] 5069-IB8S/A Safety_Input
- 

[2] 5069-OBV8S/A Safety_Output
- 

[3] 5069-IB16/A DC_Input
- 

[4] 5069-OW16/A DC_Relay_Outut
- 

[5] 5069-IF8/A Analog_Input

A1, Ethernet



5069-L330ERMS3 Capstone_Test

A2, Ethernet

- 

5069-L330ERMS3 Capstone_Test
- 

ETHERNET-MODULE Fanuc_Robot
- 

1732E-IB16M12R/A Input_RemoteIO
- 

ETHERNET-MODULE Cognex_5000
- 

<EDS not registered> Winding_Servo
- 

<EDS not registered> microScan
- 

1732E-OB16M12R/A Output_RemoteIO

General			
Vendor:	Rockwell Automation/Allen-Bradley		
Revision:	33.011	Mode:	Offline
Chassis Type:	<none>	Key Switch Position:	Offline
Slot:	0	Created:	8/11/2025 11:08:28 PM
Safety Partner Slot:	<internal>	Edited:	8/12/2025 1:34:24 AM
EtherNet/IP Mode:	A1/A2: Dual-IP		
Date/Time			
Date and Time:	<offline>		
Time Zone:	<offline>		
Daylight Saving (+00:00):	<offline>		
Enable Time Synchronization:	No		
Is the system time master:	<offline>		
Is a synchronized time slave:	<offline>		
Duplicate CST Master Detected:	<offline>		
CST Mastership disabled:	<offline>		
No CST Master:	<offline>		
Advanced Time Sync			
CIP Sync Time Synchronization:	Disabled		
Advanced			
Controller Fault Handler:	<none>	Allow Consumed Tags to Use RPI Provided by Producer:	No
Power-Up Handler:	<none>	Report Overflow Faults:	No
Match Project To Controller:	No	Enable Automatic Diagnostics:	Yes
Serial Number:	D02C9825		
SFC Execution			
Execution Control:	Execute current active steps only	Last Scan of Active Step:	Don't scan
Restart Position:	Restart at most recently executed step		
Nonvolatile Memory			
<offline>			
Safety			
Safety Application:	Unlocked		
Safety Status:	<offline>		
Safety Signature:	<none>		
Protect Signature in Run Mode:	No		
When replacing Safety I/O:	Configure Always		
Safety Network Numbers:			
5069 Backplane	4C2C_03D9_C3CA 5/22/2025 1:56:43.82 PM		
A1, Ethernet	4C2C_03BB_5D7B 5/22/2025 1:23:30.811 PM		
A2, Ethernet	4C33_0505_D74C 5/29/2025 7:24:28.876 PM		
Memory			
Standard Capacity:			
Total:	3,145,728 blocks		
Available:	2,960,538 blocks		
Used:	185,190 blocks		
Safety Capacity:			
Total:	1,572,864 blocks		
Available:	1,557,639 blocks		
Used:	15,225 blocks		
Security			
Primary Security Authority:	No Protection	Restrict Communications Except Through Selected Slots:	No
Use only the selected Security Authority for Authentication and Authorization:	No	Selected Slots:	
Secondary Security Authority:	No Protection	Changes To Detect:	16#ffff_ffff_ffff_ffff
Use only the selected Security Authority for Authentication and Authorization:	No	Audit Value:	<offline>

Permission Set: Enable Controller Web Pages: No

Port Configuration

Port A1
Enable: Yes

Port A2
Enable: Yes

Name	Value	Data Type	Scope
abortable	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>abortable - MainProgram/selfdestruct - *2(OTE), 2(XIC), 3(XIC)</i>			
AOI_Motor		AOI_Motor	Capstone_Test
Constant	No		
External Access:	Read/Write		
AOI_Motor.DDO		OM_AZ_DDO	
<i>AOI_Motor.DDO - MainProgram/MoveWinding - *8(OM_AZ_DDO)</i>			
AOI_Motor.DDO.EnableIn	1	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.DDO.EnableOut	1	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.DDO.Set_Exec	0	BOOL	
Execute AOI			
<i>AOI_Motor.DDO.Set_Exec - MainProgram/MoveWinding - *4(OTL), *5(OTU), *6(OTU)</i>			
AOI_Motor.DDO.Set_OperationType	1	INT	
Direct data operation operation type			
AOI_Motor.DDO.Set_Position	284774	DINT	
Direct data operation position			
<i>AOI_Motor.DDO.Set_Position - MainProgram/MoveWinding - *3(MUL), 4(NEQ)</i>			
AOI_Motor.DDO.Set_Velocity	750	DINT	
Direct data operation operating speed			
<i>AOI_Motor.DDO.Set_Velocity - MainProgram/MoveWinding - *3(MOV)</i>			
AOI_Motor.DDO.Set_Acceleration	300	DINT	
Direct data operation starting/changing rate			
<i>AOI_Motor.DDO.Set_Acceleration - MainProgram/MoveWinding - *3(MOV)</i>			
AOI_Motor.DDO.Set_Deceleration	750	DINT	
Direct data operation stopping deceleration			
<i>AOI_Motor.DDO.Set_Deceleration - MainProgram/MoveWinding - *3(MOV)</i>			
AOI_Motor.DDO.Set_Current	1000	INT	
Direct data operation operating current			
AOI_Motor.DDO.Set_Send	0	INT	
Direct data operation forwarding destination			
AOI_Motor.DDO.Sts_Busy	0	BOOL	
Executing status in AOI			
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/MoveWinding - 7(XIC)</i>			
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/SetWindingDia - B4, #1(Tran,Tran_041 (Condition))</i>			
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/ThugShake - B2, #1(Tran,Tran_027 (Condition)), C4, #1(Tran,Tran_029 (Condition))</i>			
AOI_Motor.DDO.Sts_Err	0	BOOL	
Error status in AOI			
<i>AOI_Motor.DDO.Sts_Err - MainProgram/MoveWinding - 4(XIO), 6(XIC)</i>			
AOI_Motor.DDO.Sts_Done	0	BOOL	
Complete status in AOI			
<i>AOI_Motor.DDO.Sts_Done - MainProgram/MoveWinding - 5(XIC)</i>			
AOI_Motor.DDO.Read_Present_alarm	16#0000	INT	
Alarm code monitor[Hex]			
AOI_Motor.DDO.Read_Command_position	284774	DINT	
Command position monitor[step]			
AOI_Motor.DDO.Read_Feedback_position	284774	DINT	
Feedback position monitor[step]			
AOI_Motor.DDO.Read_Torque_monitor	-2	INT	
Torque monitor as a percentage of the maximum holding torque(1=0.1%)			
AOI_Motor.DDO.Read_Cumulative_load_monitor	2083663	DINT	
Cumulative load monitor			
AOI_Motor.Move.EnableIn	0	BOOL	
Enable Input - System Defined Parameter			

AOI_Motor (Continued)		
AOI_Motor.Move.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Move.Set_Exe	0	BOOL
Execute AOI		
AOI_Motor.Move.Set_OpeDataNo	0	INT
Selected Operation data No.		
AOI_Motor.Move.Sts_Busy	0	BOOL
Executing status in AOI		
AOI_Motor.Move.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.Move.Sts_Done	1	BOOL
Complete status in AOI		
AOI_Motor.Move.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.Move.Read_Command_position	-2692	DINT
Command position monitor[step]		
AOI_Motor.Move.Read_Feedback_position	-2692	DINT
Feedback position monitor[step]		
AOI_Motor.Move.Read_Torque_monitor	11	INT
Torque monitor as a percentage of the maximum holding torque(1=0.1%)		
AOI_Motor.Move.Read_Cumulative_load_monitor	677105	DINT
Cumulative load monitor		
AOI_Motor.Monitor	OM_AZ_MONITOR	
<i>AOI_Motor.Monitor - MainProgram/MoveWinding - *I(OM_AZ_MONITOR)</i>		
AOI_Motor.Monitor.EnableIn	1	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Monitor.EnableOut	1	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Monitor.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.Monitor.Read_Feedback_position	284774	DINT
Feedback position monitor[step]		
AOI_Motor.Monitor.Read_Feedback_speed	0	DINT
Feedback speed monitor[Hz]		
AOI_Motor.Monitor.Read_Command_position	284774	DINT
Command position monitor[step]		
<i>CurrentWindingPos - MainProgram/MoveWinding - 2(DIV), 4(NEQ)</i>		
AOI_Motor.Monitor.Read_Torque_monitor	-2	INT
Torque monitor as a percentage of the maximum holding torque[1=0.1%]		
AOI_Motor.Monitor.Read_CST_operating_current	500	INT
Operation current monitor in alpha control mode(CST) (1=0.1%)		
AOI_Motor.Monitor.Read_Information	16#0000_0000	DINT
Information monitor before an alarm is generated.		
AOI_Motor.Monitor.Read_Driver_temperature	430	DINT
Driver temperature monitor(1=0.1deg)		
AOI_Motor.Monitor.Read_Motor_temperature	459	DINT
Motor temperature monitor(1=0.1deg)		
AOI_Motor.Monitor.Read_Cumulative_load_monitor	2083663	DINT

AOI_Motor (Continued)		
Cumulative load monitor		
AOI_Motor.Monitor.Read_Tripmeter		
317	DINT	
Travel distance monitor[x1000rev]		
AOI_Motor.ReadPosition.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.ReadPosition.EnableOut		
0	BOOL	
Enable Output - System Defined Parameter		
AOI_Motor.ReadPosition.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.ReadPosition.ReadPRMID		
16#0c01	INT	
Read parameter ID		
AOI_Motor.ReadPosition.ReadData		
5000	DINT	
Read data		
AOI_Motor.ReadPosition.Sts_Busy	1	BOOL
Executing status in AOI		
AOI_Motor.ReadPosition.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.ReadPosition.Sts_Done		
0	BOOL	
Complete status in AOI		
AOI_Motor.WritePosition.EnableIn		
0	BOOL	
Enable Input - System Defined Parameter		
AOI_Motor.WritePosition.EnableOut		
0	BOOL	
Enable Output - System Defined Parameter		
AOI_Motor.WritePosition.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.WritePosition.WritePRMID		
16#0c01	INT	
Write parameter ID		
AOI_Motor.WritePosition.WriteData		
4000	DINT	
Write data		
AOI_Motor.WritePosition.Sts_Busy		
0	BOOL	
Executing status in AOI		
AOI_Motor.WritePosition.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.WritePosition.Sts_Done		
1	BOOL	
Complete status in AOI		
AOI_Motor.Home.EnableIn	1	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Home.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Home.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.Home.Sts_Busy	0	BOOL
Executing status in AOI		
AOI_Motor.Home.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.Home.Sts_Done	0	BOOL
Complete status in AOI		
AOI_Motor.Home.Read_Present_alarm		
16#0000	INT	
Alarm code monitor[Hex]		
AOI_Motor.Home.Read_Feedback_position		
0	DINT	

AOI_Motor (Continued)			
Feedback position monitor[step]			
AOI_Motor.WriteSpeed.EnableIn	0	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.WriteSpeed.EnableOut	0	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.WriteSpeed.Set_Exec	0	BOOL	
Execute AOI			
AOI_Motor.WriteSpeed.WritePRMID	16#0c02	INT	
Write parameter ID			
AOI_Motor.WriteSpeed.WriteData	1000	DINT	
Write data			
AOI_Motor.WriteSpeed.Sts_Busy	0	BOOL	
Executing status in AOI			
AOI_Motor.WriteSpeed.Sts_Err	0	BOOL	
Error status in AOI			
AOI_Motor.WriteSpeed.Sts_Done	1	BOOL	
Complete status in AOI			
AOI_Motor.ReadSpeed.EnableIn	0	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.ReadSpeed.EnableOut	0	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.ReadSpeed.Set_Exec	0	BOOL	
Execute AOI			
AOI_Motor.ReadSpeed.ReadPRMID	16#0c02	INT	
Read parameter ID			
AOI_Motor.ReadSpeed.ReadData	1000	DINT	
Read data			
AOI_Motor.ReadSpeed.Sts_Busy	0	BOOL	
Executing status in AOI			
AOI_Motor.ReadSpeed.Sts_Err	0	BOOL	
Error status in AOI			
AOI_Motor.ReadSpeed.Sts_Done	1	BOOL	
Complete status in AOI			
 CameraFlash	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>CameraFlash - MainProgram/FillValues - #47</i>			
<i>CameraFlash - MainProgram/MainRoutine - *9(OTE)</i>			
 ClampMotorFwd	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>ClampMotorFwd - MainProgram/CONTROL - *B1, #14(Action,Action_024 (Body)), *B16, #1(Action,Action_057 (Body)), *B17, #1(Action,Action_058 (Body)), *B19, #1(Action,Action_052 (Body)), *B20, #1(Action,Action_053 (Body)), *B23, #34(Action,Action_039 (Body))</i>			
<i>ClampMotorFwd - MainProgram/FillValues - #50</i>			
<i>ClampMotorFwd - MainProgram/pause - #24, *#46, *#70</i>			
<i>ClampMotorFwd - MainProgram/StopAll - *#16</i>			
 ClampMotorRev	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>ClampMotorRev - MainProgram/CONTROL - *B1, #15(Action,Action_024 (Body)), *B17, #1(Action,Action_059 (Body)), *B18, #1(Action,Action_051 (Body)), *B35, #2(Action,Action_067 (Body)), *B36, #1(Action,Action_060 (Body))</i>			
<i>ClampMotorRev - MainProgram/FillValues - #51</i>			
<i>ClampMotorRev - MainProgram/pause - #25, *#47, *#71</i>			
<i>ClampMotorRev - MainProgram/StopAll - *#17</i>			
 Coil		Coil	Capstone_Test
Constant	No		

Coil (Continued)			
External Access:	Read/Write		
Coil.Inductance	8.92412616e-004	REAL	
mH			
Coil.Mass	2.11115021e-005	REAL	
Kg			
Coil.Diameter	0.18	REAL	
m			
Coil.Diameter - MainProgram/CONTROL - B13, #1(Action,Action_049 (Body))			
Coil.Turns_X	2	INT	
Coil.Turns_X - MainProgram/CONTROL - A34, #1(Tran,Tran_036 (Condition))			
Coil.Turns_Z	2	INT	
Coil.Turns_Z - MainProgram/CONTROL - A34, #1(Tran,Tran_036 (Condition)), B23, #16(Action,Action_039 (Body)), C30, #1(Tran,Tran_064 (Condition))			
Coil.Current	0.15	REAL	
A			
Coil.Length	0.003	REAL	
m			
Coil.Gauge	16	DINT	
Coil.Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)			
countdown		TIMER	Capstone_Test
Constant	No		
External Access:	Read/Write		
countdown - MainProgram/selfdestruct - *1(TON)			
countdown.TT	0	BOOL	
countdown.TT - MainProgram/selfdestruct - 2(XIC)			
CycleTime	0	DINT	Capstone_Test
Constant	No		
External Access:	Read/Write		
CycleTime - MainProgram/CONTROL - A30, #1(Tran,Tran_065 (Condition)), F34, #1(Tran,Tran_040 (Condition))			
CycleTime - MainProgram/FillValues - #63, *#63			
Fanuc_Robot:I		AB:ETHERNET_MODULE_INT_8Bytes:I:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Fanuc_Robot:I.Data[0]	0	INT	
In From Robot			
Fanuc_Robot:I.Data[0].0	0	BOOL	
In From Robot			
RobotAtHome1 - MainProgram/CONTROL - A39, #1(Tran,Tran_037 (Condition))			
Fanuc_Robot:I.Data[0].1	0	BOOL	
In From Robot			
RobotAtCutting2 - MainProgram/CONTROL - A11, #1(Tran,Tran_021 (Condition))			
Fanuc_Robot:I.Data[0].2	0	BOOL	
In From Robot			
RobotWireInserted3 - MainProgram/CONTROL - A19, #1(Tran,Tran_051 (Condition))			
Fanuc_Robot:I.Data[1]	168	INT	
Robot Status			
Fanuc_Robot:I.Data[1].0	0	BOOL	
Cmd Enabled			
Fanuc_Robot:I.Data[1].1	0	BOOL	
System Ready			
Fanuc_Robot:I.Data[1].1 - MainProgram/Alarms - 10(ALMD)			
Fanuc_Robot:I.Data[1].1 - MainProgram/RobotAutoControl - 5(XIC)			
Fanuc_Robot:I.Data[1].2	0	BOOL	
PRG Running			
Fanuc_Robot:I.Data[1].2 - MainProgram/RobotAutoControl - 6(XIC), 7(XIO)			
Fanuc_Robot:I.Data[1].5	1	BOOL	
Fault			
Fanuc_Robot:I.Data[1].5 - MainProgram/Alarms - 10(XIC)			
Fanuc_Robot:I.Data[1].5 - MainProgram/RobotAutoControl - 7(XIO)			

Fanuc_Robot:O		AB:ETHERNET_MODULE_INT_8Bytes:O:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Fanuc_Robot:O.Data[0]	26	INT	
Out to Robot			
Fanuc_Robot:O.Data[0].0	0	BOOL	
Out to Robot			
RobotToCutting - MainProgram/CONTROL - *B1, #9(Action,Action_024 (Body)), *B11, #2(Action,Action_021 (Body)), *B11, #2(Action,Action_022 (Body))			
RobotToCutting - MainProgram/pause - #20, *#42, *#66			
RobotToCutting - MainProgram/StopAll - *#12			
Fanuc_Robot:O.Data[0].1	1	BOOL	
Out to Robot			
RobotLoaded2 - MainProgram/CONTROL - *B1, #10(Action,Action_024 (Body)), *B12, #4(Action,Action_023 (Body)), *B18, #2(Action,Action_051 (Body))			
RobotLoaded2 - MainProgram/pause - #21, *#43, *#67			
RobotLoaded2 - MainProgram/StopAll - *#13			
Fanuc_Robot:O.Data[0].2	0	BOOL	
Out to Robot			
RobotMoveNextLayer3 - MainProgram/CONTROL - *B1, #11(Action,Action_024 (Body)), *B12, #3(Action,Action_023 (Body)), *B23, #18(Action,Action_039 (Body)), *B27, #1(Action,Action_040 (Body))			
RobotMoveNextLayer3 - MainProgram/pause - #22, *#44, *#68			
RobotMoveNextLayer3 - MainProgram/StopAll - *#14			
Fanuc_Robot:O.Data[0].3	1	BOOL	
Out to Robot			
RobotDirection4 - MainProgram/CONTROL - *B1, #13(Action,Action_024 (Body)), *B12, #2(Action,Action_023 (Body)), *B23, #10(Action,Action_039 (Body)), *B23, #12(Action,Action_039 (Body))			
RobotDirection4 - MainProgram/pause - #23, *#45, *#69			
RobotDirection4 - MainProgram/StopAll - *#15			
Fanuc_Robot:O.Data[0].4	1	BOOL	
Out to Robot			
RobotDoneGetThisShitOutOfHere4 - MainProgram/CONTROL - *B1, #12(Action,Action_024 (Body)), *B38, #1(Action,Action_041 (Body)), *B38, #1(Action,Action_042 (Body))			
Fanuc_Robot:O.Data[0].5	0	BOOL	
Out to Robot			
RobotMoveBack6 - MainProgram/CONTROL - *B20, #2(Action,Action_053 (Body)), *B21, #1(Action,Action_025 (Body))			
Fanuc_Robot:O.Data[1]	143	INT	
AutoMode			
Fanuc_Robot:O.Data[1].0	1	BOOL	
IMSTP			
Fanuc_Robot:O.Data[1].0 - MainProgram/RobotAutoControl - *3(OTE)			
Fanuc_Robot:O.Data[1].1	1	BOOL	
Hold			
Fanuc_Robot:O.Data[1].1 - MainProgram/RobotAutoControl - *3(OTE)			
Fanuc_Robot:O.Data[1].2	1	BOOL	
SFSPD			
Fanuc_Robot:O.Data[1].2 - MainProgram/RobotAutoControl - *3(OTE)			
Fanuc_Robot:O.Data[1].4	0	BOOL	
Fault Reset			
Fanuc_Robot:O.Data[1].4 - MainProgram/RobotAutoControl - *11(OTE)			
Fanuc_Robot:O.Data[1].5	0	BOOL	
Start			
Fanuc_Robot:O.Data[1].5 - MainProgram/RobotAutoControl - *12(OTE)			
Fanuc_Robot:O.Data[1].7	1	BOOL	
Enable			
Fanuc_Robot:O.Data[1].7 - MainProgram/RobotAutoControl - *3(OTE)			
Heater		0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Heater - MainProgram/FillValues - #59			
Heater - MainProgram/HeatExtruder - *1(OTE), *5(OTE)			

	HeightMotorDown	0	BOOL	Capstone_Test
	Constant	No		
	External Access:	Read/Write		
	<i>HeightMotorDown - MainProgram/CONTROL - *B1, #19(Action,Action_024 (Body)), *B6, #1(Action,Action_064 (Body)), *B7, #1(Action,Action_065 (Body)), *C30, #2(Action,Action_026 (Body)), *C31, #1(Action,Action_062 (Body))</i>			
	<i>HeightMotorDown - MainProgram/FillValues - #53</i>			
	<i>HeightMotorDown - MainProgram/pause - #27, *#49, *#73</i>			
	<i>HeightMotorDown - MainProgram/StopAll - *#21</i>			
	HeightMotorUp	0	BOOL	Capstone_Test
	Constant	No		
	External Access:	Read/Write		
	<i>HeightMotorUp - MainProgram/CONTROL - *B1, #20(Action,Action_024 (Body)), *B35, #3(Action,Action_067 (Body)), *B36, #2(Action,Action_060 (Body)), *B4, #1(Action,Action_066 (Body)), *B6, #1(Action,Action_054 (Body)), *C32, #1(Action,Action_068 (Body)), *C32, #1(Action,Action_069 (Body))</i>			
	<i>HeightMotorUp - MainProgram/FillValues - #54</i>			
	<i>HeightMotorUp - MainProgram/pause - #28, *#50, *#74</i>			
	<i>HeightMotorUp - MainProgram/StopAll - *#22</i>			
	HMI		HMIControl	Capstone_Test
	Constant	No		
	External Access:	Read/Write		
	HMI.Manual.Spool.FWD1	0	BOOL	
	<i>HMI.Manual.Spool.FWD1 - MainProgram/FillValues - #40</i>			
	HMI.Manual.Spool.FWD2	0	BOOL	
	<i>HMI.Manual.Spool.FWD2 - MainProgram/FillValues - #42</i>			
	HMI.Manual.Spool.FWD3	0	BOOL	
	<i>HMI.Manual.Spool.FWD3 - MainProgram/FillValues - #44</i>			
	HMI.Manual.Spool.REV1	0	BOOL	
	<i>HMI.Manual.Spool.REV1 - MainProgram/FillValues - #41</i>			
	HMI.Manual.Spool.REV2	0	BOOL	
	<i>HMI.Manual.Spool.REV2 - MainProgram/FillValues - #43</i>			
	HMI.Manual.Spool.REV3	0	BOOL	
	<i>HMI.Manual.Spool.REV3 - MainProgram/FillValues - #45</i>			
	HMI.Manual.Cut	0	BOOL	
	<i>HMI.Manual.Cut - MainProgram/FillValues - #57</i>			
	HMI.Manual.CameraLight	0	BOOL	
	<i>HMI.Manual.CameraLight - MainProgram/FillValues - #47</i>			
	HMI.Manual.ExtrudeHeater	0	BOOL	
	<i>HMI.Manual.ExtrudeHeater - MainProgram/FillValues - #59</i>			
	HMI.Manual.ExtrudeMotor	0	BOOL	
	<i>HMI.Manual.ExtrudeMotor - MainProgram/FillValues - #58</i>			
	HMI.Manual.WindingSolenoid	0	BOOL	
	<i>HMI.Manual.WindingSolenoid - MainProgram/FillValues - #60</i>			
	HMI.Manual.ClampMotorFWD	0	BOOL	
	<i>HMI.Manual.ClampMotorFWD - MainProgram/FillValues - #50</i>			
	HMI.Manual.ClampMotorREV	0	BOOL	
	<i>HMI.Manual.ClampMotorREV - MainProgram/FillValues - #51</i>			
	HMI.Manual.HeightMotorUp	0	BOOL	
	<i>HMI.Manual.HeightMotorUp - MainProgram/FillValues - #54</i>			
	HMI.Manual.HeightMotorDown	0	BOOL	
	<i>HMI.Manual.HeightMotorDown - MainProgram/FillValues - #53</i>			
	HMI.StartPB	0	BOOL	
	<i>HMI.StartPB - MainProgram/MainRoutine - 2(XIC)</i>			
	HMI.ResetPB	0	BOOL	
	<i>HMI.ResetPB - MainProgram/MainRoutine - 0(XIC)</i>			
	HMI.StopPB	0	BOOL	
	<i>HMI.StopPB - MainProgram/MainRoutine - 2(XIO)</i>			
	<i>HMI.StopPB - MainProgram/pause - #8</i>			
	HMI.RobotFaultReset	0	BOOL	
	<i>HMI.RobotFaultReset - MainProgram/RobotAutoControl - 8(XIC)</i>			
	Input_RemoteIO:I		AB:1732_DI16:I:0	Capstone_Test
	Constant	No		

Input_RemoteIO:I (Continued)			
External Access:	Read/Write		
Input_RemoteIO:I.Data.0	0	BOOL	
StartPB			
<i>PB_Start - MainProgram/MainRoutine - 2(XIC)</i>			
Input_RemoteIO:I.Data.2	1	BOOL	
<i>Input_RemoteIO:I.Data.2 - MainProgram/FillValues - #10</i>			
Input_RemoteIO:I.Data.3	1	BOOL	
<i>Input_RemoteIO:I.Data.3 - MainProgram/FillValues - #11</i>			
Input_RemoteIO:I.Data.5	0	BOOL	
<i>Input_RemoteIO:I.Data.5 - MainProgram/FillValues - #12</i>			
Input_RemoteIO:I.Data.6	0	BOOL	
<i>WireAtEndOfLine - MainProgram/CONTROL - A12, #1(Tran,Tran_022 (Condition))</i>			
Input_RemoteIO:I.Data.9	1	BOOL	
<i>WireAtCutting - MainProgram/CONTROL - A9, #1(Tran,Tran_020 (Condition)), C10, #1(Tran,Tran_063 (Condition)), C9, #1(Tran,Tran_061 (Condition))</i>			
Local:1:I			
Constant	No	AB:5000_SDI8:I:0	Capstone_Test
External Access:	Read/Write		
Local:1:I.Pt00.Data	1	BOOL	
<i>EStop1 - SafetyProgram /MainRoutine - 1(XIO)</i>			
Local:1:I.Pt01.Data	1	BOOL	
<i>EStop2 - SafetyProgram /MainRoutine - 1(XIO)</i>			
Local:1:I.Pt02.Data	0	BOOL	
<i>DoorInterlock1 - SafetyProgram /MainRoutine - 2(XIO)</i>			
Local:1:I.Pt03.Data	0	BOOL	
<i>DoorInterlock2 - SafetyProgram /MainRoutine - 2(XIO)</i>			
Local:1:I.Pt04.Data	0	BOOL	
<i>nanoScanCh1 - SafetyProgram /MainRoutine - 3(XIO)</i>			
Local:1:I.Pt05.Data	0	BOOL	
<i>nanoScanCh2 - SafetyProgram /MainRoutine - 3(XIO)</i>			
Local:1:I.Pt06.Data	0	BOOL	
<i>PB_Reset - MainProgram/Alarms - 0(ALMD), 1(ALMD), 2(ALMD), 3(ALMD), 4(ALMD)</i>			
<i>PB_Reset - MainProgram/RobotAutoControl - 0(XIC)</i>			
<i>PB_Reset - SafetyProgram /MainRoutine - 10(XIC), 6(XIO), 8(XIC)</i>			
Local:1:I.Pt07.Data	1	BOOL	
<i>NanoScanWarning - SafetyProgram /MainRoutine - 5(XIC)</i>			
Local:2:O			
Constant	No	AB:5000_DO8:O:0	Capstone_Test
External Access:	Read/Write		
Local:2:O.Pt00		CHANNEL_DO:O:0	
Robot Safety Relay K1			
Local:2:O.Pt00.Data	1	BOOL	
Robot Safety Relay K1			
<i>Local:2:O.Pt00.Data - SafetyProgram /MainRoutine - *9(OTE), 8(XIC)</i>			
Local:2:O.Pt01		CHANNEL_DO:O:0	
Robot Safety Relay K2			
Local:2:O.Pt01.Data	1	BOOL	
Robot Safety Relay K2			
<i>Local:2:O.Pt01.Data - SafetyProgram /MainRoutine - *9(OTE), 8(XIC)</i>			
Local:2:O.Pt02.Data	0	BOOL	
Robot Reset Relay			
<i>Local:2:O.Pt02.Data - SafetyProgram /MainRoutine - *8(OTE)</i>			
Local:4:O			
Constant	No	AB:5000_DO16:O:0	Capstone_Test
External Access:	Read/Write		
Local:4:O.Pt00.Data	0	BOOL	
Top Feed Motor Fwd			
<i>Local:4:O.Pt00.Data - MainProgram/FillValues - *#40</i>			
Local:4:O.Pt01.Data	0	BOOL	
Top Feed Motor Rev			

Local:4:O (Continued)		
<i>Local:4:O.Pt01.Data - MainProgram/FillValues - *#41</i>		
Local:4:O.Pt02.Data	0	BOOL
Mid Feed Motor Fwd		
<i>Local:4:O.Pt02.Data - MainProgram/FillValues - *#42</i>		
Local:4:O.Pt03.Data	0	BOOL
Mid Feed Motor Rev		
<i>Local:4:O.Pt03.Data - MainProgram/FillValues - *#43</i>		
Local:4:O.Pt04.Data	0	BOOL
Bottom Feed Motor Fwd		
<i>Local:4:O.Pt04.Data - MainProgram/FillValues - *#44</i>		
Local:4:O.Pt05.Data	0	BOOL
Bottom Feed Motor Rev		
<i>Local:4:O.Pt05.Data - MainProgram/FillValues - *#45</i>		
Local:4:O.Pt06	CHANNEL_DO:O:0	
Camera Flash		
Local:4:O.Pt06.Data	0	BOOL
Reset Robot Safety Relay		
<i>Local:4:O.Pt06.Data - MainProgram/FillValues - *#47</i>		
Local:4:O.Pt07.Data	1	BOOL
<i>StackLightGreen - MainProgram/MainRoutine - *5(OTE)</i>		
<i>StackLightGreen - MainProgram/pause - #10, *#32, *#56</i>		
<i>StackLightGreen - MainProgram/StopAll - *#2</i>		
Local:4:O.Pt08.Data	1	BOOL
<i>StackLightYellow - MainProgram/MainRoutine - *6(OTE)</i>		
<i>StackLightYellow - MainProgram/pause - #11, *#33, *#57</i>		
<i>StackLightYellow - MainProgram/StopAll - *#3</i>		
Local:4:O.Pt09.Data	0	BOOL
<i>StackLightRed - MainProgram/MainRoutine - *7(OTE)</i>		
<i>StackLightRed - MainProgram/pause - #9, *#31, *#55</i>		
<i>StackLightRed - MainProgram/StopAll - *#1</i>		
Local:4:O.Pt10.Data	0	BOOL
<i>Local:4:O.Pt10.Data - MainProgram/FillValues - *#53</i>		
Local:4:O.Pt11.Data	0	BOOL
<i>Local:4:O.Pt11.Data - MainProgram/FillValues - *#54</i>		
Local:4:O.Pt12.Data	0	BOOL
<i>Local:4:O.Pt12.Data - MainProgram/FillValues - *#50</i>		
Local:4:O.Pt13.Data	0	BOOL
<i>Local:4:O.Pt13.Data - MainProgram/FillValues - *#51</i>		
Local:4:O.Pt14.Data	0	BOOL
<i>ResetLight - MainProgram/MainRoutine - *7(OTE)</i>		
 Local:5:I	AB:5000_AI8:I:0	Capstone_Test
Constant No		
External Access: Read/Write		
Local:5:I.Ch00.Data	164.73665	REAL
<i>Local:5:I.Ch00.Data - MainProgram/Alarms - 7(LEQ)</i>		
<i>Local:5:I.Ch00.Data - MainProgram/FillValues - #19, #20</i>		
Local:5:I.Ch01.Data	625.2037	REAL
<i>Local:5:I.Ch01.Data - MainProgram/Alarms - 8(LEQ)</i>		
Local:5:I.Ch02.Data	600.10535	REAL
<i>Local:5:I.Ch02.Data - MainProgram/Alarms - 9(LEQ)</i>		
<i>Local:5:I.Ch02.Data - MainProgram/FillValues - #31, #32</i>		
Local:5:I.Ch03.Data	-2.61334376e-003	REAL
<i>Local:5:I.Ch03.Data - MainProgram/FillValues - #16</i>		
 MANLY	0	BOOL
Constant No		
External Access: Read/Write		
<i>MANLY - MainProgram/Alarms - 5(XIC)</i>		
<i>MANLY - MainProgram/MainRoutine - 0(XIO), 4(XIO)</i>		
 microScan:SI	_0328:MICS3_CBAZXXIZ1_F211ADCF:SI:0	Capstone_Test

microScan:SI (Continued)			
Constant	No		
External Access:	Read/Write		
microScan:SI.Data[0]	1	SINT	
Mode			
microScan:SI.Data[1]	1	SINT	
Safe CutOff Path(1-8)			
microScan:SI.Data[2]	1	SINT	
NonSafe CutOff Path(1-8)			
<i>microScan:SI.Data[2] - SafetyProgram</i>  <i>/MainRoutine - 4(LES), 5(EQU)</i>			
microScan:SI.Data[3]	0	SINT	
Reserved			
microScan:SI.Data[4]	1	SINT	
Current Monitoring Case No.			
microScan:SI.Data[5]	0	SINT	
Reset Required CutOff Path(1-8)			
microScan:SI.Data[6]	0	SINT	
Reserved			
microScan:SI.Data[7]	0	SINT	
Errors			
 microScan:SO			
			_0328:MICS3_CBAZXXIZ1_DD7B81FB:SO:0
			Capstone_Test
Constant	No		
External Access:	Read/Write		
microScan:SO.Data[0]	3	SINT	
Mode			
<i>microScan:SO.Data[0] - SafetyProgram</i>  <i>/MainRoutine - *0(MOV)</i>			
microScan:SO.Data[0].0	1	BOOL	
Trigger Run-Mode			
microScan:SO.Data[1]	0	SINT	
Reserved			
microScan:SO.Data[2]	0	SINT	
Reserved			
microScan:SO.Data[3]	1	SINT	
Control Inputs			
<i>microScan:SO.Data[3] - SafetyProgram</i>  <i>/MainRoutine - *0(MOV)</i>			
microScan:SO.Data[3].0	1	BOOL	
Control input 1 (A1)			
microScan:SO.Data[4]	0	SINT	
Reserved			
microScan:SO.Data[5]	0	SINT	
Trigger Reset CutOff Path (1-8)			
microScan:SO.Data[6]	0	SINT	
Reserved			
microScan:SO.Data[7]	0	SINT	
Trigger Device Reboot			
 NewLoadedButtonHMI			
	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>NewLoadedButtonHMI - MainProgram/WindInductor - 4(XIC)</i>			
 Output_RemoteIO:O			
		AB:1732_DO16:O:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Output_RemoteIO:O.Data.0	0	BOOL	
<i>Output_RemoteIO:O.Data.0 - MainProgram/FillValues - *#57</i>			
Output_RemoteIO:O.Data.2	0	BOOL	
<i>Output_RemoteIO:O.Data.2 - MainProgram/FillValues - *#58</i>			
Output_RemoteIO:O.Data.4	0	BOOL	
<i>Output_RemoteIO:O.Data.4 - MainProgram/FillValues - *#59</i>			
Output_RemoteIO:O.Data.6	0	BOOL	
<i>Output_RemoteIO:O.Data.6 - MainProgram/FillValues - *#60</i>			

 PB_Reset	0	BOOL	Capstone_Test
AliasFor:	Local:1:I.Pt06.Data		
Base Tag:	Local:1:I.Pt06.Data		
Constant	No		
External Access:	Read/Write		
<i>PB_Reset - MainProgram/Alarms - 0(ALMD), 1(ALMD), 2(ALMD), 3(ALMD), 4(ALMD)</i>			
<i>PB_Reset - MainProgram/RobotAutoControl - 0(XIC)</i>			
<i>PB_Reset - SafetyProgram  /MainRoutine - 10(XIC), 6(XIO), 8(XIC)</i>			
 PB_Start	0	BOOL	Capstone_Test
StartPB			
AliasFor:	Input_RemoteIO:I.Data.0		
Base Tag:	Input_RemoteIO:I.Data.0		
Constant	No		
External Access:	Read/Write		
<i>PB_Start - MainProgram/MainRoutine - 2(XIC)</i>			
 Progress	0	DINT	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>Progress - MainProgram/FillValues - *#71</i>			
 ResetLight	0	BOOL	Capstone_Test
AliasFor:	Local:4:O.Pt14.Data		
Base Tag:	Local:4:O.Pt14.Data		
Constant	No		
External Access:	Read/Write		
<i>ResetLight - MainProgram/MainRoutine - *7(OTE)</i>			
 RobotAtCutting2	0	BOOL	Capstone_Test
In From Robot			
AliasFor:	Fanuc_Robot:I.Data[0].1		
Base Tag:	Fanuc_Robot:I.Data[0].1		
Constant	No		
External Access:	Read/Write		
<i>RobotAtCutting2 - MainProgram/CONTROL - A11, #1(Tran,Tran_021 (Condition))</i>			
 RobotAtHome1	0	BOOL	Capstone_Test
In From Robot			
AliasFor:	Fanuc_Robot:I.Data[0].0		
Base Tag:	Fanuc_Robot:I.Data[0].0		
Constant	No		
External Access:	Read/Write		
<i>RobotAtHome1 - MainProgram/CONTROL - A39, #1(Tran,Tran_037 (Condition))</i>			
 RobotDirection4	1	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].3		
Base Tag:	Fanuc_Robot:O.Data[0].3		
Constant	No		
External Access:	Read/Write		
<i>RobotDirection4 - MainProgram/CONTROL - *B1, #13(Action,Action_024 (Body)), *B12, #2(Action,Action_023 (Body)), *B23, #10(Action,Action_039 (Body)), *B23, #12(Action,Action_039 (Body))</i>			
<i>RobotDirection4 - MainProgram/pause - #23, *#45, *#69</i>			
<i>RobotDirection4 - MainProgram/StopAll - *#15</i>			
 RobotDoneGetThisShitOutOfHere4	1	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].4		
Base Tag:	Fanuc_Robot:O.Data[0].4		
Constant	No		
External Access:	Read/Write		

RobotDoneGetThisShitOutOfHere4 (Continued)			
<i>RobotDoneGetThisShitOutOfHere4 - MainProgram/CONTROL - *B1, #12(Action,Action_024 (Body)), *B38, #1(Action,Action_041 (Body)), *B38, #1(Action,Action_042 (Body))</i>			
 RobotLoaded2	1	BOOL	Capstone_Test
Out to Robot			
AliasFor: Fanuc_Robot:O.Data[0].1			
Base Tag: Fanuc_Robot:O.Data[0].1			
Constant No			
External Access: Read/Write			
<i>RobotLoaded2 - MainProgram/CONTROL - *B1, #10(Action,Action_024 (Body)), *B12, #4(Action,Action_023 (Body)), *B18, #2(Action,Action_051 (Body))</i>			
<i>RobotLoaded2 - MainProgram/pause - #21, *#43, *#67</i>			
<i>RobotLoaded2 - MainProgram/StopAll - *#13</i>			
 RobotMoveBack6	0	BOOL	Capstone_Test
Out to Robot			
AliasFor: Fanuc_Robot:O.Data[0].5			
Base Tag: Fanuc_Robot:O.Data[0].5			
Constant No			
External Access: Read/Write			
<i>RobotMoveBack6 - MainProgram/CONTROL - *B20, #2(Action,Action_053 (Body)), *B21, #1(Action,Action_025 (Body))</i>			
 RobotMoveNextLayer3	0	BOOL	Capstone_Test
Out to Robot			
AliasFor: Fanuc_Robot:O.Data[0].2			
Base Tag: Fanuc_Robot:O.Data[0].2			
Constant No			
External Access: Read/Write			
<i>RobotMoveNextLayer3 - MainProgram/CONTROL - *B1, #11(Action,Action_024 (Body)), *B12, #3(Action,Action_023 (Body)), *B23, #18(Action,Action_039 (Body)), *B27, #1(Action,Action_040 (Body))</i>			
<i>RobotMoveNextLayer3 - MainProgram/pause - #22, *#44, *#68</i>			
<i>RobotMoveNextLayer3 - MainProgram/StopAll - *#14</i>			
 RobotRelayReset	0	BOOL	Capstone_Test
Reset Robot Safety Relay			
AliasFor: Local:4:O.Pt06.Data			
Base Tag: Local:4:O.Pt06.Data			
Constant No			
External Access: Read/Write			
<i>Local:4:O.Pt06.Data - MainProgram/FillValues - *#47</i>			
 RobotToCutting	0	BOOL	Capstone_Test
Out to Robot			
AliasFor: Fanuc_Robot:O.Data[0].0			
Base Tag: Fanuc_Robot:O.Data[0].0			
Constant No			
External Access: Read/Write			
<i>RobotToCutting - MainProgram/CONTROL - *B1, #9(Action,Action_024 (Body)), *B11, #2(Action,Action_021 (Body)), *B11, #2(Action,Action_022 (Body))</i>			
<i>RobotToCutting - MainProgram/pause - #20, *#42, *#66</i>			
<i>RobotToCutting - MainProgram/StopAll - *#12</i>			
 RobotWireInserted3	0	BOOL	Capstone_Test
also means wire in			
AliasFor: Fanuc_Robot:I.Data[0].2			
Base Tag: Fanuc_Robot:I.Data[0].2			
Constant No			
External Access: Read/Write			
<i>RobotWireInserted3 - MainProgram/CONTROL - A19, #1(Tran,Tran_051 (Condition))</i>			
 SafetyBit	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			

SafetyBit (Continued)			
<i>SafetyBit - MainProgram/Alarms - 0(XIC), 1(XIC), 2(XIC), 3(XIC), 4(XIC)</i>			
<i>SafetyBit - MainProgram/MainRoutine - 10(XIC), 4(XIO), 5(XIO), 6(XIO), 7(XIC)</i>			
<i>SafetyBit - MainProgram/RobotAutoControl - 1(XIC), 4(XIO), 5(XIO), 6(XIO)</i>			
<i>SafetyBit - SafetyProgram  /MainRoutine - *7(OTE), 7(XIC), 9(XIO)</i>			
 selfdestruct	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>selfdestruct - MainProgram/MainRoutine - 0(XIC)</i>			
<i>selfdestruct - MainProgram/selfdestruct - 0(XIC), 2(XIO), 3(XIC)</i>			
 Spool0F	0	BOOL	Capstone_Test
Top Feed Motor Fwd			
AliasFor:	Local:4:O.Pt00.Data		
Base Tag:	Local:4:O.Pt00.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt00.Data - MainProgram/FillValues - *#40</i>			
 Spool0R	0	BOOL	Capstone_Test
Top Feed Motor Rev			
AliasFor:	Local:4:O.Pt01.Data		
Base Tag:	Local:4:O.Pt01.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt01.Data - MainProgram/FillValues - *#41</i>			
 Spool1F	0	BOOL	Capstone_Test
Mid Feed Motor Fwd			
AliasFor:	Local:4:O.Pt02.Data		
Base Tag:	Local:4:O.Pt02.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt02.Data - MainProgram/FillValues - *#42</i>			
 Spool1R	0	BOOL	Capstone_Test
Mid Feed Motor Fwd			
AliasFor:	Local:4:O.Pt02.Data		
Base Tag:	Local:4:O.Pt02.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt02.Data - MainProgram/FillValues - *#42</i>			
 Spool2F	0	BOOL	Capstone_Test
Bottom Feed Motor Fwd			
AliasFor:	Local:4:O.Pt04.Data		
Base Tag:	Local:4:O.Pt04.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt04.Data - MainProgram/FillValues - *#44</i>			
 Spool2R	0	BOOL	Capstone_Test
Bottom Feed Motor Rev			
AliasFor:	Local:4:O.Pt05.Data		
Base Tag:	Local:4:O.Pt05.Data		
Constant	No		
External Access:	Read/Write		
<i>Local:4:O.Pt05.Data - MainProgram/FillValues - *#45</i>			
 SpoolArray		Spool[3]	Capstone_Test
Constant	No		
External Access:	Read/Write		
SpoolArray[0].REV	0	BOOL	

SpoolArray (Continued)		
out		
SpoolArray[0].REV - MainProgram/CONTROL - *B1, #3(Action,Action_024 (Body))		
SpoolArray[0].REV - MainProgram/FillValues - #41		
SpoolArray[0].REV - MainProgram/pause - #14, *#36, *#60		
SpoolArray[0].REV - MainProgram/StopAll - *#6		
SpoolArray[0].FWD	0	BOOL
out		
SpoolArray[0].FWD - MainProgram/CONTROL - *B1, #2(Action,Action_024 (Body))		
SpoolArray[0].FWD - MainProgram/FillValues - #40		
SpoolArray[0].FWD - MainProgram/pause - #13, *#35, *#59		
SpoolArray[0].FWD - MainProgram/StopAll - *#5		
SpoolArray[0].Mass	164.73665	REAL
current mass reading		
SpoolArray[0].Mass - MainProgram/FillValues - *#20, *#22		
SpoolArray[0].Length	0.0	REAL
approximated legth based on Mass		
SpoolArray[0].Loaded	1	BOOL
if is in loaded state		
SpoolArray[0].Loaded - MainProgram/FillValues - *#10		
SpoolArray[0].Gauge	-1	DINT
Wire Gauge		
SpoolArray[1].REV	0	BOOL
out		
SpoolArray[1].REV - MainProgram/CONTROL - *B1, #5(Action,Action_024 (Body))		
SpoolArray[1].REV - MainProgram/FillValues - #43		
SpoolArray[1].REV - MainProgram/pause - #16, *#38, *#62		
SpoolArray[1].REV - MainProgram/StopAll - *#8		
SpoolArray[1].FWD	0	BOOL
out		
SpoolArray[1].FWD - MainProgram/CONTROL - *B1, #4(Action,Action_024 (Body))		
SpoolArray[1].FWD - MainProgram/FillValues - #42		
SpoolArray[1].FWD - MainProgram/pause - #15, *#37, *#61		
SpoolArray[1].FWD - MainProgram/StopAll - *#7		
SpoolArray[1].Mass	163.34396	REAL
current mass reading		
SpoolArray[1].Length	0.0	REAL
approximated legth based on Mass		
SpoolArray[1].Loaded	1	BOOL
if is in loaded state		
SpoolArray[1].Loaded - MainProgram/FillValues - *#11		
SpoolArray[1].Gauge	-1	DINT
Wire Gauge		
SpoolArray[2].REV	0	BOOL
out		
SpoolArray[2].REV - MainProgram/CONTROL - *B1, #7(Action,Action_024 (Body))		
SpoolArray[2].REV - MainProgram/FillValues - #45		
SpoolArray[2].REV - MainProgram/pause - #18, *#40, *#64		
SpoolArray[2].REV - MainProgram/StopAll - *#10		
SpoolArray[2].FWD	0	BOOL
out		
SpoolArray[2].FWD - MainProgram/CONTROL - *B1, #6(Action,Action_024 (Body)), *B11, #1(Action,Action_021 (Body)), *B11, #1(Action,Action_022 (Body)), *B12, #1(Action,Action_023 (Body)), *B23, #3(Action,Action_039 (Body)), *B37, #1(Action,Action_061 (Body)), *C9, #1(Action,Action_019 (Body))		
SpoolArray[2].FWD - MainProgram/FillValues - #44		
SpoolArray[2].FWD - MainProgram/pause - #17, *#39, *#63		
SpoolArray[2].FWD - MainProgram/StopAll - *#9		
SpoolArray[2].Mass	600.10535	REAL
current mass reading		
SpoolArray[2].Mass - MainProgram/FillValues - *#32, *#34		
SpoolArray[2].Length	0.0	REAL
approximated legth based on Mass		
SpoolArray[2].Loaded	0	BOOL
if is in loaded state		

SpoolArray (Continued)			
<i>SpoolArray[2].Loaded - MainProgram/FillValues - *#12</i>			
SpoolArray[2].Gauge	16	DINT	
Wire Gauge			
<i>SpoolArray[2].Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)</i>			
 StackLightGreen	1	BOOL	Capstone_Test
AliasFor: Local:4:O.Pt07.Data			
Base Tag: Local:4:O.Pt07.Data			
Constant No			
External Access: Read/Write			
<i>StackLightGreen - MainProgram/MainRoutine - *5(OTE)</i>			
<i>StackLightGreen - MainProgram/pause - #10, *#32, *#56</i>			
<i>StackLightGreen - MainProgram/StopAll - *#2</i>			
 StackLightRed	0	BOOL	Capstone_Test
AliasFor: Local:4:O.Pt09.Data			
Base Tag: Local:4:O.Pt09.Data			
Constant No			
External Access: Read/Write			
<i>StackLightRed - MainProgram/MainRoutine - *7(OTE)</i>			
<i>StackLightRed - MainProgram/pause - #9, *#31, *#55</i>			
<i>StackLightRed - MainProgram/StopAll - *#1</i>			
 StackLightYellow	1	BOOL	Capstone_Test
AliasFor: Local:4:O.Pt08.Data			
Base Tag: Local:4:O.Pt08.Data			
Constant No			
External Access: Read/Write			
<i>StackLightYellow - MainProgram/MainRoutine - *6(OTE)</i>			
<i>StackLightYellow - MainProgram/pause - #11, *#33, *#57</i>			
<i>StackLightYellow - MainProgram/StopAll - *#3</i>			
 STEP_WireLoaded	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
<i>STEP_WireLoaded - MainProgram/WindInductor - *4(OTE), 2(XIO), 3(XIO), 4(XIC)</i>			
 Temperature	-2.61334376e-003	REAL	Capstone_Test
Constant No			
External Access: Read/Write			
<i>Temperature - MainProgram/FillValues - *#16</i>			
<i>Temperature - MainProgram/HeatExtruder - 1(GEQ), 1(LEQ), 2(GEQ), 2(GEQ), 2(LES), 3(LES), 4(LES), 5(LES)</i>			
 WarningBit	1	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
<i>WarningBit - MainProgram/MainRoutine - 6(XIC)</i>			
<i>WarningBit - SafetyProgram  /MainRoutine - *5(OTE)</i>			
 Winding_Servo:I		_00BB:AZD_AEP_A3E1851B:I:0	Capstone_Test
Constant No			
External Access: Read/Write			
Winding_Servo:I.Data		INT	
<i>Winding_Servo:I.Data - MainProgram/MoveWinding - *1(OM_AZ_MONITOR), *8(OM_AZ_DDO)</i>			
Winding_Servo:I.Data[0]	16416	INT	
Remote I/O (R-OUT)			
Winding_Servo:I.Data[0].0	0	BOOL	
R-OUT0			
Winding_Servo:I.Data[1]	0	INT	
Operation data number selection_R			
Winding_Servo:I.Data[2]	612	INT	
Fixed I/O (OUT)			
Winding_Servo:I.Data[2].0	0	BOOL	

Winding_Servo:I (Continued)		
SEQ-BSY		
Winding_Servo:I.Data[2].7	0	BOOL
ALM-A		
Winding_Servo:I.Data[2].7 - MainProgram/Alarms - 6(XIC)		
Winding_Servo:I.Data[3]	0	INT
Present Alarm		
Winding_Servo:I.Data[4]	22630	INT
Feedback position (lower)		
Winding_Servo:I.Data[5]	4	INT
Feedback position (upper)		
Winding_Servo:I.Data[6]	0	INT
Feedback speed [Hz] (lower)		
Winding_Servo:I.Data[7]	0	INT
Feedback speed [Hz] (upper)		
Winding_Servo:I.Data[8]	22630	INT
Command position (lower)		
Winding_Servo:I.Data[9]	4	INT
Command position (upper)		
Winding_Servo:I.Data[10]	-6	INT
Torque monitor		
Winding_Servo:I.Data[11]	500	INT
CST operating current		
Winding_Servo:I.Data[12]	0	INT
Information (lower)		
Winding_Servo:I.Data[13]	0	INT
Information (upper)		
Winding_Servo:I.Data[14]	0	INT
Reserved		
Winding_Servo:I.Data[15]	3074	INT
Read parameter ID_R		
Winding_Servo:I.Data[16]	0	INT
Read/write status		
Winding_Servo:I.Data[17]	0	INT
Write parameter ID_R		
Winding_Servo:I.Data[18]	-21072	INT
Read data (lower)		
Winding_Servo:I.Data[19]	1	INT
Read data (upper)		
Winding_Servo:I.Data[20]	430	INT
Assignable monitor 0 (lower)		
Winding_Servo:I.Data[21]	0	INT
Assignable monitor 0 (upper)		
Winding_Servo:I.Data[22]	459	INT
Assignable monitor 1 (lower)		
Winding_Servo:I.Data[23]	0	INT
Assignable monitor 1 (upper)		
Winding_Servo:I.Data[24]	-13489	INT
Assignable monitor 2 (lower)		
Winding_Servo:I.Data[25]	31	INT
Assignable monitor 2 (upper)		
Winding_Servo:I.Data[26]	317	INT
Assignable monitor 3 (lower)		
Winding_Servo:I.Data[27]	0	INT
Assignable monitor 3 (upper)		
Winding_Servo:O		
Constant	No	_00BB:AZD_AEP_60D82B42:O:0
External Access:	Read/Write	Capstone_Test
Winding_Servo:O - MainProgram/MoveWinding - *8(OM_AZ_DDO)		
Winding_Servo:O.Data[0]	0	INT
Remote I/O (R-IN)		
Winding_Servo:O.Data[0].0	0	BOOL
R-IN0		

Winding_Servo:O (Continued)			
Winding_Servo:O.Data[1]	0	INT	
Operation data number selection			
Winding_Servo:O.Data[2]	512	INT	
Fixed I/O (IN)			
Winding_Servo:O.Data[2].0	0	BOOL	
FW-JOG			
Winding_Servo:O.Data[2].5	0	BOOL	
STOP			
<i>Winding_Servo:O.Data[2].5 - MainProgram/MainRoutine - *7(OTE)</i>			
Winding_Servo:O.Data[2].6	0	BOOL	
FREE			
<i>Winding_Servo:O.Data[2].6 - MainProgram/FillValues - *#7</i>			
Winding_Servo:O.Data[2].7	0	BOOL	
ALM-RST			
<i>Winding_Servo:O.Data[2].7 - MainProgram/Alarms - 6(ALMD)</i>			
<i>Winding_Servo:O.Data[2].7 - MainProgram/MoveWinding - *6(OTE)</i>			
Winding_Servo:O.Data[3]	1	INT	
Direct data operation type			
Winding_Servo:O.Data[4]	22630	INT	
Direct data operation position (lower)			
Winding_Servo:O.Data[5]	4	INT	
Direct data operation position (upper)			
Winding_Servo:O.Data[6]	750	INT	
Direct data operation operating speed (lower)			
Winding_Servo:O.Data[7]	0	INT	
Direct data operation operating speed (upper)			
Winding_Servo:O.Data[8]	300	INT	
Direct data operation starting/changing rate (lower)			
Winding_Servo:O.Data[9]	0	INT	
Direct data operation starting/changing rate (upper)			
Winding_Servo:O.Data[10]	750	INT	
Direct data operation stopping deceleration (lower)			
Winding_Servo:O.Data[11]	0	INT	
Direct data operation stopping deceleration (upper)			
Winding_Servo:O.Data[12]	1000	INT	
Direct data operation operating current			
Winding_Servo:O.Data[13]	0	INT	
Direct data operation forwarding destination			
Winding_Servo:O.Data[14]	0	INT	
Reserved			
Winding_Servo:O.Data[15]	3074	INT	
Read Parameter ID			
Winding_Servo:O.Data[16]	0	INT	
Write Request			
Winding_Servo:O.Data[17]	3074	INT	
Write Parameter ID			
Winding_Servo:O.Data[18]	1000	INT	
Write data (lower)			
Winding_Servo:O.Data[19]	0	INT	
Write data (upper)			
 WindingSolenoid	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>WindingSolenoid - MainProgram/CONTROL - *B14, #1(Action,Action_034 (Body))</i>			
<i>WindingSolenoid - MainProgram/FillValues - #60</i>			
<i>WindingSolenoid - MainProgram/MainRoutine - 8(XIC)</i>			
<i>WindingSolenoid - MainProgram/pause - #12, *#34, *#58</i>			
<i>WindingSolenoid - MainProgram/SetWindingDia - *B1, #1(Action,Action_045 (Body)), *B2, #2(Action,Action_031 (Body)), *B4, #1(Action,Action_046 (Body))</i>			
<i>WindingSolenoid - MainProgram/StopAll - *#4</i>			
 WireAtCutting	1	BOOL	Capstone_Test

WireAtCutting (Continued)

AliasFor: Input_RemoteIO:I.Data.9
Base Tag: Input_RemoteIO:I.Data.9
Constant No
External Access: Read/Write
WireAtCutting - MainProgram/CONTROL - A9, #1(Tran,Tran_020 (Condition)), C10, #1(Tran,Tran_063 (Condition)), C9, #1(Tran,Tran_061 (Condition))

WireAtEndOfLine 0 BOOL Capstone_Test

AliasFor: Input_RemoteIO:I.Data.6
Base Tag: Input_RemoteIO:I.Data.6
Constant No
External Access: Read/Write
WireAtEndOfLine - MainProgram/CONTROL - A12, #1(Tran,Tran_022 (Condition))

General

Class: Standard

Configuration

Type:	Continuous	Disable automatic output processing to reduce task overhead:	No
Watchdog:	500.000 ms	Inhibit task:	No

Program Schedule

Scheduled

MainProgram

Unscheduled

Monitor

Scan Times(Elapsed Time)

Max:	1.076000 ms	Last:	0.137000 ms
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Interval Times(Elapsed Times Between Triggers)

Max:	1.084000 ms	Min:	0.028000 ms
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Task overlap count: 0

General

Class: Standard

Configuration

Main:  MainRoutine Inhibit program: No
Fault: None

Monitor

Scan Times(Execution Time)
Max: 272 us Last: 104 us

Name	Value	Data Type	Scope
abort	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>abort - MainProgram/selfdestruct - *3(OTE), 0(XIO)</i>			
accel	300	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>accel - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>accel - MainProgram/ThugShake - *B2, #3(Action,Action_027 (Body)), *B4, #2(Action,Action_030 (Body))</i>			
Action_019		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_019 - MainProgram/CONTROL - *C9(Action,Action_019)</i>			
Action_021		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_021 - MainProgram/CONTROL - *B11(Action,Action_021)</i>			
Action_022		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_022 - MainProgram/CONTROL - *B11(Action,Action_022)</i>			
Action_023		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_023 - MainProgram/CONTROL - *B12(Action,Action_023)</i>			
Action_024		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_024 - MainProgram/CONTROL - *B1(Action,Action_024)</i>			
Action_025		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_025 - MainProgram/CONTROL - *B21(Action,Action_025)</i>			
Action_026		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_026 - MainProgram/CONTROL - *C30(Action,Action_026)</i>			
Action_027		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_027 - MainProgram/ThugShake - *B2(Action,Action_027)</i>			
Action_028		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_028 - MainProgram/ThugShake - *B1(Action,Action_028)</i>			
Action_029		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_029 - MainProgram/ThugShake - *B3(Action,Action_029)</i>			
Action_030		SFC_ACTION	MainProgram
Constant	No		

Action_030 (Continued)				
External Access:	Read/Write			
<i>Action_030 - MainProgram/ThugShake - *B4(Action,Action_030)</i>				
Action_031			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_031 - MainProgram/SetWindingDia - *B2(Action,Action_031)</i>				
Action_032			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_032 - MainProgram/SetWindingDia - *B3(Action,Action_032)</i>				
Action_034			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_034 - MainProgram/CONTROL - *B14(Action,Action_034)</i>				
Action_039			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_039 - MainProgram/CONTROL - *B23(Action,Action_039)</i>				
Action_040			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_040 - MainProgram/CONTROL - *B27(Action,Action_040)</i>				
Action_041			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_041 - MainProgram/CONTROL - *B38(Action,Action_041)</i>				
Action_042			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_042 - MainProgram/CONTROL - *B38(Action,Action_042)</i>				
Action_043			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_043 - MainProgram/CONTROL - *B15(Action,Action_043)</i>				
Action_044			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_044 - MainProgram/SetWindingDia - *B3(Action,Action_044)</i>				
Action_045			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_045 - MainProgram/SetWindingDia - *B1(Action,Action_045)</i>				
Action_046			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_046 - MainProgram/SetWindingDia - *B4(Action,Action_046)</i>				
Action_047			SFC_ACTION	MainProgram
Constant	No			
External Access:	Read/Write			
<i>Action_047 - MainProgram/SetWindingDia - *B6(Action,Action_047)</i>				

Action_048		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_048 - MainProgram/SetWindingDia - *B7(Action,Action_048)</i>			
Action_049		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_049 - MainProgram/CONTROL - *B13(Action,Action_049)</i>			
Action_051		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_051 - MainProgram/CONTROL - *B18(Action,Action_051)</i>			
Action_052		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_052 - MainProgram/CONTROL - *B19(Action,Action_052)</i>			
Action_053		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_053 - MainProgram/CONTROL - *B20(Action,Action_053)</i>			
Action_054		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_054 - MainProgram/CONTROL - *B6(Action,Action_054)</i>			
Action_056		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_056 - MainProgram/SetWindingDia - *B5(Action,Action_056)</i>			
Action_057		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_057 - MainProgram/CONTROL - *B16(Action,Action_057)</i>			
Action_058		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_058 - MainProgram/CONTROL - *B17(Action,Action_058)</i>			
Action_059		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_059 - MainProgram/CONTROL - *B17(Action,Action_059)</i>			
Action_060		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_060 - MainProgram/CONTROL - *B36(Action,Action_060)</i>			
Action_061		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_061 - MainProgram/CONTROL - *B37(Action,Action_061)</i>			
Action_062		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_062 - MainProgram/CONTROL - *C31(Action,Action_062)</i>			

Action_063		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_063 - MainProgram/CONTROL - *B39(Action,Action_063)</i>			
Action_064		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_064 - MainProgram/CONTROL - *B6(Action,Action_064)</i>			
Action_065		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_065 - MainProgram/CONTROL - *B7(Action,Action_065)</i>			
Action_066		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_066 - MainProgram/CONTROL - *B4(Action,Action_066)</i>			
Action_067		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_067 - MainProgram/CONTROL - *B35(Action,Action_067)</i>			
Action_068		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_068 - MainProgram/CONTROL - *C32(Action,Action_068)</i>			
Action_069		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_069 - MainProgram/CONTROL - *C32(Action,Action_069)</i>			
Action_070		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_070 - MainProgram/CONTROL - *B35(Action,Action_070)</i>			
AdhesiveHeaterDutyCycle	0.65	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AdhesiveHeaterDutyCycle - MainProgram/HeatExtruder - 4(MUL)</i>			
AdhesiveHeaterOnTime	13000	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AdhesiveHeaterOnTime - MainProgram/HeatExtruder - *4(MUL), 5(LEQ)</i>			
AdhesiveTimer		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AdhesiveTimer - MainProgram/HeatExtruder - *4(TON)</i>			
AdhesiveTimer.PRE	20000	DINT	
<i>AdhesiveTimer.PRE - MainProgram/HeatExtruder - 4(MUL)</i>			
AdhesiveTimer.ACC	0	DINT	
<i>AdhesiveTimer.ACC - MainProgram/HeatExtruder - 5(LEQ)</i>			
AdhesiveTimer.DN	0	BOOL	
<i>AdhesiveTimer.DN - MainProgram/HeatExtruder - 4(XIO)</i>			
Alarm_DOOR		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_DOOR - MainProgram/Alarms - *4(ALMD)</i>			

Alarm_ESTOP		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_ESTOP - MainProgram/Alarms - *1(ALMD)</i>			
Alarm_Manual		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Manual - MainProgram/Alarms - *5(ALMD)</i>			
Alarm_MIRCOSCAN		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_MIRCOSCAN - MainProgram/Alarms - *3(ALMD)</i>			
Alarm_NANOSCAN		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_NANOSCAN - MainProgram/Alarms - *2(ALMD)</i>			
Alarm_RobotFault		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_RobotFault - MainProgram/Alarms - *10(ALMD)</i>			
Alarm_Safety		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Safety - MainProgram/Alarms - *0(ALMD)</i>			
Alarm_Servo		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Servo - MainProgram/Alarms - *6(ALMD)</i>			
Alarm_Spool1		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Spool1 - MainProgram/Alarms - *7(ALMD)</i>			
Alarm_Spool2		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Spool2 - MainProgram/Alarms - *8(ALMD)</i>			
Alarm_Spool3		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Spool3 - MainProgram/Alarms - *9(ALMD)</i>			
AutoONS	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AutoONS - MainProgram/RobotAutoControl - *4(ONS)</i>			
CoilIndex_X	1	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>CoilIndex_X - MainProgram/CONTROL - *B1, #16(Action,Action_024 (Body)), *B23, #25(Action,Action_039 (Body)), A34, #1(Tran,Tran_036 (Condition)), B23, #17(Action,Action_039 (Body)), B23, #25(Action,Action_039 (Body)), B23, #29(Action,Action_039 (Body)), B23, #8(Action,Action_039 (Body))</i>			
<i>CoilIndex_X - MainProgram/StopAll - *#18</i>			
CoilIndex_Z	1	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>CoilIndex_Z - MainProgram/CONTROL - *B1, #17(Action,Action_024 (Body)), *B23, #21(Action,Action_039 (Body)), *B23, #24(Action,Action_039 (Body)), A34, #1(Tran,Tran_036 (Condition)), B23, #16(Action,Action_039 (Body)), B23, #17(Action,Action_039 (Body)), B23, #21(Action,Action_039 (Body)), B23, #33(Action,Action_039 (Body)), C30, #1(Tran,Tran_064 (Condition))</i>			
<i>CoilIndex_Z - MainProgram/StopAll - *#19</i>			
CoilPickupDeviation	10	DINT	MainProgram
Constant	No		
External Access:	Read/Write		

CoilPickupDeviation (Continued)			
<i>CoilPickupDeviation - MainProgram/CONTROL - B36, #3(Action,Action_060 (Body)), B39, #1(Action,Action_063 (Body))</i>			
CurrentTime	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>CurrentTime - MainProgram/FillValues - #63, #64, *#65</i>			
CurrentWindingPos	284774	DINT	MainProgram
Command position monitor[step]			
AliasFor:	AOI_Motor.Monitor.Read_Command_position(C)		
Base Tag:	AOI_Motor.Monitor.Read_Command_position(C)		
Constant	No		
External Access:	Read/Write		
<i>CurrentWindingPos - MainProgram/MoveWinding - 2(DIV), 4(NEQ)</i>			
decel	750	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>decel - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>decel - MainProgram/ThugShake - *B2, #4(Action,Action_027 (Body)), *B4, #3(Action,Action_030 (Body))</i>			
DoTheThugShake	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>DoTheThugShake - MainProgram/CONTROL - *B15, #1(Action,Action_043 (Body)), A15, #1(Tran,Tran_039 (Condition))</i>			
<i>DoTheThugShake - MainProgram/SetWindingDia - *B3, #1(Action,Action_044 (Body)), *B5, #1(Action,Action_056 (Body)), B3, #1(Tran,Tran_044 (Condition)), B5, #1(Tran,Tran_057 (Condition))</i>			
<i>DoTheThugShake - MainProgram/ThugShake - *B4, #1(Action,Action_030 (Body)), B1, #1(Tran,Tran_026 (Condition))</i>			
ENABLE	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ENABLE - MainProgram/CONTROL - A8, #1(Tran,Tran_019 (Condition))</i>			
<i>ENABLE - MainProgram/MainRoutine - 0(XIO)</i>			
<i>ENABLE - MainProgram/WindInductor - *4(OTE), 2(XIO), 3(XIO)</i>			
ExtendCutting	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ExtendCutting - MainProgram/CONTROL - *B35, #1(Action,Action_067 (Body)), *B35, #1(Action,Action_070 (Body))</i>			
<i>ExtendCutting - MainProgram/FillValues - #57</i>			
ExtruderMotor	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ExtruderMotor - MainProgram/CONTROL - *B23, #30(Action,Action_039 (Body)), *B35, #2(Action,Action_070 (Body)), *C30, #1(Action,Action_026 (Body))</i>			
<i>ExtruderMotor - MainProgram/FillValues - #58</i>			
ExtruderOn	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ExtruderOn - MainProgram/CONTROL - *B20, #5(Action,Action_053 (Body))</i>			
<i>ExtruderOn - MainProgram/HeatExtruder - *0(OTE), 0(XIC), 1(XIC), 2(XIC), 3(XIC), 4(XIC), 5(XIC)</i>			
GoodToRun	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>GoodToRun - MainProgram/MainRoutine - *2(OTE), 2(XIC), 4(XIC)</i>			
<i>GoodToRun - MainProgram/pause - #2, #5</i>			
<i>GoodToRun - MainProgram/WindInductor - 0(XIC)</i>			
GoodToRunFalling		FBD_ONESHOT	MainProgram

GoodToRunFalling (Continued)			
Constant	No		
External Access:	Read/Write		
<i>GoodToRunFalling - MainProgram/pause - *#3</i>			
GoodToRunFalling.InputBit	1	BOOL	
<i>GoodToRunFalling.InputBit - MainProgram/pause - *#2</i>			
GoodToRunFalling.OutputBit	0	BOOL	
<i>GoodToRunFalling.OutputBit - MainProgram/pause - #8</i>			
GoodToRunRising			
Constant	No	FBD_ONESHOT	MainProgram
External Access:	Read/Write		
<i>GoodToRunRising - MainProgram/pause - *#6</i>			
GoodToRunRising.InputBit	1	BOOL	
<i>GoodToRunRising.InputBit - MainProgram/pause - *#5</i>			
GoodToRunRising.OutputBit	0	BOOL	
<i>GoodToRunRising.OutputBit - MainProgram/pause - #54</i>			
HeaterAtTemp			
Constant	0	BOOL	MainProgram
External Access:	No		
<i>HeaterAtTemp - MainProgram/CONTROL - A20, #1(Tran,Tran_053 (Condition))</i>			
<i>HeaterAtTemp - MainProgram/HeatExtruder - *2(OTE), 2(XIC)</i>			
HeaterAtTempTimer			
Constant	No	TIMER	MainProgram
External Access:	Read/Write		
<i>HeaterAtTempTimer - MainProgram/HeatExtruder - *3(TON)</i>			
HeaterAtTempTimer.DN	0	BOOL	
<i>HeaterAtTempTimer.DN - MainProgram/HeatExtruder - 2(XIC)</i>			
InductorSent			
Constant	0	BOOL	MainProgram
<i>PC sets this true when the inductor data has been sent</i>			
External Access:	No		
<i>InductorSent - MainProgram/CONTROL - *B1, #18(Action,Action_024 (Body))</i>			
<i>InductorSent - MainProgram/pause - #26, *#48, *#72</i>			
<i>InductorSent - MainProgram/StopAll - *#20</i>			
<i>InductorSent - MainProgram/WindInductor - 1(XIC)</i>			
InitialLayers			
Constant	20	DINT	MainProgram
External Access:	No		
<i>InitialLayers - MainProgram/CONTROL - *B12, #6(Action,Action_023 (Body))</i>			
<i>InitialLayers - MainProgram/FillValues - #69, #69</i>			
Layers			
Constant	20	DINT	MainProgram
External Access:	No		
<i>Layers - MainProgram/CONTROL - *B12, #5(Action,Action_023 (Body)), B12, #6(Action,Action_023 (Body))</i>			
<i>Layers - MainProgram/FillValues - #69</i>			
LoadCommand			
Constant	0	BOOL	MainProgram
External Access:	No		
<i>LoadCommand - MainProgram/WindInductor - *3(OTE)</i>			
OldTime			
Constant	0	DINT	MainProgram
External Access:	No		
<i>OldTime - MainProgram/FillValues - #63, *#64</i>			
prog			
Constant	0.0	REAL	MainProgram
Constant	No		

prog (Continued)			
External Access:	Read/Write		
<i>prog - MainProgram/FillValues - #70, *#69</i>			
proggy	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>proggy - MainProgram/FillValues - #71, *#70</i>			
Ready	0	BOOL	MainProgram
Bit for PC "I am ready for the next inductor"			
Constant	No		
External Access:	Read/Write		
<i>Ready - MainProgram/WindInductor - *0(OTE), 0(XIC), 1(XIC)</i>			
RequiredTemp	22.5	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>RequiredTemp - MainProgram/HeatExtruder - 1(LEQ), 2(GEQ)</i>			
RobotAutoAutoTimer		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>RobotAutoAutoTimer - MainProgram/RobotAutoControl - *0(TON)</i>			
RobotAutoAutoTimer.DN	1	BOOL	
<i>RobotAutoAutoTimer.DN - MainProgram/RobotAutoControl - 2(XIC)</i>			
RobotAutoStep	2	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
RobotAutoStep.0	0	BOOL	
Fault Reset			
<i>RobotAutoStep.0 - MainProgram/RobotAutoControl - *4(OTE), 4(XIC), 5(XIC), 8(XIC)</i>			
RobotAutoStep.1	1	BOOL	
Start Program			
<i>RobotAutoStep.1 - MainProgram/RobotAutoControl - *5(OTE), 4(XIO), 5(XIC), 6(XIC), 9(XIC)</i>			
RobotAutoStep.2	0	BOOL	
Robot Program Running			
<i>RobotAutoStep.2 - MainProgram/RobotAutoControl - *6(OTE), 5(XIO), 6(XIC), 7(XIC)</i>			
RobotAutoStep.3	0	BOOL	
Robot Program Done			
<i>RobotAutoStep.3 - MainProgram/RobotAutoControl - *7(OTE), 10(XIC), 6(XIO), 7(XIC)</i>			
RobotEnable	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>RobotEnable - MainProgram/FillValues - *#3</i>			
<i>RobotEnable - MainProgram/pause - #29, *#51, *#75</i>			
<i>RobotEnable - MainProgram/RobotAutoControl - *0(OTL), *1(OTU), 0(XIC), 3(XIC)</i>			
RobotStartAuto	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>RobotStartAuto - MainProgram/RobotAutoControl - *2(OTE), 4(XIC)</i>			
SettingDia	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>SettingDia - MainProgram/CONTROL - A14, #1(Tran,Tran_049 (Condition))</i>			
<i>SettingDia - MainProgram/SetWindingDia - *B1, #2(Action,Action_045 (Body)), *B2, #1(Action,Action_031 (Body)), *B7, #2(Action,Action_048 (Body))</i>			
SolenoidAlarm		TIMER	MainProgram
Constant	No		

SolenoidAlarm (Continued)			
External Access:	Read/Write		
<i>SolenoidAlarm - MainProgram/MainRoutine - *8(TON)</i>			
SolenoidAlarm.DN	0	BOOL	
<i>SolenoidAlarm.DN - MainProgram/MainRoutine - 9(XIC)</i>			
speed	750	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>speed - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
StartHeating	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>StartHeating - MainProgram/HeatExtruder - 0(XIC)</i>			
Step_021		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_021 - MainProgram/CONTROL - *A1(Step,Step_021)</i>			
<i>Step_021 - MainProgram/MainRoutine - 0(SFR)</i>			
Step_021.T	201	DINT	
<i>Step_021.T - MainProgram/CONTROL - A4, #1(Tran,Tran_073 (Condition))</i>			
Step_022		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
Step_022.T	0	DINT	
<i>Step_022.T - MainProgram/WindInductor - 4(LES)</i>			
Step_023		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_023 - MainProgram/CONTROL - *A11(Step,Step_023)</i>			
Step_024		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_024 - MainProgram/CONTROL - *A11(Step,Step_024)</i>			
Step_025		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_025 - MainProgram/CONTROL - *A12(Step,Step_025)</i>			
Step_028		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_028 - MainProgram/CONTROL - *A21(Step,Step_028)</i>			
Step_028.T	1501	DINT	
<i>Step_028.T - MainProgram/CONTROL - A21, #1(Tran,Tran_042 (Condition))</i>			
Step_029		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_029 - MainProgram/CONTROL - *A28(Step,Step_029)</i>			
Step_029.T	4017	DINT	
<i>Step_029.T - MainProgram/CONTROL - A28, #1(Tran,Tran_038 (Condition))</i>			
Step_030		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_030 - MainProgram/ThugShake - *B1(Step,Step_030)</i>			

Step_031		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_031 - MainProgram/ThugShake - *B2(Step,Step_031)</i>			
Step_031.T	140	DINT	
<i>Step_031.T - MainProgram/ThugShake - B2, #1(Tran,Tran_027 (Condition))</i>			
Step_032		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_032 - MainProgram/ThugShake - *B3(Step,Step_032)</i>			
Step_032.T	0	DINT	
<i>Step_032.T - MainProgram/ThugShake - C4, #1(Tran,Tran_029 (Condition))</i>			
Step_033		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_033 - MainProgram/ThugShake - *B4(Step,Step_033)</i>			
Step_034		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_034 - MainProgram/SetWindingDia - *B1(Step,Step_034)</i>			
Step_037		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_037 - MainProgram/CONTROL - *A23(Step,Step_037)</i>			
Step_037.T	11	DINT	
<i>Step_037.T - MainProgram/CONTROL - A28, #1(Tran,Tran_025 (Condition))</i>			
Step_038		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_038 - MainProgram/CONTROL - *A38(Step,Step_038)</i>			
Step_040		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_040 - MainProgram/CONTROL - *C30(Step,Step_040)</i>			
Step_040.T	901	DINT	
<i>Step_040.T - MainProgram/CONTROL - C31, #1(Tran,Tran_066 (Condition))</i>			
Step_041		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_041 - MainProgram/CONTROL - *A14(Step,Step_041)</i>			
Step_041.T	101	DINT	
<i>Step_041.T - MainProgram/CONTROL - A14, #1(Tran,Tran_045 (Condition))</i>			
Step_042		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_042 - MainProgram/SetWindingDia - *B2(Step,Step_042)</i>			
Step_042.T	501	DINT	
<i>Step_042.T - MainProgram/SetWindingDia - B2, #1(Tran,Tran_035 (Condition))</i>			
Step_043		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_043 - MainProgram/SetWindingDia - *B3(Step,Step_043)</i>			
Step_043.T	4120	DINT	
<i>Step_043.T - MainProgram/SetWindingDia - B4, #1(Tran,Tran_041 (Condition))</i>			

Step_045		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_045 - MainProgram/CONTROL - *A15(Step,Step_045)</i>			
Step_046		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_046 - MainProgram/SetWindingDia - *B3(Step,Step_046)</i>			
Step_047		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_047 - MainProgram/SetWindingDia - *B4(Step,Step_047)</i>			
Step_047.T	101	DINT	
<i>Step_047.T - MainProgram/SetWindingDia - B5, #1(Tran,Tran_046 (Condition))</i>			
Step_048		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_048 - MainProgram/SetWindingDia - *B6(Step,Step_048)</i>			
Step_048.T	629	DINT	
<i>Step_048.T - MainProgram/SetWindingDia - B6, #1(Tran,Tran_047 (Condition))</i>			
Step_049		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_049 - MainProgram/SetWindingDia - *B7(Step,Step_049)</i>			
Step_050		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_050 - MainProgram/CONTROL - *A13(Step,Step_050)</i>			
Step_050.T	101	DINT	
<i>Step_050.T - MainProgram/CONTROL - A14, #1(Tran,Tran_049 (Condition))</i>			
Step_052		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_052 - MainProgram/CONTROL - *A18(Step,Step_052)</i>			
Step_053		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_053 - MainProgram/CONTROL - *A19(Step,Step_053)</i>			
Step_053.T	1001	DINT	
<i>Step_053.T - MainProgram/CONTROL - A19, #1(Tran,Tran_052 (Condition))</i>			
Step_054		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_054 - MainProgram/CONTROL - *A20(Step,Step_054)</i>			
Step_054.T	180000	DINT	
<i>Step_054.T - MainProgram/CONTROL - A20, #1(Tran,Tran_053 (Condition))</i>			
Step_056		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_056 - MainProgram/CONTROL - *A37(Step,Step_056)</i>			
Step_056.T	1500	DINT	
<i>Step_056.T - MainProgram/CONTROL - A38, #1(Tran,Tran_055 (Condition))</i>			
Step_057		SFC_STEP	MainProgram
Constant	No		

Step_057 (Continued)			
External Access:	Read/Write		
<i>Step_057 - MainProgram/SetWindingDia - *B5(Step,Step_057)</i>			
Step_059		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_059 - MainProgram/CONTROL - *A16(Step,Step_059)</i>			
Step_059.T	2001	DINT	
<i>Step_059.T - MainProgram/CONTROL - A16, #1(Tran,Tran_058 (Condition))</i>			
Step_060		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_060 - MainProgram/CONTROL - *A17(Step,Step_060)</i>			
Step_060.T	101	DINT	
<i>Step_060.T - MainProgram/CONTROL - A17, #1(Tran,Tran_059 (Condition))</i>			
Step_061		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_061 - MainProgram/CONTROL - *A17(Step,Step_061)</i>			
Step_061.T	901	DINT	
<i>Step_061.T - MainProgram/CONTROL - A18, #1(Tran,Tran_060 (Condition))</i>			
Step_062		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_062 - MainProgram/CONTROL - *A35(Step,Step_062)</i>			
Step_062.T	101	DINT	
<i>Step_062.T - MainProgram/CONTROL - A35, #1(Tran,Tran_062 (Condition))</i>			
Step_063		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_063 - MainProgram/CONTROL - *A8(Step,Step_063)</i>			
Step_064		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_064 - MainProgram/CONTROL - *C9(Step,Step_064)</i>			
Step_065		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_065 - MainProgram/CONTROL - *A30(Step,Step_065)</i>			
Step_066		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_066 - MainProgram/CONTROL - *C31(Step,Step_066)</i>			
Step_066.T	101	DINT	
<i>Step_066.T - MainProgram/CONTROL - C31, #1(Tran,Tran_067 (Condition))</i>			
Step_067		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_067 - MainProgram/CONTROL - *A29(Step,Step_067)</i>			
Step_067.T	1	DINT	
<i>Step_067.T - MainProgram/CONTROL - A30, #1(Tran,Tran_065 (Condition))</i>			
Step_068		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		

Step_068 (Continued)			
<i>Step_068 - MainProgram/CONTROL - *A33(Step,Step_068)</i>			
Step_068.T	0	DINT	
<i>Step_068.T - MainProgram/CONTROL - F34, #1(Tran,Tran_040 (Condition))</i>			
Step_069		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_069 - MainProgram/CONTROL - *A6(Step,Step_069)</i>			
Step_069.T	101	DINT	
<i>Step_069.T - MainProgram/CONTROL - A6, #1(Tran,Tran_072 (Condition))</i>			
Step_070		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_070 - MainProgram/CONTROL - *A36(Step,Step_070)</i>			
Step_071		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_071 - MainProgram/CONTROL - *A39(Step,Step_071)</i>			
Step_072		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_072 - MainProgram/CONTROL - *A6(Step,Step_072)</i>			
Step_072.T	251	DINT	
<i>Step_072.T - MainProgram/CONTROL - A7, #1(Tran,Tran_071 (Condition))</i>			
Step_073		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_073 - MainProgram/CONTROL - *A7(Step,Step_073)</i>			
Step_074		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_074 - MainProgram/CONTROL - *A4(Step,Step_074)</i>			
Step_074.T	1201	DINT	
<i>Step_074.T - MainProgram/CONTROL - A5, #1(Tran,Tran_068 (Condition))</i>			
Step_075		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_075 - MainProgram/CONTROL - *A35(Step,Step_075)</i>			
Step_075.T	701	DINT	
<i>Step_075.T - MainProgram/CONTROL - A36, #1(Tran,Tran_074 (Condition))</i>			
Step_076		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_076 - MainProgram/CONTROL - *C32(Step,Step_076)</i>			
Step_076.T	501	DINT	
<i>Step_076.T - MainProgram/CONTROL - C32, #1(Tran,Tran_075 (Condition))</i>			
Step_077		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_077 - MainProgram/CONTROL - *C32(Step,Step_077)</i>			
Step_077.T	101	DINT	
<i>Step_077.T - MainProgram/CONTROL - C33, #1(Tran,Tran_076 (Condition))</i>			
STEP_CoilKnown	0	BOOL	MainProgram
Constant	No		

STEP_CoilKnown (Continued)			
External Access:	Read/Write		
<i>STEP_CoilKnown - MainProgram/WindInductor - *1(OTE), 0(XIO), 1(XIC), 2(XIC), 3(XIC)</i>			
Step_GettingSpoolCurrent	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_GettingSpoolCurrent - MainProgram/WindInductor - *3(OTE), 0(XIO), 1(XIO), 3(XIC), 4(XIC)</i>			
STEP_OperatorSwapsSpool	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>STEP_OperatorSwapsSpool - MainProgram/WindInductor - *2(OTE), 0(XIO), 1(XIO), 2(XIC), 4(XIC)</i>			
StopHeating	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>StopHeating - MainProgram/CONTROL - *B20, #4(Action,Action_053 (Body)), *B35, #3(Action,Action_070 (Body))</i>			
<i>StopHeating - MainProgram/HeatExtruder - 0(XIO), 2(XIO), 3(XIO)</i>			
SystemState	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>SystemState - MainProgram/MainRoutine - *1(MOV)</i>			
SystemState.0	0	BOOL	
<i>SystemState.0 - MainProgram/pause - #55, *#9</i>			
SystemState.1	0	BOOL	
<i>SystemState.1 - MainProgram/pause - #56, *#10</i>			
SystemState.2	0	BOOL	
<i>SystemState.2 - MainProgram/pause - #57, *#11</i>			
SystemState.3	0	BOOL	
<i>SystemState.3 - MainProgram/pause - #58, *#12</i>			
SystemState.4	0	BOOL	
<i>SystemState.4 - MainProgram/pause - #59, *#13</i>			
SystemState.5	0	BOOL	
<i>SystemState.5 - MainProgram/pause - #60, *#14</i>			
SystemState.6	0	BOOL	
<i>SystemState.6 - MainProgram/pause - #61, *#15</i>			
SystemState.7	0	BOOL	
<i>SystemState.7 - MainProgram/pause - #62, *#16</i>			
SystemState.8	0	BOOL	
<i>SystemState.8 - MainProgram/pause - #63, *#17</i>			
SystemState.9	0	BOOL	
<i>SystemState.9 - MainProgram/pause - #64, *#18</i>			
SystemState.10	0	BOOL	
<i>SystemState.10 - MainProgram/pause - #65, *#19</i>			
SystemState.11	0	BOOL	
<i>SystemState.11 - MainProgram/pause - #66, *#20</i>			
SystemState.12	0	BOOL	
<i>SystemState.12 - MainProgram/pause - #67, *#21</i>			
SystemState.13	0	BOOL	
<i>SystemState.13 - MainProgram/pause - #68, *#22</i>			
SystemState.14	0	BOOL	
<i>SystemState.14 - MainProgram/pause - #69, *#23</i>			
SystemState.15	0	BOOL	
<i>SystemState.15 - MainProgram/pause - #70, *#24</i>			
SystemState.16	0	BOOL	
<i>SystemState.16 - MainProgram/pause - #71, *#25</i>			
SystemState.17	0	BOOL	
<i>SystemState.17 - MainProgram/pause - #72, *#26</i>			
SystemState.18	0	BOOL	
<i>SystemState.18 - MainProgram/pause - #73, *#27</i>			
SystemState.19	0	BOOL	
<i>SystemState.19 - MainProgram/pause - #74, *#28</i>			

SystemState (Continued)			
SystemState.20	0	BOOL	
<i>SystemState.20 - MainProgram/pause - #75, *#29</i>			
TempDint	100	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>TempDint - MainProgram/CONTROL - *B1, #22(Action,Action_024 (Body)), *B37, #3(Action,Action_061 (Body)), B1, #23(Action,Action_024 (Body)), B37, #4(Action,Action_061 (Body))</i>			
<i>TempDint - MainProgram/SetWindingDia - *B6, #1(Action,Action_047 (Body)), B6, #2(Action,Action_047 (Body))</i>			
ThugShakeIndex	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ThugShakeIndex - MainProgram/ThugShake - *B1, #1(Action,Action_028 (Body)), *B2, #1(Action,Action_027 (Body)), B2, #1(Action,Action_027 (Body)), B4, #1(Tran,Tran_028 (Condition)), C4, #1(Tran,Tran_029 (Condition))</i>			
timeout		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>timeout - MainProgram/selfdestruct - *0(TON)</i>			
timeout.DN	0	BOOL	
<i>timeout.DN - MainProgram/selfdestruct - 0(XIC), 1(XIC)</i>			
Tran_019	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_019 - MainProgram/CONTROL - *A8(Tran,Tran_019), *A8, #1(Tran,Tran_019 (Condition))</i>			
Tran_020	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_020 - MainProgram/CONTROL - *A9(Tran,Tran_020), *A9, #1(Tran,Tran_020 (Condition))</i>			
Tran_021	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_021 - MainProgram/CONTROL - *A11(Tran,Tran_021), *A11, #1(Tran,Tran_021 (Condition))</i>			
Tran_022	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_022 - MainProgram/CONTROL - *A12(Tran,Tran_022), *A12, #1(Tran,Tran_022 (Condition))</i>			
Tran_025	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_025 - MainProgram/CONTROL - *A28(Tran,Tran_025), *A28, #1(Tran,Tran_025 (Condition))</i>			
Tran_026	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_026 - MainProgram/ThugShake - *B1(Tran,Tran_026), *B1, #1(Tran,Tran_026 (Condition))</i>			
Tran_027	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_027 - MainProgram/ThugShake - *B2(Tran,Tran_027), *B2, #1(Tran,Tran_027 (Condition))</i>			
Tran_028	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_028 - MainProgram/ThugShake - *B4(Tran,Tran_028), *B4, #1(Tran,Tran_028 (Condition))</i>			

Tran_029	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_029 - MainProgram/ThugShake - *C4(Tran,Tran_029), *C4, #1(Tran,Tran_029 (Condition))</i>			
Tran_030	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_030 - MainProgram/ThugShake - *B4(Tran,Tran_030), *B4, #1(Tran,Tran_030 (Condition))</i>			
Tran_031	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_031 - MainProgram/SetWindingDia - *B1(Tran,Tran_031), *B1, #1(Tran,Tran_031 (Condition))</i>			
Tran_034	1	BOOL	MainProgram
Ready to wind			
Constant	No		
External Access:	Read/Write		
<i>Tran_034 - MainProgram/CONTROL - *A13(Tran,Tran_034), *A13, #1(Tran,Tran_034 (Condition))</i>			
Tran_035	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_035 - MainProgram/SetWindingDia - *B2(Tran,Tran_035), *B2, #1(Tran,Tran_035 (Condition))</i>			
Tran_036	1	BOOL	MainProgram
Done Winding			
Constant	No		
External Access:	Read/Write		
<i>Tran_036 - MainProgram/CONTROL - *A34(Tran,Tran_036), *A34, #1(Tran,Tran_036 (Condition))</i>			
Tran_037	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_037 - MainProgram/CONTROL - *A39(Tran,Tran_037), *A39, #1(Tran,Tran_037 (Condition))</i>			
Tran_038	1	BOOL	MainProgram
robot takes ~ 500ms and wait until winding is done moving			
Constant	No		
External Access:	Read/Write		
<i>Tran_038 - MainProgram/CONTROL - *A28(Tran,Tran_038), *A28, #1(Tran,Tran_038 (Condition))</i>			
Tran_039	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_039 - MainProgram/CONTROL - *A15(Tran,Tran_039), *A15, #1(Tran,Tran_039 (Condition))</i>			
Tran_040	1	BOOL	MainProgram
Still winding, allow time for done winding to be true before going here			
Constant	No		
External Access:	Read/Write		
<i>Tran_040 - MainProgram/CONTROL - *F34(Tran,Tran_040), *F34, #1(Tran,Tran_040 (Condition))</i>			
Tran_041	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_041 - MainProgram/SetWindingDia - *B4(Tran,Tran_041), *B4, #1(Tran,Tran_041 (Condition))</i>			
Tran_042	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_042 - MainProgram/CONTROL - *A21(Tran,Tran_042), *A21, #1(Tran,Tran_042 (Condition))</i>			

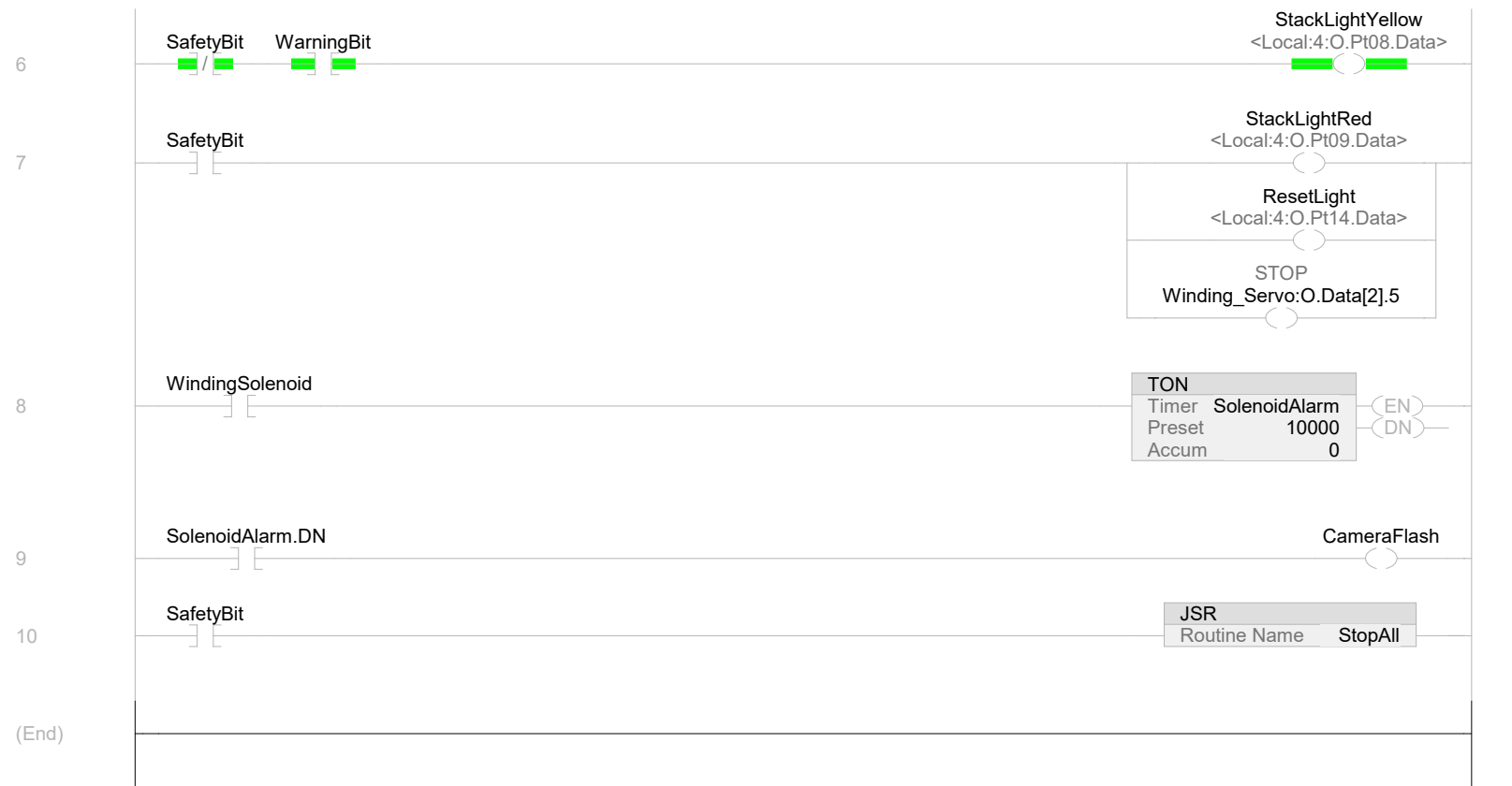
Tran_044	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_044 - MainProgram/SetWindingDia - *B3(Tran,Tran_044), *B3, #1(Tran,Tran_044 (Condition))</i>			
Tran_045	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_045 - MainProgram/CONTROL - *A14(Tran,Tran_045), *A14, #1(Tran,Tran_045 (Condition))</i>			
Tran_046	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_046 - MainProgram/SetWindingDia - *B5(Tran,Tran_046), *B5, #1(Tran,Tran_046 (Condition))</i>			
Tran_047	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_047 - MainProgram/SetWindingDia - *B6(Tran,Tran_047), *B6, #1(Tran,Tran_047 (Condition))</i>			
Tran_048	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_048 - MainProgram/SetWindingDia - *B7(Tran,Tran_048), *B7, #1(Tran,Tran_048 (Condition))</i>			
Tran_049	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_049 - MainProgram/CONTROL - *A14(Tran,Tran_049), *A14, #1(Tran,Tran_049 (Condition))</i>			
Tran_051	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_051 - MainProgram/CONTROL - *A19(Tran,Tran_051), *A19, #1(Tran,Tran_051 (Condition))</i>			
Tran_052	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_052 - MainProgram/CONTROL - *A19(Tran,Tran_052), *A19, #1(Tran,Tran_052 (Condition))</i>			
Tran_053	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_053 - MainProgram/CONTROL - *A20(Tran,Tran_053), *A20, #1(Tran,Tran_053 (Condition))</i>			
Tran_054	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_054 - MainProgram/CONTROL - *A30(Tran,Tran_054), *A30, #1(Tran,Tran_054 (Condition))</i>			
Tran_055	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_055 - MainProgram/CONTROL - *A38(Tran,Tran_055), *A38, #1(Tran,Tran_055 (Condition))</i>			
Tran_057	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_057 - MainProgram/SetWindingDia - *B5(Tran,Tran_057), *B5, #1(Tran,Tran_057 (Condition))</i>			
Tran_058	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_058 - MainProgram/CONTROL - *A16(Tran,Tran_058), *A16, #1(Tran,Tran_058 (Condition))</i>			

Tran_059	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_059 - MainProgram/CONTROL - *A17(Tran,Tran_059), *A17, #1(Tran,Tran_059 (Condition))</i>			
Tran_060	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_060 - MainProgram/CONTROL - *A18(Tran,Tran_060), *A18, #1(Tran,Tran_060 (Condition))</i>			
Tran_061	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_061 - MainProgram/CONTROL - *C9(Tran,Tran_061), *C9, #1(Tran,Tran_061 (Condition))</i>			
Tran_062	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_062 - MainProgram/CONTROL - *A35(Tran,Tran_062), *A35, #1(Tran,Tran_062 (Condition))</i>			
Tran_063	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_063 - MainProgram/CONTROL - *C10(Tran,Tran_063), *C10, #1(Tran,Tran_063 (Condition))</i>			
Tran_064	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_064 - MainProgram/CONTROL - *C30(Tran,Tran_064), *C30, #1(Tran,Tran_064 (Condition))</i>			
Tran_065	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_065 - MainProgram/CONTROL - *A30(Tran,Tran_065), *A30, #1(Tran,Tran_065 (Condition))</i>			
Tran_066	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_066 - MainProgram/CONTROL - *C31(Tran,Tran_066), *C31, #1(Tran,Tran_066 (Condition))</i>			
Tran_067	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_067 - MainProgram/CONTROL - *C31(Tran,Tran_067), *C31, #1(Tran,Tran_067 (Condition))</i>			
Tran_068	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_068 - MainProgram/CONTROL - *A5(Tran,Tran_068), *A5, #1(Tran,Tran_068 (Condition))</i>			
Tran_069	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_069 - MainProgram/CONTROL - *A37(Tran,Tran_069), *A37, #1(Tran,Tran_069 (Condition))</i>			
Tran_070	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_070 - MainProgram/CONTROL - *A39(Tran,Tran_070), *A39, #1(Tran,Tran_070 (Condition))</i>			
Tran_071	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_071 - MainProgram/CONTROL - *A7(Tran,Tran_071), *A7, #1(Tran,Tran_071 (Condition))</i>			

Tran_072	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_072 - MainProgram/CONTROL - *A6(Tran,Tran_072), *A6, #1(Tran,Tran_072 (Condition))</i>			
Tran_073	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_073 - MainProgram/CONTROL - *A4(Tran,Tran_073), *A4, #1(Tran,Tran_073 (Condition))</i>			
Tran_074	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_074 - MainProgram/CONTROL - *A36(Tran,Tran_074), *A36, #1(Tran,Tran_074 (Condition))</i>			
Tran_075	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_075 - MainProgram/CONTROL - *C32(Tran,Tran_075), *C32, #1(Tran,Tran_075 (Condition))</i>			
Tran_076	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_076 - MainProgram/CONTROL - *C33(Tran,Tran_076), *C33, #1(Tran,Tran_076 (Condition))</i>			
tRobotFaultReset		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>tRobotFaultReset - MainProgram/RobotAutoControl - *8(TON)</i>			
tRobotFaultReset.TT	0	BOOL	
<i>tRobotFaultReset.TT - MainProgram/RobotAutoControl - 11(XIC)</i>			
tRobotProgDone		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>tRobotProgDone - MainProgram/RobotAutoControl - *10(TON)</i>			
tRobotProgDone.DN	0	BOOL	
<i>tRobotProgDone.DN - MainProgram/RobotAutoControl - 3(XIO)</i>			
tRobotStart		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>tRobotStart - MainProgram/RobotAutoControl - *9(TON)</i>			
tRobotStart.TT	0	BOOL	
<i>tRobotStart.TT - MainProgram/RobotAutoControl - 12(XIC)</i>			
TurnsToStepsConstant	2857.25	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>TurnsToStepsConstant - MainProgram/MoveWinding - 2(DIV), 3(MUL)</i>			
Winding		ServoMotor	MainProgram
Constant	No		
External Access:	Read/Write		
Winding.POS	99.667	REAL	
request angle relative to current pos			
<i>Winding.POS - MainProgram/MoveWinding - *0(SBR), 3(MUL)</i>			
Winding.SPEED	750	DINT	
deg/s			
<i>Winding.SPEED - MainProgram/MoveWinding - *0(SBR), 3(MOV)</i>			
Winding.ACCEL	300	DINT	
deg/s/s			
<i>Winding.ACCEL - MainProgram/MoveWinding - *0(SBR), 3(MOV)</i>			
Winding.DECCEL	750	DINT	

Winding (Continued)			
deg/s/s			
<i>Winding.DECCEL - MainProgram/MoveWinding - *0(SBR), 3(MOV)</i>			
WindingCurrentDia	170	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingCurrentDia - MainProgram/CONTROL - *B13, #1(Action,Action_049 (Body)), *B36, #3(Action,Action_060 (Body)), *B39, #1(Action,Action_063 (Body)), A37, #1(Tran,Tran_069 (Condition)), B36, #3(Action,Action_060 (Body)), B39, #1(Action,Action_063 (Body))</i>			
<i>WindingCurrentDia - MainProgram/SetWindingDia - B1, #1(Tran,Tran_031 (Condition)), B3, #1(Action,Action_032 (Body)), B7, #1(Action,Action_048 (Body))</i>			
WindingFree	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingFree - MainProgram/FillValues - #7</i>			
WindingMoving	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingMoving - MainProgram/CONTROL - A28, #1(Tran,Tran_038 (Condition)), A34, #1(Tran,Tran_036 (Condition)), A38, #1(Tran,Tran_055 (Condition)), A39, #1(Tran,Tran_070 (Condition)), A4, #1(Tran,Tran_073 (Condition))</i>			
<i>WindingMoving - MainProgram/MoveWinding - *7(OTE)</i>			
<i>WindingMoving - MainProgram/SetWindingDia - B6, #1(Tran,Tran_047 (Condition))</i>			
WindingOldDia	170	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingOldDia - MainProgram/CONTROL - A37, #1(Tran,Tran_069 (Condition))</i>			
<i>WindingOldDia - MainProgram/SetWindingDia - *B7, #1(Action,Action_048 (Body)), B1, #1(Tran,Tran_031 (Condition)), B3, #1(Action,Action_032 (Body))</i>			
WindingTurns	99.667	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingTurns - MainProgram/CONTROL - *B1, #23(Action,Action_024 (Body)), *B23, #6(Action,Action_039 (Body)), *B37, #4(Action,Action_061 (Body)), *B37, #5(Action,Action_061 (Body)), B1, #22(Action,Action_024 (Body)), B23, #6(Action,Action_039 (Body)), B37, #3(Action,Action_061 (Body)), B37, #5(Action,Action_061 (Body))</i>			
<i>WindingTurns - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>WindingTurns - MainProgram/MoveWinding - *2(DIV)</i>			
<i>WindingTurns - MainProgram/SetWindingDia - *B3, #1(Action,Action_032 (Body)), *B6, #2(Action,Action_047 (Body)), B3, #1(Action,Action_032 (Body)), B6, #1(Action,Action_047 (Body))</i>			
<i>WindingTurns - MainProgram/ThugShake - *B2, #2(Action,Action_027 (Body)), *B3, #1(Action,Action_029 (Body)), B2, #2(Action,Action_027 (Body)), B3, #1(Action,Action_029 (Body))</i>			
WireInRobot	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WireInRobot - MainProgram/CONTROL - *B1, #8(Action,Action_024 (Body))</i>			
<i>WireInRobot - MainProgram/pause - #19, *#41, *#65</i>			
<i>WireInRobot - MainProgram/StopAll - *#11</i>			
wireLoadDist	3000	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>wireLoadDist - MainProgram/FillValues - *#67</i>			





Name	Value	Data Type	Scope
acel	300	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>acel - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>acel - MainProgram/ThugShake - *B2, #3(Action,Action_027 (Body)), *B4, #2(Action,Action_030 (Body))</i>			
CameraFlash	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>CameraFlash - MainProgram/FillValues - #47</i>			
<i>CameraFlash - MainProgram/MainRoutine - *9(OTE)</i>			
decel	750	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>decel - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>decel - MainProgram/ThugShake - *B2, #4(Action,Action_027 (Body)), *B4, #3(Action,Action_030 (Body))</i>			
ENABLE	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ENABLE - MainProgram/CONTROL - A8, #1(Tran,Tran_019 (Condition))</i>			
<i>ENABLE - MainProgram/MainRoutine - 0(XIO)</i>			
<i>ENABLE - MainProgram/WindInductor - *4(OTE), 2(XIO), 3(XIO)</i>			
GoodToRun	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>GoodToRun - MainProgram/MainRoutine - *2(OTE), 2(XIC), 4(XIC)</i>			
<i>GoodToRun - MainProgram/pause - #2, #5</i>			
<i>GoodToRun - MainProgram/WindInductor - 0(XIC)</i>			
HMI		HMIControl	Capstone_Test
Constant	No		
External Access:	Read/Write		
HMI.Manual.Spool.FWD1	0	BOOL	
<i>HMI.Manual.Spool.FWD1 - MainProgram/FillValues - #40</i>			
HMI.Manual.Spool.FWD2	0	BOOL	
<i>HMI.Manual.Spool.FWD2 - MainProgram/FillValues - #42</i>			
HMI.Manual.Spool.FWD3	0	BOOL	
<i>HMI.Manual.Spool.FWD3 - MainProgram/FillValues - #44</i>			
HMI.Manual.Spool.REV1	0	BOOL	
<i>HMI.Manual.Spool.REV1 - MainProgram/FillValues - #41</i>			
HMI.Manual.Spool.REV2	0	BOOL	
<i>HMI.Manual.Spool.REV2 - MainProgram/FillValues - #43</i>			
HMI.Manual.Spool.REV3	0	BOOL	
<i>HMI.Manual.Spool.REV3 - MainProgram/FillValues - #45</i>			
HMI.Manual.Cut	0	BOOL	
<i>HMI.Manual.Cut - MainProgram/FillValues - #57</i>			
HMI.Manual.CameraLight	0	BOOL	
<i>HMI.Manual.CameraLight - MainProgram/FillValues - #47</i>			
HMI.Manual.ExtrudeHeater	0	BOOL	
<i>HMI.Manual.ExtrudeHeater - MainProgram/FillValues - #59</i>			
HMI.Manual.ExtrudeMotor	0	BOOL	
<i>HMI.Manual.ExtrudeMotor - MainProgram/FillValues - #58</i>			
HMI.Manual.WindingSolenoid	0	BOOL	
<i>HMI.Manual.WindingSolenoid - MainProgram/FillValues - #60</i>			
HMI.Manual.ClampMotorFWD	0	BOOL	
<i>HMI.Manual.ClampMotorFWD - MainProgram/FillValues - #50</i>			
HMI.Manual.ClampMotorREV	0	BOOL	
<i>HMI.Manual.ClampMotorREV - MainProgram/FillValues - #51</i>			
HMI.Manual.HeightMotorUp	0	BOOL	
<i>HMI.Manual.HeightMotorUp - MainProgram/FillValues - #54</i>			

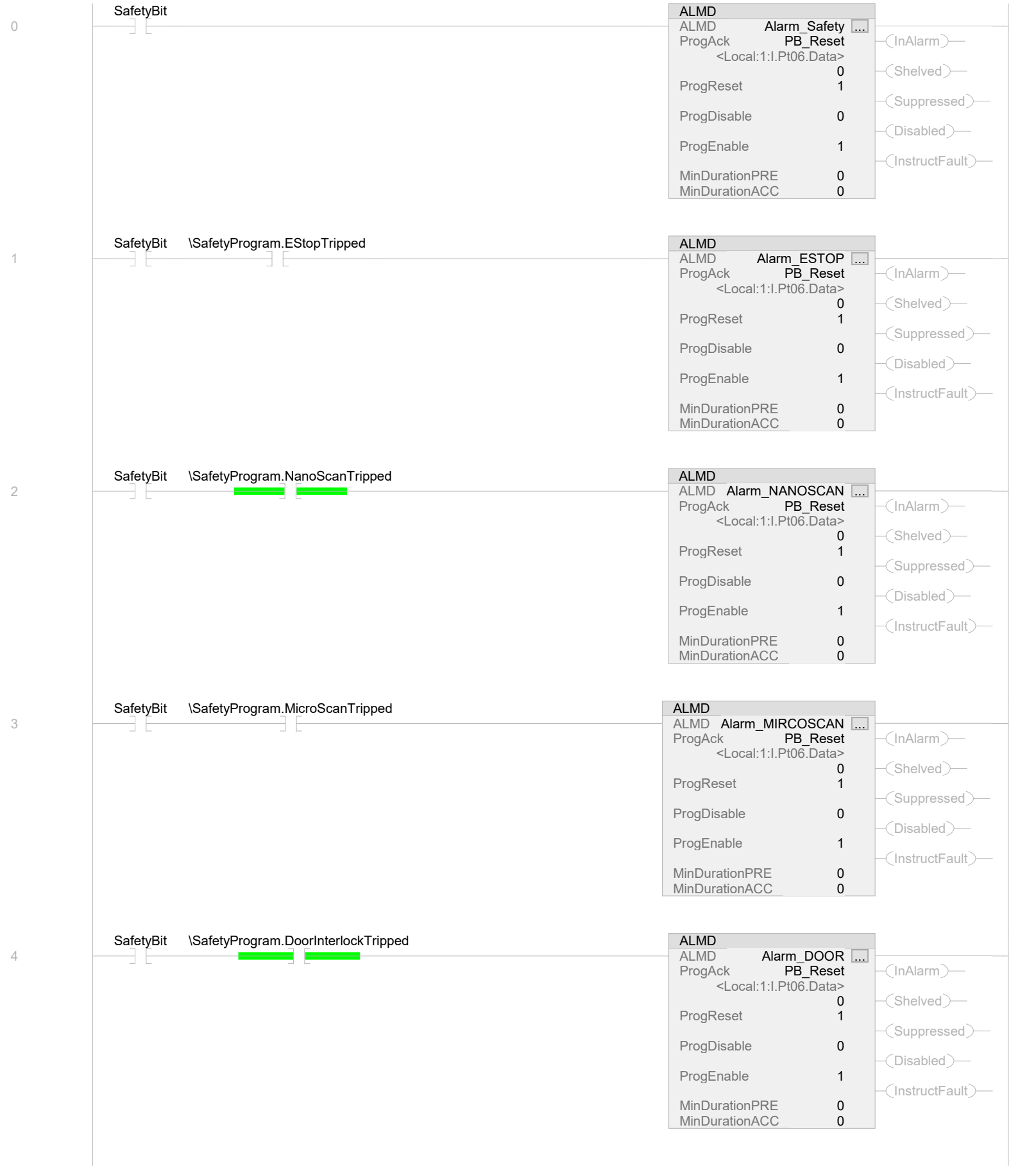
HMI (Continued)			
HMI.Manual.HeightMotorDown	0	BOOL	
HMI.Manual.HeightMotorDown - MainProgram/FillValues - #53			
HMI.StartPB	0	BOOL	
HMI.StartPB - MainProgram/MainRoutine - 2(XIC)			
HMI.ResetPB	0	BOOL	
HMI.ResetPB - MainProgram/MainRoutine - 0(XIC)			
HMI.StopPB	0	BOOL	
HMI.StopPB - MainProgram/MainRoutine - 2(XIO)			
HMI.StopPB - MainProgram/pause - #8			
HMI.RobotFaultReset	0	BOOL	
HMI.RobotFaultReset - MainProgram/RobotAutoControl - 8(XIC)			
 MANLY	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
MANLY - MainProgram/Alarms - 5(XIC)			
MANLY - MainProgram/MainRoutine - 0(XIO), 4(XIO)			
 PB_Start	0	BOOL	Capstone_Test
StartPB			
AliasFor: Input_RemoteIO:I.Data.0			
Base Tag: Input_RemoteIO:I.Data.0			
Constant No			
External Access: Read/Write			
PB_Start - MainProgram/MainRoutine - 2(XIC)			
 ResetLight	0	BOOL	Capstone_Test
AliasFor: Local:4:O.Pt14.Data			
Base Tag: Local:4:O.Pt14.Data			
Constant No			
External Access: Read/Write			
ResetLight - MainProgram/MainRoutine - *7(OTE)			
 SafetyBit	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
SafetyBit - MainProgram/Alarms - 0(XIC), 1(XIC), 2(XIC), 3(XIC), 4(XIC)			
SafetyBit - MainProgram/MainRoutine - 10(XIC), 4(XIO), 5(XIO), 6(XIO), 7(XIC)			
SafetyBit - MainProgram/RobotAutoControl - 1(XIC), 4(XIO), 5(XIO), 6(XIO)			
SafetyBit - SafetyProgram  /MainRoutine - *7(OTE), 7(XIC), 9(XIO)			
 selfdestruct	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
selfdestruct - MainProgram/MainRoutine - 0(XIC)			
selfdestruct - MainProgram/selfdestruct - 0(XIC), 2(XIO), 3(XIC)			
SolenoidAlarm		TIMER	MainProgram
Constant No			
External Access: Read/Write			
SolenoidAlarm - MainProgram/MainRoutine - *8(TON)			
SolenoidAlarm.DN	0	BOOL	
SolenoidAlarm.DN - MainProgram/MainRoutine - 9(XIC)			
speed	750	DINT	MainProgram
Constant No			
External Access: Read/Write			
speed - MainProgram/MainRoutine - 1(JSR), 4(JSR)			
 StackLightGreen	1	BOOL	Capstone_Test
AliasFor: Local:4:O.Pt07.Data			
Base Tag: Local:4:O.Pt07.Data			
Constant No			

StackLightGreen (Continued)			
External Access:	Read/Write		
StackLightGreen - MainProgram/MainRoutine - *5(OTE)			
StackLightGreen - MainProgram/pause - #10, *#32, *#56			
StackLightGreen - MainProgram/StopAll - *#2			
StackLightRed	0	BOOL	Capstone_Test
AliasFor:	Local:4:O.Pt09.Data		
Base Tag:	Local:4:O.Pt09.Data		
Constant	No		
External Access:	Read/Write		
StackLightRed - MainProgram/MainRoutine - *7(OTE)			
StackLightRed - MainProgram/pause - #9, *#31, *#55			
StackLightRed - MainProgram/StopAll - *#1			
StackLightYellow	1	BOOL	Capstone_Test
AliasFor:	Local:4:O.Pt08.Data		
Base Tag:	Local:4:O.Pt08.Data		
Constant	No		
External Access:	Read/Write		
StackLightYellow - MainProgram/MainRoutine - *6(OTE)			
StackLightYellow - MainProgram/pause - #11, *#33, *#57			
StackLightYellow - MainProgram/StopAll - *#3			
Step_021		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
Step_021 - MainProgram/CONTROL - *A1(Step,Step_021)			
Step_021 - MainProgram/MainRoutine - 0(SFR)			
Step_021.T	201	DINT	
Step_021.T - MainProgram/CONTROL - A4, #1(Tran,Tran_073 (Condition))			
SystemState	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
SystemState - MainProgram/MainRoutine - *I(MOV)			
SystemState.0	0	BOOL	
SystemState.0 - MainProgram/pause - #55, *#9			
SystemState.1	0	BOOL	
SystemState.1 - MainProgram/pause - #56, *#10			
SystemState.2	0	BOOL	
SystemState.2 - MainProgram/pause - #57, *#11			
SystemState.3	0	BOOL	
SystemState.3 - MainProgram/pause - #58, *#12			
SystemState.4	0	BOOL	
SystemState.4 - MainProgram/pause - #59, *#13			
SystemState.5	0	BOOL	
SystemState.5 - MainProgram/pause - #60, *#14			
SystemState.6	0	BOOL	
SystemState.6 - MainProgram/pause - #61, *#15			
SystemState.7	0	BOOL	
SystemState.7 - MainProgram/pause - #62, *#16			
SystemState.8	0	BOOL	
SystemState.8 - MainProgram/pause - #63, *#17			
SystemState.9	0	BOOL	
SystemState.9 - MainProgram/pause - #64, *#18			
SystemState.10	0	BOOL	
SystemState.10 - MainProgram/pause - #65, *#19			
SystemState.11	0	BOOL	
SystemState.11 - MainProgram/pause - #66, *#20			
SystemState.12	0	BOOL	
SystemState.12 - MainProgram/pause - #67, *#21			
SystemState.13	0	BOOL	
SystemState.13 - MainProgram/pause - #68, *#22			

SystemState (Continued)			
SystemState.14	0	BOOL	
SystemState.14 - MainProgram/pause - #69, *#23			
SystemState.15	0	BOOL	
SystemState.15 - MainProgram/pause - #70, *#24			
SystemState.16	0	BOOL	
SystemState.16 - MainProgram/pause - #71, *#25			
SystemState.17	0	BOOL	
SystemState.17 - MainProgram/pause - #72, *#26			
SystemState.18	0	BOOL	
SystemState.18 - MainProgram/pause - #73, *#27			
SystemState.19	0	BOOL	
SystemState.19 - MainProgram/pause - #74, *#28			
SystemState.20	0	BOOL	
SystemState.20 - MainProgram/pause - #75, *#29			
 WarningBit	1	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
WarningBit - MainProgram/MainRoutine - 6(XIC)			
WarningBit - SafetyProgram  /MainRoutine - *5(OTE)			
 Winding_Servo:O		_00BB:AZD_AEP_60D82B42:O:0	Capstone_Test
Constant No			
External Access: Read/Write			
Winding_Servo:O - MainProgram/MoveWinding - *8(OM_AZ_DDO)			
Winding_Servo:O.Data[0]	0	INT	
Remote I/O (R-IN)			
Winding_Servo:O.Data[0].0	0	BOOL	
R-IN0			
Winding_Servo:O.Data[1]	0	INT	
Operation data number selection			
Winding_Servo:O.Data[2]	512	INT	
Fixed I/O (IN)			
Winding_Servo:O.Data[2].0	0	BOOL	
FW-JOG			
Winding_Servo:O.Data[2].5	0	BOOL	
STOP			
Winding_Servo:O.Data[2].5 - MainProgram/MainRoutine - *7(OTE)			
Winding_Servo:O.Data[2].6	0	BOOL	
FREE			
Winding_Servo:O.Data[2].6 - MainProgram/FillValues - *#7			
Winding_Servo:O.Data[2].7	0	BOOL	
ALM-RST			
Winding_Servo:O.Data[2].7 - MainProgram/Alarms - 6(ALMD)			
Winding_Servo:O.Data[2].7 - MainProgram/MoveWinding - *6(OTE)			
Winding_Servo:O.Data[3]	1	INT	
Direct data operation type			
Winding_Servo:O.Data[4]	22630	INT	
Direct data operation position (lower)			
Winding_Servo:O.Data[5]	4	INT	
Direct data operation position (upper)			
Winding_Servo:O.Data[6]	750	INT	
Direct data operation operating speed (lower)			
Winding_Servo:O.Data[7]	0	INT	
Direct data operation operating speed (upper)			
Winding_Servo:O.Data[8]	300	INT	
Direct data operation starting/changing rate (lower)			
Winding_Servo:O.Data[9]	0	INT	
Direct data operation starting/changing rate (upper)			
Winding_Servo:O.Data[10]	750	INT	
Direct data operation stopping deceleration (lower)			
Winding_Servo:O.Data[11]	0	INT	
Direct data operation stopping deceleration (upper)			

Winding_Servo:O (Continued)			
Winding_Servo:O.Data[12]	1000	INT	
Direct data operation operating current			
Winding_Servo:O.Data[13]	0	INT	
Direct data operation forwarding destination			
Winding_Servo:O.Data[14]	0	INT	
Reserved			
Winding_Servo:O.Data[15]	3074	INT	
Read Parameter ID			
Winding_Servo:O.Data[16]	0	INT	
Write Request			
Winding_Servo:O.Data[17]	3074	INT	
Write Parameter ID			
Winding_Servo:O.Data[18]	1000	INT	
Write data (lower)			
Winding_Servo:O.Data[19]	0	INT	
Write data (upper)			
WindingSolenoid	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
WindingSolenoid - MainProgram/CONTROL - *B14, #1(Action,Action_034 (Body))			
WindingSolenoid - MainProgram/FillValues - #60			
WindingSolenoid - MainProgram/MainRoutine - 8(XIC)			
WindingSolenoid - MainProgram/pause - #12, *#34, *#58			
WindingSolenoid - MainProgram/SetWindingDia - *B1, #1(Action,Action_045 (Body)), *B2, #2(Action,Action_031 (Body)), *B4, #1(Action,Action_046 (Body))			
WindingSolenoid - MainProgram/StopAll - *#4			
WindingTurns	99.667	REAL	MainProgram
Constant No			
External Access: Read/Write			
WindingTurns - MainProgram/CONTROL - *B1, #23(Action,Action_024 (Body)), *B23, #6(Action,Action_039 (Body)), *B37, #4(Action,Action_061 (Body)), *B37, #5(Action,Action_061 (Body)), B1, #22(Action,Action_024 (Body)), B23, #6(Action,Action_039 (Body)), B37, #3(Action,Action_061 (Body)), B37, #5(Action,Action_061 (Body))			
WindingTurns - MainProgram/MainRoutine - 1(JSR), 4(JSR)			
WindingTurns - MainProgram/MoveWinding - *2(DIV)			
WindingTurns - MainProgram/SetWindingDia - *B3, #1(Action,Action_032 (Body)), *B6, #2(Action,Action_047 (Body)), B3, #1(Action,Action_032 (Body)), B6, #1(Action,Action_047 (Body))			
WindingTurns - MainProgram/ThugShake - *B2, #2(Action,Action_027 (Body)), *B3, #1(Action,Action_029 (Body)), B2, #2(Action,Action_027 (Body)), B3, #1(Action,Action_029 (Body))			

General			
Type:	 Ladder Diagram	Class:	Standard
In Program:	 MainProgram	Number of Rungs:	11







Name	Value	Data Type	Scope
Alarm_DOOR		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_DOOR - MainProgram/Alarms - *4(ALMD)</i>			
Alarm_ESTOP		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_ESTOP - MainProgram/Alarms - *1(ALMD)</i>			
Alarm_Manual		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Manual - MainProgram/Alarms - *5(ALMD)</i>			
Alarm_MIRCOSCAN		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_MIRCOSCAN - MainProgram/Alarms - *3(ALMD)</i>			
Alarm_NANOSCAN		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_NANOSCAN - MainProgram/Alarms - *2(ALMD)</i>			
Alarm_RobotFault		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_RobotFault - MainProgram/Alarms - *10(ALMD)</i>			
Alarm_Safety		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Safety - MainProgram/Alarms - *0(ALMD)</i>			
Alarm_Servo		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Servo - MainProgram/Alarms - *6(ALMD)</i>			
Alarm_Spool1		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Spool1 - MainProgram/Alarms - *7(ALMD)</i>			
Alarm_Spool2		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Spool2 - MainProgram/Alarms - *8(ALMD)</i>			
Alarm_Spool3		ALARM_DIGITAL	MainProgram
External Access:	Read/Write		
<i>Alarm_Spool3 - MainProgram/Alarms - *9(ALMD)</i>			
DoorInterlockTripped	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
<i>DoorInterlockTripped - SafetyProgram ▢/MainRoutine - *2(OTE), 6(XIC), 7(XIC)</i>			
<i>\SafetyProgram.DoorInterlockTripped - MainProgram/Alarms - 4(XIC)</i>			
EStopTripped	0	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
<i>EStopTripped - SafetyProgram ▢/MainRoutine - *1(OTE), 6(XIC), 7(XIC)</i>			
<i>\SafetyProgram.EStopTripped - MainProgram/Alarms - 1(XIC)</i>			
Fanuc_Robot:I		AB:ETHERNET_MODULE_INT_8Bytes:I:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Fanuc_Robot:I.Data[0]	0	INT	
In From Robot			
Fanuc_Robot:I.Data[0].0	0	BOOL	

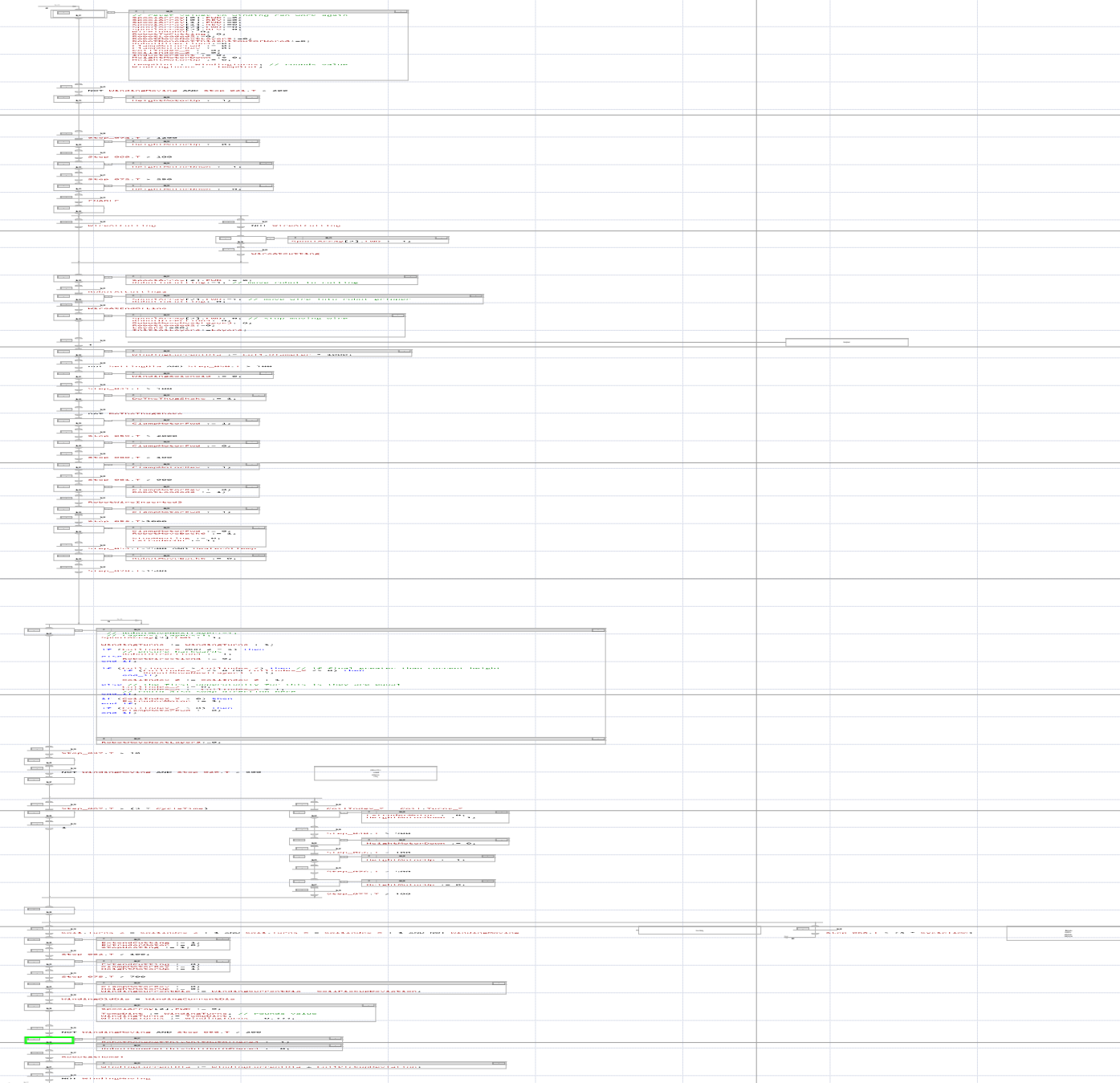
Fanuc_Robot:I (Continued)			
In From Robot			
RobotAtHome1 - MainProgram/CONTROL - A39, #1(Tran,Tran_037 (Condition))			
Fanuc_Robot:I.Data[0].1	0	BOOL	
In From Robot			
RobotAtCutting2 - MainProgram/CONTROL - A11, #1(Tran,Tran_021 (Condition))			
Fanuc_Robot:I.Data[0].2	0	BOOL	
In From Robot			
RobotWireInserted3 - MainProgram/CONTROL - A19, #1(Tran,Tran_051 (Condition))			
Fanuc_Robot:I.Data[1]	168	INT	
Robot Status			
Fanuc_Robot:I.Data[1].0	0	BOOL	
Cmd Enabled			
Fanuc_Robot:I.Data[1].1	0	BOOL	
System Ready			
Fanuc_Robot:I.Data[1].1 - MainProgram/Alarms - 10(ALMD)			
Fanuc_Robot:I.Data[1].1 - MainProgram/RobotAutoControl - 5(XIC)			
Fanuc_Robot:I.Data[1].2	0	BOOL	
PRG Running			
Fanuc_Robot:I.Data[1].2 - MainProgram/RobotAutoControl - 6(XIC), 7(XIO)			
Fanuc_Robot:I.Data[1].5	1	BOOL	
Fault			
Fanuc_Robot:I.Data[1].5 - MainProgram/Alarms - 10(XIC)			
Fanuc_Robot:I.Data[1].5 - MainProgram/RobotAutoControl - 7(XIO)			
Local:5:I		AB:5000_AI8:I:0	Capstone_Test
Constant No			
External Access: Read/Write			
Local:5:I.Ch00.Data	164.73665	REAL	
Local:5:I.Ch00.Data - MainProgram/Alarms - 7(LEQ)			
Local:5:I.Ch00.Data - MainProgram/FillValues - #19, #20			
Local:5:I.Ch01.Data	625.2037	REAL	
Local:5:I.Ch01.Data - MainProgram/Alarms - 8(LEQ)			
Local:5:I.Ch02.Data	600.10535	REAL	
Local:5:I.Ch02.Data - MainProgram/Alarms - 9(LEQ)			
Local:5:I.Ch02.Data - MainProgram/FillValues - #31, #32			
Local:5:I.Ch03.Data	-2.61334376e-003	REAL	
Local:5:I.Ch03.Data - MainProgram/FillValues - #16			
MANLY	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
MANLY - MainProgram/Alarms - 5(XIC)			
MANLY - MainProgram/MainRoutine - 0(XIO), 4(XIO)			
MicroScanTripped	0	BOOL	SafetyProgram
Constant No			
External Access: Read/Write			
MicroScanTripped - SafetyProgram /MainRoutine - *4(OTE), 6(XIC), 7(XIC)			
\SafetyProgram.MicroScanTripped - MainProgram/Alarms - 3(XIC)			
NanoScanTripped	1	BOOL	SafetyProgram
Constant No			
External Access: Read/Write			
NanoScanTripped - SafetyProgram /MainRoutine - *3(OTE), 6(XIC), 7(XIC)			
\SafetyProgram.NanoScanTripped - MainProgram/Alarms - 2(XIC)			
PB_Reset	0	BOOL	Capstone_Test
AliasFor: Local:1:I.Pt06.Data			
Base Tag: Local:1:I.Pt06.Data			
Constant No			
External Access: Read/Write			
PB_Reset - MainProgram/Alarms - 0(ALMD), 1(ALMD), 2(ALMD), 3(ALMD), 4(ALMD)			
PB_Reset - MainProgram/RobotAutoControl - 0(XIC)			

PB_Reset (Continued)			
PB_Reset - SafetyProgram  /MainRoutine - 10(XIC), 6(XIO), 8(XIC)			
 SafetyBit	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
SafetyBit - MainProgram/Alarms - 0(XIC), 1(XIC), 2(XIC), 3(XIC), 4(XIC)			
SafetyBit - MainProgram/MainRoutine - 10(XIC), 4(XIO), 5(XIO), 6(XIO), 7(XIC)			
SafetyBit - MainProgram/RobotAutoControl - 1(XIC), 4(XIO), 5(XIO), 6(XIO)			
SafetyBit - SafetyProgram  /MainRoutine - *7(OTE), 7(XIC), 9(XIO)			
 Winding_Servo:I		_00BB:AZD_AEP_A3E1851B:I:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Winding_Servo:I.Data		INT	
Winding_Servo:I.Data - MainProgram/MoveWinding - *1(OM_AZ_MONITOR), *8(OM_AZ_DDO)			
Winding_Servo:I.Data[0]	16416	INT	
Remote I/O (R-OUT)			
Winding_Servo:I.Data[0].0	0	BOOL	
R-OUT0			
Winding_Servo:I.Data[1]	0	INT	
Operation data number selection_R			
Winding_Servo:I.Data[2]	612	INT	
Fixed I/O (OUT)			
Winding_Servo:I.Data[2].0	0	BOOL	
SEQ-BSY			
Winding_Servo:I.Data[2].7	0	BOOL	
ALM-A			
Winding_Servo:I.Data[2].7 - MainProgram/Alarms - 6(XIC)			
Winding_Servo:I.Data[3]	0	INT	
Present Alarm			
Winding_Servo:I.Data[4]	22630	INT	
Feedback position (lower)			
Winding_Servo:I.Data[5]	4	INT	
Feedback position (upper)			
Winding_Servo:I.Data[6]	0	INT	
Feedback speed [Hz] (lower)			
Winding_Servo:I.Data[7]	0	INT	
Feedback speed [Hz] (upper)			
Winding_Servo:I.Data[8]	22630	INT	
Command position (lower)			
Winding_Servo:I.Data[9]	4	INT	
Command position (upper)			
Winding_Servo:I.Data[10]	-6	INT	
Torque monitor			
Winding_Servo:I.Data[11]	500	INT	
CST operating current			
Winding_Servo:I.Data[12]	0	INT	
Information (lower)			
Winding_Servo:I.Data[13]	0	INT	
Information (upper)			
Winding_Servo:I.Data[14]	0	INT	
Reserved			
Winding_Servo:I.Data[15]	3074	INT	
Read parameter ID_R			
Winding_Servo:I.Data[16]	0	INT	
Read/write status			
Winding_Servo:I.Data[17]	0	INT	
Write parameter ID_R			
Winding_Servo:I.Data[18]	-21072	INT	
Read data (lower)			
Winding_Servo:I.Data[19]	1	INT	
Read data (upper)			
Winding_Servo:I.Data[20]	430	INT	

Winding_Servo:I (Continued)		
Assignable monitor 0 (lower)		
Winding_Servo:I.Data[21]	0	INT
Assignable monitor 0 (upper)		
Winding_Servo:I.Data[22]	459	INT
Assignable monitor 1 (lower)		
Winding_Servo:I.Data[23]	0	INT
Assignable monitor 1 (upper)		
Winding_Servo:I.Data[24]	-13489	INT
Assignable monitor 2 (lower)		
Winding_Servo:I.Data[25]	31	INT
Assignable monitor 2 (upper)		
Winding_Servo:I.Data[26]	317	INT
Assignable monitor 3 (lower)		
Winding_Servo:I.Data[27]	0	INT
Assignable monitor 3 (upper)		
Winding_Servo:O		
Constant	No	
External Access:	Read/Write	
Winding_Servo:O - MainProgram/MoveWinding - *8(OM_AZ_DDO)		
Winding_Servo:O.Data[0]	0	INT
Remote I/O (R-IN)		
Winding_Servo:O.Data[0].0	0	BOOL
R-IN0		
Winding_Servo:O.Data[1]	0	INT
Operation data number selection		
Winding_Servo:O.Data[2]	512	INT
Fixed I/O (IN)		
Winding_Servo:O.Data[2].0	0	BOOL
FW-JOG		
Winding_Servo:O.Data[2].5	0	BOOL
STOP		
Winding_Servo:O.Data[2].5 - MainProgram/MainRoutine - *7(OTE)		
Winding_Servo:O.Data[2].6	0	BOOL
FREE		
Winding_Servo:O.Data[2].6 - MainProgram/FillValues - *#7		
Winding_Servo:O.Data[2].7	0	BOOL
ALM-RST		
Winding_Servo:O.Data[2].7 - MainProgram/Alarms - 6(ALMD)		
Winding_Servo:O.Data[2].7 - MainProgram/MoveWinding - *6(OTE)		
Winding_Servo:O.Data[3]	1	INT
Direct data operation type		
Winding_Servo:O.Data[4]	22630	INT
Direct data operation position (lower)		
Winding_Servo:O.Data[5]	4	INT
Direct data operation position (upper)		
Winding_Servo:O.Data[6]	750	INT
Direct data operation operating speed (lower)		
Winding_Servo:O.Data[7]	0	INT
Direct data operation operating speed (upper)		
Winding_Servo:O.Data[8]	300	INT
Direct data operation starting/changing rate (lower)		
Winding_Servo:O.Data[9]	0	INT
Direct data operation starting/changing rate (upper)		
Winding_Servo:O.Data[10]	750	INT
Direct data operation stopping deceleration (lower)		
Winding_Servo:O.Data[11]	0	INT
Direct data operation stopping deceleration (upper)		
Winding_Servo:O.Data[12]	1000	INT
Direct data operation operating current		
Winding_Servo:O.Data[13]	0	INT
Direct data operation forwarding destination		
Winding_Servo:O.Data[14]	0	INT

Winding_Servo:O (Continued)		
Reserved		
Winding_Servo:O.Data[15]	3074	INT
Read Parameter ID		
Winding_Servo:O.Data[16]	0	INT
Write Request		
Winding_Servo:O.Data[17]	3074	INT
Write Parameter ID		
Winding_Servo:O.Data[18]	1000	INT
Write data (lower)		
Winding_Servo:O.Data[19]	0	INT
Write data (upper)		

General			
Type:	 Ladder Diagram	Class:	Standard
In Program:	 MainProgram	Number of Rungs:	11



Name	Value	Data Type	Scope
Action_019		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_019 - MainProgram/CONTROL - *C9(Action,Action_019)</i>			
Action_021		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_021 - MainProgram/CONTROL - *B11(Action,Action_021)</i>			
Action_022		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_022 - MainProgram/CONTROL - *B11(Action,Action_022)</i>			
Action_023		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_023 - MainProgram/CONTROL - *B12(Action,Action_023)</i>			
Action_024		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_024 - MainProgram/CONTROL - *B1(Action,Action_024)</i>			
Action_025		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_025 - MainProgram/CONTROL - *B21(Action,Action_025)</i>			
Action_026		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_026 - MainProgram/CONTROL - *C30(Action,Action_026)</i>			
Action_034		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_034 - MainProgram/CONTROL - *B14(Action,Action_034)</i>			
Action_039		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_039 - MainProgram/CONTROL - *B23(Action,Action_039)</i>			
Action_040		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_040 - MainProgram/CONTROL - *B27(Action,Action_040)</i>			
Action_041		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_041 - MainProgram/CONTROL - *B38(Action,Action_041)</i>			
Action_042		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_042 - MainProgram/CONTROL - *B38(Action,Action_042)</i>			
Action_043		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		

Action_043 (Continued)		
<i>Action_043 - MainProgram/CONTROL - *B15(Action,Action_043)</i>		
Action_049	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_049 - MainProgram/CONTROL - *B13(Action,Action_049)</i>		
Action_051	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_051 - MainProgram/CONTROL - *B18(Action,Action_051)</i>		
Action_052	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_052 - MainProgram/CONTROL - *B19(Action,Action_052)</i>		
Action_053	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_053 - MainProgram/CONTROL - *B20(Action,Action_053)</i>		
Action_054	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_054 - MainProgram/CONTROL - *B6(Action,Action_054)</i>		
Action_057	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_057 - MainProgram/CONTROL - *B16(Action,Action_057)</i>		
Action_058	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_058 - MainProgram/CONTROL - *B17(Action,Action_058)</i>		
Action_059	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_059 - MainProgram/CONTROL - *B17(Action,Action_059)</i>		
Action_060	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_060 - MainProgram/CONTROL - *B36(Action,Action_060)</i>		
Action_061	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_061 - MainProgram/CONTROL - *B37(Action,Action_061)</i>		
Action_062	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_062 - MainProgram/CONTROL - *C31(Action,Action_062)</i>		
Action_063	SFC_ACTION	MainProgram
Constant	No	
External Access:	Read/Write	
<i>Action_063 - MainProgram/CONTROL - *B39(Action,Action_063)</i>		
Action_064	SFC_ACTION	MainProgram

Action_064 (Continued)			
Constant	No		
External Access:	Read/Write		
<i>Action_064 - MainProgram/CONTROL - *B6(Action,Action_064)</i>			
Action_065			
Constant	No	SFC_ACTION	MainProgram
External Access:	Read/Write		
<i>Action_065 - MainProgram/CONTROL - *B7(Action,Action_065)</i>			
Action_066			
Constant	No	SFC_ACTION	MainProgram
External Access:	Read/Write		
<i>Action_066 - MainProgram/CONTROL - *B4(Action,Action_066)</i>			
Action_067			
Constant	No	SFC_ACTION	MainProgram
External Access:	Read/Write		
<i>Action_067 - MainProgram/CONTROL - *B35(Action,Action_067)</i>			
Action_068			
Constant	No	SFC_ACTION	MainProgram
External Access:	Read/Write		
<i>Action_068 - MainProgram/CONTROL - *C32(Action,Action_068)</i>			
Action_069			
Constant	No	SFC_ACTION	MainProgram
External Access:	Read/Write		
<i>Action_069 - MainProgram/CONTROL - *C32(Action,Action_069)</i>			
Action_070			
Constant	No	SFC_ACTION	MainProgram
External Access:	Read/Write		
<i>Action_070 - MainProgram/CONTROL - *B35(Action,Action_070)</i>			
ClampMotorFwd			
Constant	0	BOOL	Capstone_Test
External Access:	No		
<i>ClampMotorFwd - MainProgram/CONTROL - *B1, #14(Action,Action_024 (Body)), *B16, #1(Action,Action_057 (Body)), *B17, #1(Action,Action_058 (Body)), *B19, #1(Action,Action_052 (Body)), *B20, #1(Action,Action_053 (Body)), *B23, #34(Action,Action_039 (Body))</i>			
<i>ClampMotorFwd - MainProgram/FillValues - #50</i>			
<i>ClampMotorFwd - MainProgram/pause - #24, *#46, *#70</i>			
<i>ClampMotorFwd - MainProgram/StopAll - *#16</i>			
ClampMotorRev			
Constant	0	BOOL	Capstone_Test
External Access:	No		
<i>ClampMotorRev - MainProgram/CONTROL - *B1, #15(Action,Action_024 (Body)), *B17, #1(Action,Action_059 (Body)), *B18, #1(Action,Action_051 (Body)), *B35, #2(Action,Action_067 (Body)), *B36, #1(Action,Action_060 (Body))</i>			
<i>ClampMotorRev - MainProgram/FillValues - #51</i>			
<i>ClampMotorRev - MainProgram/pause - #25, *#47, *#71</i>			
<i>ClampMotorRev - MainProgram/StopAll - *#17</i>			
Coil			
Constant	No	Coil	Capstone_Test
External Access:	Read/Write		
Coil.Inductance	8.92412616e-004	REAL	
mH			
Coil.Mass	2.11115021e-005	REAL	
Kg			
Coil.Diameter	0.18	REAL	
m			
<i>Coil.Diameter - MainProgram/CONTROL - B13, #1(Action,Action_049 (Body))</i>			
Coil.Turns_X	2	INT	

Coil (Continued)			
<i>Coil.Turns_X - MainProgram/CONTROL - A34, #1(Tran,Tran_036 (Condition))</i>			
Coil.Turns_Z	2	INT	
<i>Coil.Turns_Z - MainProgram/CONTROL - A34, #1(Tran,Tran_036 (Condition)), B23, #16(Action,Action_039 (Body)), C30, #1(Tran,Tran_064 (Condition))</i>			
Coil.Current	0.15	REAL	
<i>A</i>			
Coil.Length	0.003	REAL	
<i>m</i>			
Coil.Gauge	16	DINT	
<i>Coil.Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)</i>			
CoilIndex_X	1	DINT	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>CoilIndex_X - MainProgram/CONTROL - *B1, #16(Action,Action_024 (Body)), *B23, #25(Action,Action_039 (Body)), A34, #1(Tran,Tran_036 (Condition)), B23, #17(Action,Action_039 (Body)), B23, #25(Action,Action_039 (Body)), B23, #29(Action,Action_039 (Body)), B23, #8(Action,Action_039 (Body))</i>			
<i>CoilIndex_X - MainProgram/StopAll - *#18</i>			
CoilIndex_Z	1	DINT	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>CoilIndex_Z - MainProgram/CONTROL - *B1, #17(Action,Action_024 (Body)), *B23, #21(Action,Action_039 (Body)), *B23, #24(Action,Action_039 (Body)), A34, #1(Tran,Tran_036 (Condition)), B23, #16(Action,Action_039 (Body)), B23, #17(Action,Action_039 (Body)), B23, #21(Action,Action_039 (Body)), B23, #33(Action,Action_039 (Body)), C30, #1(Tran,Tran_064 (Condition))</i>			
<i>CoilIndex_Z - MainProgram/StopAll - *#19</i>			
CoilPickupDeviation	10	DINT	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>CoilPickupDeviation - MainProgram/CONTROL - B36, #3(Action,Action_060 (Body)), B39, #1(Action,Action_063 (Body))</i>			
CycleTime	0	DINT	Capstone_Test
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>CycleTime - MainProgram/CONTROL - A30, #1(Tran,Tran_065 (Condition)), F34, #1(Tran,Tran_040 (Condition))</i>			
<i>CycleTime - MainProgram/FillValues - #63, *#63</i>			
DoTheThugShake	0	BOOL	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>DoTheThugShake - MainProgram/CONTROL - *B15, #1(Action,Action_043 (Body)), A15, #1(Tran,Tran_039 (Condition))</i>			
<i>DoTheThugShake - MainProgram/SetWindingDia - *B3, #1(Action,Action_044 (Body)), *B5, #1(Action,Action_056 (Body)), B3, #1(Tran,Tran_044 (Condition)), B5, #1(Tran,Tran_057 (Condition))</i>			
<i>DoTheThugShake - MainProgram/ThugShake - *B4, #1(Action,Action_030 (Body)), B1, #1(Tran,Tran_026 (Condition))</i>			
ENABLE	0	BOOL	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>ENABLE - MainProgram/CONTROL - A8, #1(Tran,Tran_019 (Condition))</i>			
<i>ENABLE - MainProgram/MainRoutine - 0(XIO)</i>			
<i>ENABLE - MainProgram/WindInductor - *4(OTE), 2(XIO), 3(XIO)</i>			
ExtendCutting	0	BOOL	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			
<i>ExtendCutting - MainProgram/CONTROL - *B35, #1(Action,Action_067 (Body)), *B35, #1(Action,Action_070 (Body))</i>			
<i>ExtendCutting - MainProgram/FillValues - #57</i>			
ExtruderMotor	0	BOOL	MainProgram
<i>Constant No</i>			
<i>External Access: Read/Write</i>			

ExtruderMotor (Continued)			
<i>ExtruderMotor - MainProgram/CONTROL - *B23, #30(Action,Action_039 (Body)), *B35, #2(Action,Action_070 (Body)), *C30, #1(Action,Action_026 (Body))</i>			
<i>ExtruderMotor - MainProgram/FillValues - #58</i>			
ExtruderOn	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ExtruderOn - MainProgram/CONTROL - *B20, #5(Action,Action_053 (Body))</i>			
<i>ExtruderOn - MainProgram/HeatExtruder - *0(OTE), 0(XIC), 1(XIC), 2(XIC), 3(XIC), 4(XIC), 5(XIC)</i>			
HeaterAtTemp	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>HeaterAtTemp - MainProgram/CONTROL - A20, #1(Tran,Tran_053 (Condition))</i>			
<i>HeaterAtTemp - MainProgram/HeatExtruder - *2(OTE), 2(XIC)</i>			
HeightMotorDown	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>HeightMotorDown - MainProgram/CONTROL - *B1, #19(Action,Action_024 (Body)), *B6, #1(Action,Action_064 (Body)), *B7, #1(Action,Action_065 (Body)), *C30, #2(Action,Action_026 (Body)), *C31, #1(Action,Action_062 (Body))</i>			
<i>HeightMotorDown - MainProgram/FillValues - #53</i>			
<i>HeightMotorDown - MainProgram/pause - #27, *#49, *#73</i>			
<i>HeightMotorDown - MainProgram/StopAll - *#21</i>			
HeightMotorUp	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>HeightMotorUp - MainProgram/CONTROL - *B1, #20(Action,Action_024 (Body)), *B35, #3(Action,Action_067 (Body)), *B36, #2(Action,Action_060 (Body)), *B4, #1(Action,Action_066 (Body)), *B6, #1(Action,Action_054 (Body)), *C32, #1(Action,Action_068 (Body)), *C32, #1(Action,Action_069 (Body))</i>			
<i>HeightMotorUp - MainProgram/FillValues - #54</i>			
<i>HeightMotorUp - MainProgram/pause - #28, *#50, *#74</i>			
<i>HeightMotorUp - MainProgram/StopAll - *#22</i>			
InductorSent	0	BOOL	MainProgram
PC sets this true when the inductor data has been sent			
Constant	No		
External Access:	Read/Write		
<i>InductorSent - MainProgram/CONTROL - *B1, #18(Action,Action_024 (Body))</i>			
<i>InductorSent - MainProgram/pause - #26, *#48, *#72</i>			
<i>InductorSent - MainProgram/StopAll - *#20</i>			
<i>InductorSent - MainProgram/WindInductor - 1(XIC)</i>			
InitialLayers	20	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>InitialLayers - MainProgram/CONTROL - *B12, #6(Action,Action_023 (Body))</i>			
<i>InitialLayers - MainProgram/FillValues - #69, #69</i>			
Layers	20	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Layers - MainProgram/CONTROL - *B12, #5(Action,Action_023 (Body)), B12, #6(Action,Action_023 (Body))</i>			
<i>Layers - MainProgram/FillValues - #69</i>			
RobotAtCutting2	0	BOOL	Capstone_Test
In From Robot			
AliasFor:	Fanuc_Robot:I.Data[0].1		
Base Tag:	Fanuc_Robot:I.Data[0].1		
Constant	No		
External Access:	Read/Write		
<i>RobotAtCutting2 - MainProgram/CONTROL - A11, #1(Tran,Tran_021 (Condition))</i>			

RobotAtHome1	0	BOOL	Capstone_Test
In From Robot			
AliasFor:	Fanuc_Robot:I.Data[0].0		
Base Tag:	Fanuc_Robot:I.Data[0].0		
Constant	No		
External Access:	Read/Write		
<i>RobotAtHome1 - MainProgram/CONTROL - A39, #1(Tran,Tran_037 (Condition))</i>			
RobotDirection4	1	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].3		
Base Tag:	Fanuc_Robot:O.Data[0].3		
Constant	No		
External Access:	Read/Write		
<i>RobotDirection4 - MainProgram/CONTROL - *B1, #13(Action,Action_024 (Body)), *B12, #2(Action,Action_023 (Body)), *B23, #10(Action,Action_039 (Body)), *B23, #12(Action,Action_039 (Body))</i>			
<i>RobotDirection4 - MainProgram/pause - #23, *#45, *#69</i>			
<i>RobotDirection4 - MainProgram/StopAll - *#15</i>			
RobotDoneGetThisShitOutOfHere4	1	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].4		
Base Tag:	Fanuc_Robot:O.Data[0].4		
Constant	No		
External Access:	Read/Write		
<i>RobotDoneGetThisShitOutOfHere4 - MainProgram/CONTROL - *B1, #12(Action,Action_024 (Body)), *B38, #1(Action,Action_041 (Body)), *B38, #1(Action,Action_042 (Body))</i>			
RobotLoaded2	1	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].1		
Base Tag:	Fanuc_Robot:O.Data[0].1		
Constant	No		
External Access:	Read/Write		
<i>RobotLoaded2 - MainProgram/CONTROL - *B1, #10(Action,Action_024 (Body)), *B12, #4(Action,Action_023 (Body)), *B18, #2(Action,Action_051 (Body))</i>			
<i>RobotLoaded2 - MainProgram/pause - #21, *#43, *#67</i>			
<i>RobotLoaded2 - MainProgram/StopAll - *#13</i>			
RobotMoveBack6	0	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].5		
Base Tag:	Fanuc_Robot:O.Data[0].5		
Constant	No		
External Access:	Read/Write		
<i>RobotMoveBack6 - MainProgram/CONTROL - *B20, #2(Action,Action_053 (Body)), *B21, #1(Action,Action_025 (Body))</i>			
RobotMoveNextLayer3	0	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].2		
Base Tag:	Fanuc_Robot:O.Data[0].2		
Constant	No		
External Access:	Read/Write		
<i>RobotMoveNextLayer3 - MainProgram/CONTROL - *B1, #11(Action,Action_024 (Body)), *B12, #3(Action,Action_023 (Body)), *B23, #18(Action,Action_039 (Body)), *B27, #1(Action,Action_040 (Body))</i>			
<i>RobotMoveNextLayer3 - MainProgram/pause - #22, *#44, *#68</i>			
<i>RobotMoveNextLayer3 - MainProgram/StopAll - *#14</i>			
RobotToCutting	0	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].0		
Base Tag:	Fanuc_Robot:O.Data[0].0		
Constant	No		

RobotToCutting (Continued)			
External Access:	Read/Write		
<i>RobotToCutting - MainProgram/CONTROL - *B1, #9(Action,Action_024 (Body)), *B11, #2(Action,Action_021 (Body)), *B11, #2(Action,Action_022 (Body))</i>			
<i>RobotToCutting - MainProgram/pause - #20, *#42, *#66</i>			
<i>RobotToCutting - MainProgram/StopAll - *#12</i>			
RobotWireInserted3	0	BOOL	Capstone_Test
also means wire in			
AliasFor:	Fanuc_Robot:I.Data[0].2		
Base Tag:	Fanuc_Robot:I.Data[0].2		
Constant	No		
External Access:	Read/Write		
<i>RobotWireInserted3 - MainProgram/CONTROL - A19, #1(Tran,Tran_051 (Condition))</i>			
SettingDia	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>SettingDia - MainProgram/CONTROL - A14, #1(Tran,Tran_049 (Condition))</i>			
<i>SettingDia - MainProgram/SetWindingDia - *B1, #2(Action,Action_045 (Body)), *B2, #1(Action,Action_031 (Body)), *B7, #2(Action,Action_048 (Body))</i>			
SpoolArray		Spool[3]	Capstone_Test
Constant	No		
External Access:	Read/Write		
SpoolArray[0].REV	0	BOOL	
out			
<i>SpoolArray[0].REV - MainProgram/CONTROL - *B1, #3(Action,Action_024 (Body))</i>			
<i>SpoolArray[0].REV - MainProgram/FillValues - #41</i>			
<i>SpoolArray[0].REV - MainProgram/pause - #14, *#36, *#60</i>			
<i>SpoolArray[0].REV - MainProgram/StopAll - *#6</i>			
SpoolArray[0].FWD	0	BOOL	
out			
<i>SpoolArray[0].FWD - MainProgram/CONTROL - *B1, #2(Action,Action_024 (Body))</i>			
<i>SpoolArray[0].FWD - MainProgram/FillValues - #40</i>			
<i>SpoolArray[0].FWD - MainProgram/pause - #13, *#35, *#59</i>			
<i>SpoolArray[0].FWD - MainProgram/StopAll - *#5</i>			
SpoolArray[0].Mass	164.73665	REAL	
current mass reading			
<i>SpoolArray[0].Mass - MainProgram/FillValues - *#20, *#22</i>			
SpoolArray[0].Length	0.0	REAL	
approximated legth based on Mass			
SpoolArray[0].Loaded	1	BOOL	
if is in loaded state			
<i>SpoolArray[0].Loaded - MainProgram/FillValues - *#10</i>			
SpoolArray[0].Gauge	-1	DINT	
Wire Gauge			
SpoolArray[1].REV	0	BOOL	
out			
<i>SpoolArray[1].REV - MainProgram/CONTROL - *B1, #5(Action,Action_024 (Body))</i>			
<i>SpoolArray[1].REV - MainProgram/FillValues - #43</i>			
<i>SpoolArray[1].REV - MainProgram/pause - #16, *#38, *#62</i>			
<i>SpoolArray[1].REV - MainProgram/StopAll - *#8</i>			
SpoolArray[1].FWD	0	BOOL	
out			
<i>SpoolArray[1].FWD - MainProgram/CONTROL - *B1, #4(Action,Action_024 (Body))</i>			
<i>SpoolArray[1].FWD - MainProgram/FillValues - #42</i>			
<i>SpoolArray[1].FWD - MainProgram/pause - #15, *#37, *#61</i>			
<i>SpoolArray[1].FWD - MainProgram/StopAll - *#7</i>			
SpoolArray[1].Mass	163.34396	REAL	
current mass reading			
SpoolArray[1].Length	0.0	REAL	
approximated legth based on Mass			
SpoolArray[1].Loaded	1	BOOL	

SpoolArray (Continued)		
if is in loaded state		
<i>SpoolArray[1].Loaded - MainProgram/FillValues - *#11</i>		
SpoolArray[1].Gauge	-1	DINT
Wire Gauge		
SpoolArray[2].REV	0	BOOL
out		
<i>SpoolArray[2].REV - MainProgram/CONTROL - *B1, #7(Action,Action_024 (Body))</i>		
<i>SpoolArray[2].REV - MainProgram/FillValues - #45</i>		
<i>SpoolArray[2].REV - MainProgram/pause - #18, *#40, *#64</i>		
<i>SpoolArray[2].REV - MainProgram/StopAll - *#10</i>		
SpoolArray[2].FWD	0	BOOL
out		
<i>SpoolArray[2].FWD - MainProgram/CONTROL - *B1, #6(Action,Action_024 (Body)), *B11, #1(Action,Action_021 (Body)), *B11, #1(Action,Action_022 (Body)), *B12, #1(Action,Action_023 (Body)), *B23, #3(Action,Action_039 (Body)), *B37, #1(Action,Action_061 (Body)), *C9, #1(Action,Action_019 (Body))</i>		
<i>SpoolArray[2].FWD - MainProgram/FillValues - #44</i>		
<i>SpoolArray[2].FWD - MainProgram/pause - #17, *#39, *#63</i>		
<i>SpoolArray[2].FWD - MainProgram/StopAll - *#9</i>		
SpoolArray[2].Mass	600.10535	REAL
current mass reading		
<i>SpoolArray[2].Mass - MainProgram/FillValues - *#32, *#34</i>		
SpoolArray[2].Length	0.0	REAL
approximated legth based on Mass		
SpoolArray[2].Loaded	0	BOOL
if is in loaded state		
<i>SpoolArray[2].Loaded - MainProgram/FillValues - *#12</i>		
SpoolArray[2].Gauge	16	DINT
Wire Gauge		
<i>SpoolArray[2].Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)</i>		
Step_021		
Constant	No	SFC_STEP
External Access:		Read/Write
<i>Step_021 - MainProgram/CONTROL - *A1(Step,Step_021)</i>		
<i>Step_021 - MainProgram/MainRoutine - 0(SFR)</i>		
Step_021.T	201	DINT
<i>Step_021.T - MainProgram/CONTROL - A4, #1(Tran,Tran_073 (Condition))</i>		
Step_023		
Constant	No	SFC_STEP
External Access:		Read/Write
<i>Step_023 - MainProgram/CONTROL - *A11(Step,Step_023)</i>		
Step_024		
Constant	No	SFC_STEP
External Access:		Read/Write
<i>Step_024 - MainProgram/CONTROL - *A11(Step,Step_024)</i>		
Step_025		
Constant	No	SFC_STEP
External Access:		Read/Write
<i>Step_025 - MainProgram/CONTROL - *A12(Step,Step_025)</i>		
Step_028		
Constant	No	SFC_STEP
External Access:		Read/Write
<i>Step_028 - MainProgram/CONTROL - *A21(Step,Step_028)</i>		
Step_028.T	1501	DINT
<i>Step_028.T - MainProgram/CONTROL - A21, #1(Tran,Tran_042 (Condition))</i>		
Step_029		
Constant	No	SFC_STEP
External Access:		Read/Write

Step_029 (Continued)			
<i>Step_029 - MainProgram/CONTROL - *A28(Step,Step_029)</i>			
Step_029.T	4017	DINT	
<i>Step_029.T - MainProgram/CONTROL - A28, #1(Tran,Tran_038 (Condition))</i>			
Step_037		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_037 - MainProgram/CONTROL - *A23(Step,Step_037)</i>			
Step_037.T	11	DINT	
<i>Step_037.T - MainProgram/CONTROL - A28, #1(Tran,Tran_025 (Condition))</i>			
Step_038		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_038 - MainProgram/CONTROL - *A38(Step,Step_038)</i>			
Step_040		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_040 - MainProgram/CONTROL - *C30(Step,Step_040)</i>			
Step_040.T	901	DINT	
<i>Step_040.T - MainProgram/CONTROL - C31, #1(Tran,Tran_066 (Condition))</i>			
Step_041		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_041 - MainProgram/CONTROL - *A14(Step,Step_041)</i>			
Step_041.T	101	DINT	
<i>Step_041.T - MainProgram/CONTROL - A14, #1(Tran,Tran_045 (Condition))</i>			
Step_045		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_045 - MainProgram/CONTROL - *A15(Step,Step_045)</i>			
Step_050		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_050 - MainProgram/CONTROL - *A13(Step,Step_050)</i>			
Step_050.T	101	DINT	
<i>Step_050.T - MainProgram/CONTROL - A14, #1(Tran,Tran_049 (Condition))</i>			
Step_052		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_052 - MainProgram/CONTROL - *A18(Step,Step_052)</i>			
Step_053		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_053 - MainProgram/CONTROL - *A19(Step,Step_053)</i>			
Step_053.T	1001	DINT	
<i>Step_053.T - MainProgram/CONTROL - A19, #1(Tran,Tran_052 (Condition))</i>			
Step_054		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_054 - MainProgram/CONTROL - *A20(Step,Step_054)</i>			
Step_054.T	180000	DINT	
<i>Step_054.T - MainProgram/CONTROL - A20, #1(Tran,Tran_053 (Condition))</i>			
Step_056		SFC_STEP	MainProgram
Constant No			

Step_056 (Continued)		
External Access:	Read/Write	
<i>Step_056 - MainProgram/CONTROL - *A37(Step,Step_056)</i>		
Step_056.T	1500	DINT
<i>Step_056.T - MainProgram/CONTROL - A38, #1(Tran,Tran_055 (Condition))</i>		
Step_059		SFC_STEP MainProgram
Constant	No	
External Access:	Read/Write	
<i>Step_059 - MainProgram/CONTROL - *A16(Step,Step_059)</i>		
Step_059.T	2001	DINT
<i>Step_059.T - MainProgram/CONTROL - A16, #1(Tran,Tran_058 (Condition))</i>		
Step_060		SFC_STEP MainProgram
Constant	No	
External Access:	Read/Write	
<i>Step_060 - MainProgram/CONTROL - *A17(Step,Step_060)</i>		
Step_060.T	101	DINT
<i>Step_060.T - MainProgram/CONTROL - A17, #1(Tran,Tran_059 (Condition))</i>		
Step_061		SFC_STEP MainProgram
Constant	No	
External Access:	Read/Write	
<i>Step_061 - MainProgram/CONTROL - *A17(Step,Step_061)</i>		
Step_061.T	901	DINT
<i>Step_061.T - MainProgram/CONTROL - A18, #1(Tran,Tran_060 (Condition))</i>		
Step_062		SFC_STEP MainProgram
Constant	No	
External Access:	Read/Write	
<i>Step_062 - MainProgram/CONTROL - *A35(Step,Step_062)</i>		
Step_062.T	101	DINT
<i>Step_062.T - MainProgram/CONTROL - A35, #1(Tran,Tran_062 (Condition))</i>		
Step_063		SFC_STEP MainProgram
Constant	No	
External Access:	Read/Write	
<i>Step_063 - MainProgram/CONTROL - *A8(Step,Step_063)</i>		
Step_064		SFC_STEP MainProgram
Constant	No	
External Access:	Read/Write	
<i>Step_064 - MainProgram/CONTROL - *C9(Step,Step_064)</i>		
Step_065		SFC_STEP MainProgram
Constant	No	
External Access:	Read/Write	
<i>Step_065 - MainProgram/CONTROL - *A30(Step,Step_065)</i>		
Step_066		SFC_STEP MainProgram
Constant	No	
External Access:	Read/Write	
<i>Step_066 - MainProgram/CONTROL - *C31(Step,Step_066)</i>		
Step_066.T	101	DINT
<i>Step_066.T - MainProgram/CONTROL - C31, #1(Tran,Tran_067 (Condition))</i>		
Step_067		SFC_STEP MainProgram
Constant	No	
External Access:	Read/Write	
<i>Step_067 - MainProgram/CONTROL - *A29(Step,Step_067)</i>		
Step_067.T	1	DINT
<i>Step_067.T - MainProgram/CONTROL - A30, #1(Tran,Tran_065 (Condition))</i>		
Step_068		SFC_STEP MainProgram

Step_068 (Continued)		
Constant	No	
External Access:	Read/Write	
<i>Step_068 - MainProgram/CONTROL - *A33(Step,Step_068)</i>		
Step_068.T	0	DINT
<i>Step_068.T - MainProgram/CONTROL - F34, #1(Tran,Tran_040 (Condition))</i>		
Step_069		
Constant	No	SFC_STEP
External Access:	Read/Write	MainProgram
<i>Step_069 - MainProgram/CONTROL - *A6(Step,Step_069)</i>		
Step_069.T	101	DINT
<i>Step_069.T - MainProgram/CONTROL - A6, #1(Tran,Tran_072 (Condition))</i>		
Step_070		
Constant	No	SFC_STEP
External Access:	Read/Write	MainProgram
<i>Step_070 - MainProgram/CONTROL - *A36(Step,Step_070)</i>		
Step_071		
Constant	No	SFC_STEP
External Access:	Read/Write	MainProgram
<i>Step_071 - MainProgram/CONTROL - *A39(Step,Step_071)</i>		
Step_072		
Constant	No	SFC_STEP
External Access:	Read/Write	MainProgram
<i>Step_072 - MainProgram/CONTROL - *A6(Step,Step_072)</i>		
Step_072.T	251	DINT
<i>Step_072.T - MainProgram/CONTROL - A7, #1(Tran,Tran_071 (Condition))</i>		
Step_073		
Constant	No	SFC_STEP
External Access:	Read/Write	MainProgram
<i>Step_073 - MainProgram/CONTROL - *A7(Step,Step_073)</i>		
Step_074		
Constant	No	SFC_STEP
External Access:	Read/Write	MainProgram
<i>Step_074 - MainProgram/CONTROL - *A4(Step,Step_074)</i>		
Step_074.T	1201	DINT
<i>Step_074.T - MainProgram/CONTROL - A5, #1(Tran,Tran_068 (Condition))</i>		
Step_075		
Constant	No	SFC_STEP
External Access:	Read/Write	MainProgram
<i>Step_075 - MainProgram/CONTROL - *A35(Step,Step_075)</i>		
Step_075.T	701	DINT
<i>Step_075.T - MainProgram/CONTROL - A36, #1(Tran,Tran_074 (Condition))</i>		
Step_076		
Constant	No	SFC_STEP
External Access:	Read/Write	MainProgram
<i>Step_076 - MainProgram/CONTROL - *C32(Step,Step_076)</i>		
Step_076.T	501	DINT
<i>Step_076.T - MainProgram/CONTROL - C32, #1(Tran,Tran_075 (Condition))</i>		
Step_077		
Constant	No	SFC_STEP
External Access:	Read/Write	MainProgram
<i>Step_077 - MainProgram/CONTROL - *C32(Step,Step_077)</i>		
Step_077.T	101	DINT
<i>Step_077.T - MainProgram/CONTROL - C33, #1(Tran,Tran_076 (Condition))</i>		

StopHeating	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>StopHeating - MainProgram/CONTROL - *B20, #4(Action,Action_053 (Body)), *B35, #3(Action,Action_070 (Body))</i>			
<i>StopHeating - MainProgram/HeatExtruder - 0(XIO), 2(XIO), 3(XIO)</i>			
TempDint	100	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>TempDint - MainProgram/CONTROL - *B1, #22(Action,Action_024 (Body)), *B37, #3(Action,Action_061 (Body)), B1, #23(Action,Action_024 (Body)), B37, #4(Action,Action_061 (Body))</i>			
<i>TempDint - MainProgram/SetWindingDia - *B6, #1(Action,Action_047 (Body)), B6, #2(Action,Action_047 (Body))</i>			
Tran_019	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_019 - MainProgram/CONTROL - *A8(Tran,Tran_019), *A8, #1(Tran,Tran_019 (Condition))</i>			
Tran_020	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_020 - MainProgram/CONTROL - *A9(Tran,Tran_020), *A9, #1(Tran,Tran_020 (Condition))</i>			
Tran_021	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_021 - MainProgram/CONTROL - *A11(Tran,Tran_021), *A11, #1(Tran,Tran_021 (Condition))</i>			
Tran_022	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_022 - MainProgram/CONTROL - *A12(Tran,Tran_022), *A12, #1(Tran,Tran_022 (Condition))</i>			
Tran_025	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_025 - MainProgram/CONTROL - *A28(Tran,Tran_025), *A28, #1(Tran,Tran_025 (Condition))</i>			
Tran_034	1	BOOL	MainProgram
Ready to wind			
Constant	No		
External Access:	Read/Write		
<i>Tran_034 - MainProgram/CONTROL - *A13(Tran,Tran_034), *A13, #1(Tran,Tran_034 (Condition))</i>			
Tran_036	1	BOOL	MainProgram
Done Winding			
Constant	No		
External Access:	Read/Write		
<i>Tran_036 - MainProgram/CONTROL - *A34(Tran,Tran_036), *A34, #1(Tran,Tran_036 (Condition))</i>			
Tran_037	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_037 - MainProgram/CONTROL - *A39(Tran,Tran_037), *A39, #1(Tran,Tran_037 (Condition))</i>			
Tran_038	1	BOOL	MainProgram
robot takes ~ 500ms and wait until winding is done moving			
Constant	No		
External Access:	Read/Write		
<i>Tran_038 - MainProgram/CONTROL - *A28(Tran,Tran_038), *A28, #1(Tran,Tran_038 (Condition))</i>			
Tran_039	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		

Tran_039 (Continued)

*Tran_039 - MainProgram/CONTROL - *A15(Tran,Tran_039), *A15, #1(Tran,Tran_039 (Condition))*

Tran_040 1 BOOL MainProgram

Still winding, allow time for done winding to be true before going here

Constant No

External Access: Read/Write

*Tran_040 - MainProgram/CONTROL - *F34(Tran,Tran_040), *F34, #1(Tran,Tran_040 (Condition))*

Tran_042 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_042 - MainProgram/CONTROL - *A21(Tran,Tran_042), *A21, #1(Tran,Tran_042 (Condition))*

Tran_045 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_045 - MainProgram/CONTROL - *A14(Tran,Tran_045), *A14, #1(Tran,Tran_045 (Condition))*

Tran_049 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_049 - MainProgram/CONTROL - *A14(Tran,Tran_049), *A14, #1(Tran,Tran_049 (Condition))*

Tran_051 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_051 - MainProgram/CONTROL - *A19(Tran,Tran_051), *A19, #1(Tran,Tran_051 (Condition))*

Tran_052 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_052 - MainProgram/CONTROL - *A19(Tran,Tran_052), *A19, #1(Tran,Tran_052 (Condition))*

Tran_053 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_053 - MainProgram/CONTROL - *A20(Tran,Tran_053), *A20, #1(Tran,Tran_053 (Condition))*

Tran_054 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_054 - MainProgram/CONTROL - *A30(Tran,Tran_054), *A30, #1(Tran,Tran_054 (Condition))*

Tran_055 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_055 - MainProgram/CONTROL - *A38(Tran,Tran_055), *A38, #1(Tran,Tran_055 (Condition))*

Tran_058 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_058 - MainProgram/CONTROL - *A16(Tran,Tran_058), *A16, #1(Tran,Tran_058 (Condition))*

Tran_059 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_059 - MainProgram/CONTROL - *A17(Tran,Tran_059), *A17, #1(Tran,Tran_059 (Condition))*

Tran_060 1 BOOL MainProgram

Constant No

External Access: Read/Write

*Tran_060 - MainProgram/CONTROL - *A18(Tran,Tran_060), *A18, #1(Tran,Tran_060 (Condition))*

Tran_061	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_061 - MainProgram/CONTROL - *C9(Tran,Tran_061), *C9, #1(Tran,Tran_061 (Condition))</i>			
Tran_062	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_062 - MainProgram/CONTROL - *A35(Tran,Tran_062), *A35, #1(Tran,Tran_062 (Condition))</i>			
Tran_063	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_063 - MainProgram/CONTROL - *C10(Tran,Tran_063), *C10, #1(Tran,Tran_063 (Condition))</i>			
Tran_064	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_064 - MainProgram/CONTROL - *C30(Tran,Tran_064), *C30, #1(Tran,Tran_064 (Condition))</i>			
Tran_065	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_065 - MainProgram/CONTROL - *A30(Tran,Tran_065), *A30, #1(Tran,Tran_065 (Condition))</i>			
Tran_066	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_066 - MainProgram/CONTROL - *C31(Tran,Tran_066), *C31, #1(Tran,Tran_066 (Condition))</i>			
Tran_067	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_067 - MainProgram/CONTROL - *C31(Tran,Tran_067), *C31, #1(Tran,Tran_067 (Condition))</i>			
Tran_068	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_068 - MainProgram/CONTROL - *A5(Tran,Tran_068), *A5, #1(Tran,Tran_068 (Condition))</i>			
Tran_069	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_069 - MainProgram/CONTROL - *A37(Tran,Tran_069), *A37, #1(Tran,Tran_069 (Condition))</i>			
Tran_070	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_070 - MainProgram/CONTROL - *A39(Tran,Tran_070), *A39, #1(Tran,Tran_070 (Condition))</i>			
Tran_071	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_071 - MainProgram/CONTROL - *A7(Tran,Tran_071), *A7, #1(Tran,Tran_071 (Condition))</i>			
Tran_072	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_072 - MainProgram/CONTROL - *A6(Tran,Tran_072), *A6, #1(Tran,Tran_072 (Condition))</i>			
Tran_073	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_073 - MainProgram/CONTROL - *A4(Tran,Tran_073), *A4, #1(Tran,Tran_073 (Condition))</i>			

Tran_074	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_074 - MainProgram/CONTROL - *A36(Tran,Tran_074), *A36, #1(Tran,Tran_074 (Condition))</i>			
Tran_075	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_075 - MainProgram/CONTROL - *C32(Tran,Tran_075), *C32, #1(Tran,Tran_075 (Condition))</i>			
Tran_076	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_076 - MainProgram/CONTROL - *C33(Tran,Tran_076), *C33, #1(Tran,Tran_076 (Condition))</i>			
WindingCurrentDia	170	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingCurrentDia - MainProgram/CONTROL - *B13, #1(Action,Action_049 (Body)), *B36, #3(Action,Action_060 (Body)), *B39, #1(Action,Action_063 (Body)), A37, #1(Tran,Tran_069 (Condition)), B36, #3(Action,Action_060 (Body)), B39, #1(Action,Action_063 (Body))</i>			
<i>WindingCurrentDia - MainProgram/SetWindingDia - B1, #1(Tran,Tran_031 (Condition)), B3, #1(Action,Action_032 (Body)), B7, #1(Action,Action_048 (Body))</i>			
WindingMoving	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingMoving - MainProgram/CONTROL - A28, #1(Tran,Tran_038 (Condition)), A34, #1(Tran,Tran_036 (Condition)), A38, #1(Tran,Tran_055 (Condition)), A39, #1(Tran,Tran_070 (Condition)), A4, #1(Tran,Tran_073 (Condition))</i>			
<i>WindingMoving - MainProgram/MoveWinding - *7(OTE)</i>			
<i>WindingMoving - MainProgram/SetWindingDia - B6, #1(Tran,Tran_047 (Condition))</i>			
WindingOldDia	170	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingOldDia - MainProgram/CONTROL - A37, #1(Tran,Tran_069 (Condition))</i>			
<i>WindingOldDia - MainProgram/SetWindingDia - *B7, #1(Action,Action_048 (Body)), B1, #1(Tran,Tran_031 (Condition)), B3, #1(Action,Action_032 (Body))</i>			
WindingSolenoid	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>WindingSolenoid - MainProgram/CONTROL - *B14, #1(Action,Action_034 (Body))</i>			
<i>WindingSolenoid - MainProgram/FillValues - #60</i>			
<i>WindingSolenoid - MainProgram/MainRoutine - 8(XIC)</i>			
<i>WindingSolenoid - MainProgram/pause - #12, *#34, *#58</i>			
<i>WindingSolenoid - MainProgram/SetWindingDia - *B1, #1(Action,Action_045 (Body)), *B2, #2(Action,Action_031 (Body)), *B4, #1(Action,Action_046 (Body))</i>			
<i>WindingSolenoid - MainProgram/StopAll - *#4</i>			
WindingTurns	99.667	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingTurns - MainProgram/CONTROL - *B1, #23(Action,Action_024 (Body)), *B23, #6(Action,Action_039 (Body)), *B37, #4(Action,Action_061 (Body)), *B37, #5(Action,Action_061 (Body)), B1, #22(Action,Action_024 (Body)), B23, #6(Action,Action_039 (Body)), B37, #3(Action,Action_061 (Body)), B37, #5(Action,Action_061 (Body))</i>			
<i>WindingTurns - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>WindingTurns - MainProgram/MoveWinding - *2(DIV)</i>			
<i>WindingTurns - MainProgram/SetWindingDia - *B3, #1(Action,Action_032 (Body)), *B6, #2(Action,Action_047 (Body)), B3, #1(Action,Action_032 (Body)), B6, #1(Action,Action_047 (Body))</i>			
<i>WindingTurns - MainProgram/ThugShake - *B2, #2(Action,Action_027 (Body)), *B3, #1(Action,Action_029 (Body)), B2, #2(Action,Action_027 (Body)), B3, #1(Action,Action_029 (Body))</i>			
WireAtCutting	1	BOOL	Capstone_Test
AliasFor:	Input_RemoteIO:I.Data.9		

WireAtCutting (Continued)			
Base Tag:	Input_RemoteIO:I.Data.9		
Constant	No		
External Access:	Read/Write		
<i>WireAtCutting - MainProgram/CONTROL - A9, #1(Tran,Tran_020 (Condition)), C10, #1(Tran,Tran_063 (Condition)), C9, #1(Tran,Tran_061 (Condition))</i>			
 WireAtEndOfLine	0	BOOL	Capstone_Test
AliasFor:	Input_RemoteIO:I.Data.6		
Base Tag:	Input_RemoteIO:I.Data.6		
Constant	No		
External Access:	Read/Write		
<i>WireAtEndOfLine - MainProgram/CONTROL - A12, #1(Tran,Tran_022 (Condition))</i>			
WireInRobot	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WireInRobot - MainProgram/CONTROL - *B1, #8(Action,Action_024 (Body))</i>			
<i>WireInRobot - MainProgram/pause - #19, *#41, *#65</i>			
<i>WireInRobot - MainProgram/StopAll - *#11</i>			

General

Type:	 Sequential Function Chart	Class:	Standard
In Program:	 MainProgram	Number of Steps:	Original 35

Auto-Naming

Step:	Step_078	Action:	Action_071
Transition:	Tran_077	Stop:	Stop_000

Sheet Layout

Sheet Size:	Letter - 8.5 x 11 in	Orientation:	Landscape
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```
1
2 if (S:FS) then
3     RobotEnable := 1;
4 end_if;
5
6
7 Winding_Servo:O.Data[2].6 := WindingFree;
8
9 //Wire Sensors
10 SpoolArray[0].Loaded := Input_RemoteIO:I.Data.2;
11 SpoolArray[1].Loaded := Input_RemoteIO:I.Data.3; // at cutting
12 SpoolArray[2].Loaded := Input_RemoteIO:I.Data.5;
13
14
15 //Extruder Block Temp
16 Temperature := Local:5:I.Ch03.Data;
17
18 //Load Cells and Calibration Factors
19 if Local:5:I.Ch00.Data>5 then
20     SpoolArray[0].Mass:=Local:5:I.Ch00.Data;
21 else
22     SpoolArray[0].Mass:=0;
23 end_if;
24 /* gone I guess
25 if Local:5:I.Ch01.Data>5 then
26     SpoolArray[1].Mass:=Local:5:I.Ch01.Data;
27 else
28     SpoolArray[1].Mass:=0;
29 end_if;
30 */
31 if Local:5:I.Ch02.Data>5 then
32     SpoolArray[2].Mass:=Local:5:I.Ch02.Data;
33 else
34     SpoolArray[2].Mass:=0;
35 end_if;
36
37 ////OUTPUTS
38
39 // FEED MOTORS. Took these out because FUCK FUKCK FUCK UFKC
40 Local:4:O.Pt00.Data := HMI.Manual.Spool.FWD1 OR SpoolArray[0].FWD;
41 Local:4:O.Pt01.Data := HMI.Manual.Spool.REV1 OR SpoolArray[0].REV;
42 Local:4:O.Pt02.Data := HMI.Manual.Spool.FWD2 OR SpoolArray[1].FWD;
43 Local:4:O.Pt03.Data := HMI.Manual.Spool.REV2 OR SpoolArray[1].REV;
44 Local:4:O.Pt04.Data := HMI.Manual.Spool.FWD3 OR SpoolArray[2].FWD;
45 Local:4:O.Pt05.Data := HMI.Manual.Spool.REV3 OR SpoolArray[2].REV;
46
47 Local:4:O.Pt06.Data := HMI.Manual.CameraLight OR CameraFlash; //CAMERA LIGHT
48
49 //Wire Clamp
50 Local:4:O.Pt12.Data := HMI.Manual.ClampMotorFWD OR ClampMotorFwd;
51 Local:4:O.Pt13.Data := HMI.Manual.ClampMotorREV OR ClampMotorRev;
52 //Limit Clamp
53 Local:4:O.Pt10.Data := HMI.Manual.HeightMotorDown OR HeightMotorDown;
54 Local:4:O.Pt11.Data := HMI.Manual.HeightMotorUp OR HeightMotorUp;
55
56 //Remote Out
57 Output_RemoteIO:O.Data.0 := HMI.Manual.Cut OR ExtendCutting;
58 Output_RemoteIO:O.Data.2 := HMI.Manual.ExtrudeMotor OR ExtruderMotor;
59 Output_RemoteIO:O.Data.4 := Heater OR HMI.Manual.ExtrudeHeater; //Extruder Heater
```

```
60 Output_RemoteIO:0.Data.6 := HMI.Manual.WindingSolenoid OR WindingSolenoid; //Winding Solenoid
61
62 // calculate cycle time
63 CycleTime := ((CurrentTime - OldTime) + CycleTime) * 0.5; // calculate cycle time, and average it with
    previous for a more stable value
64 OldTime := CurrentTime;
65 GSV(WallClockTime, ,DateTime, CurrentTime);
66
67 wireLoadDist := 3000;
68
69 prog:=((InitialLayers-Layers)/(InitialLayers));
70 proggy:=20*prog;
71 Progress:=5*proggy;
72
73
```

Name	Value	Data Type	Scope
 CameraFlash	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>CameraFlash - MainProgram/FillValues - #47</i>			
<i>CameraFlash - MainProgram/MainRoutine - *9(OTE)</i>			
 ClampMotorFwd	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>ClampMotorFwd - MainProgram/CONTROL - *B1, #14(Action,Action_024 (Body)), *B16, #1(Action,Action_057 (Body)), *B17, #1(Action,Action_058 (Body)), *B19, #1(Action,Action_052 (Body)), *B20, #1(Action,Action_053 (Body)), *B23, #34(Action,Action_039 (Body))</i>			
<i>ClampMotorFwd - MainProgram/FillValues - #50</i>			
<i>ClampMotorFwd - MainProgram/pause - #24, *#46, *#70</i>			
<i>ClampMotorFwd - MainProgram/StopAll - *#16</i>			
 ClampMotorRev	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>ClampMotorRev - MainProgram/CONTROL - *B1, #15(Action,Action_024 (Body)), *B17, #1(Action,Action_059 (Body)), *B18, #1(Action,Action_051 (Body)), *B35, #2(Action,Action_067 (Body)), *B36, #1(Action,Action_060 (Body))</i>			
<i>ClampMotorRev - MainProgram/FillValues - #51</i>			
<i>ClampMotorRev - MainProgram/pause - #25, *#47, *#71</i>			
<i>ClampMotorRev - MainProgram/StopAll - *#17</i>			
CurrentTime	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>CurrentTime - MainProgram/FillValues - #63, #64, *#65</i>			
 CycleTime	0	DINT	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>CycleTime - MainProgram/CONTROL - A30, #1(Tran,Tran_065 (Condition)), F34, #1(Tran,Tran_040 (Condition))</i>			
<i>CycleTime - MainProgram/FillValues - #63, *#63</i>			
ExtendCutting	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ExtendCutting - MainProgram/CONTROL - *B35, #1(Action,Action_067 (Body)), *B35, #1(Action,Action_070 (Body))</i>			
<i>ExtendCutting - MainProgram/FillValues - #57</i>			
ExtruderMotor	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ExtruderMotor - MainProgram/CONTROL - *B23, #30(Action,Action_039 (Body)), *B35, #2(Action,Action_070 (Body)), *C30, #1(Action,Action_026 (Body))</i>			
<i>ExtruderMotor - MainProgram/FillValues - #58</i>			
 Heater	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>Heater - MainProgram/FillValues - #59</i>			
<i>Heater - MainProgram/HeatExtruder - *1(OTE), *5(OTE)</i>			
 HeightMotorDown	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>HeightMotorDown - MainProgram/CONTROL - *B1, #19(Action,Action_024 (Body)), *B6, #1(Action,Action_064 (Body)), *B7, #1(Action,Action_065 (Body)), *C30, #2(Action,Action_026 (Body)), *C31, #1(Action,Action_062 (Body))</i>			
<i>HeightMotorDown - MainProgram/FillValues - #53</i>			
<i>HeightMotorDown - MainProgram/pause - #27, *#49, *#73</i>			
<i>HeightMotorDown - MainProgram/StopAll - *#21</i>			

 HeightMotorUp	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>HeightMotorUp - MainProgram/CONTROL - *B1, #20(Action,Action_024 (Body)), *B35, #3(Action,Action_067 (Body)), *B36, #2(Action,Action_060 (Body)), *B4, #1(Action,Action_066 (Body)), *B6, #1(Action,Action_054 (Body)), *C32, #1(Action,Action_068 (Body)), *C32, #1(Action,Action_069 (Body))</i>			
<i>HeightMotorUp - MainProgram/FillValues - #54</i>			
<i>HeightMotorUp - MainProgram/pause - #28, *#50, *#74</i>			
<i>HeightMotorUp - MainProgram/StopAll - *#22</i>			
 HMI		HMIControl	Capstone_Test
Constant	No		
External Access:	Read/Write		
HMI.Manual.Spool.FWD1	0	BOOL	
<i>HMI.Manual.Spool.FWD1 - MainProgram/FillValues - #40</i>			
HMI.Manual.Spool.FWD2	0	BOOL	
<i>HMI.Manual.Spool.FWD2 - MainProgram/FillValues - #42</i>			
HMI.Manual.Spool.FWD3	0	BOOL	
<i>HMI.Manual.Spool.FWD3 - MainProgram/FillValues - #44</i>			
HMI.Manual.Spool.REV1	0	BOOL	
<i>HMI.Manual.Spool.REV1 - MainProgram/FillValues - #41</i>			
HMI.Manual.Spool.REV2	0	BOOL	
<i>HMI.Manual.Spool.REV2 - MainProgram/FillValues - #43</i>			
HMI.Manual.Spool.REV3	0	BOOL	
<i>HMI.Manual.Spool.REV3 - MainProgram/FillValues - #45</i>			
HMI.Manual.Cut	0	BOOL	
<i>HMI.Manual.Cut - MainProgram/FillValues - #57</i>			
HMI.Manual.CameraLight	0	BOOL	
<i>HMI.Manual.CameraLight - MainProgram/FillValues - #47</i>			
HMI.Manual.ExtrudeHeater	0	BOOL	
<i>HMI.Manual.ExtrudeHeater - MainProgram/FillValues - #59</i>			
HMI.Manual.ExtrudeMotor	0	BOOL	
<i>HMI.Manual.ExtrudeMotor - MainProgram/FillValues - #58</i>			
HMI.Manual.WindingSolenoid	0	BOOL	
<i>HMI.Manual.WindingSolenoid - MainProgram/FillValues - #60</i>			
HMI.Manual.ClampMotorFWD	0	BOOL	
<i>HMI.Manual.ClampMotorFWD - MainProgram/FillValues - #50</i>			
HMI.Manual.ClampMotorREV	0	BOOL	
<i>HMI.Manual.ClampMotorREV - MainProgram/FillValues - #51</i>			
HMI.Manual.HeightMotorUp	0	BOOL	
<i>HMI.Manual.HeightMotorUp - MainProgram/FillValues - #54</i>			
HMI.Manual.HeightMotorDown	0	BOOL	
<i>HMI.Manual.HeightMotorDown - MainProgram/FillValues - #53</i>			
HMI.StartPB	0	BOOL	
<i>HMI.StartPB - MainProgram/MainRoutine - 2(XIC)</i>			
HMI.ResetPB	0	BOOL	
<i>HMI.ResetPB - MainProgram/MainRoutine - 0(XIC)</i>			
HMI.StopPB	0	BOOL	
<i>HMI.StopPB - MainProgram/MainRoutine - 2(XIO)</i>			
<i>HMI.StopPB - MainProgram/pause - #8</i>			
HMI.RobotFaultReset	0	BOOL	
<i>HMI.RobotFaultReset - MainProgram/RobotAutoControl - 8(XIC)</i>			
InitialLayers	20	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>InitialLayers - MainProgram/CONTROL - *B12, #6(Action,Action_023 (Body))</i>			
<i>InitialLayers - MainProgram/FillValues - #69, #69</i>			
 Input_RemoteIO:I		AB:1732_DI16:I:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Input_RemoteIO:I.Data.0	0	BOOL	
<i>StartPB</i>			

Input_RemoteIO:I (Continued)			
<i>PB_Start - MainProgram/MainRoutine - 2(XIC)</i>			
Input_RemoteIO:I.Data.2	1	BOOL	
<i>Input_RemoteIO:I.Data.2 - MainProgram/FillValues - #10</i>			
Input_RemoteIO:I.Data.3	1	BOOL	
<i>Input_RemoteIO:I.Data.3 - MainProgram/FillValues - #11</i>			
Input_RemoteIO:I.Data.5	0	BOOL	
<i>Input_RemoteIO:I.Data.5 - MainProgram/FillValues - #12</i>			
Input_RemoteIO:I.Data.6	0	BOOL	
<i>WireAtEndOfLine - MainProgram/CONTROL - A12, #1(Tran,Tran_022 (Condition))</i>			
Input_RemoteIO:I.Data.9	1	BOOL	
<i>WireAtCutting - MainProgram/CONTROL - A9, #1(Tran,Tran_020 (Condition)), C10, #1(Tran,Tran_063 (Condition)), C9, #1(Tran,Tran_061 (Condition))</i>			
Layers	20	DINT	MainProgram
<i>Constant</i>			
<i>No</i>			
<i>External Access:</i>			
<i>Read/Write</i>			
<i>Layers - MainProgram/CONTROL - *B12, #5(Action,Action_023 (Body)), B12, #6(Action,Action_023 (Body))</i>			
<i>Layers - MainProgram/FillValues - #69</i>			
Local:4:O		AB:5000_DO16:O:0	Capstone_Test
<i>Constant</i>			
<i>No</i>			
<i>External Access:</i>			
<i>Read/Write</i>			
Local:4:O.Pt00.Data	0	BOOL	
<i>Top Feed Motor Fwd</i>			
<i>Local:4:O.Pt00.Data - MainProgram/FillValues - *#40</i>			
Local:4:O.Pt01.Data	0	BOOL	
<i>Top Feed Motor Rev</i>			
<i>Local:4:O.Pt01.Data - MainProgram/FillValues - *#41</i>			
Local:4:O.Pt02.Data	0	BOOL	
<i>Mid Feed Motor Fwd</i>			
<i>Local:4:O.Pt02.Data - MainProgram/FillValues - *#42</i>			
Local:4:O.Pt03.Data	0	BOOL	
<i>Mid Feed Motor Rev</i>			
<i>Local:4:O.Pt03.Data - MainProgram/FillValues - *#43</i>			
Local:4:O.Pt04.Data	0	BOOL	
<i>Bottom Feed Motor Fwd</i>			
<i>Local:4:O.Pt04.Data - MainProgram/FillValues - *#44</i>			
Local:4:O.Pt05.Data	0	BOOL	
<i>Bottom Feed Motor Rev</i>			
<i>Local:4:O.Pt05.Data - MainProgram/FillValues - *#45</i>			
Local:4:O.Pt06		CHANNEL_DO:O:0	
<i>Camera Flash</i>			
Local:4:O.Pt06.Data	0	BOOL	
<i>Reset Robot Safety Relay</i>			
<i>Local:4:O.Pt06.Data - MainProgram/FillValues - *#47</i>			
Local:4:O.Pt07.Data	1	BOOL	
<i>StackLightGreen - MainProgram/MainRoutine - *5(OTE)</i>			
<i>StackLightGreen - MainProgram/pause - #10, *#32, *#56</i>			
<i>StackLightGreen - MainProgram/StopAll - *#2</i>			
Local:4:O.Pt08.Data	1	BOOL	
<i>StackLightYellow - MainProgram/MainRoutine - *6(OTE)</i>			
<i>StackLightYellow - MainProgram/pause - #11, *#33, *#57</i>			
<i>StackLightYellow - MainProgram/StopAll - *#3</i>			
Local:4:O.Pt09.Data	0	BOOL	
<i>StackLightRed - MainProgram/MainRoutine - *7(OTE)</i>			
<i>StackLightRed - MainProgram/pause - #9, *#31, *#55</i>			
<i>StackLightRed - MainProgram/StopAll - *#1</i>			
Local:4:O.Pt10.Data	0	BOOL	
<i>Local:4:O.Pt10.Data - MainProgram/FillValues - *#53</i>			
Local:4:O.Pt11.Data	0	BOOL	
<i>Local:4:O.Pt11.Data - MainProgram/FillValues - *#54</i>			
Local:4:O.Pt12.Data	0	BOOL	
<i>Local:4:O.Pt12.Data - MainProgram/FillValues - *#50</i>			

Local:4:O (Continued)			
Local:4:O.Pt13.Data	0	BOOL	
<i>Local:4:O.Pt13.Data - MainProgram/FillValues - *#51</i>			
Local:4:O.Pt14.Data	0	BOOL	
<i>ResetLight - MainProgram/MainRoutine - *7(OTE)</i>			
Local:5:I			
Constant	No	AB:5000_AI8:I:0	Capstone_Test
External Access:	Read/Write		
Local:5:I.Ch00.Data	164.73665	REAL	
<i>Local:5:I.Ch00.Data - MainProgram/Alarms - 7(LEQ)</i>			
<i>Local:5:I.Ch00.Data - MainProgram/FillValues - #19, #20</i>			
Local:5:I.Ch01.Data	625.2037	REAL	
<i>Local:5:I.Ch01.Data - MainProgram/Alarms - 8(LEQ)</i>			
Local:5:I.Ch02.Data	600.10535	REAL	
<i>Local:5:I.Ch02.Data - MainProgram/Alarms - 9(LEQ)</i>			
<i>Local:5:I.Ch02.Data - MainProgram/FillValues - #31, #32</i>			
Local:5:I.Ch03.Data	-2.61334376e-003	REAL	
<i>Local:5:I.Ch03.Data - MainProgram/FillValues - #16</i>			
OldTime			
Constant	0	DINT	MainProgram
External Access:	No		
<i>OldTime - MainProgram/FillValues - #63, *#64</i>			
Output_RemoteIO:O			
Constant	No	AB:1732_DO16:O:0	Capstone_Test
External Access:	Read/Write		
Output_RemoteIO:O.Data.0	0	BOOL	
<i>Output_RemoteIO:O.Data.0 - MainProgram/FillValues - *#57</i>			
Output_RemoteIO:O.Data.2	0	BOOL	
<i>Output_RemoteIO:O.Data.2 - MainProgram/FillValues - *#58</i>			
Output_RemoteIO:O.Data.4	0	BOOL	
<i>Output_RemoteIO:O.Data.4 - MainProgram/FillValues - *#59</i>			
Output_RemoteIO:O.Data.6	0	BOOL	
<i>Output_RemoteIO:O.Data.6 - MainProgram/FillValues - *#60</i>			
prog			
Constant	0.0	REAL	MainProgram
External Access:	No		
<i>prog - MainProgram/FillValues - #70, *#69</i>			
proggy			
Constant	0	DINT	MainProgram
External Access:	No		
<i>proggy - MainProgram/FillValues - #71, *#70</i>			
Progress			
Constant	0	DINT	Capstone_Test
External Access:	No		
<i>Progress - MainProgram/FillValues - *#71</i>			
RobotEnable			
Constant	1	BOOL	MainProgram
External Access:	No		
<i>RobotEnable - MainProgram/FillValues - *#3</i>			
<i>RobotEnable - MainProgram/pause - #29, *#51, *#75</i>			
<i>RobotEnable - MainProgram/RobotAutoControl - *0(OTL), *1(OTU), 0(XIC), 3(XIC)</i>			
SpoolArray			
Constant	No	Spool[3]	Capstone_Test
External Access:	Read/Write		
SpoolArray[0].REV	0	BOOL	
<i>out</i>			

SpoolArray (Continued)		
<i>SpoolArray[0].REV - MainProgram/CONTROL - *B1, #3(Action,Action_024 (Body))</i>		
<i>SpoolArray[0].REV - MainProgram/FillValues - #41</i>		
<i>SpoolArray[0].REV - MainProgram/pause - #14, *#36, *#60</i>		
<i>SpoolArray[0].REV - MainProgram/StopAll - *#6</i>		
SpoolArray[0].FWD	0	BOOL
out		
<i>SpoolArray[0].FWD - MainProgram/CONTROL - *B1, #2(Action,Action_024 (Body))</i>		
<i>SpoolArray[0].FWD - MainProgram/FillValues - #40</i>		
<i>SpoolArray[0].FWD - MainProgram/pause - #13, *#35, *#59</i>		
<i>SpoolArray[0].FWD - MainProgram/StopAll - *#5</i>		
SpoolArray[0].Mass	164.73665	REAL
current mass reading		
<i>SpoolArray[0].Mass - MainProgram/FillValues - *#20, *#22</i>		
SpoolArray[0].Length	0.0	REAL
approximated legth based on Mass		
SpoolArray[0].Loaded	1	BOOL
if is in loaded state		
<i>SpoolArray[0].Loaded - MainProgram/FillValues - *#10</i>		
SpoolArray[0].Gauge	-1	DINT
Wire Gauge		
SpoolArray[1].REV	0	BOOL
out		
<i>SpoolArray[1].REV - MainProgram/CONTROL - *B1, #5(Action,Action_024 (Body))</i>		
<i>SpoolArray[1].REV - MainProgram/FillValues - #43</i>		
<i>SpoolArray[1].REV - MainProgram/pause - #16, *#38, *#62</i>		
<i>SpoolArray[1].REV - MainProgram/StopAll - *#8</i>		
SpoolArray[1].FWD	0	BOOL
out		
<i>SpoolArray[1].FWD - MainProgram/CONTROL - *B1, #4(Action,Action_024 (Body))</i>		
<i>SpoolArray[1].FWD - MainProgram/FillValues - #42</i>		
<i>SpoolArray[1].FWD - MainProgram/pause - #15, *#37, *#61</i>		
<i>SpoolArray[1].FWD - MainProgram/StopAll - *#7</i>		
SpoolArray[1].Mass	163.34396	REAL
current mass reading		
SpoolArray[1].Length	0.0	REAL
approximated legth based on Mass		
SpoolArray[1].Loaded	1	BOOL
if is in loaded state		
<i>SpoolArray[1].Loaded - MainProgram/FillValues - *#11</i>		
SpoolArray[1].Gauge	-1	DINT
Wire Gauge		
SpoolArray[2].REV	0	BOOL
out		
<i>SpoolArray[2].REV - MainProgram/CONTROL - *B1, #7(Action,Action_024 (Body))</i>		
<i>SpoolArray[2].REV - MainProgram/FillValues - #45</i>		
<i>SpoolArray[2].REV - MainProgram/pause - #18, *#40, *#64</i>		
<i>SpoolArray[2].REV - MainProgram/StopAll - *#10</i>		
SpoolArray[2].FWD	0	BOOL
out		
<i>SpoolArray[2].FWD - MainProgram/CONTROL - *B1, #6(Action,Action_024 (Body)), *B11, #1(Action,Action_021 (Body)), *B11, #1(Action,Action_022 (Body)), *B12, #1(Action,Action_023 (Body)), *B23, #3(Action,Action_039 (Body)), *B37, #1(Action,Action_061 (Body)), *C9, #1(Action,Action_019 (Body))</i>		
<i>SpoolArray[2].FWD - MainProgram/FillValues - #44</i>		
<i>SpoolArray[2].FWD - MainProgram/pause - #17, *#39, *#63</i>		
<i>SpoolArray[2].FWD - MainProgram/StopAll - *#9</i>		
SpoolArray[2].Mass	600.10535	REAL
current mass reading		
<i>SpoolArray[2].Mass - MainProgram/FillValues - *#32, *#34</i>		
SpoolArray[2].Length	0.0	REAL
approximated legth based on Mass		
SpoolArray[2].Loaded	0	BOOL
if is in loaded state		
<i>SpoolArray[2].Loaded - MainProgram/FillValues - *#12</i>		

SpoolArray (Continued)			
SpoolArray[2].Gauge	16	DINT	
Wire Gauge			
<i>SpoolArray[2].Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)</i>			
Temperature	-2.61334376e-003	REAL	Capstone_Test
Constant No			
External Access: Read/Write			
<i>Temperature - MainProgram/FillValues - *#16</i>			
<i>Temperature - MainProgram/HeatExtruder - 1(GEQ), 1(LEQ), 2(GEQ), 2(GEQ), 2(LES), 3(LES), 4(LES), 5(LES)</i>			
Winding_Servo:O		_00BB:AZD_AEP_60D82B42:O:0	Capstone_Test
Constant No			
External Access: Read/Write			
<i>Winding_Servo:O - MainProgram/MoveWinding - *8(OM_AZ_DDO)</i>			
Winding_Servo:O.Data[0]	0	INT	
Remote I/O (R-IN)			
Winding_Servo:O.Data[0].0	0	BOOL	
R-IN0			
Winding_Servo:O.Data[1]	0	INT	
Operation data number selection			
Winding_Servo:O.Data[2]	512	INT	
Fixed I/O (IN)			
Winding_Servo:O.Data[2].0	0	BOOL	
FW-JOG			
Winding_Servo:O.Data[2].5	0	BOOL	
STOP			
<i>Winding_Servo:O.Data[2].5 - MainProgram/MainRoutine - *7(OTE)</i>			
Winding_Servo:O.Data[2].6	0	BOOL	
FREE			
<i>Winding_Servo:O.Data[2].6 - MainProgram/FillValues - *#7</i>			
Winding_Servo:O.Data[2].7	0	BOOL	
ALM-RST			
<i>Winding_Servo:O.Data[2].7 - MainProgram/Alarms - 6(ALMD)</i>			
<i>Winding_Servo:O.Data[2].7 - MainProgram/MoveWinding - *6(OTE)</i>			
Winding_Servo:O.Data[3]	1	INT	
Direct data operation type			
Winding_Servo:O.Data[4]	22630	INT	
Direct data operation position (lower)			
Winding_Servo:O.Data[5]	4	INT	
Direct data operation position (upper)			
Winding_Servo:O.Data[6]	750	INT	
Direct data operation operating speed (lower)			
Winding_Servo:O.Data[7]	0	INT	
Direct data operation operating speed (upper)			
Winding_Servo:O.Data[8]	300	INT	
Direct data operation starting/changing rate (lower)			
Winding_Servo:O.Data[9]	0	INT	
Direct data operation starting/changing rate (upper)			
Winding_Servo:O.Data[10]	750	INT	
Direct data operation stopping deceleration (lower)			
Winding_Servo:O.Data[11]	0	INT	
Direct data operation stopping deceleration (upper)			
Winding_Servo:O.Data[12]	1000	INT	
Direct data operation operating current			
Winding_Servo:O.Data[13]	0	INT	
Direct data operation forwarding destination			
Winding_Servo:O.Data[14]	0	INT	
Reserved			
Winding_Servo:O.Data[15]	3074	INT	
Read Parameter ID			
Winding_Servo:O.Data[16]	0	INT	
Write Request			
Winding_Servo:O.Data[17]	3074	INT	

Winding_Servo:O (Continued)			
Write Parameter ID			
Winding_Servo:O.Data[18]	1000	INT	
Write data (lower)			
Winding_Servo:O.Data[19]	0	INT	
Write data (upper)			
WindingFree			
Constant	0	BOOL	MainProgram
No			
External Access: Read/Write			
<i>WindingFree - MainProgram/FillValues - #7</i>			
WindingSolenoid			
Constant	0	BOOL	Capstone_Test
No			
External Access: Read/Write			
<i>WindingSolenoid - MainProgram/CONTROL - *B14, #1(Action,Action_034 (Body))</i>			
<i>WindingSolenoid - MainProgram/FillValues - #60</i>			
<i>WindingSolenoid - MainProgram/MainRoutine - 8(XIC)</i>			
<i>WindingSolenoid - MainProgram/pause - #12, *#34, *#58</i>			
<i>WindingSolenoid - MainProgram/SetWindingDia - *B1, #1(Action,Action_045 (Body)), *B2, #2(Action,Action_031 (Body)), *B4, #1(Action,Action_046 (Body))</i>			
<i>WindingSolenoid - MainProgram/StopAll - *#4</i>			
wireLoadDist			
Constant	3000	DINT	MainProgram
No			
External Access: Read/Write			
<i>wireLoadDist - MainProgram/FillValues - *#67</i>			

General

Progrsm that fills various values throughout the program

Type:  Structured Text

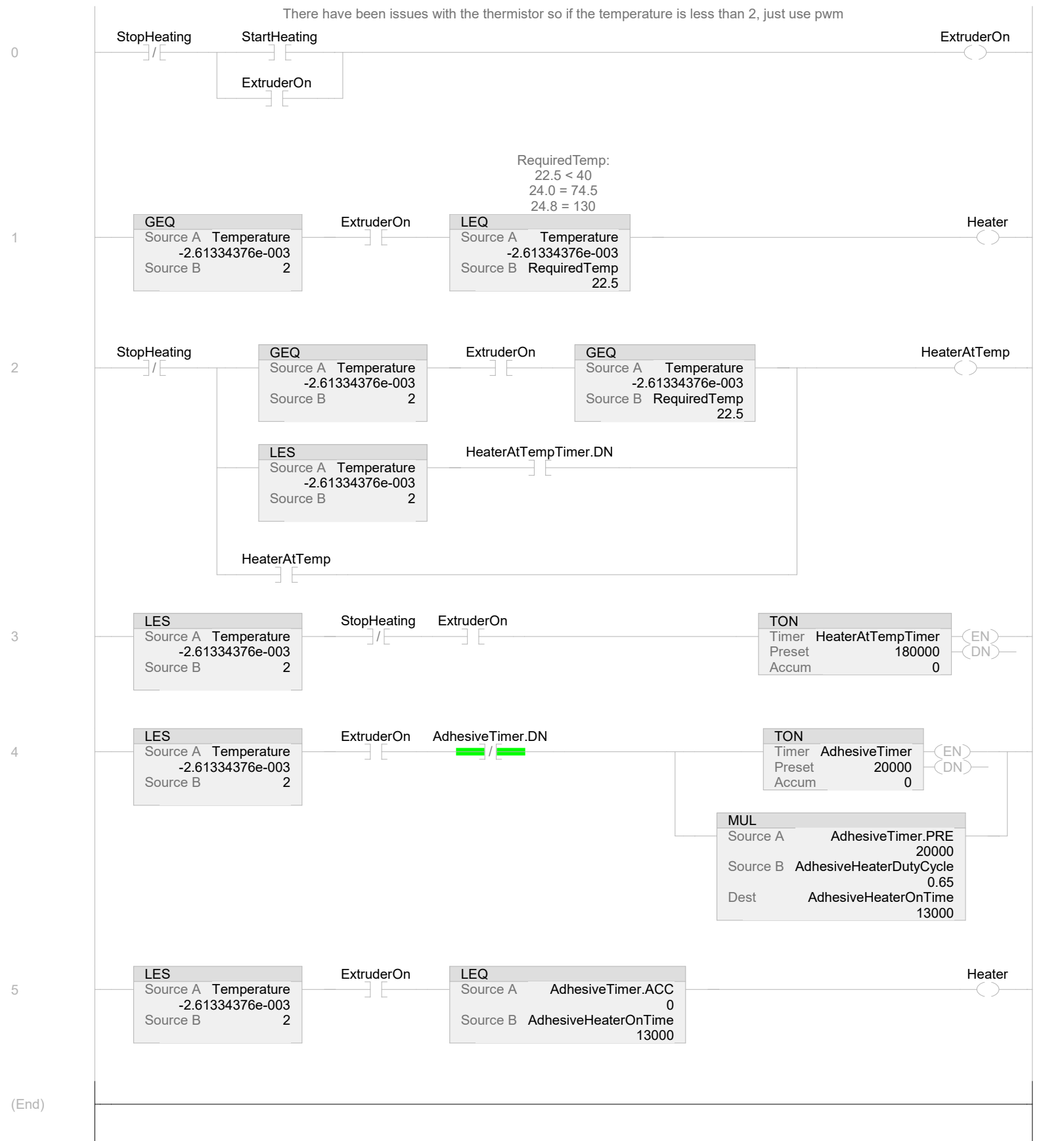
In Program:  MainProgram

Class:

Standard

Number of Lines:

Original 73

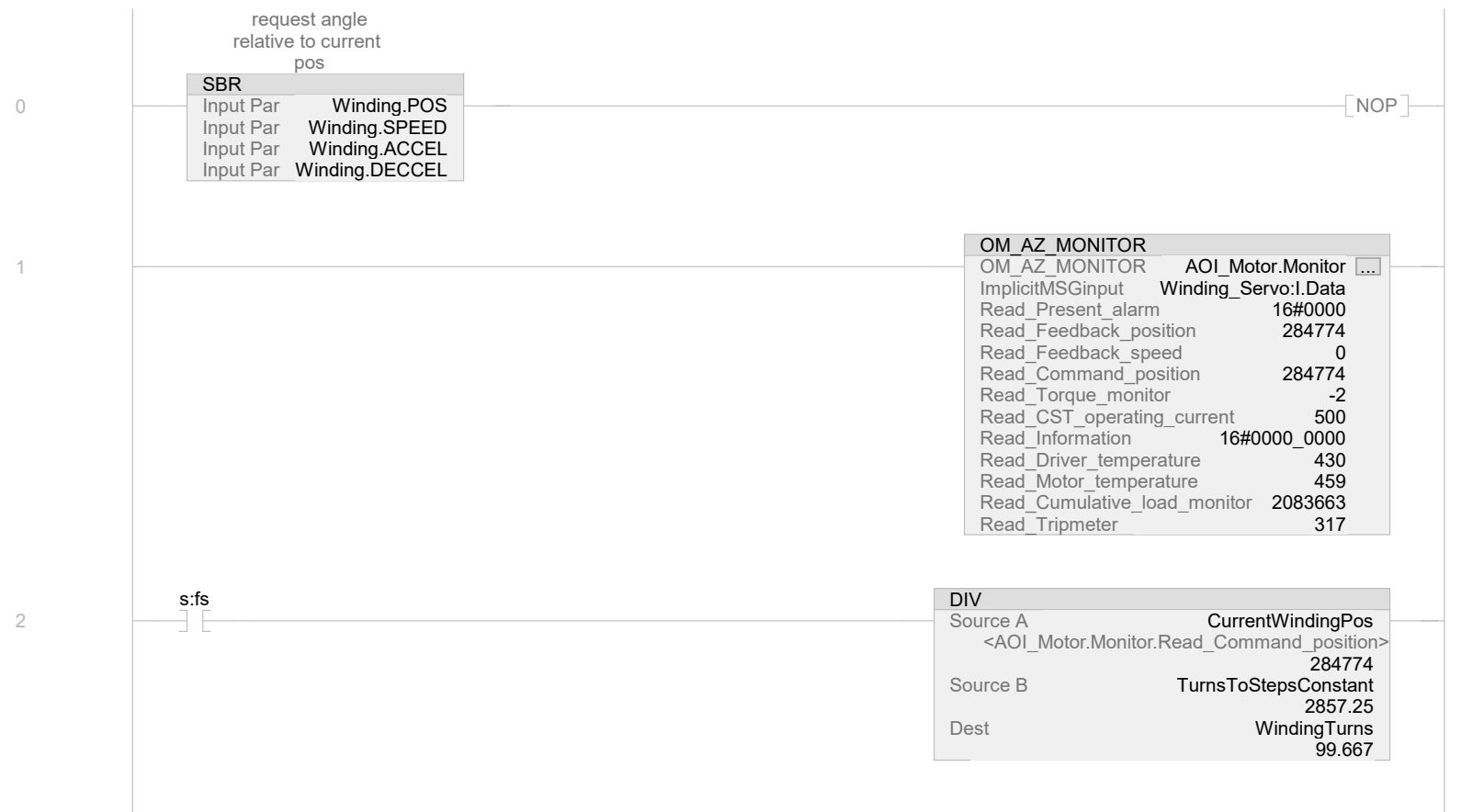


Name	Value	Data Type	Scope
AdhesiveHeaterDutyCycle	0.65	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AdhesiveHeaterDutyCycle - MainProgram/HeatExtruder - 4(MUL)</i>			
AdhesiveHeaterOnTime	13000	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AdhesiveHeaterOnTime - MainProgram/HeatExtruder - *4(MUL), 5(LEQ)</i>			
AdhesiveTimer		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AdhesiveTimer - MainProgram/HeatExtruder - *4(TON)</i>			
AdhesiveTimer.PRE	20000	DINT	
<i>AdhesiveTimer.PRE - MainProgram/HeatExtruder - 4(MUL)</i>			
AdhesiveTimer.ACC	0	DINT	
<i>AdhesiveTimer.ACC - MainProgram/HeatExtruder - 5(LEQ)</i>			
AdhesiveTimer.DN	0	BOOL	
<i>AdhesiveTimer.DN - MainProgram/HeatExtruder - 4(XIO)</i>			
ExtruderOn	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ExtruderOn - MainProgram/CONTROL - *B20, #5(Action,Action_053 (Body))</i>			
<i>ExtruderOn - MainProgram/HeatExtruder - *0(OTE), 0(XIC), 1(XIC), 2(XIC), 3(XIC), 4(XIC), 5(XIC)</i>			
Heater	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>Heater - MainProgram/FillValues - #59</i>			
<i>Heater - MainProgram/HeatExtruder - *1(OTE), *5(OTE)</i>			
HeaterAtTemp	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>HeaterAtTemp - MainProgram/CONTROL - A20, #1(Tran,Tran_053 (Condition))</i>			
<i>HeaterAtTemp - MainProgram/HeatExtruder - *2(OTE), 2(XIC)</i>			
HeaterAtTempTimer		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>HeaterAtTempTimer - MainProgram/HeatExtruder - *3(TON)</i>			
HeaterAtTempTimer.DN	0	BOOL	
<i>HeaterAtTempTimer.DN - MainProgram/HeatExtruder - 2(XIC)</i>			
RequiredTemp	22.5	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>RequiredTemp - MainProgram/HeatExtruder - 1(LEQ), 2(GEQ)</i>			
StartHeating	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>StartHeating - MainProgram/HeatExtruder - 0(XIC)</i>			
StopHeating	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>StopHeating - MainProgram/CONTROL - *B20, #4(Action,Action_053 (Body)), *B35, #3(Action,Action_070 (Body))</i>			
<i>StopHeating - MainProgram/HeatExtruder - 0(XIO), 2(XIO), 3(XIO)</i>			
Temperature	-2.61334376e-003	REAL	Capstone_Test

Temperature (Continued)

Constant	No
External Access:	Read/Write
<i>Temperature - MainProgram/FillValues - *#16</i>	
<i>Temperature - MainProgram/HeatExtruder - 1(GEQ), 1(LEQ), 2(GEQ), 2(GEQ), 2(LES), 3(LES), 4(LES), 5(LES)</i>	

General			
Type:	 Ladder Diagram	Class:	Standard
In Program:	 MainProgram	Number of Rungs:	6



Ratios:
of gear teeth on winding: 90
of gear teeth on motor: 35
motor steps / revolution: 1000?
of solenoid alignment holes: 30
pos -> steps: $1000 * (90/35)$
2571.4286

Direct data
operation
position

MUL	
Source A	Winding.POS 99.667
Source B	TurnsToStepsConstant 2857.25
Dest	AOI_Motor.DDO.Set_Position 284774

Direct data
operation
operating speed

MOV	
Source	Winding.SPEED 750
Dest	AOI_Motor.DDO.Set_Velocity 750

Direct data
operation
starting/changing
rate

MOV	
Source	Winding.ACCEL 300
Dest	AOI_Motor.DDO.Set_Acceleration 300

Direct data
operation
stopping
deceleration

MOV	
Source	Winding.DECCEL 750
Dest	AOI_Motor.DDO.Set_Deceleration 750

possible breakpoint: if oldpos gets set before this line runs
Command position
monitor[step]

S:FS

NEQ	
Source A	CurrentWindingPos <AOI_Motor.Monitor.Read_Command_position> 284774
Source B	AOI_Motor.DDO.Set_Position 284774

Error status in AOI
AOI_Motor.DDO.Sts_Err

Execute AOI
AOI_Motor.DDO.Set_Exe
(L)

Complete status in
AOI
AOI_Motor.DDO.Sts_Done

S:FS

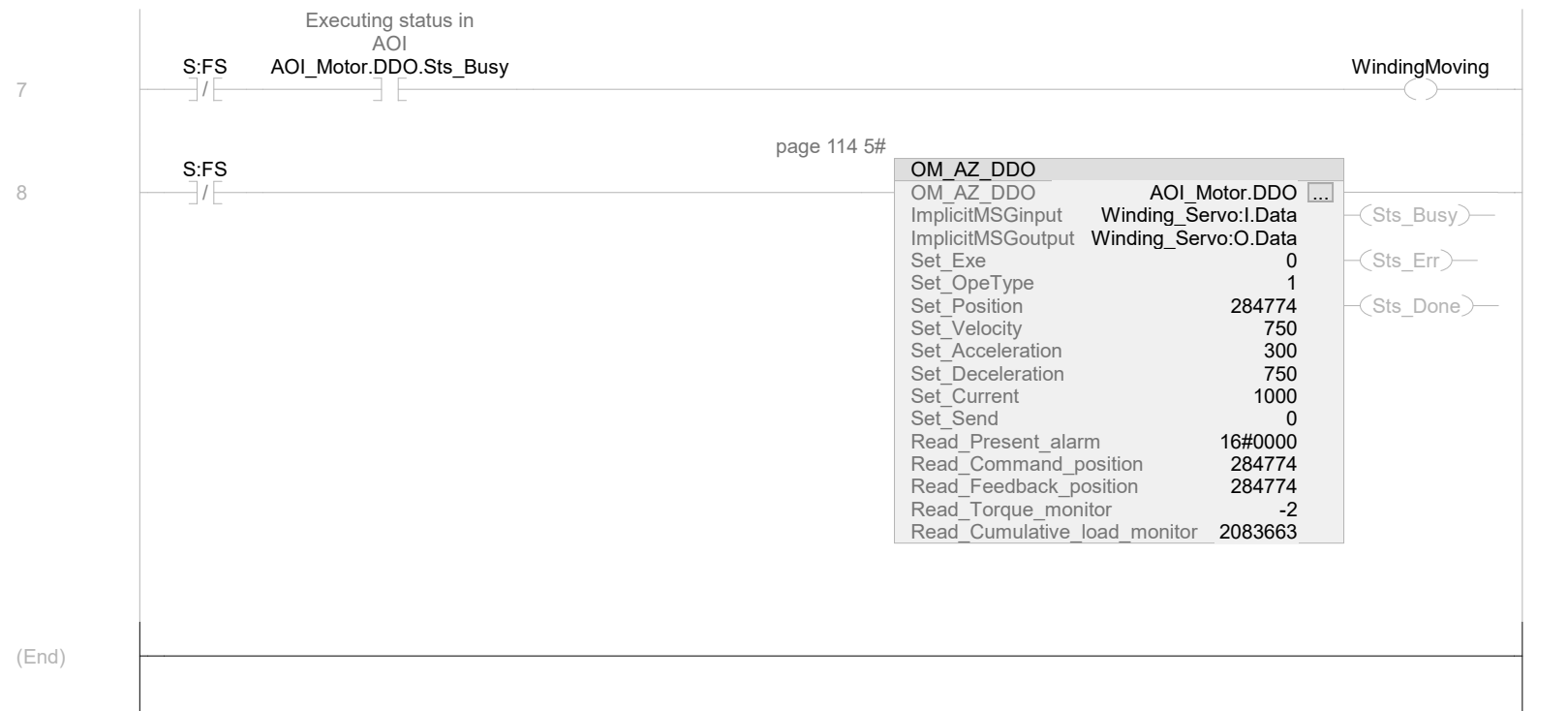
Execute AOI
AOI_Motor.DDO.Set_Exe
(U)

Error status in AOI
AOI_Motor.DDO.Sts_Err

S:FS

Execute AOI
AOI_Motor.DDO.Set_Exe
(U)

ALM-RST
Winding_Servo:O.Data[2].7



Name	Value	Data Type	Scope
AOI_Motor		AOI_Motor	Capstone_Test
Constant	No		
External Access:	Read/Write		
AOI_Motor.DDO		OM_AZ_DDO	
<i>AOI_Motor.DDO - MainProgram/MoveWinding - *8(OM_AZ_DDO)</i>			
AOI_Motor.DDO.EnableIn	1	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.DDO.EnableOut	1	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.DDO.Set_Exec	0	BOOL	
Execute AOI			
<i>AOI_Motor.DDO.Set_Exec - MainProgram/MoveWinding - *4(OTL), *5(OTU), *6(OTU)</i>			
AOI_Motor.DDO.Set_OpeType	1	INT	
Direct data operation operation type			
AOI_Motor.DDO.Set_Position	284774	DINT	
Direct data operation position			
<i>AOI_Motor.DDO.Set_Position - MainProgram/MoveWinding - *3(MUL), 4(NEQ)</i>			
AOI_Motor.DDO.Set_Velocity	750	DINT	
Direct data operation operating speed			
<i>AOI_Motor.DDO.Set_Velocity - MainProgram/MoveWinding - *3(MOV)</i>			
AOI_Motor.DDO.Set_Acceleration	300	DINT	
Direct data operation starting/changing rate			
<i>AOI_Motor.DDO.Set_Acceleration - MainProgram/MoveWinding - *3(MOV)</i>			
AOI_Motor.DDO.Set_Deceleration	750	DINT	
Direct data operation stopping deceleration			
<i>AOI_Motor.DDO.Set_Deceleration - MainProgram/MoveWinding - *3(MOV)</i>			
AOI_Motor.DDO.Set_Current	1000	INT	
Direct data operation operating current			
AOI_Motor.DDO.Set_Send	0	INT	
Direct data operation forwarding destination			
AOI_Motor.DDO.Sts_Busy	0	BOOL	
Executing status in AOI			
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/MoveWinding - 7(XIC)</i>			
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/SetWindingDia - B4, #1(Tran,Tran_041 (Condition))</i>			
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/ThugShake - B2, #1(Tran,Tran_027 (Condition)), C4, #1(Tran,Tran_029 (Condition))</i>			
AOI_Motor.DDO.Sts_Err	0	BOOL	
Error status in AOI			
<i>AOI_Motor.DDO.Sts_Err - MainProgram/MoveWinding - 4(XIO), 6(XIC)</i>			
AOI_Motor.DDO.Sts_Done	0	BOOL	
Complete status in AOI			
<i>AOI_Motor.DDO.Sts_Done - MainProgram/MoveWinding - 5(XIC)</i>			
AOI_Motor.DDO.Read_Present_alarm	16#0000	INT	
Alarm code monitor[Hex]			
AOI_Motor.DDO.Read_Command_position	284774	DINT	
Command position monitor[step]			
AOI_Motor.DDO.Read_Feedback_position	284774	DINT	
Feedback position monitor[step]			
AOI_Motor.DDO.Read_Torque_monitor	-2	INT	
Torque monitor as a percentage of the maximum holding torque(1=0.1%)			
AOI_Motor.DDO.Read_Cumulative_load_monitor	2083663	DINT	
Cumulative load monitor			
AOI_Motor.Move.EnableIn	0	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.Move.EnableOut	0	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.Move.Set_Exec	0	BOOL	
Execute AOI			
AOI_Motor.Move.Set_OpeDataNo	0	INT	

AOI_Motor (Continued)		
Selected Operation data No.		
AOI_Motor.Move.Sts_Busy	0	BOOL
Executing status in AOI		
AOI_Motor.Move.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.Move.Sts_Done	1	BOOL
Complete status in AOI		
AOI_Motor.Move.Read_Present_alarm		
16#0000		INT
Alarm code monitor[Hex]		
AOI_Motor.Move.Read_Command_position		
-2692		DINT
Command position monitor[step]		
AOI_Motor.Move.Read_Feedback_position		
-2692		DINT
Feedback position monitor[step]		
AOI_Motor.Move.Read_Torque_monitor		
11		INT
Torque monitor as a percentage of the maximum holding torque(1=0.1%)		
AOI_Motor.Move.Read_Cumulative_load_monitor		
677105		DINT
Cumulative load monitor		
AOI_Motor.Monitor		OM_AZ_MONITOR
<i>AOI_Motor.Monitor - MainProgram/MoveWinding - *I(OM_AZ_MONITOR)</i>		
AOI_Motor.Monitor.EnableIn	1	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Monitor.EnableOut	1	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Monitor.Read_Present_alarm		
16#0000		INT
Alarm code monitor[Hex]		
AOI_Motor.Monitor.Read_Feedback_position		
284774		DINT
Feedback position monitor[step]		
AOI_Motor.Monitor.Read_Feedback_speed		
0		DINT
Feedback speed monitor[Hz]		
AOI_Motor.Monitor.Read_Command_position		
284774		DINT
Command position monitor[step]		
<i>CurrentWindingPos - MainProgram/MoveWinding - 2(DIV), 4(NEQ)</i>		
AOI_Motor.Monitor.Read_Torque_monitor		
-2		INT
Torque monitor as a percentage of the maximum holding torque[1=0.1%]		
AOI_Motor.Monitor.Read_CST_operating_current		
500		INT
Operation current monitor in alpha control mode(CST) (1=0.1%)		
AOI_Motor.Monitor.Read_Information		
16#0000_0000		DINT
Information monitor before an alarm is generated.		
AOI_Motor.Monitor.Read_Driver_temperature		
430		DINT
Driver temperature monitor(1=0.1deg)		
AOI_Motor.Monitor.Read_Motor_temperature		
459		DINT
Motor temperature monitor(1=0.1deg)		
AOI_Motor.Monitor.Read_Cumulative_load_monitor		
2083663		DINT
Cumulative load monitor		
AOI_Motor.Monitor.Read_Tripmeter		
317		DINT
Travel distance monitor[x1000rev]		
AOI_Motor.ReadPosition.EnableIn	0	BOOL

AOI_Motor (Continued)		
Enable Input - System Defined Parameter		
AOI_Motor.ReadPosition.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.ReadPosition.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.ReadPosition.ReadPRMID	16#0c01	INT
Read parameter ID		
AOI_Motor.ReadPosition.ReadData	5000	DINT
Read data		
AOI_Motor.ReadPosition.Sts_Busy	1	BOOL
Executing status in AOI		
AOI_Motor.ReadPosition.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.ReadPosition.Sts_Done	0	BOOL
Complete status in AOI		
AOI_Motor.WritePosition.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.WritePosition.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.WritePosition.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.WritePosition.WritePRMID	16#0c01	INT
Write parameter ID		
AOI_Motor.WritePosition.WriteData	4000	DINT
Write data		
AOI_Motor.WritePosition.Sts_Busy	0	BOOL
Executing status in AOI		
AOI_Motor.WritePosition.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.WritePosition.Sts_Done	1	BOOL
Complete status in AOI		
AOI_Motor.Home.EnableIn	1	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Home.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Home.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.Home.Sts_Busy	0	BOOL
Executing status in AOI		
AOI_Motor.Home.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.Home.Sts_Done	0	BOOL
Complete status in AOI		
AOI_Motor.Home.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.Home.Read_Feedback_position	0	DINT
Feedback position monitor[step]		
AOI_Motor.WriteSpeed.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.WriteSpeed.EnableOut	0	BOOL

AOI_Motor (Continued)			
Enable Output - System Defined Parameter			
AOI_Motor.WriteSpeed.Set_Exec	0	BOOL	
Execute AOI			
AOI_Motor.WriteSpeed.WritePRMID	16#0c02	INT	
Write parameter ID			
AOI_Motor.WriteSpeed.WriteData	1000	DINT	
Write data			
AOI_Motor.WriteSpeed.Sts_Busy	0	BOOL	
Executing status in AOI			
AOI_Motor.WriteSpeed.Sts_Err	0	BOOL	
Error status in AOI			
AOI_Motor.WriteSpeed.Sts_Done	1	BOOL	
Complete status in AOI			
AOI_Motor.ReadSpeed.EnableIn	0	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.ReadSpeed.EnableOut	0	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.ReadSpeed.Set_Exec	0	BOOL	
Execute AOI			
AOI_Motor.ReadSpeed.ReadPRMID	16#0c02	INT	
Read parameter ID			
AOI_Motor.ReadSpeed.ReadData	1000	DINT	
Read data			
AOI_Motor.ReadSpeed.Sts_Busy	0	BOOL	
Executing status in AOI			
AOI_Motor.ReadSpeed.Sts_Err	0	BOOL	
Error status in AOI			
AOI_Motor.ReadSpeed.Sts_Done	1	BOOL	
Complete status in AOI			
CurrentWindingPos	284774	DINT	MainProgram
Command position monitor[step]			
AliasFor:	AOI_Motor.Monitor.Read_Command_position(C)		
Base Tag:	AOI_Motor.Monitor.Read_Command_position(C)		
Constant	No		
External Access:	Read/Write		
CurrentWindingPos - MainProgram/MoveWinding - 2(DIV), 4(NEQ)			
TurnsToStepsConstant	2857.25	REAL	MainProgram
Constant			
External Access:	Read/Write		
TurnsToStepsConstant - MainProgram/MoveWinding - 2(DIV), 3(MUL)			
Winding		ServoMotor	MainProgram
Constant			
External Access:	Read/Write		
Winding.POS	99.667	REAL	
request angle relative to current pos			
Winding.POS - MainProgram/MoveWinding - *0(SBR), 3(MUL)			
Winding.SPEED	750	DINT	
deg/s			
Winding.SPEED - MainProgram/MoveWinding - *0(SBR), 3(MOV)			
Winding.ACCEL	300	DINT	
deg/s/s			
Winding.ACCEL - MainProgram/MoveWinding - *0(SBR), 3(MOV)			
Winding.DECCEL	750	DINT	
deg/s/s			
Winding.DECCEL - MainProgram/MoveWinding - *0(SBR), 3(MOV)			
Winding_Servo:I		_00BB:AZD_AEP_A3E1851B:I:0	Capstone_Test
Constant			
	No		

Winding_Servo:I (Continued)		
External Access:	Read/Write	
Winding_Servo:I.Data		INT
Winding_Servo:I.Data - MainProgram/MoveWinding - *1(OM_AZ_MONITOR), *8(OM_AZ_DDO)		
Winding_Servo:I.Data[0]	16416	INT
Remote I/O (R-OUT)		
Winding_Servo:I.Data[0].0	0	BOOL
R-OUT0		
Winding_Servo:I.Data[1]	0	INT
Operation data number selection_R		
Winding_Servo:I.Data[2]	612	INT
Fixed I/O (OUT)		
Winding_Servo:I.Data[2].0	0	BOOL
SEQ-BSY		
Winding_Servo:I.Data[2].7	0	BOOL
ALM-A		
Winding_Servo:I.Data[2].7 - MainProgram/Alarms - 6(XIC)		
Winding_Servo:I.Data[3]	0	INT
Present Alarm		
Winding_Servo:I.Data[4]	22630	INT
Feedback position (lower)		
Winding_Servo:I.Data[5]	4	INT
Feedback position (upper)		
Winding_Servo:I.Data[6]	0	INT
Feedback speed [Hz] (lower)		
Winding_Servo:I.Data[7]	0	INT
Feedback speed [Hz] (upper)		
Winding_Servo:I.Data[8]	22630	INT
Command position (lower)		
Winding_Servo:I.Data[9]	4	INT
Command position (upper)		
Winding_Servo:I.Data[10]	-6	INT
Torque monitor		
Winding_Servo:I.Data[11]	500	INT
CST operating current		
Winding_Servo:I.Data[12]	0	INT
Information (lower)		
Winding_Servo:I.Data[13]	0	INT
Information (upper)		
Winding_Servo:I.Data[14]	0	INT
Reserved		
Winding_Servo:I.Data[15]	3074	INT
Read parameter ID_R		
Winding_Servo:I.Data[16]	0	INT
Read/write status		
Winding_Servo:I.Data[17]	0	INT
Write parameter ID_R		
Winding_Servo:I.Data[18]	-21072	INT
Read data (lower)		
Winding_Servo:I.Data[19]	1	INT
Read data (upper)		
Winding_Servo:I.Data[20]	430	INT
Assignable monitor 0 (lower)		
Winding_Servo:I.Data[21]	0	INT
Assignable monitor 0 (upper)		
Winding_Servo:I.Data[22]	459	INT
Assignable monitor 1 (lower)		
Winding_Servo:I.Data[23]	0	INT
Assignable monitor 1 (upper)		
Winding_Servo:I.Data[24]	-13489	INT
Assignable monitor 2 (lower)		
Winding_Servo:I.Data[25]	31	INT
Assignable monitor 2 (upper)		
Winding_Servo:I.Data[26]	317	INT

Winding_Servo:I (Continued)			
Assignable monitor 3 (lower)			
Winding_Servo:I.Data[27]	0	INT	
Assignable monitor 3 (upper)			
Winding_Servo:O			
Constant	No		_00BB:AZD_AEP_60D82B42:O:0
External Access:	Read/Write		Capstone_Test
Winding_Servo:O - MainProgram/MoveWinding - *8(OM_AZ_DDO)			
Winding_Servo:O.Data[0]	0	INT	
Remote I/O (R-IN)			
Winding_Servo:O.Data[0].0	0	BOOL	
R-IN0			
Winding_Servo:O.Data[1]	0	INT	
Operation data number selection			
Winding_Servo:O.Data[2]	512	INT	
Fixed I/O (IN)			
Winding_Servo:O.Data[2].0	0	BOOL	
FW-JOG			
Winding_Servo:O.Data[2].5	0	BOOL	
STOP			
Winding_Servo:O.Data[2].5 - MainProgram/MainRoutine - *7(OTE)			
Winding_Servo:O.Data[2].6	0	BOOL	
FREE			
Winding_Servo:O.Data[2].6 - MainProgram/FillValues - *#7			
Winding_Servo:O.Data[2].7	0	BOOL	
ALM-RST			
Winding_Servo:O.Data[2].7 - MainProgram/Alarms - 6(ALMD)			
Winding_Servo:O.Data[2].7 - MainProgram/MoveWinding - *6(OTE)			
Winding_Servo:O.Data[3]	1	INT	
Direct data operation type			
Winding_Servo:O.Data[4]	22630	INT	
Direct data operation position (lower)			
Winding_Servo:O.Data[5]	4	INT	
Direct data operation position (upper)			
Winding_Servo:O.Data[6]	750	INT	
Direct data operation operating speed (lower)			
Winding_Servo:O.Data[7]	0	INT	
Direct data operation operating speed (upper)			
Winding_Servo:O.Data[8]	300	INT	
Direct data operation starting/changing rate (lower)			
Winding_Servo:O.Data[9]	0	INT	
Direct data operation starting/changing rate (upper)			
Winding_Servo:O.Data[10]	750	INT	
Direct data operation stopping deceleration (lower)			
Winding_Servo:O.Data[11]	0	INT	
Direct data operation stopping deceleration (upper)			
Winding_Servo:O.Data[12]	1000	INT	
Direct data operation operating current			
Winding_Servo:O.Data[13]	0	INT	
Direct data operation forwarding destination			
Winding_Servo:O.Data[14]	0	INT	
Reserved			
Winding_Servo:O.Data[15]	3074	INT	
Read Parameter ID			
Winding_Servo:O.Data[16]	0	INT	
Write Request			
Winding_Servo:O.Data[17]	3074	INT	
Write Parameter ID			
Winding_Servo:O.Data[18]	1000	INT	
Write data (lower)			
Winding_Servo:O.Data[19]	0	INT	
Write data (upper)			

WindingMoving	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingMoving - MainProgram/CONTROL - A28, #1(Tran,Tran_038 (Condition)), A34, #1(Tran,Tran_036 (Condition)), A38, #1(Tran,Tran_055 (Condition)), A39, #1(Tran,Tran_070 (Condition)), A4, #1(Tran,Tran_073 (Condition))</i>			
<i>WindingMoving - MainProgram/MoveWinding - *7(OTE)</i>			
<i>WindingMoving - MainProgram/SetWindingDia - B6, #1(Tran,Tran_047 (Condition))</i>			
WindingTurns	99.667	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingTurns - MainProgram/CONTROL - *B1, #23(Action,Action_024 (Body)), *B23, #6(Action,Action_039 (Body)), *B37, #4(Action,Action_061 (Body)), *B37, #5(Action,Action_061 (Body)), B1, #22(Action,Action_024 (Body)), B23, #6(Action,Action_039 (Body)), B37, #3(Action,Action_061 (Body)), B37, #5(Action,Action_061 (Body))</i>			
<i>WindingTurns - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>WindingTurns - MainProgram/MoveWinding - *2(DIV)</i>			
<i>WindingTurns - MainProgram/SetWindingDia - *B3, #1(Action,Action_032 (Body)), *B6, #2(Action,Action_047 (Body)), B3, #1(Action,Action_032 (Body)), B6, #1(Action,Action_047 (Body))</i>			
<i>WindingTurns - MainProgram/ThugShake - *B2, #2(Action,Action_027 (Body)), *B3, #1(Action,Action_029 (Body)), B2, #2(Action,Action_027 (Body)), B3, #1(Action,Action_029 (Body))</i>			

General

INPUTS:

pos
speed
accel
deccel

Type:  Ladder Diagram
In Program:  MainProgram

Class: Standard
Number of Rungs: 9

```
1
2 GoodToRunFalling.InputBit := GoodToRun;
3 OSFI(GoodToRunFalling);
4
5 GoodToRunRising.InputBit := GoodToRun;
6 OSRI(GoodToRunRising);
7
8 if (GoodToRunFalling.OutputBit OR HMI.StopPB) then
9     SystemState.00 := StackLightRed;
10    SystemState.01 := StackLightGreen;
11    SystemState.02 := StackLightYellow;
12    SystemState.03 := WindingSolenoid;
13    SystemState.04 := SpoolArray[0].FWD;
14    SystemState.05 := SpoolArray[0].REV;
15    SystemState.06 := SpoolArray[1].FWD;
16    SystemState.07 := SpoolArray[1].REV;
17    SystemState.08 := SpoolArray[2].FWD;
18    SystemState.09 := SpoolArray[2].REV;
19    SystemState.10 := WireInRobot;
20    SystemState.11 := RobotToCutting;
21    SystemState.12 := RobotLoaded2;
22    SystemState.13 := RobotMoveNextLayer3;
23    SystemState.14 := RobotDirection4;
24    SystemState.15 := ClampMotorFwd;
25    SystemState.16 := ClampMotorRev;
26    SystemState.17 := InductorSent;
27    SystemState.18 := HeightMotorDown;
28    SystemState.19 := HeightMotorUp;
29    SystemState.20 := RobotEnable;
30
31    StackLightRed := 1;
32    StackLightGreen := 0;
33    StackLightYellow := 0;
34    WindingSolenoid := 0;
35    SpoolArray[0].FWD := 0;
36    SpoolArray[0].REV := 0;
37    SpoolArray[1].FWD := 0;
38    SpoolArray[1].REV := 0;
39    SpoolArray[2].FWD := 0;
40    SpoolArray[2].REV := 0;
41    WireInRobot := 0;
42    RobotToCutting := 0;
43    RobotLoaded2 := 0;
44    RobotMoveNextLayer3 := 0;
45    RobotDirection4 := 0;
46    ClampMotorFwd := 0;
47    ClampMotorRev := 0;
48    InductorSent := 0;
49    HeightMotorDown := 0;
50    HeightMotorUp := 0;
51    RobotEnable := 0;
52 end_if;
53
54 if (GoodToRunRising.OutputBit or S:FS) then
55     StackLightRed := SystemState.00;
56     StackLightGreen := SystemState.01;
57     StackLightYellow := SystemState.02;
58     WindingSolenoid := SystemState.03;
59     SpoolArray[0].FWD := SystemState.04;
```

```
60     SpoolArray[0].REV := SystemState.05;
61     SpoolArray[1].FWD := SystemState.06;
62     SpoolArray[1].REV := SystemState.07;
63     SpoolArray[2].FWD := SystemState.08;
64     SpoolArray[2].REV := SystemState.09;
65     WireInRobot := SystemState.10;
66     RobotToCutting := SystemState.11;
67     RobotLoaded2 := SystemState.12;
68     RobotMoveNextLayer3 := SystemState.13;
69     RobotDirection4 := SystemState.14;
70     ClampMotorFwd := SystemState.15;
71     ClampMotorRev := SystemState.16;
72     InductorSent := SystemState.17;
73     HeightMotorDown := SystemState.18;
74     HeightMotorUp := SystemState.19;
75     RobotEnable := SystemState.20;
76 end_if;
```

Name	Value	Data Type	Scope
ClampMotorFwd	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>ClampMotorFwd - MainProgram/CONTROL - *B1, #14(Action,Action_024 (Body)), *B16, #1(Action,Action_057 (Body)), *B17, #1(Action,Action_058 (Body)), *B19, #1(Action,Action_052 (Body)), *B20, #1(Action,Action_053 (Body)), *B23, #34(Action,Action_039 (Body))</i>			
<i>ClampMotorFwd - MainProgram/FillValues - #50</i>			
<i>ClampMotorFwd - MainProgram/pause - #24, *#46, *#70</i>			
<i>ClampMotorFwd - MainProgram/StopAll - *#16</i>			
ClampMotorRev	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>ClampMotorRev - MainProgram/CONTROL - *B1, #15(Action,Action_024 (Body)), *B17, #1(Action,Action_059 (Body)), *B18, #1(Action,Action_051 (Body)), *B35, #2(Action,Action_067 (Body)), *B36, #1(Action,Action_060 (Body))</i>			
<i>ClampMotorRev - MainProgram/FillValues - #51</i>			
<i>ClampMotorRev - MainProgram/pause - #25, *#47, *#71</i>			
<i>ClampMotorRev - MainProgram/StopAll - *#17</i>			
GoodToRun	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>GoodToRun - MainProgram/MainRoutine - *2(OTE), 2(XIC), 4(XIC)</i>			
<i>GoodToRun - MainProgram/pause - #2, #5</i>			
<i>GoodToRun - MainProgram/WindInductor - 0(XIC)</i>			
GoodToRunFalling		FBD_ONESHOT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>GoodToRunFalling - MainProgram/pause - *#3</i>			
GoodToRunFalling.InputBit	1	BOOL	
<i>GoodToRunFalling.InputBit - MainProgram/pause - *#2</i>			
GoodToRunFalling.OutputBit	0	BOOL	
<i>GoodToRunFalling.OutputBit - MainProgram/pause - #8</i>			
GoodToRunRising		FBD_ONESHOT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>GoodToRunRising - MainProgram/pause - *#6</i>			
GoodToRunRising.InputBit	1	BOOL	
<i>GoodToRunRising.InputBit - MainProgram/pause - *#5</i>			
GoodToRunRising.OutputBit	0	BOOL	
<i>GoodToRunRising.OutputBit - MainProgram/pause - #54</i>			
HeightMotorDown	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>HeightMotorDown - MainProgram/CONTROL - *B1, #19(Action,Action_024 (Body)), *B6, #1(Action,Action_064 (Body)), *B7, #1(Action,Action_065 (Body)), *C30, #2(Action,Action_026 (Body)), *C31, #1(Action,Action_062 (Body))</i>			
<i>HeightMotorDown - MainProgram/FillValues - #53</i>			
<i>HeightMotorDown - MainProgram/pause - #27, *#49, *#73</i>			
<i>HeightMotorDown - MainProgram/StopAll - *#21</i>			
HeightMotorUp	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>HeightMotorUp - MainProgram/CONTROL - *B1, #20(Action,Action_024 (Body)), *B35, #3(Action,Action_067 (Body)), *B36, #2(Action,Action_060 (Body)), *B4, #1(Action,Action_066 (Body)), *B6, #1(Action,Action_054 (Body)), *C32, #1(Action,Action_068 (Body)), *C32, #1(Action,Action_069 (Body))</i>			
<i>HeightMotorUp - MainProgram/FillValues - #54</i>			
<i>HeightMotorUp - MainProgram/pause - #28, *#50, *#74</i>			
<i>HeightMotorUp - MainProgram/StopAll - *#22</i>			
HMI		HMIControl	Capstone_Test

HMI (Continued)			
Constant	No		
External Access:	Read/Write		
HMI.Manual.Spool.FWD1	0	BOOL	
HMI.Manual.Spool.FWD1 - MainProgram/FillValues - #40			
HMI.Manual.Spool.FWD2	0	BOOL	
HMI.Manual.Spool.FWD2 - MainProgram/FillValues - #42			
HMI.Manual.Spool.FWD3	0	BOOL	
HMI.Manual.Spool.FWD3 - MainProgram/FillValues - #44			
HMI.Manual.Spool.REV1	0	BOOL	
HMI.Manual.Spool.REV1 - MainProgram/FillValues - #41			
HMI.Manual.Spool.REV2	0	BOOL	
HMI.Manual.Spool.REV2 - MainProgram/FillValues - #43			
HMI.Manual.Spool.REV3	0	BOOL	
HMI.Manual.Spool.REV3 - MainProgram/FillValues - #45			
HMI.Manual.Cut	0	BOOL	
HMI.Manual.Cut - MainProgram/FillValues - #57			
HMI.Manual.CameraLight	0	BOOL	
HMI.Manual.CameraLight - MainProgram/FillValues - #47			
HMI.Manual.ExtrudeHeater	0	BOOL	
HMI.Manual.ExtrudeHeater - MainProgram/FillValues - #59			
HMI.Manual.ExtrudeMotor	0	BOOL	
HMI.Manual.ExtrudeMotor - MainProgram/FillValues - #58			
HMI.Manual.WindingSolenoid	0	BOOL	
HMI.Manual.WindingSolenoid - MainProgram/FillValues - #60			
HMI.Manual.ClampMotorFWD	0	BOOL	
HMI.Manual.ClampMotorFWD - MainProgram/FillValues - #50			
HMI.Manual.ClampMotorREV	0	BOOL	
HMI.Manual.ClampMotorREV - MainProgram/FillValues - #51			
HMI.Manual.HeightMotorUp	0	BOOL	
HMI.Manual.HeightMotorUp - MainProgram/FillValues - #54			
HMI.Manual.HeightMotorDown	0	BOOL	
HMI.Manual.HeightMotorDown - MainProgram/FillValues - #53			
HMI.StartPB	0	BOOL	
HMI.StartPB - MainProgram/MainRoutine - 2(XIC)			
HMI.ResetPB	0	BOOL	
HMI.ResetPB - MainProgram/MainRoutine - 0(XIC)			
HMI.StopPB	0	BOOL	
HMI.StopPB - MainProgram/MainRoutine - 2(XIO)			
HMI.StopPB - MainProgram/pause - #8			
HMI.RobotFaultReset	0	BOOL	
HMI.RobotFaultReset - MainProgram/RobotAutoControl - 8(XIC)			
InductorSent	0	BOOL	MainProgram
PC sets this true when the inductor data has been sent			
Constant	No		
External Access:	Read/Write		
InductorSent - MainProgram/CONTROL - *B1, #18(Action,Action_024 (Body))			
InductorSent - MainProgram/pause - #26, *#48, *#72			
InductorSent - MainProgram/StopAll - *#20			
InductorSent - MainProgram/WindInductor - 1(XIC)			
RobotDirection4	1	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].3		
Base Tag:	Fanuc_Robot:O.Data[0].3		
Constant	No		
External Access:	Read/Write		
RobotDirection4 - MainProgram/CONTROL - *B1, #13(Action,Action_024 (Body)), *B12, #2(Action,Action_023 (Body)), *B23, #10(Action,Action_039 (Body)), *B23, #12(Action,Action_039 (Body))			
RobotDirection4 - MainProgram/pause - #23, *#45, *#69			
RobotDirection4 - MainProgram/StopAll - *#15			
RobotEnable	1	BOOL	MainProgram

RobotEnable (Continued)				
Constant	No			
External Access:	Read/Write			
RobotEnable - MainProgram/FillValues - *#3				
RobotEnable - MainProgram/pause - #29, *#51, *#75				
RobotEnable - MainProgram/RobotAutoControl - *0(OTL), *1(OTU), 0(XIC), 3(XIC)				
RobotLoaded2	1	BOOL		Capstone_Test
Out to Robot				
AliasFor:	Fanuc_Robot:O.Data[0].1			
Base Tag:	Fanuc_Robot:O.Data[0].1			
Constant	No			
External Access:	Read/Write			
RobotLoaded2 - MainProgram/CONTROL - *B1, #10(Action,Action_024 (Body)), *B12, #4(Action,Action_023 (Body)), *B18, #2(Action,Action_051 (Body))				
RobotLoaded2 - MainProgram/pause - #21, *#43, *#67				
RobotLoaded2 - MainProgram/StopAll - *#13				
RobotMoveNextLayer3	0	BOOL		Capstone_Test
Out to Robot				
AliasFor:	Fanuc_Robot:O.Data[0].2			
Base Tag:	Fanuc_Robot:O.Data[0].2			
Constant	No			
External Access:	Read/Write			
RobotMoveNextLayer3 - MainProgram/CONTROL - *B1, #11(Action,Action_024 (Body)), *B12, #3(Action,Action_023 (Body)), *B23, #18(Action,Action_039 (Body)), *B27, #1(Action,Action_040 (Body))				
RobotMoveNextLayer3 - MainProgram/pause - #22, *#44, *#68				
RobotMoveNextLayer3 - MainProgram/StopAll - *#14				
RobotToCutting	0	BOOL		Capstone_Test
Out to Robot				
AliasFor:	Fanuc_Robot:O.Data[0].0			
Base Tag:	Fanuc_Robot:O.Data[0].0			
Constant	No			
External Access:	Read/Write			
RobotToCutting - MainProgram/CONTROL - *B1, #9(Action,Action_024 (Body)), *B11, #2(Action,Action_021 (Body)), *B11, #2(Action,Action_022 (Body))				
RobotToCutting - MainProgram/pause - #20, *#42, *#66				
RobotToCutting - MainProgram/StopAll - *#12				
SpoolArray		Spool[3]		Capstone_Test
Constant	No			
External Access:	Read/Write			
SpoolArray[0].REV	0	BOOL		
out				
SpoolArray[0].REV - MainProgram/CONTROL - *B1, #3(Action,Action_024 (Body))				
SpoolArray[0].REV - MainProgram/FillValues - #41				
SpoolArray[0].REV - MainProgram/pause - #14, *#36, *#60				
SpoolArray[0].REV - MainProgram/StopAll - *#6				
SpoolArray[0].FWD	0	BOOL		
out				
SpoolArray[0].FWD - MainProgram/CONTROL - *B1, #2(Action,Action_024 (Body))				
SpoolArray[0].FWD - MainProgram/FillValues - #40				
SpoolArray[0].FWD - MainProgram/pause - #13, *#35, *#59				
SpoolArray[0].FWD - MainProgram/StopAll - *#5				
SpoolArray[0].Mass	164.73665	REAL		
current mass reading				
SpoolArray[0].Mass - MainProgram/FillValues - *#20, *#22				
SpoolArray[0].Length	0.0	REAL		
approximated legth based on Mass				
SpoolArray[0].Loaded	1	BOOL		
if is in loaded state				
SpoolArray[0].Loaded - MainProgram/FillValues - *#10				
SpoolArray[0].Gauge	-1	DINT		

SpoolArray (Continued)			
Wire Gauge			
SpoolArray[1].REV	0	BOOL	
out			
SpoolArray[1].REV - MainProgram/CONTROL - *B1, #5(Action,Action_024 (Body))			
SpoolArray[1].REV - MainProgram/FillValues - #43			
SpoolArray[1].REV - MainProgram/pause - #16, *#38, *#62			
SpoolArray[1].REV - MainProgram/StopAll - *#8			
SpoolArray[1].FWD	0	BOOL	
out			
SpoolArray[1].FWD - MainProgram/CONTROL - *B1, #4(Action,Action_024 (Body))			
SpoolArray[1].FWD - MainProgram/FillValues - #42			
SpoolArray[1].FWD - MainProgram/pause - #15, *#37, *#61			
SpoolArray[1].FWD - MainProgram/StopAll - *#7			
SpoolArray[1].Mass	163.34396	REAL	
current mass reading			
SpoolArray[1].Length	0.0	REAL	
approximated legth based on Mass			
SpoolArray[1].Loaded	1	BOOL	
if is in loaded state			
SpoolArray[1].Loaded - MainProgram/FillValues - *#11			
SpoolArray[1].Gauge	-1	DINT	
Wire Gauge			
SpoolArray[2].REV	0	BOOL	
out			
SpoolArray[2].REV - MainProgram/CONTROL - *B1, #7(Action,Action_024 (Body))			
SpoolArray[2].REV - MainProgram/FillValues - #45			
SpoolArray[2].REV - MainProgram/pause - #18, *#40, *#64			
SpoolArray[2].REV - MainProgram/StopAll - *#10			
SpoolArray[2].FWD	0	BOOL	
out			
SpoolArray[2].FWD - MainProgram/CONTROL - *B1, #6(Action,Action_024 (Body)), *B11, #1(Action,Action_021 (Body)), *B11, #1(Action,Action_022 (Body)), *B12, #1(Action,Action_023 (Body)), *B23, #3(Action,Action_039 (Body)), *B37, #1(Action,Action_061 (Body)), *C9, #1(Action,Action_019 (Body))			
SpoolArray[2].FWD - MainProgram/FillValues - #44			
SpoolArray[2].FWD - MainProgram/pause - #17, *#39, *#63			
SpoolArray[2].FWD - MainProgram/StopAll - *#9			
SpoolArray[2].Mass	600.10535	REAL	
current mass reading			
SpoolArray[2].Mass - MainProgram/FillValues - *#32, *#34			
SpoolArray[2].Length	0.0	REAL	
approximated legth based on Mass			
SpoolArray[2].Loaded	0	BOOL	
if is in loaded state			
SpoolArray[2].Loaded - MainProgram/FillValues - *#12			
SpoolArray[2].Gauge	16	DINT	
Wire Gauge			
SpoolArray[2].Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)			
StackLightGreen	1	BOOL	Capstone_Test
AliasFor:	Local:4:O.Pt07.Data		
Base Tag:	Local:4:O.Pt07.Data		
Constant	No		
External Access:	Read/Write		
StackLightGreen - MainProgram/MainRoutine - *5(OTE)			
StackLightGreen - MainProgram/pause - #10, *#32, *#56			
StackLightGreen - MainProgram/StopAll - *#2			
StackLightRed	0	BOOL	Capstone_Test
AliasFor:	Local:4:O.Pt09.Data		
Base Tag:	Local:4:O.Pt09.Data		
Constant	No		
External Access:	Read/Write		
StackLightRed - MainProgram/MainRoutine - *7(OTE)			

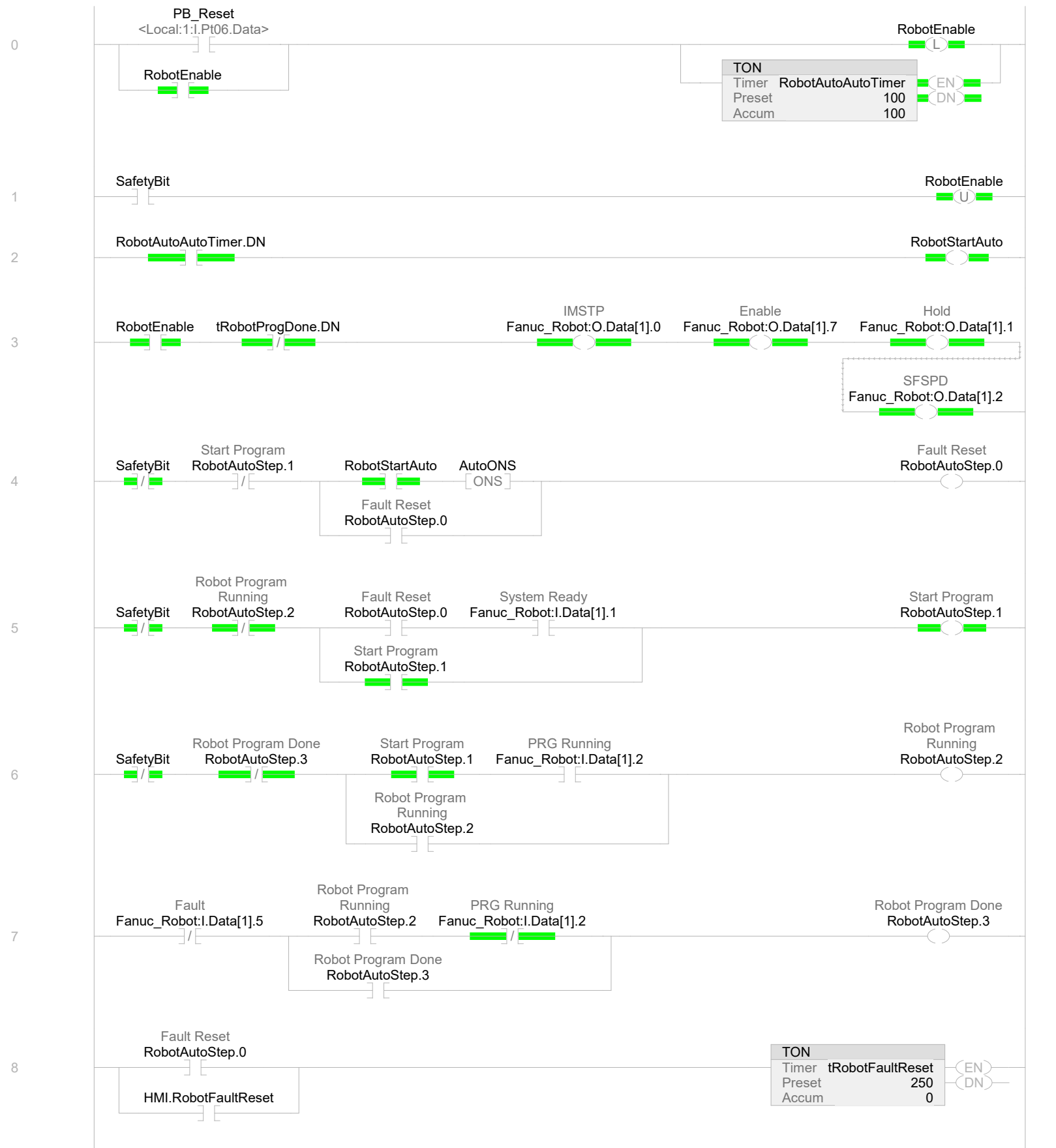
StackLightRed (Continued)			
StackLightRed - MainProgram/pause - #9, *#31, *#55			
StackLightRed - MainProgram/StopAll - *#1			
StackLightYellow	1	BOOL	Capstone_Test
AliasFor: Local:4:O.Pt08.Data			
Base Tag: Local:4:O.Pt08.Data			
Constant No			
External Access: Read/Write			
StackLightYellow - MainProgram/MainRoutine - *6(OTE)			
StackLightYellow - MainProgram/pause - #11, *#33, *#57			
StackLightYellow - MainProgram/StopAll - *#3			
SystemState	0	DINT	MainProgram
Constant No			
External Access: Read/Write			
SystemState - MainProgram/MainRoutine - *1(MOV)			
SystemState.0	0	BOOL	
SystemState.0 - MainProgram/pause - #55, *#9			
SystemState.1	0	BOOL	
SystemState.1 - MainProgram/pause - #56, *#10			
SystemState.2	0	BOOL	
SystemState.2 - MainProgram/pause - #57, *#11			
SystemState.3	0	BOOL	
SystemState.3 - MainProgram/pause - #58, *#12			
SystemState.4	0	BOOL	
SystemState.4 - MainProgram/pause - #59, *#13			
SystemState.5	0	BOOL	
SystemState.5 - MainProgram/pause - #60, *#14			
SystemState.6	0	BOOL	
SystemState.6 - MainProgram/pause - #61, *#15			
SystemState.7	0	BOOL	
SystemState.7 - MainProgram/pause - #62, *#16			
SystemState.8	0	BOOL	
SystemState.8 - MainProgram/pause - #63, *#17			
SystemState.9	0	BOOL	
SystemState.9 - MainProgram/pause - #64, *#18			
SystemState.10	0	BOOL	
SystemState.10 - MainProgram/pause - #65, *#19			
SystemState.11	0	BOOL	
SystemState.11 - MainProgram/pause - #66, *#20			
SystemState.12	0	BOOL	
SystemState.12 - MainProgram/pause - #67, *#21			
SystemState.13	0	BOOL	
SystemState.13 - MainProgram/pause - #68, *#22			
SystemState.14	0	BOOL	
SystemState.14 - MainProgram/pause - #69, *#23			
SystemState.15	0	BOOL	
SystemState.15 - MainProgram/pause - #70, *#24			
SystemState.16	0	BOOL	
SystemState.16 - MainProgram/pause - #71, *#25			
SystemState.17	0	BOOL	
SystemState.17 - MainProgram/pause - #72, *#26			
SystemState.18	0	BOOL	
SystemState.18 - MainProgram/pause - #73, *#27			
SystemState.19	0	BOOL	
SystemState.19 - MainProgram/pause - #74, *#28			
SystemState.20	0	BOOL	
SystemState.20 - MainProgram/pause - #75, *#29			
WindingSolenoid	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
WindingSolenoid - MainProgram/CONTROL - *B14, #1(Action,Action_034 (Body))			

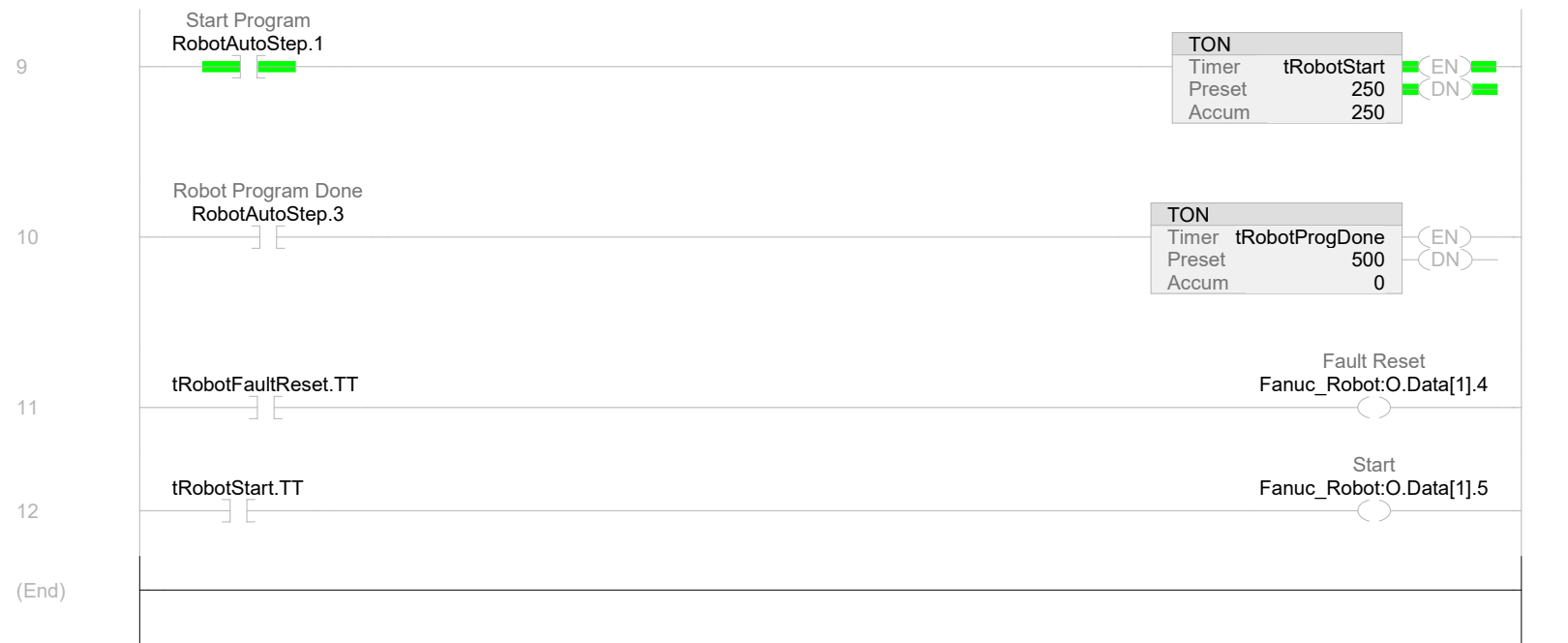
WindingSolenoid (Continued)

WindingSolenoid - MainProgram/FillValues - #60
WindingSolenoid - MainProgram/MainRoutine - 8(XIC)
WindingSolenoid - MainProgram/pause - #12, *#34, *#58
WindingSolenoid - MainProgram/SetWindingDia - *B1, #1(Action,Action_045 (Body)), *B2, #2(Action,Action_031 (Body)), *B4, #1(Action,Action_046 (Body))
WindingSolenoid - MainProgram/StopAll - *#4

WireInRobot	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
WireInRobot - MainProgram/CONTROL - *B1, #8(Action,Action_024 (Body))			
WireInRobot - MainProgram/pause - #19, *#41, *#65			
WireInRobot - MainProgram/StopAll - *#11			

General			
Type:	 Structured Text	Class:	Standard
In Program:	 MainProgram	Number of Lines:	Original 76





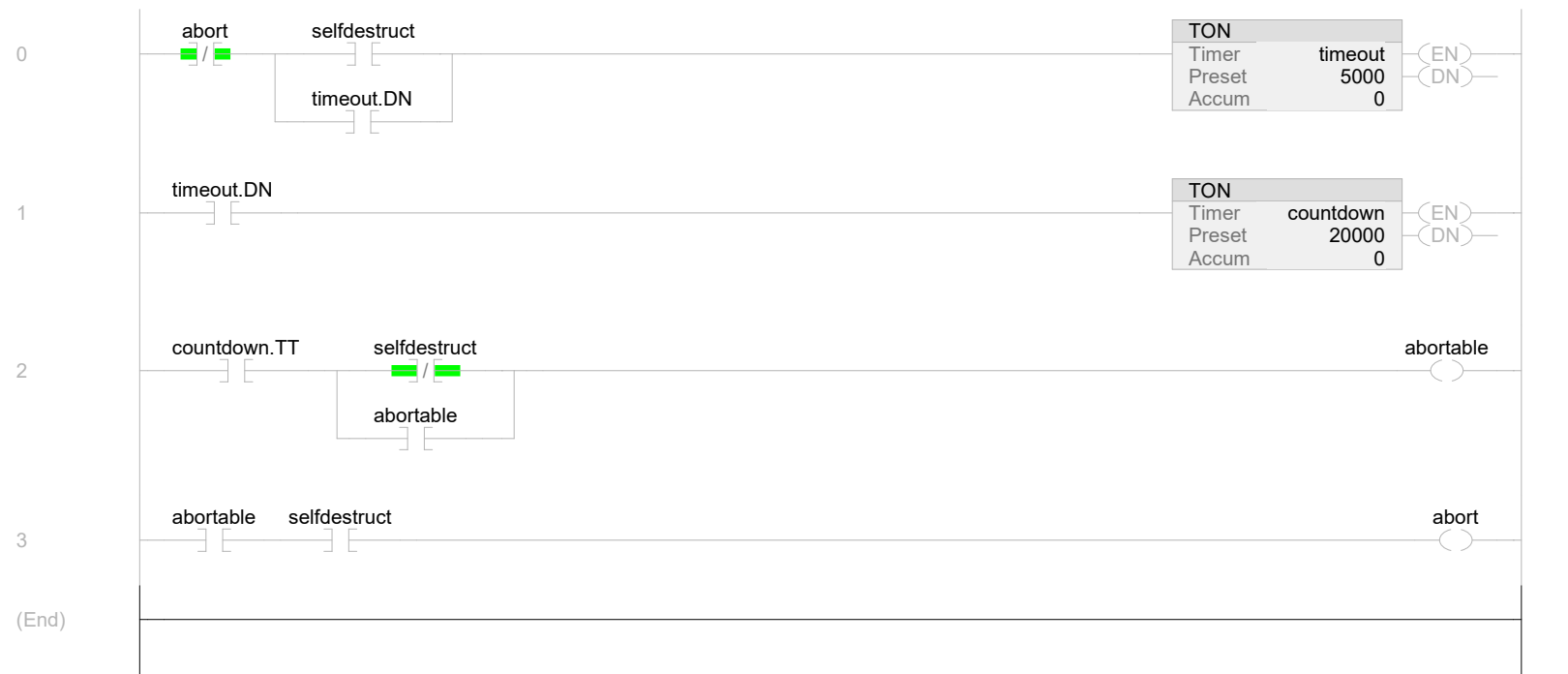
Name	Value	Data Type	Scope
AutoONS	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>AutoONS - MainProgram/RobotAutoControl - *4(ONS)</i>			
Fanuc_Robot:I		AB:ETHERNET_MODULE_INT_8Bytes:I:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Fanuc_Robot:I.Data[0]	0	INT	
In From Robot			
Fanuc_Robot:I.Data[0].0	0	BOOL	
In From Robot			
<i>RobotAtHome1 - MainProgram/CONTROL - A39, #1(Tran,Tran_037 (Condition))</i>			
Fanuc_Robot:I.Data[0].1	0	BOOL	
In From Robot			
<i>RobotAtCutting2 - MainProgram/CONTROL - A11, #1(Tran,Tran_021 (Condition))</i>			
Fanuc_Robot:I.Data[0].2	0	BOOL	
In From Robot			
<i>RobotWireInserted3 - MainProgram/CONTROL - A19, #1(Tran,Tran_051 (Condition))</i>			
Fanuc_Robot:I.Data[1]	168	INT	
Robot Status			
Fanuc_Robot:I.Data[1].0	0	BOOL	
Cmd Enabled			
Fanuc_Robot:I.Data[1].1	0	BOOL	
System Ready			
<i>Fanuc_Robot:I.Data[1].1 - MainProgram/Alarms - 10(ALMD)</i>			
<i>Fanuc_Robot:I.Data[1].1 - MainProgram/RobotAutoControl - 5(XIC)</i>			
Fanuc_Robot:I.Data[1].2	0	BOOL	
PRG Running			
<i>Fanuc_Robot:I.Data[1].2 - MainProgram/RobotAutoControl - 6(XIC), 7(XIO)</i>			
Fanuc_Robot:I.Data[1].5	1	BOOL	
Fault			
<i>Fanuc_Robot:I.Data[1].5 - MainProgram/Alarms - 10(XIC)</i>			
<i>Fanuc_Robot:I.Data[1].5 - MainProgram/RobotAutoControl - 7(XIO)</i>			
Fanuc_Robot:O		AB:ETHERNET_MODULE_INT_8Bytes:O:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
Fanuc_Robot:O.Data[0]	26	INT	
Out to Robot			
Fanuc_Robot:O.Data[0].0	0	BOOL	
Out to Robot			
<i>RobotToCutting - MainProgram/CONTROL - *B1, #9(Action,Action_024 (Body)), *B11, #2(Action,Action_021 (Body)), *B11, #2(Action,Action_022 (Body))</i>			
<i>RobotToCutting - MainProgram/pause - #20, *#42, *#66</i>			
<i>RobotToCutting - MainProgram/StopAll - *#12</i>			
Fanuc_Robot:O.Data[0].1	1	BOOL	
Out to Robot			
<i>RobotLoaded2 - MainProgram/CONTROL - *B1, #10(Action,Action_024 (Body)), *B12, #4(Action,Action_023 (Body)), *B18, #2(Action,Action_051 (Body))</i>			
<i>RobotLoaded2 - MainProgram/pause - #21, *#43, *#67</i>			
<i>RobotLoaded2 - MainProgram/StopAll - *#13</i>			
Fanuc_Robot:O.Data[0].2	0	BOOL	
Out to Robot			
<i>RobotMoveNextLayer3 - MainProgram/CONTROL - *B1, #11(Action,Action_024 (Body)), *B12, #3(Action,Action_023 (Body)), *B23, #18(Action,Action_039 (Body)), *B27, #1(Action,Action_040 (Body))</i>			
<i>RobotMoveNextLayer3 - MainProgram/pause - #22, *#44, *#68</i>			
<i>RobotMoveNextLayer3 - MainProgram/StopAll - *#14</i>			
Fanuc_Robot:O.Data[0].3	1	BOOL	
Out to Robot			

Fanuc_Robot:O (Continued)			
<i>RobotDirection4 - MainProgram/CONTROL - *B1, #13(Action,Action_024 (Body)), *B12, #2(Action,Action_023 (Body)), *B23, #10(Action,Action_039 (Body)), *B23, #12(Action,Action_039 (Body))</i>			
<i>RobotDirection4 - MainProgram/pause - #23, *#45, *#69</i>			
<i>RobotDirection4 - MainProgram/StopAll - *#15</i>			
Fanuc_Robot:O.Data[0].4	1	BOOL	
Out to Robot			
<i>RobotDoneGetThisShitOutOfHere4 - MainProgram/CONTROL - *B1, #12(Action,Action_024 (Body)), *B38, #1(Action,Action_041 (Body)), *B38, #1(Action,Action_042 (Body))</i>			
Fanuc_Robot:O.Data[0].5	0	BOOL	
Out to Robot			
<i>RobotMoveBack6 - MainProgram/CONTROL - *B20, #2(Action,Action_053 (Body)), *B21, #1(Action,Action_025 (Body))</i>			
Fanuc_Robot:O.Data[1]	143	INT	
AutoMode			
Fanuc_Robot:O.Data[1].0	1	BOOL	
IMSTP			
<i>Fanuc_Robot:O.Data[1].0 - MainProgram/RobotAutoControl - *3(OTE)</i>			
Fanuc_Robot:O.Data[1].1	1	BOOL	
Hold			
<i>Fanuc_Robot:O.Data[1].1 - MainProgram/RobotAutoControl - *3(OTE)</i>			
Fanuc_Robot:O.Data[1].2	1	BOOL	
SFSPD			
<i>Fanuc_Robot:O.Data[1].2 - MainProgram/RobotAutoControl - *3(OTE)</i>			
Fanuc_Robot:O.Data[1].4	0	BOOL	
Fault Reset			
<i>Fanuc_Robot:O.Data[1].4 - MainProgram/RobotAutoControl - *11(OTE)</i>			
Fanuc_Robot:O.Data[1].5	0	BOOL	
Start			
<i>Fanuc_Robot:O.Data[1].5 - MainProgram/RobotAutoControl - *12(OTE)</i>			
Fanuc_Robot:O.Data[1].7	1	BOOL	
Enable			
<i>Fanuc_Robot:O.Data[1].7 - MainProgram/RobotAutoControl - *3(OTE)</i>			
HMI		HMIControl	Capstone_Test
Constant No			
External Access: Read/Write			
HMI.Manual.Spool.FWD1	0	BOOL	
<i>HMI.Manual.Spool.FWD1 - MainProgram/FillValues - #40</i>			
HMI.Manual.Spool.FWD2	0	BOOL	
<i>HMI.Manual.Spool.FWD2 - MainProgram/FillValues - #42</i>			
HMI.Manual.Spool.FWD3	0	BOOL	
<i>HMI.Manual.Spool.FWD3 - MainProgram/FillValues - #44</i>			
HMI.Manual.Spool.REV1	0	BOOL	
<i>HMI.Manual.Spool.REV1 - MainProgram/FillValues - #41</i>			
HMI.Manual.Spool.REV2	0	BOOL	
<i>HMI.Manual.Spool.REV2 - MainProgram/FillValues - #43</i>			
HMI.Manual.Spool.REV3	0	BOOL	
<i>HMI.Manual.Spool.REV3 - MainProgram/FillValues - #45</i>			
HMI.Manual.Cut	0	BOOL	
<i>HMI.Manual.Cut - MainProgram/FillValues - #57</i>			
HMI.Manual.CameraLight	0	BOOL	
<i>HMI.Manual.CameraLight - MainProgram/FillValues - #47</i>			
HMI.Manual.ExtrudeHeater	0	BOOL	
<i>HMI.Manual.ExtrudeHeater - MainProgram/FillValues - #59</i>			
HMI.Manual.ExtrudeMotor	0	BOOL	
<i>HMI.Manual.ExtrudeMotor - MainProgram/FillValues - #58</i>			
HMI.Manual.WindingSolenoid	0	BOOL	
<i>HMI.Manual.WindingSolenoid - MainProgram/FillValues - #60</i>			
HMI.Manual.ClampMotorFWD	0	BOOL	
<i>HMI.Manual.ClampMotorFWD - MainProgram/FillValues - #50</i>			
HMI.Manual.ClampMotorREV	0	BOOL	
<i>HMI.Manual.ClampMotorREV - MainProgram/FillValues - #51</i>			
HMI.Manual.HeightMotorUp	0	BOOL	
<i>HMI.Manual.HeightMotorUp - MainProgram/FillValues - #54</i>			

HMI (Continued)			
HMI.Manual.HeightMotorDown	0	BOOL	Capstone_Test
HMI.Manual.HeightMotorDown - MainProgram/FillValues - #53			
HMI.StartPB	0	BOOL	
HMI.StartPB - MainProgram/MainRoutine - 2(XIC)			
HMI.ResetPB	0	BOOL	
HMI.ResetPB - MainProgram/MainRoutine - 0(XIC)			
HMI.StopPB	0	BOOL	
HMI.StopPB - MainProgram/MainRoutine - 2(XIO)			
HMI.StopPB - MainProgram/pause - #8			
HMI.RobotFaultReset	0	BOOL	
HMI.RobotFaultReset - MainProgram/RobotAutoControl - 8(XIC)			
 PB_Reset	0	BOOL	Capstone_Test
AliasFor:	Local:1:I.Pt06.Data		
Base Tag:	Local:1:I.Pt06.Data		
Constant	No		
External Access:	Read/Write		
PB_Reset - MainProgram/Alarms - 0(ALMD), 1(ALMD), 2(ALMD), 3(ALMD), 4(ALMD)			
PB_Reset - MainProgram/RobotAutoControl - 0(XIC)			
PB_Reset - SafetyProgram  /MainRoutine - 10(XIC), 6(XIO), 8(XIC)			
RobotAutoAutoTimer		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
RobotAutoAutoTimer - MainProgram/RobotAutoControl - *0(TON)			
RobotAutoAutoTimer.DN	1	BOOL	
RobotAutoAutoTimer.DN - MainProgram/RobotAutoControl - 2(XIC)			
RobotAutoStep	2	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
RobotAutoStep.0	0	BOOL	
Fault Reset			
RobotAutoStep.0 - MainProgram/RobotAutoControl - *4(OTE), 4(XIC), 5(XIC), 8(XIC)			
RobotAutoStep.1	1	BOOL	
Start Program			
RobotAutoStep.1 - MainProgram/RobotAutoControl - *5(OTE), 4(XIO), 5(XIC), 6(XIC), 9(XIC)			
RobotAutoStep.2	0	BOOL	
Robot Program Running			
RobotAutoStep.2 - MainProgram/RobotAutoControl - *6(OTE), 5(XIO), 6(XIC), 7(XIC)			
RobotAutoStep.3	0	BOOL	
Robot Program Done			
RobotAutoStep.3 - MainProgram/RobotAutoControl - *7(OTE), 10(XIC), 6(XIO), 7(XIC)			
RobotEnable	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
RobotEnable - MainProgram/FillValues - *#3			
RobotEnable - MainProgram/pause - #29, *#51, *#75			
RobotEnable - MainProgram/RobotAutoControl - *0(OTL), *1(OTU), 0(XIC), 3(XIC)			
RobotStartAuto	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
RobotStartAuto - MainProgram/RobotAutoControl - *2(OTE), 4(XIC)			
 SafetyBit	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
SafetyBit - MainProgram/Alarms - 0(XIC), 1(XIC), 2(XIC), 3(XIC), 4(XIC)			
SafetyBit - MainProgram/MainRoutine - 10(XIC), 4(XIO), 5(XIO), 6(XIO), 7(XIC)			
SafetyBit - MainProgram/RobotAutoControl - 1(XIC), 4(XIO), 5(XIO), 6(XIO)			
SafetyBit - SafetyProgram  /MainRoutine - *7(OTE), 7(XIC), 9(XIO)			

tRobotFaultReset		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>tRobotFaultReset - MainProgram/RobotAutoControl - *8(TON)</i>			
tRobotFaultReset.TT		BOOL	
	0		
<i>tRobotFaultReset.TT - MainProgram/RobotAutoControl - 11(XIC)</i>			
tRobotProgDone		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>tRobotProgDone - MainProgram/RobotAutoControl - *10(TON)</i>			
tRobotProgDone.DN		BOOL	
	0		
<i>tRobotProgDone.DN - MainProgram/RobotAutoControl - 3(XIO)</i>			
tRobotStart		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>tRobotStart - MainProgram/RobotAutoControl - *9(TON)</i>			
tRobotStart.TT		BOOL	
	0		
<i>tRobotStart.TT - MainProgram/RobotAutoControl - 12(XIC)</i>			

General			
Type:	 Ladder Diagram	Class:	Standard
In Program:	 MainProgram	Number of Rungs:	13



Name	Value	Data Type	Scope
abort	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>abort - MainProgram/selfdestruct - *3(OTE), 0(XIO)</i>			
abortable	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>abortable - MainProgram/selfdestruct - *2(OTE), 2(XIC), 3(XIC)</i>			
countdown		TIMER	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>countdown - MainProgram/selfdestruct - *1(TON)</i>			
countdown.TT	0	BOOL	
<i>countdown.TT - MainProgram/selfdestruct - 2(XIC)</i>			
selfdestruct	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>selfdestruct - MainProgram/MainRoutine - 0(XIC)</i>			
<i>selfdestruct - MainProgram/selfdestruct - 0(XIC), 2(XIO), 3(XIC)</i>			
timeout		TIMER	MainProgram
Constant	No		
External Access:	Read/Write		
<i>timeout - MainProgram/selfdestruct - *0(TON)</i>			
timeout.DN	0	BOOL	
<i>timeout.DN - MainProgram/selfdestruct - 0(XIC), 1(XIC)</i>			

General			
Type:	 Ladder Diagram	Class:	Standard
In Program:	 MainProgram	Number of Rungs:	4

sets the diameter for winding mechanism



Name	Value	Data Type	Scope
Action_031		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_031 - MainProgram/SetWindingDia - *B2(Action,Action_031)</i>			
Action_032		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_032 - MainProgram/SetWindingDia - *B3(Action,Action_032)</i>			
Action_044		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_044 - MainProgram/SetWindingDia - *B3(Action,Action_044)</i>			
Action_045		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_045 - MainProgram/SetWindingDia - *B1(Action,Action_045)</i>			
Action_046		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_046 - MainProgram/SetWindingDia - *B4(Action,Action_046)</i>			
Action_047		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_047 - MainProgram/SetWindingDia - *B6(Action,Action_047)</i>			
Action_048		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_048 - MainProgram/SetWindingDia - *B7(Action,Action_048)</i>			
Action_056		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_056 - MainProgram/SetWindingDia - *B5(Action,Action_056)</i>			
AOI_Motor		AOI_Motor	Capstone_Test
Constant	No		
External Access:	Read/Write		
AOI_Motor.DDO		OM_AZ_DDO	
<i>AOI_Motor.DDO - MainProgram/MoveWinding - *8(OM_AZ_DDO)</i>			
AOI_Motor.DDO.EnableIn	1	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.DDO.EnableOut	1	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.DDO.Set_Exec	0	BOOL	
Execute AOI			
<i>AOI_Motor.DDO.Set_Exec - MainProgram/MoveWinding - *4(OTL), *5(OTU), *6(OTU)</i>			
AOI_Motor.DDO.Set_OpeType	1	INT	
Direct data operation operation type			
AOI_Motor.DDO.Set_Position	284774	DINT	
Direct data operation position			
<i>AOI_Motor.DDO.Set_Position - MainProgram/MoveWinding - *3(MUL), 4(NEQ)</i>			
AOI_Motor.DDO.Set_Velocity	750	DINT	
Direct data operation operating speed			
<i>AOI_Motor.DDO.Set_Velocity - MainProgram/MoveWinding - *3(MOV)</i>			
AOI_Motor.DDO.Set_Acceleration	300	DINT	
Direct data operation starting/changing rate			
<i>AOI_Motor.DDO.Set_Acceleration - MainProgram/MoveWinding - *3(MOV)</i>			

AOI_Motor (Continued)		
AOI_Motor.DDO.Set_Deceleration	750	DINT
Direct data operation stopping deceleration		
<i>AOI_Motor.DDO.Set_Deceleration - MainProgram/MoveWinding - *3(MOV)</i>		
AOI_Motor.DDO.Set_Current	1000	INT
Direct data operation operating current		
AOI_Motor.DDO.Set_Send	0	INT
Direct data operation forwarding destination		
AOI_Motor.DDO.Sts_Busy	0	BOOL
Executing status in AOI		
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/MoveWinding - 7(XIC)</i>		
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/SetWindingDia - B4, #1(Tran,Tran_041 (Condition))</i>		
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/ThugShake - B2, #1(Tran,Tran_027 (Condition)), C4, #1(Tran,Tran_029 (Condition))</i>		
AOI_Motor.DDO.Sts_Err	0	BOOL
Error status in AOI		
<i>AOI_Motor.DDO.Sts_Err - MainProgram/MoveWinding - 4(XIO), 6(XIC)</i>		
AOI_Motor.DDO.Sts_Done	0	BOOL
Complete status in AOI		
<i>AOI_Motor.DDO.Sts_Done - MainProgram/MoveWinding - 5(XIC)</i>		
AOI_Motor.DDO.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.DDO.Read_Command_position	284774	DINT
Command position monitor[step]		
AOI_Motor.DDO.Read_Feedback_position	284774	DINT
Feedback position monitor[step]		
AOI_Motor.DDO.Read_Torque_monitor	-2	INT
Torque monitor as a percentage of the maximum holding torque(1=0.1%)		
AOI_Motor.DDO.Read_Cumulative_load_monitor	2083663	DINT
Cumulative load monitor		
AOI_Motor.Move.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Move.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Move.Set_Exe	0	BOOL
Execute AOI		
AOI_Motor.Move.Set_OpeDataNo	0	INT
Selected Operation data No.		
AOI_Motor.Move.Sts_Busy	0	BOOL
Executing status in AOI		
AOI_Motor.Move.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.Move.Sts_Done	1	BOOL
Complete status in AOI		
AOI_Motor.Move.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.Move.Read_Command_position	-2692	DINT
Command position monitor[step]		
AOI_Motor.Move.Read_Feedback_position	-2692	DINT
Feedback position monitor[step]		
AOI_Motor.Move.Read_Torque_monitor	11	INT
Torque monitor as a percentage of the maximum holding torque(1=0.1%)		
AOI_Motor.Move.Read_Cumulative_load_monitor	677105	DINT
Cumulative load monitor		
AOI_Motor.Monitor		OM_AZ_MONITOR

AOI_Motor (Continued)*AOI_Motor.Monitor - MainProgram/MoveWinding - *1(OM_AZ_MONITOR)*

AOI_Motor.Monitor.EnableIn	1	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Monitor.EnableOut	1	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Monitor.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.Monitor.Read_Feedback_position	284774	DINT
Feedback position monitor[step]		
AOI_Motor.Monitor.Read_Feedback_speed	0	DINT
Feedback speed monitor[Hz]		
AOI_Motor.Monitor.Read_Command_position	284774	DINT
Command position monitor[step]		
<i>CurrentWindingPos - MainProgram/MoveWinding - 2(DIV), 4(NEQ)</i>		
AOI_Motor.Monitor.Read_Torque_monitor	-2	INT
Torque monitor as a percentage of the maximum holding torque[1=0.1%]		
AOI_Motor.Monitor.Read_CST_operating_current	500	INT
Operation current monitor in alpha control mode(CST) (1=0.1%)		
AOI_Motor.Monitor.Read_Information	16#0000_0000	DINT
Information monitor before an alarm is generated.		
AOI_Motor.Monitor.Read_Driver_temperature	430	DINT
Driver temperature monitor(1=0.1deg)		
AOI_Motor.Monitor.Read_Motor_temperature	459	DINT
Motor temperature monitor(1=0.1deg)		
AOI_Motor.Monitor.Read_Cumulative_load_monitor	2083663	DINT
Cumulative load monitor		
AOI_Motor.Monitor.Read_Tripmeter	317	DINT
Travel distance monitor[x1000rev]		
AOI_Motor.ReadPosition.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.ReadPosition.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.ReadPosition.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.ReadPosition.ReadPRMID	16#0c01	INT
Read parameter ID		
AOI_Motor.ReadPosition.ReadData	5000	DINT
Read data		
AOI_Motor.ReadPosition.Sts_Busy	1	BOOL
Executing status in AOI		
AOI_Motor.ReadPosition.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.ReadPosition.Sts_Done	0	BOOL
Complete status in AOI		
AOI_Motor.WritePosition.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.WritePosition.EnableOut		

AOI_Motor (Continued)		
0	BOOL	Enable Output - System Defined Parameter
AOI_Motor.WritePosition.Set_Exec 0	BOOL	Execute AOI
AOI_Motor.WritePosition.WritePRMID		
16#0c01	INT	Write parameter ID
AOI_Motor.WritePosition.WriteData		
4000	DINT	Write data
AOI_Motor.WritePosition.Sts_Busy		
0	BOOL	Executing status in AOI
AOI_Motor.WritePosition.Sts_Err 0	BOOL	Error status in AOI
AOI_Motor.WritePosition.Sts_Done		
1	BOOL	Complete status in AOI
AOI_Motor.Home.EnableIn 1	BOOL	Enable Input - System Defined Parameter
AOI_Motor.Home.EnableOut 0	BOOL	Enable Output - System Defined Parameter
AOI_Motor.Home.Set_Exec 0	BOOL	Execute AOI
AOI_Motor.Home.Sts_Busy 0	BOOL	Executing status in AOI
AOI_Motor.Home.Sts_Err 0	BOOL	Error status in AOI
AOI_Motor.Home.Sts_Done 0	BOOL	Complete status in AOI
AOI_Motor.Home.Read_Present_alarm		
16#0000	INT	Alarm code monitor[Hex]
AOI_Motor.Home.Read_Feedback_position		
0	DINT	Feedback position monitor[step]
AOI_Motor.WriteSpeed.EnableIn 0	BOOL	Enable Input - System Defined Parameter
AOI_Motor.WriteSpeed.EnableOut		
0	BOOL	Enable Output - System Defined Parameter
AOI_Motor.WriteSpeed.Set_Exec 0	BOOL	Execute AOI
AOI_Motor.WriteSpeed.WritePRMID		
16#0c02	INT	Write parameter ID
AOI_Motor.WriteSpeed.WriteData 1000	DINT	Write data
AOI_Motor.WriteSpeed.Sts_Busy 0	BOOL	Executing status in AOI
AOI_Motor.WriteSpeed.Sts_Err 0	BOOL	Error status in AOI
AOI_Motor.WriteSpeed.Sts_Done 1	BOOL	Complete status in AOI
AOI_Motor.ReadSpeed.EnableIn 0	BOOL	Enable Input - System Defined Parameter
AOI_Motor.ReadSpeed.EnableOut 0	BOOL	Enable Output - System Defined Parameter
AOI_Motor.ReadSpeed.Set_Exec 0	BOOL	Execute AOI
AOI_Motor.ReadSpeed.ReadPRMID		
16#0c02	INT	Read parameter ID

AOI_Motor (Continued)			
AOI_Motor.ReadSpeed.ReadData	1000	DINT	
Read data			
AOI_Motor.ReadSpeed.Sts_Busy	0	BOOL	
Executing status in AOI			
AOI_Motor.ReadSpeed.Sts_Err	0	BOOL	
Error status in AOI			
AOI_Motor.ReadSpeed.Sts_Done	1	BOOL	
Complete status in AOI			
DoTheThugShake	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
<i>DoTheThugShake - MainProgram/CONTROL - *B15, #1(Action,Action_043 (Body)), A15, #1(Tran,Tran_039 (Condition))</i>			
<i>DoTheThugShake - MainProgram/SetWindingDia - *B3, #1(Action,Action_044 (Body)), *B5, #1(Action,Action_056 (Body)), B3, #1(Tran,Tran_044 (Condition)), B5, #1(Tran,Tran_057 (Condition))</i>			
<i>DoTheThugShake - MainProgram/ThugShake - *B4, #1(Action,Action_030 (Body)), B1, #1(Tran,Tran_026 (Condition))</i>			
SettingDia	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
<i>SettingDia - MainProgram/CONTROL - A14, #1(Tran,Tran_049 (Condition))</i>			
<i>SettingDia - MainProgram/SetWindingDia - *B1, #2(Action,Action_045 (Body)), *B2, #1(Action,Action_031 (Body)), *B7, #2(Action,Action_048 (Body))</i>			
Step_034		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_034 - MainProgram/SetWindingDia - *B1(Step,Step_034)</i>			
Step_042		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_042 - MainProgram/SetWindingDia - *B2(Step,Step_042)</i>			
Step_042.T	501	DINT	
<i>Step_042.T - MainProgram/SetWindingDia - B2, #1(Tran,Tran_035 (Condition))</i>			
Step_043		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_043 - MainProgram/SetWindingDia - *B3(Step,Step_043)</i>			
Step_043.T	4120	DINT	
<i>Step_043.T - MainProgram/SetWindingDia - B4, #1(Tran,Tran_041 (Condition))</i>			
Step_046		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_046 - MainProgram/SetWindingDia - *B3(Step,Step_046)</i>			
Step_047		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_047 - MainProgram/SetWindingDia - *B4(Step,Step_047)</i>			
Step_047.T	101	DINT	
<i>Step_047.T - MainProgram/SetWindingDia - B5, #1(Tran,Tran_046 (Condition))</i>			
Step_048		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
<i>Step_048 - MainProgram/SetWindingDia - *B6(Step,Step_048)</i>			
Step_048.T	629	DINT	
<i>Step_048.T - MainProgram/SetWindingDia - B6, #1(Tran,Tran_047 (Condition))</i>			
Step_049		SFC_STEP	MainProgram

Step_049 (Continued)			
Constant	No		
External Access:	Read/Write		
<i>Step_049 - MainProgram/SetWindingDia - *B7(Step,Step_049)</i>			
Step_057			
		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_057 - MainProgram/SetWindingDia - *B5(Step,Step_057)</i>			
TempDint			
	100	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>TempDint - MainProgram/CONTROL - *B1, #22(Action,Action_024 (Body)), *B37, #3(Action,Action_061 (Body)), B1, #23(Action,Action_024 (Body)), B37, #4(Action,Action_061 (Body))</i>			
<i>TempDint - MainProgram/SetWindingDia - *B6, #1(Action,Action_047 (Body)), B6, #2(Action,Action_047 (Body))</i>			
Tran_031			
	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_031 - MainProgram/SetWindingDia - *B1(Tran,Tran_031), *B1, #1(Tran,Tran_031 (Condition))</i>			
Tran_035			
	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_035 - MainProgram/SetWindingDia - *B2(Tran,Tran_035), *B2, #1(Tran,Tran_035 (Condition))</i>			
Tran_041			
	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_041 - MainProgram/SetWindingDia - *B4(Tran,Tran_041), *B4, #1(Tran,Tran_041 (Condition))</i>			
Tran_044			
	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_044 - MainProgram/SetWindingDia - *B3(Tran,Tran_044), *B3, #1(Tran,Tran_044 (Condition))</i>			
Tran_046			
	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_046 - MainProgram/SetWindingDia - *B5(Tran,Tran_046), *B5, #1(Tran,Tran_046 (Condition))</i>			
Tran_047			
	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_047 - MainProgram/SetWindingDia - *B6(Tran,Tran_047), *B6, #1(Tran,Tran_047 (Condition))</i>			
Tran_048			
	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_048 - MainProgram/SetWindingDia - *B7(Tran,Tran_048), *B7, #1(Tran,Tran_048 (Condition))</i>			
Tran_057			
	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_057 - MainProgram/SetWindingDia - *B5(Tran,Tran_057), *B5, #1(Tran,Tran_057 (Condition))</i>			
WindingCurrentDia			
	170	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingCurrentDia - MainProgram/CONTROL - *B13, #1(Action,Action_049 (Body)), *B36, #3(Action,Action_060 (Body)), *B39, #1(Action,Action_063 (Body)), A37, #1(Tran,Tran_069 (Condition)), B36, #3(Action,Action_060 (Body)), B39, #1(Action,Action_063 (Body))</i>			
<i>WindingCurrentDia - MainProgram/SetWindingDia - B1, #1(Tran,Tran_031 (Condition)), B3, #1(Action,Action_032 (Body)), B7, #1(Action,Action_048 (Body))</i>			

WindingMoving	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingMoving - MainProgram/CONTROL - A28, #1(Tran,Tran_038 (Condition)), A34, #1(Tran,Tran_036 (Condition)), A38, #1(Tran,Tran_055 (Condition)), A39, #1(Tran,Tran_070 (Condition)), A4, #1(Tran,Tran_073 (Condition))</i> <i>WindingMoving - MainProgram/MoveWinding - *7(OTE)</i> <i>WindingMoving - MainProgram/SetWindingDia - B6, #1(Tran,Tran_047 (Condition))</i>			
WindingOldDia	170	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingOldDia - MainProgram/CONTROL - A37, #1(Tran,Tran_069 (Condition))</i> <i>WindingOldDia - MainProgram/SetWindingDia - *B7, #1(Action,Action_048 (Body)), B1, #1(Tran,Tran_031 (Condition)), B3, #1(Action,Action_032 (Body))</i>			
WindingSolenoid	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>WindingSolenoid - MainProgram/CONTROL - *B14, #1(Action,Action_034 (Body))</i> <i>WindingSolenoid - MainProgram/FillValues - #60</i> <i>WindingSolenoid - MainProgram/MainRoutine - 8(XIC)</i> <i>WindingSolenoid - MainProgram/pause - #12, *#34, *#58</i> <i>WindingSolenoid - MainProgram/SetWindingDia - *B1, #1(Action,Action_045 (Body)), *B2, #2(Action,Action_031 (Body)), *B4, #1(Action,Action_046 (Body))</i> <i>WindingSolenoid - MainProgram/StopAll - *#4</i>			
WindingTurns	99.667	REAL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WindingTurns - MainProgram/CONTROL - *B1, #23(Action,Action_024 (Body)), *B23, #6(Action,Action_039 (Body)), *B37, #4(Action,Action_061 (Body)), *B37, #5(Action,Action_061 (Body)), B1, #22(Action,Action_024 (Body)), B23, #6(Action,Action_039 (Body)), B37, #3(Action,Action_061 (Body)), B37, #5(Action,Action_061 (Body))</i> <i>WindingTurns - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i> <i>WindingTurns - MainProgram/MoveWinding - *2(DIV)</i> <i>WindingTurns - MainProgram/SetWindingDia - *B3, #1(Action,Action_032 (Body)), *B6, #2(Action,Action_047 (Body)), B3, #1(Action,Action_032 (Body)), B6, #1(Action,Action_047 (Body))</i> <i>WindingTurns - MainProgram/ThugShake - *B2, #2(Action,Action_027 (Body)), *B3, #1(Action,Action_029 (Body)), B2, #2(Action,Action_027 (Body)), B3, #1(Action,Action_029 (Body))</i>			

General

sets the diameter for winding mechanism			
Type:	 Sequential Function Chart	Class:	Standard
In Program:	 MainProgram	Number of Steps:	Original 8

Auto-Naming

Step:	Step_078	Action:	Action_071
Transition:	Tran_077	Stop:	Stop_000

Sheet Layout

Sheet Size:	Letter - 8.5 x 11 in	Orientation:	Landscape
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```
1 StackLightRed:=1;
2 StackLightGreen:=0;
3 StackLightYellow:=0;
4 WindingSolenoid:=0;
5 SpoolArray[0].FWD:=0;
6 SpoolArray[0].REV:=0;
7 SpoolArray[1].FWD:=0;
8 SpoolArray[1].REV:=0;
9 SpoolArray[2].FWD:=0;
10 SpoolArray[2].REV:=0;
11 WireInRobot:=0;
12 RobotToCutting:=0;
13 RobotLoaded2:=0;
14 RobotMoveNextLayer3:=0;
15 RobotDirection4:=0;
16 ClampMotorFwd := 0;
17 ClampMotorRev := 0;
18 CoilIndex_X := 0;
19 CoilIndex_Z := 0;
20 InductorSent := 0;
21 HeightMotorDown := 0;
22 HeightMotorUp := 0;
```

Name	Value	Data Type	Scope
 ClampMotorFwd	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
ClampMotorFwd - MainProgram/CONTROL - *B1, #14(Action,Action_024 (Body)), *B16, #1(Action,Action_057 (Body)), *B17, #1(Action,Action_058 (Body)), *B19, #1(Action,Action_052 (Body)), *B20, #1(Action,Action_053 (Body)), *B23, #34(Action,Action_039 (Body))			
ClampMotorFwd - MainProgram/FillValues - #50			
ClampMotorFwd - MainProgram/pause - #24, *#46, *#70			
ClampMotorFwd - MainProgram/StopAll - *#16			
 ClampMotorRev	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
ClampMotorRev - MainProgram/CONTROL - *B1, #15(Action,Action_024 (Body)), *B17, #1(Action,Action_059 (Body)), *B18, #1(Action,Action_051 (Body)), *B35, #2(Action,Action_067 (Body)), *B36, #1(Action,Action_060 (Body))			
ClampMotorRev - MainProgram/FillValues - #51			
ClampMotorRev - MainProgram/pause - #25, *#47, *#71			
ClampMotorRev - MainProgram/StopAll - *#17			
CoilIndex_X	1	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
CoilIndex_X - MainProgram/CONTROL - *B1, #16(Action,Action_024 (Body)), *B23, #25(Action,Action_039 (Body)), A34, #1(Tran,Tran_036 (Condition)), B23, #17(Action,Action_039 (Body)), B23, #25(Action,Action_039 (Body)), B23, #29(Action,Action_039 (Body)), B23, #8(Action,Action_039 (Body))			
CoilIndex_X - MainProgram/StopAll - *#18			
CoilIndex_Z	1	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
CoilIndex_Z - MainProgram/CONTROL - *B1, #17(Action,Action_024 (Body)), *B23, #21(Action,Action_039 (Body)), *B23, #24(Action,Action_039 (Body)), A34, #1(Tran,Tran_036 (Condition)), B23, #16(Action,Action_039 (Body)), B23, #17(Action,Action_039 (Body)), B23, #21(Action,Action_039 (Body)), B23, #33(Action,Action_039 (Body)), C30, #1(Tran,Tran_064 (Condition))			
CoilIndex_Z - MainProgram/StopAll - *#19			
 HeightMotorDown	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
HeightMotorDown - MainProgram/CONTROL - *B1, #19(Action,Action_024 (Body)), *B6, #1(Action,Action_064 (Body)), *B7, #1(Action,Action_065 (Body)), *C30, #2(Action,Action_026 (Body)), *C31, #1(Action,Action_062 (Body))			
HeightMotorDown - MainProgram/FillValues - #53			
HeightMotorDown - MainProgram/pause - #27, *#49, *#73			
HeightMotorDown - MainProgram/StopAll - *#21			
 HeightMotorUp	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
HeightMotorUp - MainProgram/CONTROL - *B1, #20(Action,Action_024 (Body)), *B35, #3(Action,Action_067 (Body)), *B36, #2(Action,Action_060 (Body)), *B4, #1(Action,Action_066 (Body)), *B6, #1(Action,Action_054 (Body)), *C32, #1(Action,Action_068 (Body)), *C32, #1(Action,Action_069 (Body))			
HeightMotorUp - MainProgram/FillValues - #54			
HeightMotorUp - MainProgram/pause - #28, *#50, *#74			
HeightMotorUp - MainProgram/StopAll - *#22			
InductorSent	0	BOOL	MainProgram
PC sets this true when the inductor data has been sent			
Constant	No		
External Access:	Read/Write		
InductorSent - MainProgram/CONTROL - *B1, #18(Action,Action_024 (Body))			
InductorSent - MainProgram/pause - #26, *#48, *#72			
InductorSent - MainProgram/StopAll - *#20			
InductorSent - MainProgram/WindInductor - I(XIC)			
 RobotDirection4	1	BOOL	Capstone_Test

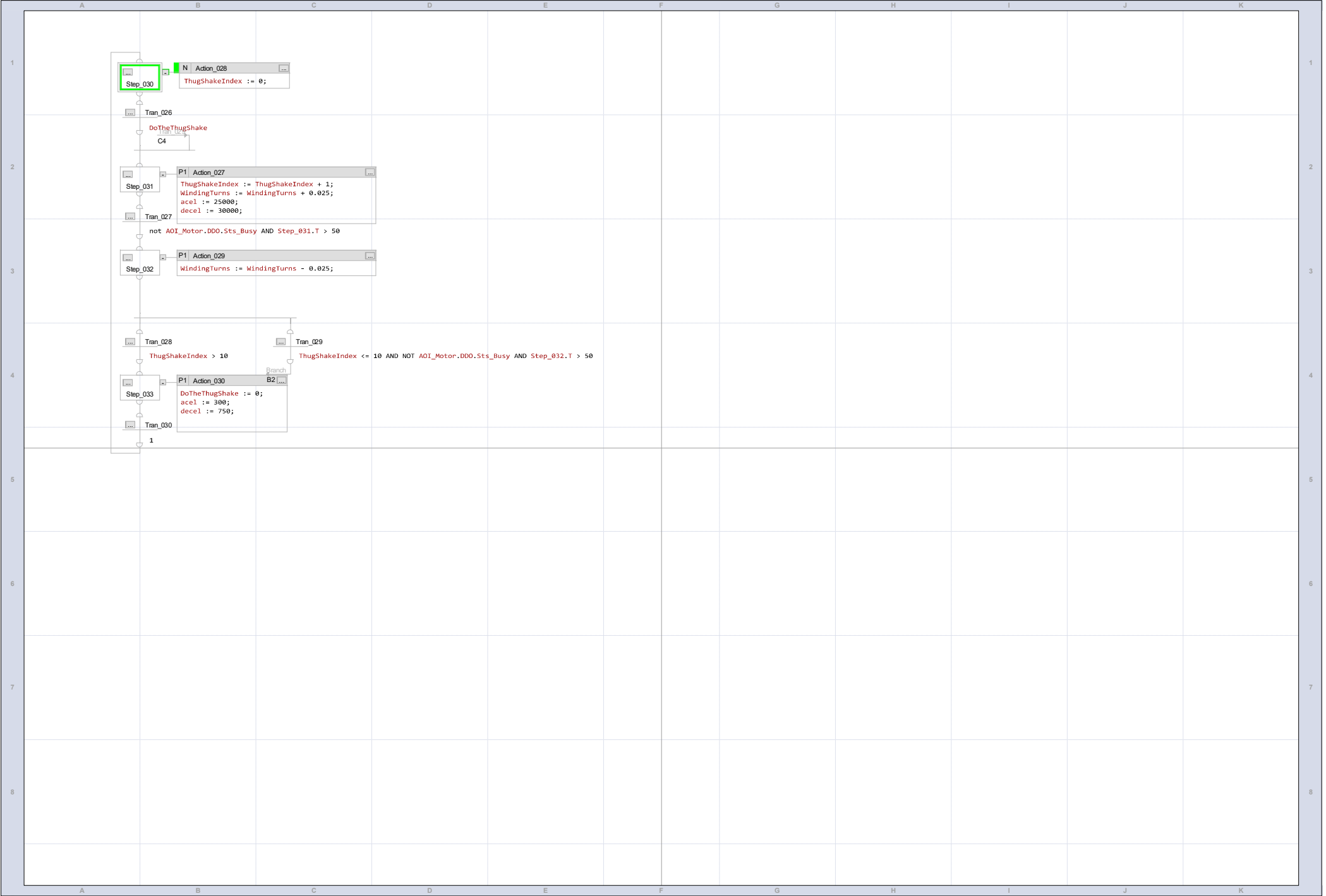
RobotDirection4 (Continued)			
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].3		
Base Tag:	Fanuc_Robot:O.Data[0].3		
Constant	No		
External Access:	Read/Write		
<i>RobotDirection4 - MainProgram/CONTROL - *B1, #13(Action,Action_024 (Body)), *B12, #2(Action,Action_023 (Body)), *B23, #10(Action,Action_039 (Body)), *B23, #12(Action,Action_039 (Body))</i>			
<i>RobotDirection4 - MainProgram/pause - #23, *#45, *#69</i>			
<i>RobotDirection4 - MainProgram/StopAll - *#15</i>			
RobotLoaded2	1	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].1		
Base Tag:	Fanuc_Robot:O.Data[0].1		
Constant	No		
External Access:	Read/Write		
<i>RobotLoaded2 - MainProgram/CONTROL - *B1, #10(Action,Action_024 (Body)), *B12, #4(Action,Action_023 (Body)), *B18, #2(Action,Action_051 (Body))</i>			
<i>RobotLoaded2 - MainProgram/pause - #21, *#43, *#67</i>			
<i>RobotLoaded2 - MainProgram/StopAll - *#13</i>			
RobotMoveNextLayer3	0	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].2		
Base Tag:	Fanuc_Robot:O.Data[0].2		
Constant	No		
External Access:	Read/Write		
<i>RobotMoveNextLayer3 - MainProgram/CONTROL - *B1, #11(Action,Action_024 (Body)), *B12, #3(Action,Action_023 (Body)), *B23, #18(Action,Action_039 (Body)), *B27, #1(Action,Action_040 (Body))</i>			
<i>RobotMoveNextLayer3 - MainProgram/pause - #22, *#44, *#68</i>			
<i>RobotMoveNextLayer3 - MainProgram/StopAll - *#14</i>			
RobotToCutting	0	BOOL	Capstone_Test
Out to Robot			
AliasFor:	Fanuc_Robot:O.Data[0].0		
Base Tag:	Fanuc_Robot:O.Data[0].0		
Constant	No		
External Access:	Read/Write		
<i>RobotToCutting - MainProgram/CONTROL - *B1, #9(Action,Action_024 (Body)), *B11, #2(Action,Action_021 (Body)), *B11, #2(Action,Action_022 (Body))</i>			
<i>RobotToCutting - MainProgram/pause - #20, *#42, *#66</i>			
<i>RobotToCutting - MainProgram/StopAll - *#12</i>			
SpoolArray		Spool[3]	Capstone_Test
Constant	No		
External Access:	Read/Write		
SpoolArray[0].REV	0	BOOL	
out			
<i>SpoolArray[0].REV - MainProgram/CONTROL - *B1, #3(Action,Action_024 (Body))</i>			
<i>SpoolArray[0].REV - MainProgram/FillValues - #41</i>			
<i>SpoolArray[0].REV - MainProgram/pause - #14, *#36, *#60</i>			
<i>SpoolArray[0].REV - MainProgram/StopAll - *#6</i>			
SpoolArray[0].FWD	0	BOOL	
out			
<i>SpoolArray[0].FWD - MainProgram/CONTROL - *B1, #2(Action,Action_024 (Body))</i>			
<i>SpoolArray[0].FWD - MainProgram/FillValues - #40</i>			
<i>SpoolArray[0].FWD - MainProgram/pause - #13, *#35, *#59</i>			
<i>SpoolArray[0].FWD - MainProgram/StopAll - *#5</i>			
SpoolArray[0].Mass	164.73665	REAL	
current mass reading			
<i>SpoolArray[0].Mass - MainProgram/FillValues - *#20, *#22</i>			
SpoolArray[0].Length	0.0	REAL	
approximated legth based on Mass			

SpoolArray (Continued)			
SpoolArray[0].Loaded	1	BOOL	
if is in loaded state			
<i>SpoolArray[0].Loaded - MainProgram/FillValues - *#10</i>			
SpoolArray[0].Gauge	-1	DINT	
Wire Gauge			
SpoolArray[1].REV	0	BOOL	
out			
<i>SpoolArray[1].REV - MainProgram/CONTROL - *B1, #5(Action,Action_024 (Body))</i>			
<i>SpoolArray[1].REV - MainProgram/FillValues - #43</i>			
<i>SpoolArray[1].REV - MainProgram/pause - #16, *#38, *#62</i>			
<i>SpoolArray[1].REV - MainProgram/StopAll - *#8</i>			
SpoolArray[1].FWD	0	BOOL	
out			
<i>SpoolArray[1].FWD - MainProgram/CONTROL - *B1, #4(Action,Action_024 (Body))</i>			
<i>SpoolArray[1].FWD - MainProgram/FillValues - #42</i>			
<i>SpoolArray[1].FWD - MainProgram/pause - #15, *#37, *#61</i>			
<i>SpoolArray[1].FWD - MainProgram/StopAll - *#7</i>			
SpoolArray[1].Mass	163.34396	REAL	
current mass reading			
SpoolArray[1].Length	0.0	REAL	
approximated legth based on Mass			
SpoolArray[1].Loaded	1	BOOL	
if is in loaded state			
<i>SpoolArray[1].Loaded - MainProgram/FillValues - *#11</i>			
SpoolArray[1].Gauge	-1	DINT	
Wire Gauge			
SpoolArray[2].REV	0	BOOL	
out			
<i>SpoolArray[2].REV - MainProgram/CONTROL - *B1, #7(Action,Action_024 (Body))</i>			
<i>SpoolArray[2].REV - MainProgram/FillValues - #45</i>			
<i>SpoolArray[2].REV - MainProgram/pause - #18, *#40, *#64</i>			
<i>SpoolArray[2].REV - MainProgram/StopAll - *#10</i>			
SpoolArray[2].FWD	0	BOOL	
out			
<i>SpoolArray[2].FWD - MainProgram/CONTROL - *B1, #6(Action,Action_024 (Body)), *B11, #1(Action,Action_021 (Body)), *B11, #1(Action,Action_022 (Body)), *B12, #1(Action,Action_023 (Body)), *B23, #3(Action,Action_039 (Body)), *B37, #1(Action,Action_061 (Body)), *C9, #1(Action,Action_019 (Body))</i>			
<i>SpoolArray[2].FWD - MainProgram/FillValues - #44</i>			
<i>SpoolArray[2].FWD - MainProgram/pause - #17, *#39, *#63</i>			
<i>SpoolArray[2].FWD - MainProgram/StopAll - *#9</i>			
SpoolArray[2].Mass	600.10535	REAL	
current mass reading			
<i>SpoolArray[2].Mass - MainProgram/FillValues - *#32, *#34</i>			
SpoolArray[2].Length	0.0	REAL	
approximated legth based on Mass			
SpoolArray[2].Loaded	0	BOOL	
if is in loaded state			
<i>SpoolArray[2].Loaded - MainProgram/FillValues - *#12</i>			
SpoolArray[2].Gauge	16	DINT	
Wire Gauge			
<i>SpoolArray[2].Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)</i>			
 StackLightGreen	1	BOOL	Capstone_Test
AliasFor:			
Local:4:O.Pt07.Data			
Base Tag:			
Local:4:O.Pt07.Data			
Constant			
No			
External Access:			
Read/Write			
<i>StackLightGreen - MainProgram/MainRoutine - *5(OTE)</i>			
<i>StackLightGreen - MainProgram/pause - #10, *#32, *#56</i>			
<i>StackLightGreen - MainProgram/StopAll - *#2</i>			
 StackLightRed	0	BOOL	Capstone_Test
AliasFor:			
Local:4:O.Pt09.Data			

StackLightRed (Continued)			
Base Tag:	Local:4:O.Pt09.Data		
Constant	No		
External Access:	Read/Write		
<i>StackLightRed - MainProgram/MainRoutine - *7(OTE)</i>			
<i>StackLightRed - MainProgram/pause - #9, *#31, *#55</i>			
<i>StackLightRed - MainProgram/StopAll - *#1</i>			
 StackLightYellow	1	BOOL	Capstone_Test
AliasFor:	Local:4:O.Pt08.Data		
Base Tag:	Local:4:O.Pt08.Data		
Constant	No		
External Access:	Read/Write		
<i>StackLightYellow - MainProgram/MainRoutine - *6(OTE)</i>			
<i>StackLightYellow - MainProgram/pause - #11, *#33, *#57</i>			
<i>StackLightYellow - MainProgram/StopAll - *#3</i>			
 WindingSolenoid	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
<i>WindingSolenoid - MainProgram/CONTROL - *B14, #1(Action,Action_034 (Body))</i>			
<i>WindingSolenoid - MainProgram/FillValues - #60</i>			
<i>WindingSolenoid - MainProgram/MainRoutine - 8(XIC)</i>			
<i>WindingSolenoid - MainProgram/pause - #12, *#34, *#58</i>			
<i>WindingSolenoid - MainProgram/SetWindingDia - *B1, #1(Action,Action_045 (Body)), *B2, #2(Action,Action_031 (Body)), *B4, #1(Action,Action_046 (Body))</i>			
<i>WindingSolenoid - MainProgram/StopAll - *#4</i>			
WireInRobot	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>WireInRobot - MainProgram/CONTROL - *B1, #8(Action,Action_024 (Body))</i>			
<i>WireInRobot - MainProgram/pause - #19, *#41, *#65</i>			
<i>WireInRobot - MainProgram/StopAll - *#11</i>			

General			
Type:	 Structured Text	Class:	Standard
In Program:	 MainProgram	Number of Lines:	Original 22

Shakes winding to hopefully get the solenoid to seat into a hole



Name	Value	Data Type	Scope
acel	300	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>acel - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>acel - MainProgram/ThugShake - *B2, #3(Action,Action_027 (Body)), *B4, #2(Action,Action_030 (Body))</i>			
Action_027		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_027 - MainProgram/ThugShake - *B2(Action,Action_027)</i>			
Action_028		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_028 - MainProgram/ThugShake - *B1(Action,Action_028)</i>			
Action_029		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_029 - MainProgram/ThugShake - *B3(Action,Action_029)</i>			
Action_030		SFC_ACTION	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Action_030 - MainProgram/ThugShake - *B4(Action,Action_030)</i>			
AOI_Motor		AOI_Motor	Capstone_Test
Constant	No		
External Access:	Read/Write		
AOI_Motor.DDO		OM_AZ_DDO	
<i>AOI_Motor.DDO - MainProgram/MoveWinding - *8(OM_AZ_DDO)</i>			
AOI_Motor.DDO.EnableIn	1	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.DDO.EnableOut	1	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.DDO.Set_Exec	0	BOOL	
Execute AOI			
<i>AOI_Motor.DDO.Set_Exec - MainProgram/MoveWinding - *4(OTL), *5(OTU), *6(OTU)</i>			
AOI_Motor.DDO.Set_OpType	1	INT	
Direct data operation type			
AOI_Motor.DDO.Set_Position	284774	DINT	
Direct data operation position			
<i>AOI_Motor.DDO.Set_Position - MainProgram/MoveWinding - *3(MUL), 4(NEQ)</i>			
AOI_Motor.DDO.Set_Velocity	750	DINT	
Direct data operation operating speed			
<i>AOI_Motor.DDO.Set_Velocity - MainProgram/MoveWinding - *3(MOV)</i>			
AOI_Motor.DDO.Set_Acceleration	300	DINT	
Direct data operation starting/changing rate			
<i>AOI_Motor.DDO.Set_Acceleration - MainProgram/MoveWinding - *3(MOV)</i>			
AOI_Motor.DDO.Set_Deceleration	750	DINT	
Direct data operation stopping deceleration			
<i>AOI_Motor.DDO.Set_Deceleration - MainProgram/MoveWinding - *3(MOV)</i>			
AOI_Motor.DDO.Set_Current	1000	INT	
Direct data operation operating current			
AOI_Motor.DDO.Set_Send	0	INT	
Direct data operation forwarding destination			
AOI_Motor.DDO.Sts_Busy	0	BOOL	
Executing status in AOI			
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/MoveWinding - 7(XIC)</i>			
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/SetWindingDia - B4, #1(Tran,Tran_041 (Condition))</i>			
<i>AOI_Motor.DDO.Sts_Busy - MainProgram/ThugShake - B2, #1(Tran,Tran_027 (Condition)), C4, #1(Tran,Tran_029 (Condition))</i>			
AOI_Motor.DDO.Sts_Err	0	BOOL	
Error status in AOI			

AOI_Motor (Continued)		
<i>AOI_Motor.DDO.Sts_Err - MainProgram/MoveWinding - 4(XIO), 6(XIC)</i>		
AOI_Motor.DDO.Sts_Done	0	BOOL
Complete status in AOI		
<i>AOI_Motor.DDO.Sts_Done - MainProgram/MoveWinding - 5(XIC)</i>		
AOI_Motor.DDO.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.DDO.Read_Command_position	284774	DINT
Command position monitor[step]		
AOI_Motor.DDO.Read_Feedback_position	284774	DINT
Feedback position monitor[step]		
AOI_Motor.DDO.Read_Torque_monitor	-2	INT
Torque monitor as a percentage of the maximum holding torque(1=0.1%)		
AOI_Motor.DDO.Read_Cumulative_load_monitor	2083663	DINT
Cumulative load monitor		
AOI_Motor.Move.EnableIn	0	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Move.EnableOut	0	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Move.Set_Exec	0	BOOL
Execute AOI		
AOI_Motor.Move.Set_OpDataNo	0	INT
Selected Operation data No.		
AOI_Motor.Move.Sts_Busy	0	BOOL
Executing status in AOI		
AOI_Motor.Move.Sts_Err	0	BOOL
Error status in AOI		
AOI_Motor.Move.Sts_Done	1	BOOL
Complete status in AOI		
AOI_Motor.Move.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.Move.Read_Command_position	-2692	DINT
Command position monitor[step]		
AOI_Motor.Move.Read_Feedback_position	-2692	DINT
Feedback position monitor[step]		
AOI_Motor.Move.Read_Torque_monitor	11	INT
Torque monitor as a percentage of the maximum holding torque(1=0.1%)		
AOI_Motor.Move.Read_Cumulative_load_monitor	677105	DINT
Cumulative load monitor		
AOI_Motor.Monitor	OM_AZ_MONITOR	
<i>AOI_Motor.Monitor - MainProgram/MoveWinding - *I(OM_AZ_MONITOR)</i>		
AOI_Motor.Monitor.EnableIn	1	BOOL
Enable Input - System Defined Parameter		
AOI_Motor.Monitor.EnableOut	1	BOOL
Enable Output - System Defined Parameter		
AOI_Motor.Monitor.Read_Present_alarm	16#0000	INT
Alarm code monitor[Hex]		
AOI_Motor.Monitor.Read_Feedback_position	284774	DINT
Feedback position monitor[step]		
AOI_Motor.Monitor.Read_Feedback_speed	0	DINT
Feedback speed monitor[Hz]		

AOI_Motor (Continued)		
AOI_Motor.Monitor.Read_Command_position		
284774	DINT	
Command position monitor[step]		
<i>CurrentWindingPos - MainProgram/MoveWinding - 2(DIV), 4(NEQ)</i>		
AOI_Motor.Monitor.Read_Torque_monitor		
-2	INT	
Torque monitor as a percentage of the maximum holding torque[1=0.1%]		
AOI_Motor.Monitor.Read_CST_operating_current		
500	INT	
Operation current monitor in alpha control mode(CST) (1=0.1%)		
AOI_Motor.Monitor.Read_Information		
16#0000_0000	DINT	
Information monitor before an alarm is generated.		
AOI_Motor.Monitor.Read_Driver_temperature		
430	DINT	
Driver temperature monitor(1=0.1deg)		
AOI_Motor.Monitor.Read_Motor_temperature		
459	DINT	
Motor temperature monitor(1=0.1deg)		
AOI_Motor.Monitor.Read_Cumulative_load_monitor		
2083663	DINT	
Cumulative load monitor		
AOI_Motor.Monitor.Read_Tripmeter		
317	DINT	
Travel distance monitor[x1000rev]		
AOI_Motor.ReadPosition.EnableIn 0	BOOL	
Enable Input - System Defined Parameter		
AOI_Motor.ReadPosition.EnableOut		
0	BOOL	
Enable Output - System Defined Parameter		
AOI_Motor.ReadPosition.Set_Exec 0	BOOL	
Execute AOI		
AOI_Motor.ReadPosition.ReadPRMID		
16#0c01	INT	
Read parameter ID		
AOI_Motor.ReadPosition.ReadData		
5000	DINT	
Read data		
AOI_Motor.ReadPosition.Sts_Busy 1	BOOL	
Executing status in AOI		
AOI_Motor.ReadPosition.Sts_Err 0	BOOL	
Error status in AOI		
AOI_Motor.ReadPosition.Sts_Done		
0	BOOL	
Complete status in AOI		
AOI_Motor.WritePosition.EnableIn		
0	BOOL	
Enable Input - System Defined Parameter		
AOI_Motor.WritePosition.EnableOut		
0	BOOL	
Enable Output - System Defined Parameter		
AOI_Motor.WritePosition.Set_Exec 0	BOOL	
Execute AOI		
AOI_Motor.WritePosition.WritePRMID		
16#0c01	INT	
Write parameter ID		
AOI_Motor.WritePosition.WriteData		
4000	DINT	
Write data		
AOI_Motor.WritePosition.Sts_Busy		
0	BOOL	
Executing status in AOI		
AOI_Motor.WritePosition.Sts_Err 0	BOOL	

AOI_Motor (Continued)			
Error status in AOI			
AOI_Motor.WritePosition.Sts_Done	1	BOOL	
Complete status in AOI			
AOI_Motor.Home.EnableIn	1	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.Home.EnableOut	0	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.Home.Set_Exe	0	BOOL	
Execute AOI			
AOI_Motor.Home.Sts_Busy	0	BOOL	
Executing status in AOI			
AOI_Motor.Home.Sts_Err	0	BOOL	
Error status in AOI			
AOI_Motor.Home.Sts_Done	0	BOOL	
Complete status in AOI			
AOI_Motor.Home.Read_Present_alarm	16#0000	INT	
Alarm code monitor[Hex]			
AOI_Motor.Home.Read_Feedback_position	0	DINT	
Feedback position monitor[step]			
AOI_Motor.WriteSpeed.EnableIn	0	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.WriteSpeed.EnableOut	0	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.WriteSpeed.Set_Exe	0	BOOL	
Execute AOI			
AOI_Motor.WriteSpeed.WritePRMID	16#0c02	INT	
Write parameter ID			
AOI_Motor.WriteSpeed.WriteData	1000	DINT	
Write data			
AOI_Motor.WriteSpeed.Sts_Busy	0	BOOL	
Executing status in AOI			
AOI_Motor.WriteSpeed.Sts_Err	0	BOOL	
Error status in AOI			
AOI_Motor.WriteSpeed.Sts_Done	1	BOOL	
Complete status in AOI			
AOI_Motor.ReadSpeed.EnableIn	0	BOOL	
Enable Input - System Defined Parameter			
AOI_Motor.ReadSpeed.EnableOut	0	BOOL	
Enable Output - System Defined Parameter			
AOI_Motor.ReadSpeed.Set_Exe	0	BOOL	
Execute AOI			
AOI_Motor.ReadSpeed.ReadPRMID	16#0c02	INT	
Read parameter ID			
AOI_Motor.ReadSpeed.ReadData	1000	DINT	
Read data			
AOI_Motor.ReadSpeed.Sts_Busy	0	BOOL	
Executing status in AOI			
AOI_Motor.ReadSpeed.Sts_Err	0	BOOL	
Error status in AOI			
AOI_Motor.ReadSpeed.Sts_Done	1	BOOL	
Complete status in AOI			
decel	750	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>decel - MainProgram/MainRoutine - 1(JSR), 4(JSR)</i>			
<i>decel - MainProgram/ThugShake - *B2, #4(Action,Action_027 (Body)), *B4, #3(Action,Action_030 (Body))</i>			

DoTheThugShake	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>DoTheThugShake - MainProgram/CONTROL - *B15, #1(Action,Action_043 (Body)), A15, #1(Tran,Tran_039 (Condition))</i>			
<i>DoTheThugShake - MainProgram/SetWindingDia - *B3, #1(Action,Action_044 (Body)), *B5, #1(Action,Action_056 (Body)), B3, #1(Tran,Tran_044 (Condition)), B5, #1(Tran,Tran_057 (Condition))</i>			
<i>DoTheThugShake - MainProgram/ThugShake - *B4, #1(Action,Action_030 (Body)), B1, #1(Tran,Tran_026 (Condition))</i>			
Step_030		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_030 - MainProgram/ThugShake - *B1(Step,Step_030)</i>			
Step_031		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_031 - MainProgram/ThugShake - *B2(Step,Step_031)</i>			
Step_031.T	140	DINT	
<i>Step_031.T - MainProgram/ThugShake - B2, #1(Tran,Tran_027 (Condition))</i>			
Step_032		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_032 - MainProgram/ThugShake - *B3(Step,Step_032)</i>			
Step_032.T	0	DINT	
<i>Step_032.T - MainProgram/ThugShake - C4, #1(Tran,Tran_029 (Condition))</i>			
Step_033		SFC_STEP	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Step_033 - MainProgram/ThugShake - *B4(Step,Step_033)</i>			
ThugShakeIndex	0	DINT	MainProgram
Constant	No		
External Access:	Read/Write		
<i>ThugShakeIndex - MainProgram/ThugShake - *B1, #1(Action,Action_028 (Body)), *B2, #1(Action,Action_027 (Body)), B2, #1(Action,Action_027 (Body)), B4, #1(Tran,Tran_028 (Condition)), C4, #1(Tran,Tran_029 (Condition))</i>			
Tran_026	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_026 - MainProgram/ThugShake - *B1(Tran,Tran_026), *B1, #1(Tran,Tran_026 (Condition))</i>			
Tran_027	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_027 - MainProgram/ThugShake - *B2(Tran,Tran_027), *B2, #1(Tran,Tran_027 (Condition))</i>			
Tran_028	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_028 - MainProgram/ThugShake - *B4(Tran,Tran_028), *B4, #1(Tran,Tran_028 (Condition))</i>			
Tran_029	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_029 - MainProgram/ThugShake - *C4(Tran,Tran_029), *C4, #1(Tran,Tran_029 (Condition))</i>			
Tran_030	1	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
<i>Tran_030 - MainProgram/ThugShake - *B4(Tran,Tran_030), *B4, #1(Tran,Tran_030 (Condition))</i>			
WindingTurns	99.667	REAL	MainProgram

WindingTurns (Continued)

Constant	No
External Access:	Read/Write

*WindingTurns - MainProgram/CONTROL - *B1, #23(Action,Action_024 (Body)), *B23, #6(Action,Action_039 (Body)), *B37, #4(Action,Action_061 (Body)), *B37, #5(Action,Action_061 (Body)), B1, #22(Action,Action_024 (Body)), B23, #6(Action,Action_039 (Body)), B37, #3(Action,Action_061 (Body)), B37, #5(Action,Action_061 (Body))*

WindingTurns - MainProgram/MainRoutine - 1(JSR), 4(JSR)

*WindingTurns - MainProgram/MoveWinding - *2(DIV)*

*WindingTurns - MainProgram/SetWindingDia - *B3, #1(Action,Action_032 (Body)), *B6, #2(Action,Action_047 (Body)), B3, #1(Action,Action_032 (Body)), B6, #1(Action,Action_047 (Body))*

*WindingTurns - MainProgram/ThugShake - *B2, #2(Action,Action_027 (Body)), *B3, #1(Action,Action_029 (Body)), B2, #2(Action,Action_027 (Body)), B3, #1(Action,Action_029 (Body))*

General

Shakes winding to hopefully get the solenoid to seat into a hole

Type: Sequential Function Chart
In Program: MainProgram

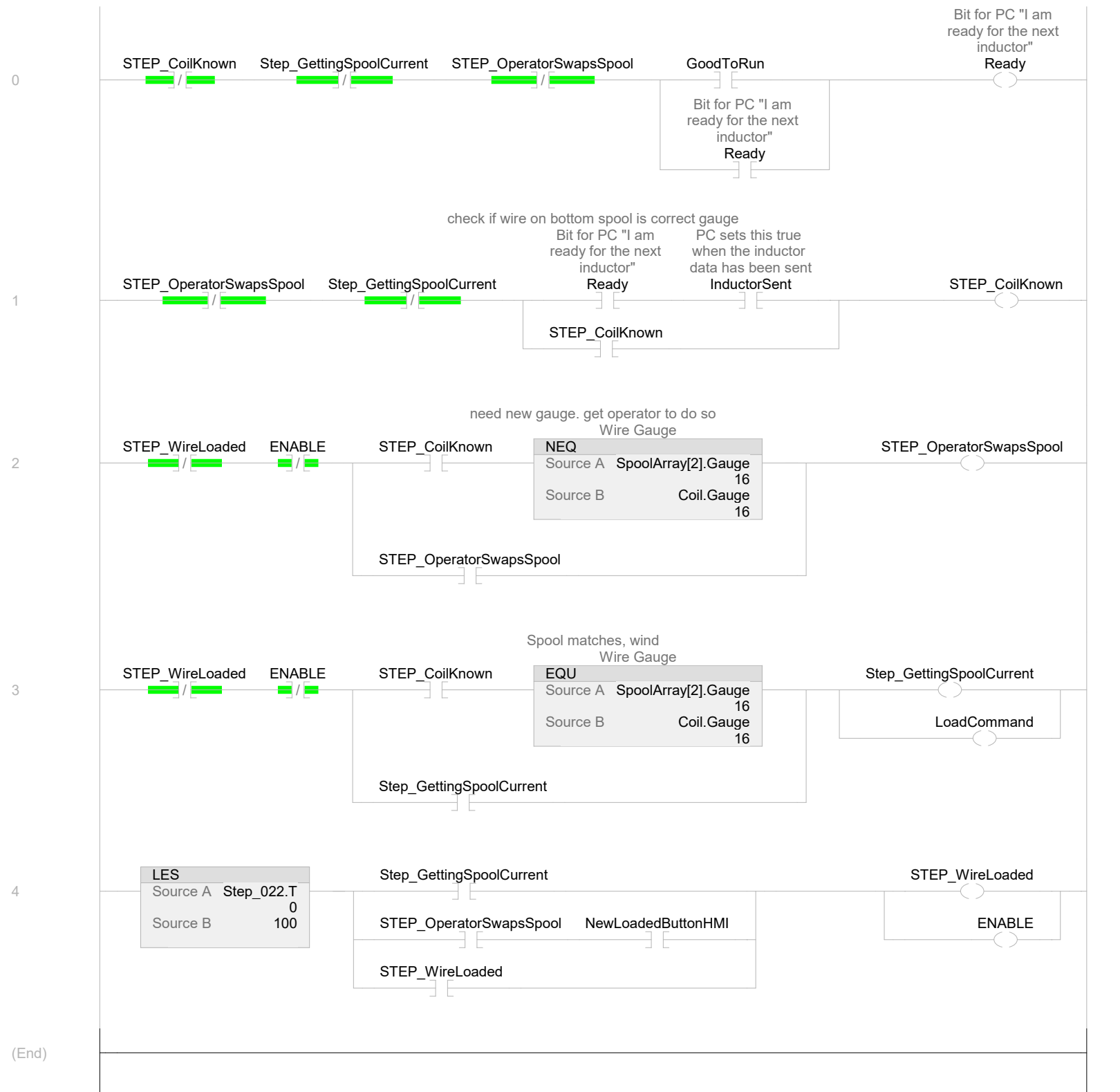
Class: Standard
Number of Steps: Original 4

Auto-Naming

Step: Step_078
Transition: Tran_077
Action: Action_071
Stop: Stop_000

Sheet Layout

Sheet Size: Letter - 8.5 x 11 in
Orientation: Landscape



Name	Value	Data Type	Scope
Coil		Coil	Capstone_Test
Constant	No		
External Access:	Read/Write		
Coil.Inductance mH	8.92412616e-004	REAL	
Coil.Mass Kg	2.11115021e-005	REAL	
Coil.Diameter m	0.18	REAL	
Coil.Diameter - MainProgram/CONTROL - B13, #1(Action,Action_049 (Body))			
Coil.Turns_X	2	INT	
Coil.Turns_X - MainProgram/CONTROL - A34, #1(Tran,Tran_036 (Condition))			
Coil.Turns_Z	2	INT	
Coil.Turns_Z - MainProgram/CONTROL - A34, #1(Tran,Tran_036 (Condition)), B23, #16(Action,Action_039 (Body)), C30, #1(Tran,Tran_064 (Condition))			
Coil.Current A	0.15	REAL	
Coil.Length m	0.003	REAL	
Coil.Gauge	16	DINT	
Coil.Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)			
ENABLE	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
ENABLE - MainProgram/CONTROL - A8, #1(Tran,Tran_019 (Condition))			
ENABLE - MainProgram/MainRoutine - 0(XIO)			
ENABLE - MainProgram/WindInductor - *4(OTE), 2(XIO), 3(XIO)			
GoodToRun	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
GoodToRun - MainProgram/MainRoutine - *2(OTE), 2(XIC), 4(XIC)			
GoodToRun - MainProgram/pause - #2, #5			
GoodToRun - MainProgram/WindInductor - 0(XIC)			
InductorSent	0	BOOL	MainProgram
PC sets this true when the inductor data has been sent			
Constant	No		
External Access:	Read/Write		
InductorSent - MainProgram/CONTROL - *B1, #18(Action,Action_024 (Body))			
InductorSent - MainProgram/pause - #26, *#48, *#72			
InductorSent - MainProgram/StopAll - *#20			
InductorSent - MainProgram/WindInductor - 1(XIC)			
LoadCommand	0	BOOL	MainProgram
Constant	No		
External Access:	Read/Write		
LoadCommand - MainProgram/WindInductor - *3(OTE)			
NewLoadedButtonHMI	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
NewLoadedButtonHMI - MainProgram/WindInductor - 4(XIC)			
Ready	0	BOOL	MainProgram
Bit for PC "I am ready for the next inductor"			
Constant	No		
External Access:	Read/Write		
Ready - MainProgram/WindInductor - *0(OTE), 0(XIC), 1(XIC)			
SpoolArray		Spool[3]	Capstone_Test
Constant	No		

SpoolArray (Continued)		
External Access:	Read/Write	
SpoolArray[0].REV	0	BOOL
out		
SpoolArray[0].REV - MainProgram/CONTROL - *B1, #3(Action,Action_024 (Body))		
SpoolArray[0].REV - MainProgram/FillValues - #41		
SpoolArray[0].REV - MainProgram/pause - #14, *#36, *#60		
SpoolArray[0].REV - MainProgram/StopAll - *#6		
SpoolArray[0].FWD	0	BOOL
out		
SpoolArray[0].FWD - MainProgram/CONTROL - *B1, #2(Action,Action_024 (Body))		
SpoolArray[0].FWD - MainProgram/FillValues - #40		
SpoolArray[0].FWD - MainProgram/pause - #13, *#35, *#59		
SpoolArray[0].FWD - MainProgram/StopAll - *#5		
SpoolArray[0].Mass	164.73665	REAL
current mass reading		
SpoolArray[0].Mass - MainProgram/FillValues - *#20, *#22		
SpoolArray[0].Length	0.0	REAL
approximated legth based on Mass		
SpoolArray[0].Loaded	1	BOOL
if is in loaded state		
SpoolArray[0].Loaded - MainProgram/FillValues - *#10		
SpoolArray[0].Gauge	-1	DINT
Wire Gauge		
SpoolArray[1].REV	0	BOOL
out		
SpoolArray[1].REV - MainProgram/CONTROL - *B1, #5(Action,Action_024 (Body))		
SpoolArray[1].REV - MainProgram/FillValues - #43		
SpoolArray[1].REV - MainProgram/pause - #16, *#38, *#62		
SpoolArray[1].REV - MainProgram/StopAll - *#8		
SpoolArray[1].FWD	0	BOOL
out		
SpoolArray[1].FWD - MainProgram/CONTROL - *B1, #4(Action,Action_024 (Body))		
SpoolArray[1].FWD - MainProgram/FillValues - #42		
SpoolArray[1].FWD - MainProgram/pause - #15, *#37, *#61		
SpoolArray[1].FWD - MainProgram/StopAll - *#7		
SpoolArray[1].Mass	163.34396	REAL
current mass reading		
SpoolArray[1].Length	0.0	REAL
approximated legth based on Mass		
SpoolArray[1].Loaded	1	BOOL
if is in loaded state		
SpoolArray[1].Loaded - MainProgram/FillValues - *#11		
SpoolArray[1].Gauge	-1	DINT
Wire Gauge		
SpoolArray[2].REV	0	BOOL
out		
SpoolArray[2].REV - MainProgram/CONTROL - *B1, #7(Action,Action_024 (Body))		
SpoolArray[2].REV - MainProgram/FillValues - #45		
SpoolArray[2].REV - MainProgram/pause - #18, *#40, *#64		
SpoolArray[2].REV - MainProgram/StopAll - *#10		
SpoolArray[2].FWD	0	BOOL
out		
SpoolArray[2].FWD - MainProgram/CONTROL - *B1, #6(Action,Action_024 (Body)), *B11, #1(Action,Action_021 (Body)), *B11, #1(Action,Action_022 (Body)), *B12, #1(Action,Action_023 (Body)), *B23, #3(Action,Action_039 (Body)), *B37, #1(Action,Action_061 (Body)), *C9, #1(Action,Action_019 (Body))		
SpoolArray[2].FWD - MainProgram/FillValues - #44		
SpoolArray[2].FWD - MainProgram/pause - #17, *#39, *#63		
SpoolArray[2].FWD - MainProgram/StopAll - *#9		
SpoolArray[2].Mass	600.10535	REAL
current mass reading		
SpoolArray[2].Mass - MainProgram/FillValues - *#32, *#34		
SpoolArray[2].Length	0.0	REAL
approximated legth based on Mass		

SpoolArray (Continued)			
SpoolArray[2].Loaded	0	BOOL	
if is in loaded state			
SpoolArray[2].Loaded - MainProgram/FillValues - *#12			
SpoolArray[2].Gauge	16	DINT	
Wire Gauge			
SpoolArray[2].Gauge - MainProgram/WindInductor - 2(NEQ), 3(EQU)			
Step_022		SFC_STEP	MainProgram
Constant No			
External Access: Read/Write			
Step_022.T	0	DINT	
Step_022.T - MainProgram/WindInductor - 4(LES)			
STEP_CoilKnown	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
STEP_CoilKnown - MainProgram/WindInductor - *1(OTE), 0(XIO), 1(XIC), 2(XIC), 3(XIC)			
Step_GettingSpoolCurrent	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
Step_GettingSpoolCurrent - MainProgram/WindInductor - *3(OTE), 0(XIO), 1(XIO), 3(XIC), 4(XIC)			
STEP_OperatorSwapsSpool	0	BOOL	MainProgram
Constant No			
External Access: Read/Write			
STEP_OperatorSwapsSpool - MainProgram/WindInductor - *2(OTE), 0(XIO), 1(XIO), 2(XIC), 4(XIC)			
STEP_WireLoaded	0	BOOL	Capstone_Test
Constant No			
External Access: Read/Write			
STEP_WireLoaded - MainProgram/WindInductor - *4(OTE), 2(XIO), 3(XIO), 4(XIC)			

General			
Type:	 Ladder Diagram	Class:	Standard
In Program:	 MainProgram	Number of Rungs:	5

General

Class: Safety

Configuration

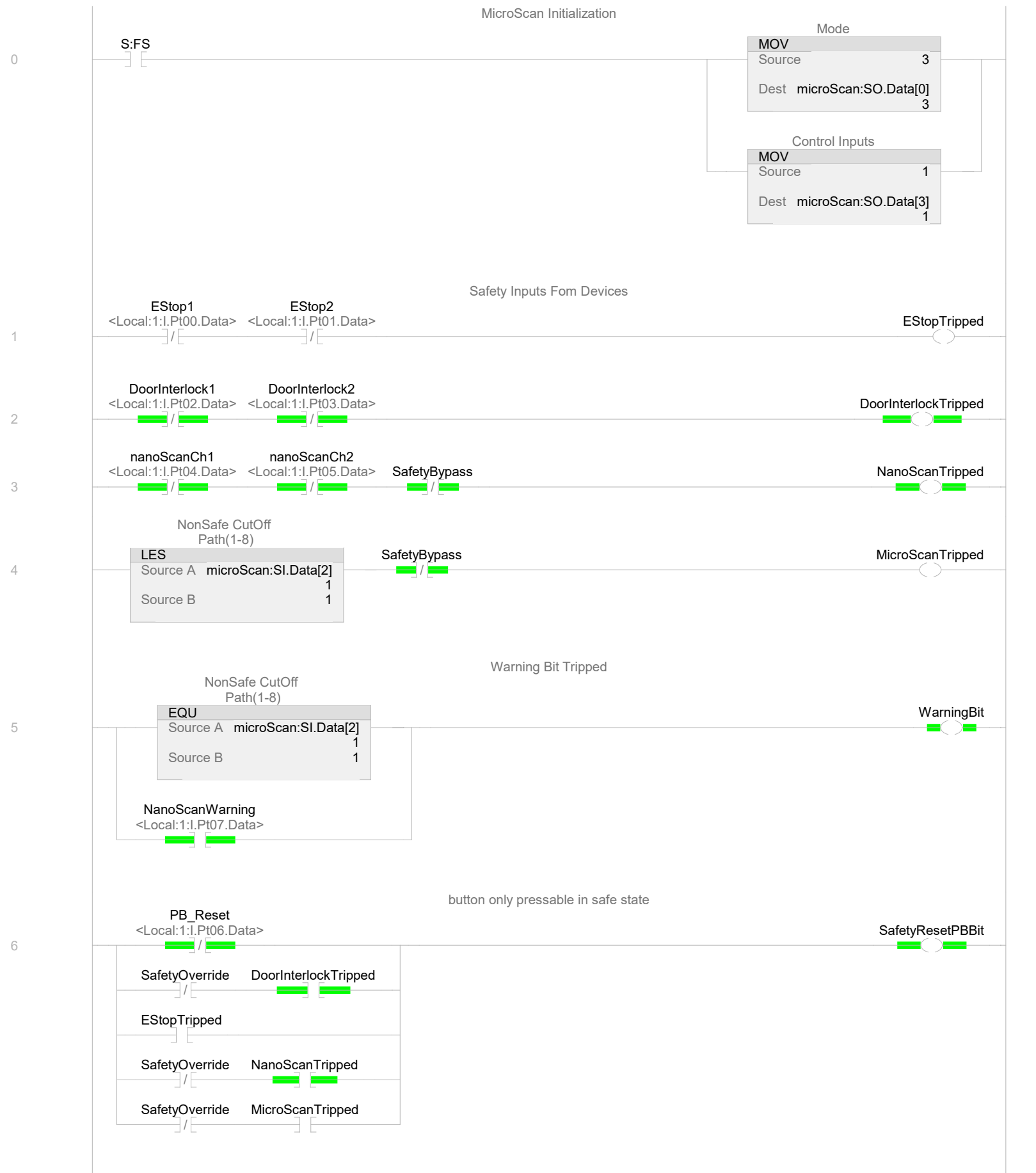
Main:  MainRoutine Inhibit program: No
Fault: None

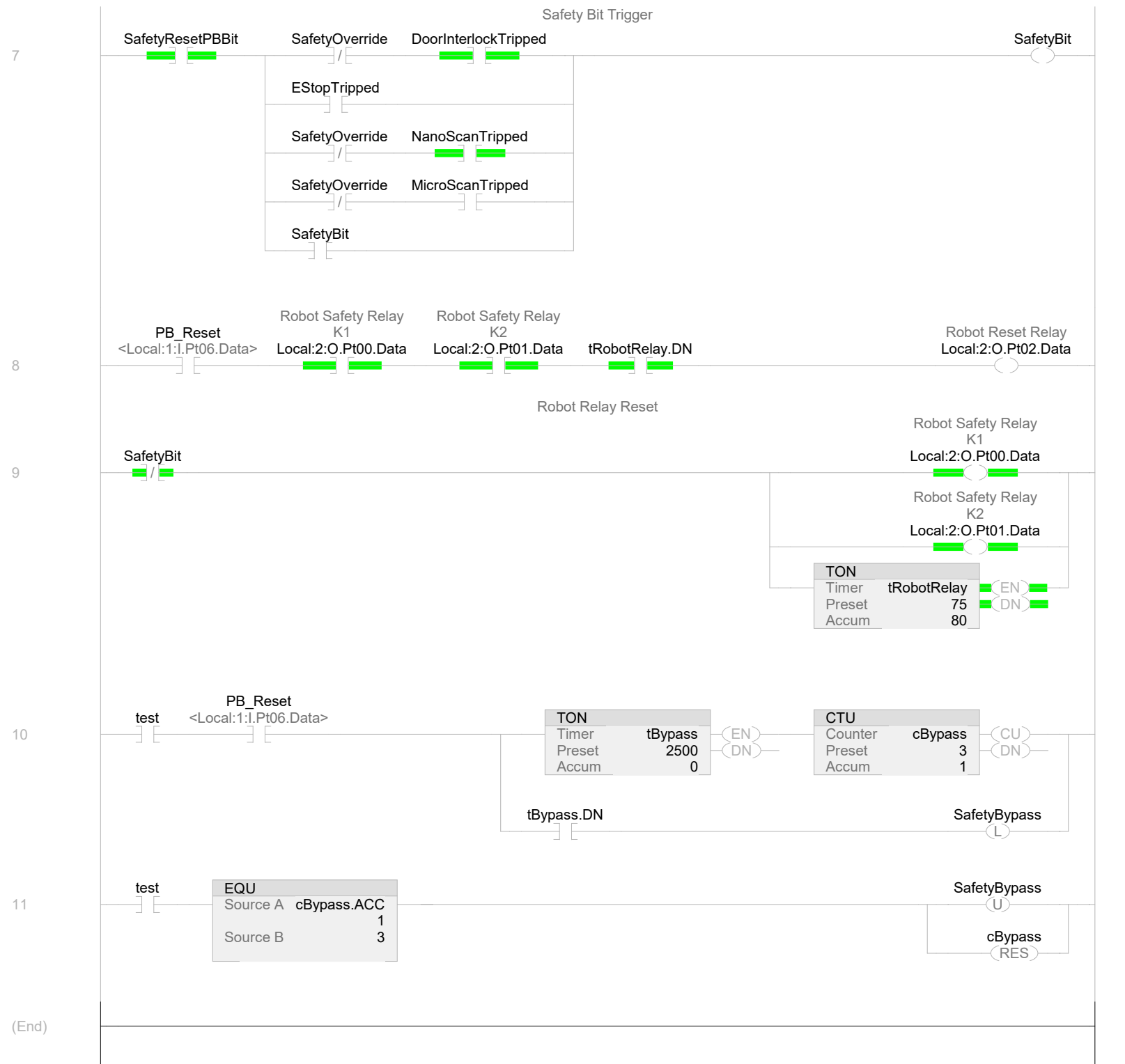
Monitor

Scan Times(Execution Time)
Max: 26 us Last: 11 us

Name	Value	Data Type	Scope
- cBypass		COUNTER	SafetyProgram
Constant	No		
External Access:	Read/Write		
cBypass - SafetyProgram /MainRoutine - *10(CTU), *11(RES)			
cBypass.ACC	1	DINT	
cBypass.ACC - SafetyProgram /MainRoutine - 11(EQU)			
- DoorInterlock1	0	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt02.Data(C)		
Base Tag:	Local:1:I.Pt02.Data(C)		
Constant	No		
External Access:	Read/Write		
DoorInterlock1 - SafetyProgram /MainRoutine - 2(XIO)			
- DoorInterlock2	0	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt03.Data(C)		
Base Tag:	Local:1:I.Pt03.Data(C)		
Constant	No		
External Access:	Read/Write		
DoorInterlock2 - SafetyProgram /MainRoutine - 2(XIO)			
- DoorInterlockTripped	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
DoorInterlockTripped - SafetyProgram /MainRoutine - *2(OTE), 6(XIC), 7(XIC)			
\SafetyProgram.DoorInterlockTripped - MainProgram/Alarms - 4(XIC)			
- EStop1	1	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt00.Data(C)		
Base Tag:	Local:1:I.Pt00.Data(C)		
Constant	No		
External Access:	Read/Write		
EStop1 - SafetyProgram /MainRoutine - 1(XIO)			
- EStop2	1	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt01.Data(C)		
Base Tag:	Local:1:I.Pt01.Data(C)		
Constant	No		
External Access:	Read/Write		
EStop2 - SafetyProgram /MainRoutine - 1(XIO)			
- EStopTripped	0	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
EStopTripped - SafetyProgram /MainRoutine - *1(OTE), 6(XIC), 7(XIC)			
\SafetyProgram.EStopTripped - MainProgram/Alarms - 1(XIC)			
- MicroScanTripped	0	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
MicroScanTripped - SafetyProgram /MainRoutine - *4(OTE), 6(XIC), 7(XIC)			
\SafetyProgram.MicroScanTripped - MainProgram/Alarms - 3(XIC)			
- nanoScanCh1	0	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt04.Data(C)		
Base Tag:	Local:1:I.Pt04.Data(C)		
Constant	No		
External Access:	Read/Write		
nanoScanCh1 - SafetyProgram /MainRoutine - 3(XIO)			
- nanoScanCh2	0	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt05.Data(C)		
Base Tag:	Local:1:I.Pt05.Data(C)		

nanoScanCh2 (Continued)			
Constant	No		
External Access:	Read/Write		
nanoScanCh2 - SafetyProgram ▣/MainRoutine - 3(XIO)			
- NanoScanTripped	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
NanoScanTripped - SafetyProgram ▣/MainRoutine - *3(OTE), 6(XIC), 7(XIC)			
\SafetyProgram.NanoScanTripped - MainProgram/Alarms - 2(XIC)			
- NanoScanWarning	1	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt07.Data(C)		
Base Tag:	Local:1:I.Pt07.Data(C)		
Constant	No		
External Access:	Read/Write		
NanoScanWarning - SafetyProgram ▣/MainRoutine - 5(XIC)			
- SafetyBypass	0	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
SafetyBypass - SafetyProgram ▣/MainRoutine - *10(OTL), *11(OTU), 3(XIO), 4(XIO)			
- SafetyOverride	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
SafetyOverride - SafetyProgram ▣/MainRoutine - 6(XIO), 6(XIO), 6(XIO), 7(XIO), 7(XIO), 7(XIO)			
- SafetyResetPBBit	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
SafetyResetPBBit - SafetyProgram ▣/MainRoutine - *6(OTE), 7(XIC)			
- tBypass		TIMER	SafetyProgram
Constant	No		
External Access:	Read/Write		
tBypass - SafetyProgram ▣/MainRoutine - *10(TON)			
tBypass.DN	0	BOOL	
tBypass.DN - SafetyProgram ▣/MainRoutine - 10(XIC)			
- test	0	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
test - SafetyProgram ▣/MainRoutine - 10(XIC), 11(XIC)			
- tRobotRelay		TIMER	SafetyProgram
Constant	No		
External Access:	Read/Write		
tRobotRelay - SafetyProgram ▣/MainRoutine - *9(TON)			
tRobotRelay.DN	1	BOOL	
tRobotRelay.DN - SafetyProgram ▣/MainRoutine - 8(XIC)			





Name	Value	Data Type	Scope
- cBypass Constant External Access: <i>cBypass - SafetyProgram</i> /MainRoutine - *10(CTU), *11(RES)	No Read/Write	COUNTER	SafetyProgram
cBypass.ACC <i>cBypass.ACC - SafetyProgram</i> /MainRoutine - 11(EQU)	1	DINT	
- DoorInterlock1 AliasFor: Base Tag: Constant External Access: <i>DoorInterlock1 - SafetyProgram</i> /MainRoutine - 2(XIO)	0 Local:1:I.Pt02.Data(C) Local:1:I.Pt02.Data(C) No Read/Write	BOOL	SafetyProgram
- DoorInterlock2 AliasFor: Base Tag: Constant External Access: <i>DoorInterlock2 - SafetyProgram</i> /MainRoutine - 2(XIO)	0 Local:1:I.Pt03.Data(C) Local:1:I.Pt03.Data(C) No Read/Write	BOOL	SafetyProgram
- DoorInterlockTripped Constant External Access: <i>DoorInterlockTripped - SafetyProgram</i> /MainRoutine - *2(OTE), 6(XIC), 7(XIC) \SafetyProgram.DoorInterlockTripped - MainProgram/Alarms - 4(XIC)	1 No Read/Write	BOOL	SafetyProgram
- EStop1 AliasFor: Base Tag: Constant External Access: <i>EStop1 - SafetyProgram</i> /MainRoutine - 1(XIO)	1 Local:1:I.Pt00.Data(C) Local:1:I.Pt00.Data(C) No Read/Write	BOOL	SafetyProgram
- EStop2 AliasFor: Base Tag: Constant External Access: <i>EStop2 - SafetyProgram</i> /MainRoutine - 1(XIO)	1 Local:1:I.Pt01.Data(C) Local:1:I.Pt01.Data(C) No Read/Write	BOOL	SafetyProgram
- EStopTripped Constant External Access: <i>EStopTripped - SafetyProgram</i> /MainRoutine - *1(OTE), 6(XIC), 7(XIC) \SafetyProgram.EStopTripped - MainProgram/Alarms - 1(XIC)	0 No Read/Write	BOOL	SafetyProgram
 Local:2:O Constant External Access:	No Read/Write	AB:5000_DO8:O:0	Capstone_Test
Local:2:O.Pt00 Robot Safety Relay K1		CHANNEL_DO:O:0	
Local:2:O.Pt00.Data Robot Safety Relay K1 <i>Local:2:O.Pt00.Data - SafetyProgram</i> /MainRoutine - *9(OTE), 8(XIC)	1	BOOL	
Local:2:O.Pt01 Robot Safety Relay K2		CHANNEL_DO:O:0	
Local:2:O.Pt01.Data Robot Safety Relay K2 <i>Local:2:O.Pt01.Data - SafetyProgram</i> /MainRoutine - *9(OTE), 8(XIC)	1	BOOL	
Local:2:O.Pt02.Data Robot Reset Relay <i>Local:2:O.Pt02.Data - SafetyProgram</i> /MainRoutine - *8(OTE)	0	BOOL	

microScan:SI		_0328:MICS3_CBAZXXIZ1_F211ADCF:SI:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
microScan:SI.Data[0]	1	SINT	
Mode			
microScan:SI.Data[1]	1	SINT	
Safe CutOff Path(1-8)			
microScan:SI.Data[2]	1	SINT	
NonSafe CutOff Path(1-8)			
microScan:SI.Data[2] - SafetyProgram /MainRoutine - 4(LES), 5(EQU)			
microScan:SI.Data[3]	0	SINT	
Reserved			
microScan:SI.Data[4]	1	SINT	
Current Monitoring Case No.			
microScan:SI.Data[5]	0	SINT	
Reset Required CutOff Path(1-8)			
microScan:SI.Data[6]	0	SINT	
Reserved			
microScan:SI.Data[7]	0	SINT	
Errors			
microScan:SO		_0328:MICS3_CBAZXXIZ1_DD7B81FB:SO:0	Capstone_Test
Constant	No		
External Access:	Read/Write		
microScan:SO.Data[0]	3	SINT	
Mode			
microScan:SO.Data[0] - SafetyProgram /MainRoutine - *0(MOV)			
microScan:SO.Data[0].0	1	BOOL	
Trigger Run-Mode			
microScan:SO.Data[1]	0	SINT	
Reserved			
microScan:SO.Data[2]	0	SINT	
Reserved			
microScan:SO.Data[3]	1	SINT	
Control Inputs			
microScan:SO.Data[3] - SafetyProgram /MainRoutine - *0(MOV)			
microScan:SO.Data[3].0	1	BOOL	
Control input 1 (A1)			
microScan:SO.Data[4]	0	SINT	
Reserved			
microScan:SO.Data[5]	0	SINT	
Trigger Reset CutOff Path (1-8)			
microScan:SO.Data[6]	0	SINT	
Reserved			
microScan:SO.Data[7]	0	SINT	
Trigger Device Reboot			
MicroScanTripped	0	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
MicroScanTripped - SafetyProgram /MainRoutine - *4(OTE), 6(XIC), 7(XIC)			
\SafetyProgram.MicroScanTripped - MainProgram/Alarms - 3(XIC)			
nanoScanCh1	0	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt04.Data(C)		
Base Tag:	Local:1:I.Pt04.Data(C)		
Constant	No		
External Access:	Read/Write		
nanoScanCh1 - SafetyProgram /MainRoutine - 3(XIO)			
nanoScanCh2	0	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt05.Data(C)		

nanoScanCh2 (Continued)			
Base Tag:	Local:1:I.Pt05.Data(C)		
Constant	No		
External Access:	Read/Write		
nanoScanCh2 - SafetyProgram ▣/MainRoutine - 3(XIO)			
- NanoScanTripped	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
NanoScanTripped - SafetyProgram ▣/MainRoutine - *3(OTE), 6(XIC), 7(XIC)			
\SafetyProgram.NanoScanTripped - MainProgram/Alarms - 2(XIC)			
- NanoScanWarning	1	BOOL	SafetyProgram
AliasFor:	Local:1:I.Pt07.Data(C)		
Base Tag:	Local:1:I.Pt07.Data(C)		
Constant	No		
External Access:	Read/Write		
NanoScanWarning - SafetyProgram ▣/MainRoutine - 5(XIC)			
▣ PB_Reset	0	BOOL	Capstone_Test
AliasFor:	Local:1:I.Pt06.Data		
Base Tag:	Local:1:I.Pt06.Data		
Constant	No		
External Access:	Read/Write		
PB_Reset - MainProgram/Alarms - 0(ALMD), 1(ALMD), 2(ALMD), 3(ALMD), 4(ALMD)			
PB_Reset - MainProgram/RobotAutoControl - 0(XIC)			
PB_Reset - SafetyProgram ▣/MainRoutine - 10(XIC), 6(XIO), 8(XIC)			
▣ SafetyBit	0	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
SafetyBit - MainProgram/Alarms - 0(XIC), 1(XIC), 2(XIC), 3(XIC), 4(XIC)			
SafetyBit - MainProgram/MainRoutine - 10(XIC), 4(XIO), 5(XIO), 6(XIO), 7(XIC)			
SafetyBit - MainProgram/RobotAutoControl - 1(XIC), 4(XIO), 5(XIO), 6(XIO)			
SafetyBit - SafetyProgram ▣/MainRoutine - *7(OTE), 7(XIC), 9(XIO)			
- SafetyBypass	0	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
SafetyBypass - SafetyProgram ▣/MainRoutine - *10(OTL), *11(OTU), 3(XIO), 4(XIO)			
- SafetyOverride	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
SafetyOverride - SafetyProgram ▣/MainRoutine - 6(XIO), 6(XIO), 6(XIO), 7(XIO), 7(XIO), 7(XIO)			
- SafetyResetPBBit	1	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
SafetyResetPBBit - SafetyProgram ▣/MainRoutine - *6(OTE), 7(XIC)			
- tBypass		TIMER	SafetyProgram
Constant	No		
External Access:	Read/Write		
tBypass - SafetyProgram ▣/MainRoutine - *10(TON)			
tBypass.DN	0	BOOL	
tBypass.DN - SafetyProgram ▣/MainRoutine - 10(XIC)			
- test	0	BOOL	SafetyProgram
Constant	No		
External Access:	Read/Write		
test - SafetyProgram ▣/MainRoutine - 10(XIC), 11(XIC)			
- tRobotRelay		TIMER	SafetyProgram

tRobotRelay (Continued)

Constant	No	
External Access:	Read/Write	
tRobotRelay - SafetyProgram ▣/MainRoutine - *9(TON)		
tRobotRelay.DN	1	BOOL
tRobotRelay.DN - SafetyProgram ▣/MainRoutine - 8(XIC)		

WarningBit	1	BOOL	Capstone_Test
Constant	No		
External Access:	Read/Write		
WarningBit - MainProgram/MainRoutine - 6(XIC)			
WarningBit - SafetyProgram ▣/MainRoutine - *5(OTE)			

General

Type:	 Ladder Diagram	Class:	Safety
In Program:	 SafetyProgram	Number of Rungs:	12

Signature Listing

 CurrentToSpoolIndex v1.0

Intakes a current and ouputs a spool index

Class: Standard

Available Languages

 Relay Ladder

CurrentToSpoolIndex

Intakes a current and ouputs a spool ind...

CurrentToSpoolIndex

CoilGauge

?

?

...

 Function Block

CurrentToSpoolIndex

Intakes a current and ouputs a spool inde

CoilGauge

...

?

 Structured Text

CurrentToSpoolIndex(CoilGauge);

Parameters

Required	Name	Data Type	Usage	Description
X	CurrentToSpoolIndex	CurrentToSpoolIndex	InOut	Intakes a current and ouputs a spool index
	EnableIn	BOOL	Input	
	EnableOut	BOOL	Output	
	Current	REAL	Input	
	SpoolIndex	DINT	Output	
	Spool1Gauge	DINT	Input	
	Spool2Gauge	DINT	Input	
	Spool3Gauge	DINT	Input	
	DN	BOOL	Output	
X	CoilGauge	DINT	InOut	

Extended Description

Execution

Condition	Description
EnableIn is true	

Revision v1.0 Notes

Name	Default	Data Type	Scope
Current	0.0	REAL	CurrentToSpoolIndex
Usage:	Input Parameter		
Required:	No		
Visible:	No		
External Access:	Read/Write		
Current - CurrentToSpoolIndex/Logic - #32			
DN	0	BOOL	CurrentToSpoolIndex
Usage:	Output Parameter		
Required:	No		
Visible:	No		
External Access:	None		
DN - CurrentToSpoolIndex/Logic - *#2, *#39			
Spool1Gauge	0	DINT	CurrentToSpoolIndex
Usage:	Input Parameter		
Required:	No		
Visible:	No		
External Access:	Read/Write		
Spool1Gauge - CurrentToSpoolIndex/Logic - #4			
Spool2Gauge	0	DINT	CurrentToSpoolIndex
Usage:	Input Parameter		
Required:	No		
Visible:	No		
External Access:	None		
Spool2Gauge - CurrentToSpoolIndex/Logic - #5			
Spool3Gauge	0	DINT	CurrentToSpoolIndex
Usage:	Input Parameter		
Required:	No		
Visible:	No		
External Access:	None		
Spool3Gauge - CurrentToSpoolIndex/Logic - #6			
SpoolIndex	0	DINT	CurrentToSpoolIndex
Usage:	Output Parameter		
Required:	No		
Visible:	No		
External Access:	Read/Write		
SpoolIndex - CurrentToSpoolIndex/Logic - #32, *#33, *#8			

Name	Default	Data Type	Scope
i	0	DINT	CurrentToSpoolIndex
Usage:	Local Tag		
External Access:	None		
<i>i</i> - <i>CurrentToSpoolIndex/Logic</i> - #19, #19, #20, #21, #21, #22, #23, #23, #24, #25, #25, #26, #27, #28, #32, #32, #33, *#11			
SpoolCurrent		REAL[3]	CurrentToSpoolIndex
Usage:	Local Tag		
External Access:	None		
<i>SpoolCurrent</i> - <i>CurrentToSpoolIndex/Logic</i> - #32, #32, #32, *#20, *#22, *#24, *#26, *#28			
SpoolGauge		DINT[3]	CurrentToSpoolIndex
Usage:	Local Tag		
External Access:	None		
<i>SpoolGauge</i> - <i>CurrentToSpoolIndex/Logic</i> - #19, #19, #21, #21, #23, #23, #25, #25, #27			
SpoolGauge[0]	0	DINT	
<i>SpoolGauge[0]</i> - <i>CurrentToSpoolIndex/Logic</i> - *#4			
SpoolGauge[1]	0	DINT	
<i>SpoolGauge[1]</i> - <i>CurrentToSpoolIndex/Logic</i> - *#5			
SpoolGauge[2]	0	DINT	
<i>SpoolGauge[2]</i> - <i>CurrentToSpoolIndex/Logic</i> - *#6			

```
1
2 DN := 0;
3
4 SpoolGauge[0] := Spool1Gauge; // stupid that you cant make an input array
5 SpoolGauge[1] := Spool2Gauge;
6 SpoolGauge[2] := Spool3Gauge;
7
8 SpoolIndex := 0;
9
10 // SpoolCurrent values taken from CEC. Table 2, halved because of table 5C
11 for i := 0 to 2 by 1 do // iderate through all 3 available spools
12
13     /* If someone would like to tell me how to continue from the next ideration like in pretty much all oop I ↗
    would apreciate +1 226-268-9249
14     if SpoolGauge[i] = -1 then
15         Continue;
16     end_if;
17     */
18
19     if 13 <= SpoolGauge[i] AND 250 > SpoolGauge[i] then // set SpoolCurrent values
20         SpoolCurrent[i] := 15 * 0.5;
21     elsif 11 <= SpoolGauge[i] AND 13 > SpoolGauge[i] then
22         SpoolCurrent[i] := 20 * 0.5;
23     elsif 9 <= SpoolGauge[i] AND 11 > SpoolGauge[i] then
24         SpoolCurrent[i] := 30 * 0.5;
25     elsif 7 <= SpoolGauge[i] AND 9 > SpoolGauge[i] then
26         SpoolCurrent[i] := 40 * 0.5;
27     elsif 7 > SpoolGauge[i] then
28         SpoolCurrent[i] := 55 * 0.5;
29     end_if;
30
31     // if spool has viable current value and is a better fit, choose it
32     if SpoolCurrent[i] > Current AND SpoolCurrent[SpoolIndex] < SpoolCurrent[i] then
33         SpoolIndex := i;
34     end_if;
35
36 end_for;
37
38
39 DN := 1;
40
```

 CurrentToSpoolIndex v1.0

Intakes a current and ouputs a spool index

Class: Standard

Available Languages

 Relay Ladder

CurrentToSpoolIndex

Intakes a current and ouputs a spool ind...

CurrentToSpoolIndex ? ...

CoilGauge ?

 Function Block

CurrentToSpoolIndex ...

Intakes a current and ouputs a spool inde

CoilGauge ?

 Structured Text

CurrentToSpoolIndex(CoilGauge);

Parameters

Required	Name	Data Type	Usage	Description
X	CurrentToSpoolIndex	CurrentToSpoolIndex	InOut	Intakes a current and ouputs a spool index
	EnableIn	BOOL	Input	
	EnableOut	BOOL	Output	
	Current	REAL	Input	
	SpoolIndex	DINT	Output	
	Spool1Gauge	DINT	Input	
	Spool2Gauge	DINT	Input	
	Spool3Gauge	DINT	Input	
	DN	BOOL	Output	
X	CoilGauge	DINT	InOut	

Extended Description

Execution

Condition	Description
EnableIn is true	

Revision v1.0 Notes

Name	Default	Data Type	Scope
Current	0.0	REAL	CurrentToSpoolIndex
Usage:	Input Parameter		
Required:	No		
Visible:	No		
External Access:	Read/Write		
<i>Current - CurrentToSpoolIndex/Logic - #32</i>			
DN	0	BOOL	CurrentToSpoolIndex
Usage:	Output Parameter		
Required:	No		
Visible:	No		
External Access:	None		
<i>DN - CurrentToSpoolIndex/Logic - *#2, *#39</i>			
Spool1Gauge	0	DINT	CurrentToSpoolIndex
Usage:	Input Parameter		
Required:	No		
Visible:	No		
External Access:	Read/Write		
<i>Spool1Gauge - CurrentToSpoolIndex/Logic - #4</i>			
Spool2Gauge	0	DINT	CurrentToSpoolIndex
Usage:	Input Parameter		
Required:	No		
Visible:	No		
External Access:	None		
<i>Spool2Gauge - CurrentToSpoolIndex/Logic - #5</i>			
Spool3Gauge	0	DINT	CurrentToSpoolIndex
Usage:	Input Parameter		
Required:	No		
Visible:	No		
External Access:	None		
<i>Spool3Gauge - CurrentToSpoolIndex/Logic - #6</i>			
SpoolIndex	0	DINT	CurrentToSpoolIndex
Usage:	Output Parameter		
Required:	No		
Visible:	No		
External Access:	Read/Write		
<i>SpoolIndex - CurrentToSpoolIndex/Logic - #32, *#33, *#8</i>			

Name	Default	Data Type	Scope
i	0	DINT	CurrentToSpoolIndex
Usage:	Local Tag		
External Access:	None		
<i>i</i> - CurrentToSpoolIndex/Logic - #19, #19, #20, #21, #21, #22, #23, #23, #24, #25, #25, #26, #27, #28, #32, #32, #33, *#11			
SpoolCurrent		REAL[3]	CurrentToSpoolIndex
Usage:	Local Tag		
External Access:	None		
<i>SpoolCurrent</i> - CurrentToSpoolIndex/Logic - #32, #32, #32, *#20, *#22, *#24, *#26, *#28			
SpoolGauge		DINT[3]	CurrentToSpoolIndex
Usage:	Local Tag		
External Access:	None		
<i>SpoolGauge</i> - CurrentToSpoolIndex/Logic - #19, #19, #21, #21, #23, #23, #25, #25, #27			
SpoolGauge[0]	0	DINT	
<i>SpoolGauge[0]</i> - CurrentToSpoolIndex/Logic - *#4			
SpoolGauge[1]	0	DINT	
<i>SpoolGauge[1]</i> - CurrentToSpoolIndex/Logic - *#5			
SpoolGauge[2]	0	DINT	
<i>SpoolGauge[2]</i> - CurrentToSpoolIndex/Logic - *#6			

```
1
2 DN := 0;
3
4 SpoolGauge[0] := Spool1Gauge; // stupid that you cant make an input array
5 SpoolGauge[1] := Spool2Gauge;
6 SpoolGauge[2] := Spool3Gauge;
7
8 SpoolIndex := 0;
9
10 // SpoolCurrent values taken from CEC. Table 2, halved because of table 5C
11 for i := 0 to 2 by 1 do // iderate through all 3 available spools
12
13     /* If someone would like to tell me how to continue from the next ideration like in pretty much all oop I ↗
    would apreciate +1 226-268-9249
14     if SpoolGauge[i] = -1 then
15         Continue;
16     end_if;
17     */
18
19     if 13 <= SpoolGauge[i] AND 250 > SpoolGauge[i] then // set SpoolCurrent values
20         SpoolCurrent[i] := 15 * 0.5;
21     elsif 11 <= SpoolGauge[i] AND 13 > SpoolGauge[i] then
22         SpoolCurrent[i] := 20 * 0.5;
23     elsif 9 <= SpoolGauge[i] AND 11 > SpoolGauge[i] then
24         SpoolCurrent[i] := 30 * 0.5;
25     elsif 7 <= SpoolGauge[i] AND 9 > SpoolGauge[i] then
26         SpoolCurrent[i] := 40 * 0.5;
27     elsif 7 > SpoolGauge[i] then
28         SpoolCurrent[i] := 55 * 0.5;
29     end_if;
30
31     // if spool has viable current value and is a better fit, choose it
32     if SpoolCurrent[i] > Current AND SpoolCurrent[SpoolIndex] < SpoolCurrent[i] then
33         SpoolIndex := i;
34     end_if;
35
36 end_for;
37
38
39 DN := 1;
40
```

OM_AZ_DDO v1.3
ORIENTALMOTOR CO.,LTD.

Class: Standard

Available Languages

Relay Ladder

OM_AZ_DDO		
OM_AZ_DDO	?	(Sts_Busy)
ImplicitMSGinput	?	(Sts_Err)
ImplicitMSGoutput	?	(Sts_Done)
Set_Exe	?	
Set_OpeType	?	
Set_Position	?	
Set_Velocity	?	
Set_Acceleration	?	
Set_Deceleration	?	
Set_Current	?	
Set_Send	?	
Read_Present_alarm	?	
Read_Command_position	?	
Read_Feedback_position	?	
Read_Torque_monitor	?	
Read_Cumulative_load_monitor	?	

Function Block

OM_AZ_DDO		
ImplicitMSGinput	?	
ImplicitMSGoutput	?	
Set_Exe	Sts_Busy	
Set_OpeType	Sts_Err	
Set_Position	Sts_Done	
Set_Velocity	Read_Present_alarm	
Set_Acceleration	Read_Command_position	
Set_Deceleration	Read_Feedback_position	
Set_Current	Read_Torque_monitor	
Set_Send	Read_Cumulative_load_monitor	

Structured Text

OM_AZ_DDO(ImplicitMSGinput, ImplicitMSGoutput);

Parameters

Required	Name	Data Type	Usage	Description
X	OM_AZ_DDO	OM_AZ_DDO	InOut	
	EnableIn	BOOL	Input	
	EnableOut	BOOL	Output	
X	ImplicitMSGinput	INT	InOut	Implicit message Input (driver->scanner)
X	ImplicitMSGoutput	INT	InOut	Implicit message Output (scanner->driver)
	Set_Exe	BOOL	Input	Execute AOI
	Set_OpeType	INT	Input	Direct data operation operation type

Set_Position	DINT	Input	Direct data operation position
Set_Velocity	DINT	Input	Direct data operation operating speed
Set_Acceleration	DINT	Input	Direct data operation starting/changing rate
Set_Deceleration	DINT	Input	Direct data operation stopping deceleration
Set_Current	INT	Input	Direct data operation operating current
Set_Send	INT	Input	Direct data operation forwarding destination
Sts_Busy	BOOL	Output	Executing status in AOI
Sts_Err	BOOL	Output	Error status in AOI
Sts_Done	BOOL	Output	Complete status in AOI
Read_Present_alarm	INT	Output	Alarm code monitor[Hex]
Read_Command_position	DINT	Output	Command position monitor[step]
Read_Feedback_position	DINT	Output	Feedback position monitor[step]
Read_Torque_monitor	INT	Output	Torque monitor as a percentage of the maximum holding torque(1=0.1%)
Read_Cumulative_load_monitor	DINT	Output	Cumulative load monitor

Extended Description

Direct data operation(DDO) is a mode that allows rewriting of data and start of operation to be executed at the same time. It is suitable to be used in frequently changed operation data such as the position (travel amount) and operating speed or to fine-tune the position. There are seven types of triggers to start operation at the same time as rewriting of data.

-By changing one of the following items: Operation type, position, operating speed, starting/changing speed rate, stopping deceleration, and operating current. The six items are collectively rewritten.

The direct data operation is executed by the TRIG of fixed I/O (IN).

The condition to execute direct data operation can be selected from the following two types of parameters ON edge, or ON level by using the TRIG-MODE of fixed I/O (IN).

- When executing "Set_Exe" the TRIG-MODE is set at ON level : The motor will start at the same time when the data of the trigger set in the "Direct data operation trigger setting" parameter is changed.

Execution

Condition Description

EnableIn is true

Revision v1.3 Notes

Implicit message based on AZD-AEP,-CEP and -KEP.

Name	Default	Data Type	Scope
ImplicitMSGInput		INT[28]	OM_AZ_DDO
Implicit message Input (driver->scanner)			
Usage:	InOut Parameter		
Required:	Yes		
Visible:	Yes		
Constant	No		
<i>ImplicitMSGInput - OM_AZ_DDO/Logic - 0(CPS), 0(CPS), 0(CPS)</i>			
ImplicitMSGInput[0]	??	INT	
Remote I/O (R-OUT)			
ImplicitMSGInput[0].0	??	BOOL	
R-OUT0			
ImplicitMSGInput[0].1	??	BOOL	
R-OUT1			
ImplicitMSGInput[0].2	??	BOOL	
R-OUT2			
ImplicitMSGInput[0].3	??	BOOL	
R-OUT3			
ImplicitMSGInput[0].4	??	BOOL	
R-OUT4			
ImplicitMSGInput[0].5	??	BOOL	
R-OUT5			
ImplicitMSGInput[0].6	??	BOOL	
R-OUT6			
ImplicitMSGInput[0].7	??	BOOL	
R-OUT7			
ImplicitMSGInput[0].8	??	BOOL	
R-OUT8			
ImplicitMSGInput[0].9	??	BOOL	
R-OUT9			
ImplicitMSGInput[0].10	??	BOOL	
R-OUT10			
ImplicitMSGInput[0].11	??	BOOL	
R-OUT11			
ImplicitMSGInput[0].12	??	BOOL	
R-OUT12			
ImplicitMSGInput[0].13	??	BOOL	
R-OUT13			
ImplicitMSGInput[0].14	??	BOOL	
R-OUT14			
ImplicitMSGInput[0].15	??	BOOL	
R-OUT15			
ImplicitMSGInput[1]	??	INT	
Operation data number selection_R			
ImplicitMSGInput[1].0	??	BOOL	
M0_R			
ImplicitMSGInput[1].1	??	BOOL	
M1_R			
ImplicitMSGInput[1].2	??	BOOL	
M2_R			
ImplicitMSGInput[1].3	??	BOOL	
M3_R			
ImplicitMSGInput[1].4	??	BOOL	
M4_R			
ImplicitMSGInput[1].5	??	BOOL	
M5_R			
ImplicitMSGInput[1].6	??	BOOL	
M6_R			
ImplicitMSGInput[1].7	??	BOOL	
M7_R			
ImplicitMSGInput[1].8	??	BOOL	
Reserved			
ImplicitMSGInput[1].9	??	BOOL	

ImplicitMSGinput (Continued)		
Reserved		
ImplicitMSGinput[1].10	??	BOOL
Reserved		
ImplicitMSGinput[1].11	??	BOOL
Reserved		
ImplicitMSGinput[1].12	??	BOOL
Reserved		
ImplicitMSGinput[1].13	??	BOOL
Reserved		
ImplicitMSGinput[1].14	??	BOOL
Reserved		
ImplicitMSGinput[1].15	??	BOOL
Reserved		
ImplicitMSGinput[2]	??	INT
Fixed I/O (OUT)		
<i>ImplicitMSGinput[2] - OM_AZ_DDO/Logic - 4(CPS), 5(CPS)</i>		
ImplicitMSGinput[2].0	??	BOOL
SEQ-BSY		
ImplicitMSGinput[2].1	??	BOOL
MOVE		
ImplicitMSGinput[2].2	??	BOOL
IN-POS		
ImplicitMSGinput[2].3	??	BOOL
START_R		
ImplicitMSGinput[2].4	??	BOOL
HOME-END		
ImplicitMSGinput[2].5	??	BOOL
READY		
ImplicitMSGinput[2].6	??	BOOL
DCMD-RDY		
<i>ImplicitMSGinput[2].6 - OM_AZ_DDO/Logic - 3(XIO)</i>		
ImplicitMSGinput[2].7	??	BOOL
ALM-A		
<i>ImplicitMSGinput[2].7 - OM_AZ_DDO/Logic - 3(XIC)</i>		
ImplicitMSGinput[2].8	??	BOOL
TRIG_R		
<i>ImplicitMSGinput[2].8 - OM_AZ_DDO/Logic - 2(XIC), 2(XIO)</i>		
ImplicitMSGinput[2].9	??	BOOL
TRIG-MODE_R		
ImplicitMSGinput[2].10	??	BOOL
SET-ERR		
ImplicitMSGinput[2].11	??	BOOL
EXE-ERR		
ImplicitMSGinput[2].12	??	BOOL
DCMD-FULL		
ImplicitMSGinput[2].13	??	BOOL
STOP_R		
<i>ImplicitMSGinput[2].13 - OM_AZ_DDO/Logic - 3(XIC)</i>		
ImplicitMSGinput[2].14	??	BOOL
ETO-MON		
<i>ImplicitMSGinput[2].14 - OM_AZ_DDO/Logic - 3(XIC)</i>		
ImplicitMSGinput[2].15	??	BOOL
TLC		
ImplicitMSGinput[3]	??	INT
Present alarm		
<i>ImplicitMSGinput[3] - OM_AZ_DDO/Logic - 0(CPS)</i>		
ImplicitMSGinput[4]	??	INT
Feedback position (lower)		
ImplicitMSGinput[5]	??	INT
Feedback position (upper)		
ImplicitMSGinput[6]	??	INT
Feedback speed [Hz] (lower)		

ImplicitMSGInput (Continued)		
ImplicitMSGInput[7]	??	INT
Feedback speed [Hz] (upper)		
ImplicitMSGInput[8]	??	INT
Command position (lower)		
ImplicitMSGInput[9]	??	INT
Command position (upper)		
ImplicitMSGInput[10]	??	INT
Torque monitor		
<i>ImplicitMSGInput[10] - OM_AZ_DDO/Logic - 0(CPS)</i>		
ImplicitMSGInput[11]	??	INT
CST operating current		
ImplicitMSGInput[12]	??	INT
Information (lower)		
ImplicitMSGInput[13]	??	INT
Information (upper)		
ImplicitMSGInput[14]	??	INT
Reserved		
ImplicitMSGInput[15]	??	INT
Read parameter ID_R		
ImplicitMSGInput[16]	??	INT
Read/write status		
ImplicitMSGInput[16].0	??	BOOL
Reserved		
ImplicitMSGInput[16].1	??	BOOL
Reserved		
ImplicitMSGInput[16].2	??	BOOL
Reserved		
ImplicitMSGInput[16].3	??	BOOL
Reserved		
ImplicitMSGInput[16].4	??	BOOL
Reserved		
ImplicitMSGInput[16].5	??	BOOL
Reserved		
ImplicitMSGInput[16].6	??	BOOL
Reserved		
ImplicitMSGInput[16].7	??	BOOL
RD-ERR		
ImplicitMSGInput[16].8	??	BOOL
WR-END		
ImplicitMSGInput[16].9	??	BOOL
SYS-BSY		
ImplicitMSGInput[16].10	??	BOOL
Reserved		
ImplicitMSGInput[16].11	??	BOOL
WR-SET-ERR		
ImplicitMSGInput[16].12	??	BOOL
WR-IF-ERR		
ImplicitMSGInput[16].13	??	BOOL
WR-NV-ERR		
ImplicitMSGInput[16].14	??	BOOL
WR-EXE-ERR		
ImplicitMSGInput[16].15	??	BOOL
WR-ERR		
ImplicitMSGInput[17]	??	INT
Write parameter ID_R		
ImplicitMSGInput[18]	??	INT
Read data (lower)		
ImplicitMSGInput[19]	??	INT
Read data (upper)		
ImplicitMSGInput[20]	??	INT
Assignable monitor 0 (lower)		
ImplicitMSGInput[21]	??	INT

ImplicitMSGinput (Continued)		
Assignable monitor 0 (upper)		
ImplicitMSGinput[22]	??	INT
Assignable monitor 1 (lower)		
ImplicitMSGinput[23]	??	INT
Assignable monitor 1 (upper)		
ImplicitMSGinput[24]	??	INT
Assignable monitor 2 (lower)		
ImplicitMSGinput[25]	??	INT
Assignable monitor 2 (upper)		
ImplicitMSGinput[26]	??	INT
Assignable monitor 3 (lower)		
ImplicitMSGinput[27]	??	INT
Assignable monitor 3 (upper)		
ImplicitMSGoutput	INT[20]	OM_AZ_DDO
Implicit message Output (scanner->driver)		
Usage:	InOut	Parameter
Required:	Yes	
Visible:	Yes	
Constant	No	
<i>ImplicitMSGoutput - OM_AZ_DDO/Logic - *3(CPS), *3(CPS), *3(CPS), *3(CPS)</i>		
ImplicitMSGoutput[0]	??	INT
Remote I/O(R-IN)		
ImplicitMSGoutput[0].0	??	BOOL
R-IN0		
ImplicitMSGoutput[0].1	??	BOOL
R-IN1		
ImplicitMSGoutput[0].2	??	BOOL
R-IN2		
ImplicitMSGoutput[0].3	??	BOOL
R-IN3		
ImplicitMSGoutput[0].4	??	BOOL
R-IN4		
ImplicitMSGoutput[0].5	??	BOOL
R-IN5		
ImplicitMSGoutput[0].6	??	BOOL
R-IN6		
ImplicitMSGoutput[0].7	??	BOOL
R-IN7		
ImplicitMSGoutput[0].8	??	BOOL
R-IN8		
ImplicitMSGoutput[0].9	??	BOOL
R-IN9		
ImplicitMSGoutput[0].10	??	BOOL
R-IN10		
ImplicitMSGoutput[0].11	??	BOOL
R-IN11		
ImplicitMSGoutput[0].12	??	BOOL
R-IN12		
ImplicitMSGoutput[0].13	??	BOOL
R-IN13		
ImplicitMSGoutput[0].14	??	BOOL
R-IN14		
ImplicitMSGoutput[0].15	??	BOOL
R-IN15		
ImplicitMSGoutput[1]	??	INT
Operation data number selection		
ImplicitMSGoutput[1].0	??	BOOL
M0		
ImplicitMSGoutput[1].1	??	BOOL
M1		
ImplicitMSGoutput[1].2	??	BOOL

ImplicitMSGGoutput (Continued)		
M2		
ImplicitMSGGoutput[1].3	??	BOOL
M3		
ImplicitMSGGoutput[1].4	??	BOOL
M4		
ImplicitMSGGoutput[1].5	??	BOOL
M5		
ImplicitMSGGoutput[1].6	??	BOOL
M6		
ImplicitMSGGoutput[1].7	??	BOOL
M7		
ImplicitMSGGoutput[1].8	??	BOOL
Reserved		
ImplicitMSGGoutput[1].9	??	BOOL
Reserved		
ImplicitMSGGoutput[1].10	??	BOOL
Reserved		
ImplicitMSGGoutput[1].11	??	BOOL
Reserved		
ImplicitMSGGoutput[1].12	??	BOOL
Reserved		
ImplicitMSGGoutput[1].13	??	BOOL
Reserved		
ImplicitMSGGoutput[1].14	??	BOOL
Reserved		
ImplicitMSGGoutput[1].15	??	BOOL
Reserved		
ImplicitMSGGoutput[2]	??	INT
Fixed I/O(IN)		
ImplicitMSGGoutput[2].0	??	BOOL
FW-JOG		
ImplicitMSGGoutput[2].1	??	BOOL
RV-JOG		
ImplicitMSGGoutput[2].2	??	BOOL
Reserved		
ImplicitMSGGoutput[2].3	??	BOOL
START		
ImplicitMSGGoutput[2].4	??	BOOL
ZHOME		
ImplicitMSGGoutput[2].5	??	BOOL
STOP		
ImplicitMSGGoutput[2].6	??	BOOL
FREE		
ImplicitMSGGoutput[2].7	??	BOOL
ALM-RST		
ImplicitMSGGoutput[2].8	??	BOOL
TRIG		
ImplicitMSGGoutput[2].8 - OM_AZ_DDO/Logic - *1(OTU), *2(OTU), *3(OTL)		
ImplicitMSGGoutput[2].9	??	BOOL
TRIG-MODE		
ImplicitMSGGoutput[2].9 - OM_AZ_DDO/Logic - *2(OTL)		
ImplicitMSGGoutput[2].10	??	BOOL
ETO-CLR		
ImplicitMSGGoutput[2].11	??	BOOL
Reserved		
ImplicitMSGGoutput[2].12	??	BOOL
FW-JOG-P		
ImplicitMSGGoutput[2].13	??	BOOL
RV-JOG-P		
ImplicitMSGGoutput[2].14	??	BOOL
FW-POS		
ImplicitMSGGoutput[2].15	??	BOOL

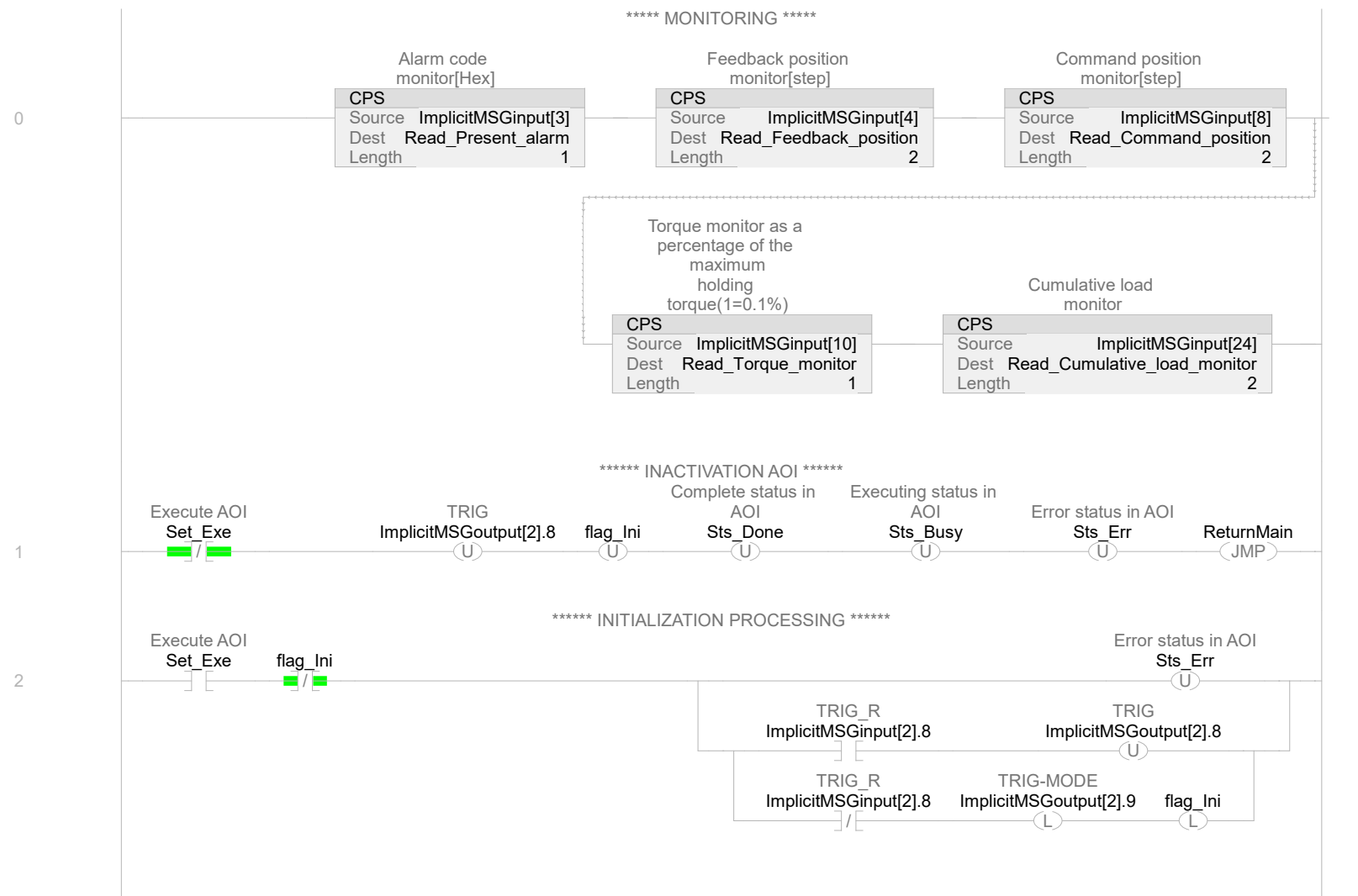
ImplicitMSGoutput (Continued)		
RV-POS		
ImplicitMSGoutput[3]	??	INT
Direct data operation operation type		
<i>ImplicitMSGoutput[3] - OM_AZ_DDO/Logic - *3(CPS)</i>		
ImplicitMSGoutput[4]	??	INT
Direct data operation position (lower)		
ImplicitMSGoutput[5]	??	INT
Direct data operation position (upper)		
ImplicitMSGoutput[6]	??	INT
Direct data operation operating speed (lower)		
ImplicitMSGoutput[7]	??	INT
Direct data operation operating speed (upper)		
ImplicitMSGoutput[8]	??	INT
Direct data operation starting/changing rate (lower)		
ImplicitMSGoutput[9]	??	INT
Direct data operation starting/changing rate (upper)		
ImplicitMSGoutput[10]	??	INT
Direct data operation stopping deceleration (lower)		
ImplicitMSGoutput[11]	??	INT
Direct data operation stopping deceleration (upper)		
ImplicitMSGoutput[12]	??	INT
Direct data operation operating current		
<i>ImplicitMSGoutput[12] - OM_AZ_DDO/Logic - *3(CPS)</i>		
ImplicitMSGoutput[13]	??	INT
Direct data operation forwarding destination		
<i>ImplicitMSGoutput[13] - OM_AZ_DDO/Logic - *3(CPS)</i>		
ImplicitMSGoutput[14]	??	INT
Reserved		
ImplicitMSGoutput[15]	??	INT
Read parameter ID		
ImplicitMSGoutput[16]	??	INT
Write request		
ImplicitMSGoutput[16].0	??	BOOL
WR-REQ		
ImplicitMSGoutput[16].1	??	BOOL
Reserved		
ImplicitMSGoutput[16].2	??	BOOL
Reserved		
ImplicitMSGoutput[16].3	??	BOOL
Reserved		
ImplicitMSGoutput[16].4	??	BOOL
Reserved		
ImplicitMSGoutput[16].5	??	BOOL
Reserved		
ImplicitMSGoutput[16].6	??	BOOL
Reserved		
ImplicitMSGoutput[16].7	??	BOOL
Reserved		
ImplicitMSGoutput[16].8	??	BOOL
Reserved		
ImplicitMSGoutput[16].9	??	BOOL
Reserved		
ImplicitMSGoutput[16].10	??	BOOL
Reserved		
ImplicitMSGoutput[16].11	??	BOOL
Reserved		
ImplicitMSGoutput[16].12	??	BOOL
Reserved		
ImplicitMSGoutput[16].13	??	BOOL
Reserved		
ImplicitMSGoutput[16].14	??	BOOL
Reserved		

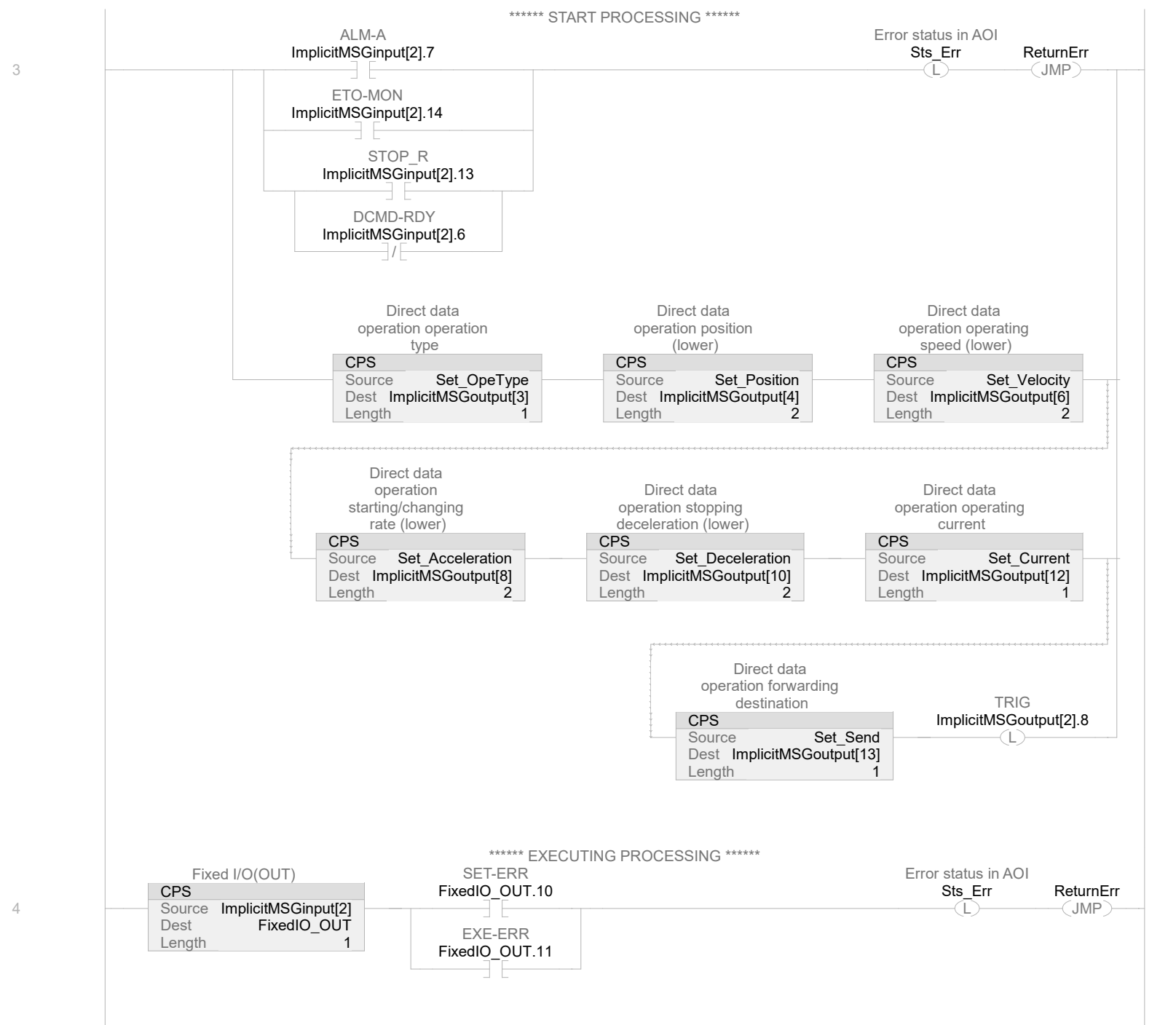
ImplicitMSGOutput (Continued)				
ImplicitMSGOutput[16].15		??	BOOL	
Reserved				
ImplicitMSGOutput[17]		??	INT	
Write parameter ID				
ImplicitMSGOutput[18]		??	INT	
Write data (lower)				
ImplicitMSGOutput[19]		??	INT	
Write data (upper)				
Read_Command_position		0	DINT	OM_AZ_DDO
Command position monitor[step]				
Usage:		Output Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read Only		
Read_Command_position - OM_AZ_DDO/Logic - *0(CPS)				
Read_Cumulative_load_monitor		0	DINT	OM_AZ_DDO
Cumulative load monitor				
Usage:		Output Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read Only		
Read_Cumulative_load_monitor - OM_AZ_DDO/Logic - *0(CPS)				
Read_Feedback_position		0	DINT	OM_AZ_DDO
Feedback position monitor[step]				
Usage:		Output Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read Only		
Read_Feedback_position - OM_AZ_DDO/Logic - *0(CPS)				
Read_Present_alarm		16#0000	INT	OM_AZ_DDO
Alarm code monitor[Hex]				
Usage:		Output Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read Only		
Read_Present_alarm - OM_AZ_DDO/Logic - *0(CPS)				
Read_Torque_monitor		0	INT	OM_AZ_DDO
Torque monitor as a percentage of the maximum holding torque(1=0.1%)				
Usage:		Output Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read Only		
Read_Torque_monitor - OM_AZ_DDO/Logic - *0(CPS)				
Set_Acceleration		1000000	DINT	OM_AZ_DDO
Direct data operation starting/changing rate				
Usage:		Input Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read/Write		
Set_Acceleration - OM_AZ_DDO/Logic - 3(CPS)				
Set_Current		1000	INT	OM_AZ_DDO
Direct data operation operating current				
Usage:		Input Parameter		
Required:		No		
Visible:		Yes		

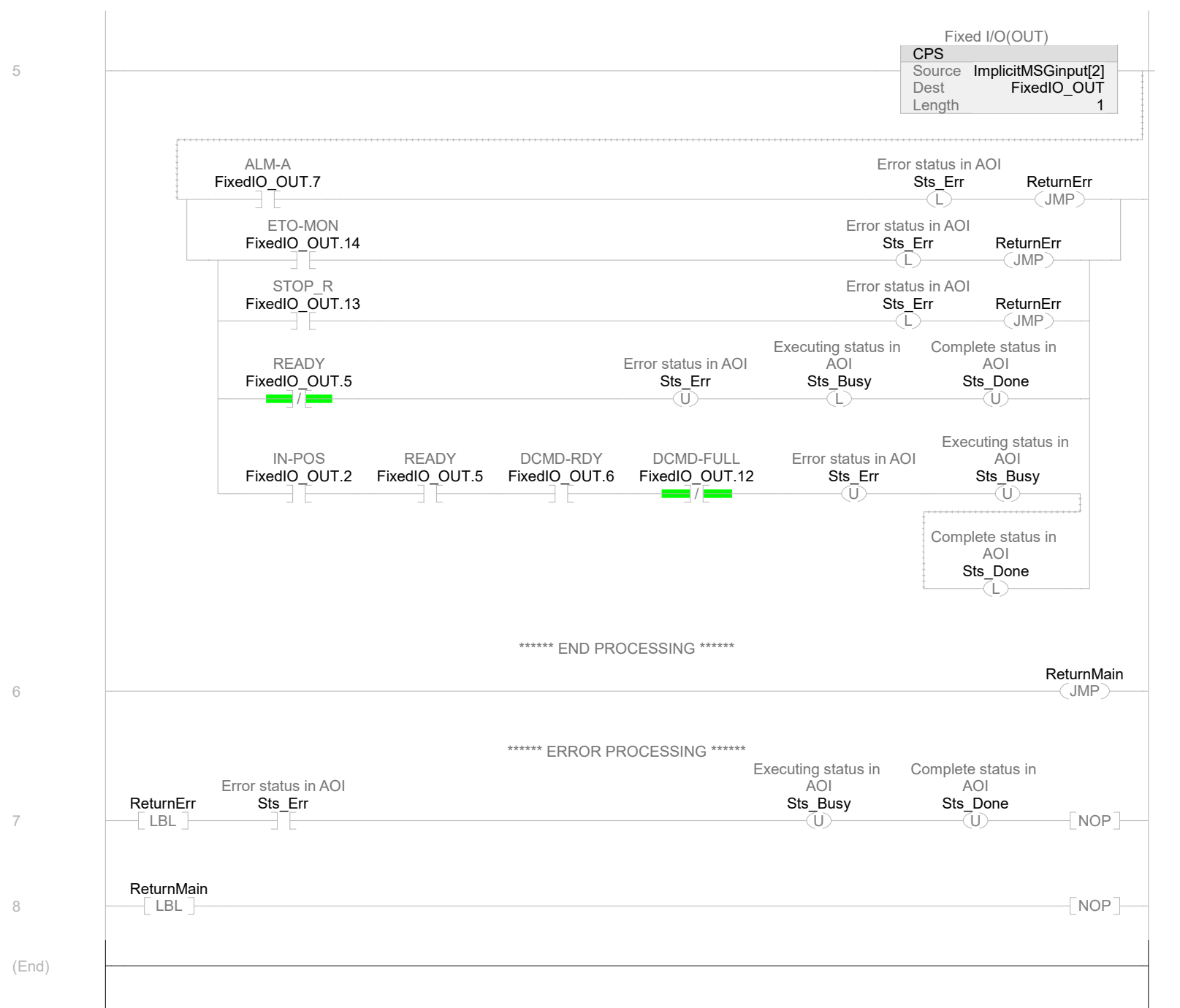
Set_Current (Continued)				
External Access:		Read/Write		
Set_Current - OM_AZ_DDO/Logic - 3(CPS)				
Set_Deceleration	1000000	DINT	OM_AZ_DDO	
Direct data operation stopping deceleration				
Usage:		Input Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read/Write		
Set_Deceleration - OM_AZ_DDO/Logic - 3(CPS)				
Set_Exec	0	BOOL	OM_AZ_DDO	
Execute AOI				
Usage:		Input Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read/Write		
Set_Exec - OM_AZ_DDO/Logic - 1(XIO), 2(XIC)				
Set_OpeType	2	INT	OM_AZ_DDO	
Direct data operation operation type				
Usage:		Input Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read/Write		
Set_OpeType - OM_AZ_DDO/Logic - 3(CPS)				
Set_Position	0	DINT	OM_AZ_DDO	
Direct data operation position				
Usage:		Input Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read/Write		
Set_Position - OM_AZ_DDO/Logic - 3(CPS)				
Set_Send	0	INT	OM_AZ_DDO	
Direct data operation forwarding destination				
Usage:		Input Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read/Write		
Set_Send - OM_AZ_DDO/Logic - 3(CPS)				
Set_Velocity	1000	DINT	OM_AZ_DDO	
Direct data operation operating speed				
Usage:		Input Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read/Write		
Set_Velocity - OM_AZ_DDO/Logic - 3(CPS)				
Sts_Busy	0	BOOL	OM_AZ_DDO	
Executing status in AOI				
Usage:		Output Parameter		
Required:		No		
Visible:		Yes		
External Access:		Read Only		
Sts_Busy - OM_AZ_DDO/Logic - *1(OTU), *5(OTL), *5(OTU), *7(OTU)				
Sts_Done	0	BOOL	OM_AZ_DDO	
Complete status in AOI				
Usage:		Output Parameter		

Sts_Done (Continued)			
Required:	No		
Visible:	Yes		
External Access:	Read Only		
Sts_Done - OM_AZ_DDO/Logic - *1(OTU), *5(OTL), *5(OTU), *7(OTU)			
Sts_Err	0	BOOL	OM_AZ_DDO
Error status in AOI			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read Only		
Sts_Err - OM_AZ_DDO/Logic - *1(OTU), *2(OTU), *3(OTL), *4(OTL), *5(OTL), *5(OTL), *5(OTU), *5(OTU), 7(XIC)			

Name	Default	Data Type	Scope
FixedIO_OUT	0	INT	OM_AZ_DDO
Fixed I/O(OUT)			
Usage:	Local Tag		
External Access:	Read/Write		
<i>FixedIO_OUT - OM_AZ_DDO/Logic - *4(CPS), *5(CPS)</i>			
FixedIO_OUT.0	0	BOOL	
SEQ-BSY			
FixedIO_OUT.1	0	BOOL	
MOVE			
FixedIO_OUT.2	0	BOOL	
IN-POS			
<i>FixedIO_OUT.2 - OM_AZ_DDO/Logic - 5(XIC)</i>			
FixedIO_OUT.3	0	BOOL	
START_R			
FixedIO_OUT.4	0	BOOL	
HOME-END			
FixedIO_OUT.5	0	BOOL	
READY			
<i>FixedIO_OUT.5 - OM_AZ_DDO/Logic - 5(XIC), 5(XIO)</i>			
FixedIO_OUT.6	0	BOOL	
DCMD-RDY			
<i>FixedIO_OUT.6 - OM_AZ_DDO/Logic - 5(XIC)</i>			
FixedIO_OUT.7	0	BOOL	
ALM-A			
<i>FixedIO_OUT.7 - OM_AZ_DDO/Logic - 5(XIC)</i>			
FixedIO_OUT.8	0	BOOL	
TRIG_R			
FixedIO_OUT.9	0	BOOL	
TRIG-MODE_R			
FixedIO_OUT.10	0	BOOL	
SET-ERR			
<i>FixedIO_OUT.10 - OM_AZ_DDO/Logic - 4(XIC)</i>			
FixedIO_OUT.11	0	BOOL	
EXE-ERR			
<i>FixedIO_OUT.11 - OM_AZ_DDO/Logic - 4(XIC)</i>			
FixedIO_OUT.12	0	BOOL	
DCMD-FULL			
<i>FixedIO_OUT.12 - OM_AZ_DDO/Logic - 5(XIO)</i>			
FixedIO_OUT.13	0	BOOL	
STOP_R			
<i>FixedIO_OUT.13 - OM_AZ_DDO/Logic - 5(XIC)</i>			
FixedIO_OUT.14	0	BOOL	
ETO-MON			
<i>FixedIO_OUT.14 - OM_AZ_DDO/Logic - 5(XIC)</i>			
FixedIO_OUT.15	0	BOOL	
TLC			
flag_Ini	0	BOOL	OM_AZ_DDO
Usage:	Local Tag		
External Access:	None		
<i>flag_Ini - OM_AZ_DDO/Logic - *1(OTU), *2(OTL), 2(XIO)</i>			







OM_AZ_MONITOR v1.3
ORIENTALMOTOR CO.,LTD.

Class: Standard

Available Languages

Relay Ladder

OM_AZ_MONITOR		
OM_AZ_MONITOR	?	...
ImplicitMSGinput	?	
Read_Present_alarm	?	
Read_Feedback_position	?	
Read_Feedback_speed	?	
Read_Command_position	?	
Read_Torque_monitor	?	
Read_CST_operating_current	?	
Read_Information	?	
Read_Driver_temperature	?	
Read_Motor_temperature	?	
Read_Cumulative_load_monitor	?	
Read_Tripmeter	?	

Function Block

OM_AZ_MONITOR		...
ImplicitMSGinput	?	
Read_Present_alarm		)
Read_Feedback_position		)
Read_Feedback_speed		)
Read_Command_position		)
Read_Torque_monitor		)
Read_CST_operating_current		)
Read_Information		)
Read_Driver_temperature		)
Read_Motor_temperature		)
Read_Cumulative_load_monitor		)
Read_Tripmeter		)

Structured Text

OM_AZ_MONITOR(ImplicitMSGinput);

Parameters

Required	Name	Data Type	Usage	Description
X	OM_AZ_MONITOR	OM_AZ_MONITOR	InOut	
	EnableIn	BOOL	Input	
	EnableOut	BOOL	Output	
X	ImplicitMSGinput	INT	InOut	Implicit message Input (driver->scanner)
	Read_Present_alarm	INT	Output	Alarm code monitor[Hex]
	Read_Feedback_position	DINT	Output	Feedback position monitor[step]
	Read_Feedback_speed	DINT	Output	Feedback speed monitor[Hz]
	Read_Command_position	DINT	Output	Command position monitor[step]

Read_Torque_monitor	INT	Output	Torque monitor as a percentage of the maximum holding torque[1=0.1%]
Read_CST_operating_current	INT	Output	Operation current monitor in alpha control mode(CST) (1=0.1%)
Read_Information	DINT	Output	Information monitor before an alarm is generated.
Read_Driver_temperature	DINT	Output	Driver temperature monitor(1=0.1deg)
Read_Motor_temperature	DINT	Output	Motor temperature monitor(1=0.1deg)
Read_Cumulative_load_monitor	DINT	Output	Cumulative load monitor
Read_Tripmeter	DINT	Output	Travel distance monitor[x1000rev]

Extended Description

This AOI is used to monitor the present alarm, feedback position, feedback speed, command position, torque monitor, CST operating current, information, driver temperature, motor temperature, cumulative load and tripmeter.
The driver temperature, motor temperature, cumulative load monitor, and tripmeter are used as assignable monitor setting parameters as an initial value. Assignable monitor settings can also be changed to check for other parameters.

Execution

Condition	Description
EnableIn is true	

Revision v1.3 Notes

Implicit message based on AZD-AEP,-CEP and -KEP.

Name	Default	Data Type	Scope
ImplicitMSGinput		INT[28]	OM_AZ_MONITOR
Implicit message Input (driver->scanner)			
Usage:	InOut Parameter		
Required:	Yes		
Visible:	Yes		
Constant	No		
<i>ImplicitMSGinput - OM_AZ_MONITOR/Logic - 0(CPS)</i>			
ImplicitMSGinput[0]	??	INT	
Remote I/O (R-OUT)			
ImplicitMSGinput[0].0	??	BOOL	
R-OUT0			
ImplicitMSGinput[0].1	??	BOOL	
R-OUT1			
ImplicitMSGinput[0].2	??	BOOL	
R-OUT2			
ImplicitMSGinput[0].3	??	BOOL	
R-OUT3			
ImplicitMSGinput[0].4	??	BOOL	
R-OUT4			
ImplicitMSGinput[0].5	??	BOOL	
R-OUT5			
ImplicitMSGinput[0].6	??	BOOL	
R-OUT6			
ImplicitMSGinput[0].7	??	BOOL	
R-OUT7			
ImplicitMSGinput[0].8	??	BOOL	
R-OUT8			
ImplicitMSGinput[0].9	??	BOOL	
R-OUT9			
ImplicitMSGinput[0].10	??	BOOL	
R-OUT10			
ImplicitMSGinput[0].11	??	BOOL	
R-OUT11			
ImplicitMSGinput[0].12	??	BOOL	
R-OUT12			
ImplicitMSGinput[0].13	??	BOOL	
R-OUT13			
ImplicitMSGinput[0].14	??	BOOL	
R-OUT14			
ImplicitMSGinput[0].15	??	BOOL	
R-OUT15			
ImplicitMSGinput[1]	??	INT	
Operation data number selection_R			
ImplicitMSGinput[1].0	??	BOOL	
M0_R			
ImplicitMSGinput[1].1	??	BOOL	
M1_R			
ImplicitMSGinput[1].2	??	BOOL	
M2_R			
ImplicitMSGinput[1].3	??	BOOL	
M3_R			
ImplicitMSGinput[1].4	??	BOOL	
M4_R			
ImplicitMSGinput[1].5	??	BOOL	
M5_R			
ImplicitMSGinput[1].6	??	BOOL	
M6_R			
ImplicitMSGinput[1].7	??	BOOL	
M7_R			
ImplicitMSGinput[1].8	??	BOOL	
Reserved			
ImplicitMSGinput[1].9	??	BOOL	

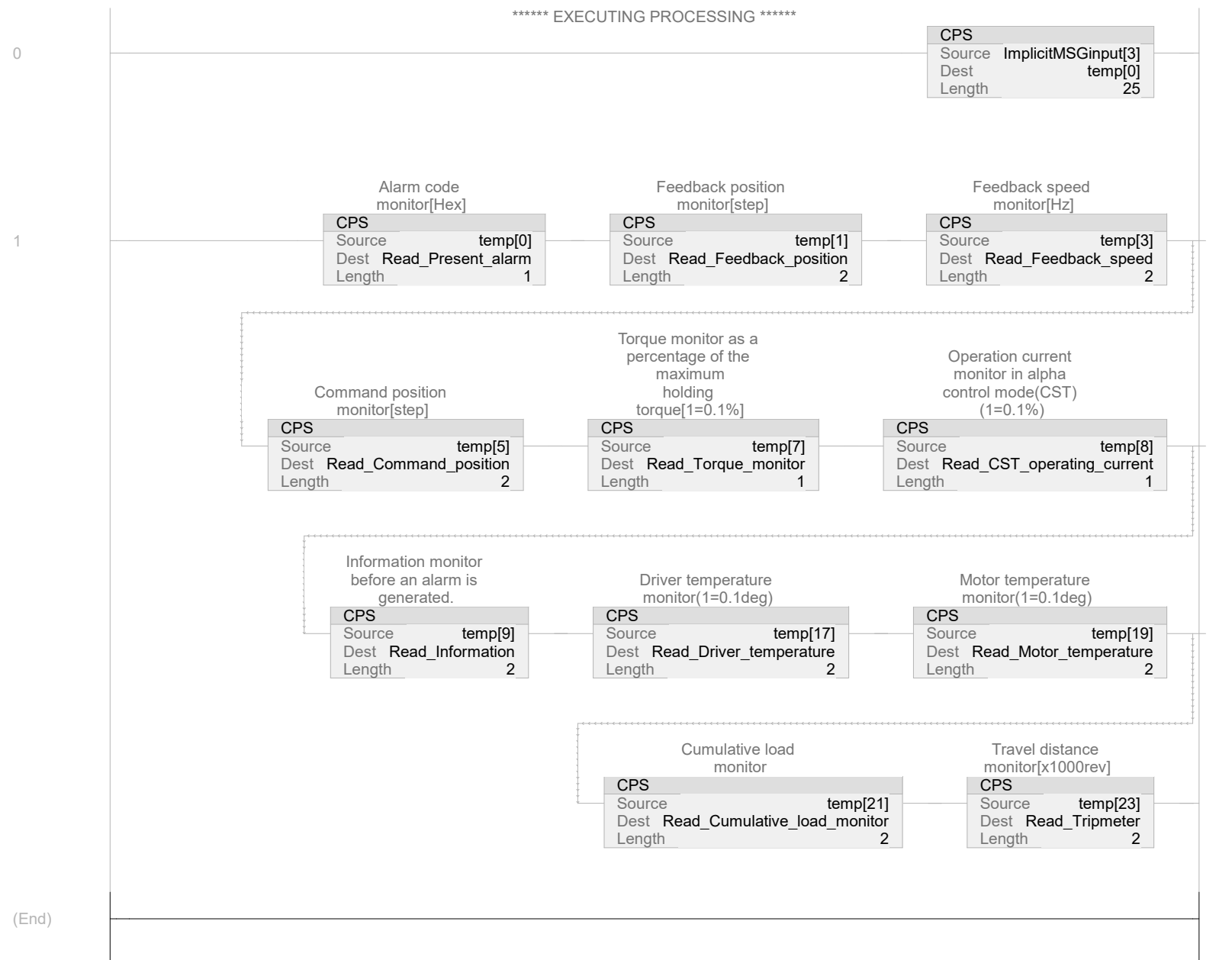
ImplicitMSGinput (Continued)		
Reserved		
ImplicitMSGinput[1].10	??	BOOL
Reserved		
ImplicitMSGinput[1].11	??	BOOL
Reserved		
ImplicitMSGinput[1].12	??	BOOL
Reserved		
ImplicitMSGinput[1].13	??	BOOL
Reserved		
ImplicitMSGinput[1].14	??	BOOL
Reserved		
ImplicitMSGinput[1].15	??	BOOL
Reserved		
ImplicitMSGinput[2]	??	INT
Fixed I/O (OUT)		
ImplicitMSGinput[2].0	??	BOOL
SEQ-BSY		
ImplicitMSGinput[2].1	??	BOOL
MOVE		
ImplicitMSGinput[2].2	??	BOOL
IN-POS		
ImplicitMSGinput[2].3	??	BOOL
START_R		
ImplicitMSGinput[2].4	??	BOOL
HOME-END		
ImplicitMSGinput[2].5	??	BOOL
READY		
ImplicitMSGinput[2].6	??	BOOL
DCMD-RDY		
ImplicitMSGinput[2].7	??	BOOL
ALM-A		
ImplicitMSGinput[2].8	??	BOOL
TRIG_R		
ImplicitMSGinput[2].9	??	BOOL
TRIG-MODE_R		
ImplicitMSGinput[2].10	??	BOOL
SET-ERR		
ImplicitMSGinput[2].11	??	BOOL
EXE-ERR		
ImplicitMSGinput[2].12	??	BOOL
DCMD-FULL		
ImplicitMSGinput[2].13	??	BOOL
STOP_R		
ImplicitMSGinput[2].14	??	BOOL
ETO-MON		
ImplicitMSGinput[2].15	??	BOOL
TLC		
ImplicitMSGinput[3]	??	INT
Present alarm		
ImplicitMSGinput[4]	??	INT
Feedback position (lower)		
ImplicitMSGinput[5]	??	INT
Feedback position (upper)		
ImplicitMSGinput[6]	??	INT
Feedback speed [Hz] (lower)		
ImplicitMSGinput[7]	??	INT
Feedback speed [Hz] (upper)		
ImplicitMSGinput[8]	??	INT
Command position (lower)		
ImplicitMSGinput[9]	??	INT
Command position (upper)		
ImplicitMSGinput[10]	??	INT

ImplicitMSGinput (Continued)		
Torque monitor		
ImplicitMSGinput[11]	??	INT
CST operating current		
ImplicitMSGinput[12]	??	INT
Information (lower)		
ImplicitMSGinput[13]	??	INT
Information (upper)		
ImplicitMSGinput[14]	??	INT
Reserved		
ImplicitMSGinput[15]	??	INT
Read parameter ID_R		
ImplicitMSGinput[16]	??	INT
Read/write status		
ImplicitMSGinput[16].0	??	BOOL
Reserved		
ImplicitMSGinput[16].1	??	BOOL
Reserved		
ImplicitMSGinput[16].2	??	BOOL
Reserved		
ImplicitMSGinput[16].3	??	BOOL
Reserved		
ImplicitMSGinput[16].4	??	BOOL
Reserved		
ImplicitMSGinput[16].5	??	BOOL
Reserved		
ImplicitMSGinput[16].6	??	BOOL
Reserved		
ImplicitMSGinput[16].7	??	BOOL
RD-ERR		
ImplicitMSGinput[16].8	??	BOOL
WR-END		
ImplicitMSGinput[16].9	??	BOOL
SYS-BSY		
ImplicitMSGinput[16].10	??	BOOL
Reserved		
ImplicitMSGinput[16].11	??	BOOL
WR-SET-ERR		
ImplicitMSGinput[16].12	??	BOOL
WR-IF-ERR		
ImplicitMSGinput[16].13	??	BOOL
WR-NV-ERR		
ImplicitMSGinput[16].14	??	BOOL
WR-EXE-ERR		
ImplicitMSGinput[16].15	??	BOOL
WR-ERR		
ImplicitMSGinput[17]	??	INT
Write parameter ID_R		
ImplicitMSGinput[18]	??	INT
Read data (lower)		
ImplicitMSGinput[19]	??	INT
Read data (upper)		
ImplicitMSGinput[20]	??	INT
Assignable monitor 0 (lower)		
ImplicitMSGinput[21]	??	INT
Assignable monitor 0 (upper)		
ImplicitMSGinput[22]	??	INT
Assignable monitor 1 (lower)		
ImplicitMSGinput[23]	??	INT
Assignable monitor 1 (upper)		
ImplicitMSGinput[24]	??	INT
Assignable monitor 2 (lower)		
ImplicitMSGinput[25]	??	INT

ImplicitMSGinput (Continued)			
Assignable monitor 2 (upper)			
ImplicitMSGinput[26]	??	INT	
Assignable monitor 3 (lower)			
ImplicitMSGinput[27]	??	INT	
Assignable monitor 3 (upper)			
Read_Command_position	0	DINT	OM_AZ_MONITOR
Command position monitor[step]			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
Read_Command_position - OM_AZ_MONITOR/Logic - *1(CPS)			
Read_CST_operating_current	0	INT	OM_AZ_MONITOR
Operation current monitor in alpha control mode(CST) (1=0.1%)			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
Read_CST_operating_current - OM_AZ_MONITOR/Logic - *1(CPS)			
Read_Cumulative_load_monitor	0	DINT	OM_AZ_MONITOR
Cumulative load monitor			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
Read_Cumulative_load_monitor - OM_AZ_MONITOR/Logic - *1(CPS)			
Read_Driver_temperature	0	DINT	OM_AZ_MONITOR
Driver temperature monitor(1=0.1deg)			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
Read_Driver_temperature - OM_AZ_MONITOR/Logic - *1(CPS)			
Read_Feedback_position	0	DINT	OM_AZ_MONITOR
Feedback position monitor[step]			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
Read_Feedback_position - OM_AZ_MONITOR/Logic - *1(CPS)			
Read_Feedback_speed	0	DINT	OM_AZ_MONITOR
Feedback speed monitor[Hz]			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
Read_Feedback_speed - OM_AZ_MONITOR/Logic - *1(CPS)			
Read_Information	16#0000_0000	DINT	OM_AZ_MONITOR
Information monitor before an alarm is generated.			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
Read_Information - OM_AZ_MONITOR/Logic - *1(CPS)			

Read_Motor_temperature Motor temperature monitor(1=0.1deg) Usage: Output Parameter Required: No Visible: Yes External Access: Read/Write Read_Motor_temperature - OM_AZ_MONITOR/Logic - *1(CPS)	0	DINT	OM_AZ_MONITOR
Read_Present_alarm Alarm code monitor[Hex] Usage: Output Parameter Required: No Visible: Yes External Access: Read/Write Read_Present_alarm - OM_AZ_MONITOR/Logic - *1(CPS)	16#0000	INT	OM_AZ_MONITOR
Read_Torque_monitor Torque monitor as a percentage of the maximum holding torque[1=0.1%] Usage: Output Parameter Required: No Visible: Yes External Access: Read/Write Read_Torque_monitor - OM_AZ_MONITOR/Logic - *1(CPS)	0	INT	OM_AZ_MONITOR
Read_Tripmeter Travel distance monitor[x1000rev] Usage: Output Parameter Required: No Visible: Yes External Access: Read/Write Read_Tripmeter - OM_AZ_MONITOR/Logic - *1(CPS)	0	DINT	OM_AZ_MONITOR

Name	Default	Data Type	Scope
temp		INT[25]	OM_AZ_MONITOR
Usage:	Local Tag		
External Access:	None		
<i>temp</i> - OM_AZ_MONITOR/Logic - *0(CPS), 1(CPS), 1(CPS), 1(CPS), 1(CPS), 1(CPS), 1(CPS), 1(CPS), 1(CPS), 1(CPS)			
temp[0]	0	INT	
<i>temp[0]</i> - OM_AZ_MONITOR/Logic - 1(CPS)			
temp[7]	0	INT	
<i>temp[7]</i> - OM_AZ_MONITOR/Logic - 1(CPS)			
temp[8]	0	INT	
<i>temp[8]</i> - OM_AZ_MONITOR/Logic - 1(CPS)			



OM_AZ_MOVE v1.3 U1.0
ORIENTALMOTOR CO.,LTD.

Class: Standard

Available Languages

Relay Ladder

OM_AZ_MOVE		
OM_AZ_MOVE	?	(Sts_Busy)
ImplicitMSGinput	?	(Sts_Err)
ImplicitMSGoutput	?	(Sts_Done)
Set_Exe	?	
Set_OpeDataNo	?	
Read_Present_alarm	?	
Read_Command_position	?	
Read_Feedback_position	?	
Read_Torque_monitor	?	
Read_Cumulative_load_monitor	?	

Function Block

OM_AZ_MOVE		
ImplicitMSGinput	?	
ImplicitMSGoutput	?	
Set_Exe	Sts_Busy	
Set_OpeDataNo	Sts_Err	
	Sts_Done	
	Read_Present_alarm	
	Read_Command_position	
	Read_Feedback_position	
	Read_Torque_monitor	
	Read_Cumulative_load_monitor	

Structured Text

OM_AZ_MOVE(ImplicitMSGinput, ImplicitMSGoutput);

Parameters

Required	Name	Data Type	Usage	Description
X	OM_AZ_MOVE	OM_AZ_MOVE	InOut	
	EnableIn	BOOL	Input	
	EnableOut	BOOL	Output	
X	ImplicitMSGinput	INT	InOut	Implicit message Input (driver->scanner)
X	ImplicitMSGoutput	INT	InOut	Implicit message Output (scanner->driver)
	Set_Exe	BOOL	Input	Execute AOI
	Set_OpeDataNo	INT	Input	Selected Operation data No.
	Sts_Busy	BOOL	Output	Executing status in AOI
	Sts_Err	BOOL	Output	Error status in AOI
	Sts_Done	BOOL	Output	Complete status in AOI
	Read_Present_alarm	INT	Output	Alarm code monitor[Hex]
	Read_Command_position	DINT	Output	Command position monitor[step]
	Read_Feedback_position	DINT	Output	Feedback position monitor[step]

Read_Torque_monitor	INT	Output	Torque monitor as a percentage of the maximum holding torque(1=0.1%)
Read_Cumulative_load_monitor	DINT	Output	Cumulative load monitor

Extended Description

This AOI is used to select and operate a pre-programmed data operation. This is used when the same operation is used repeatedly.

Execution

Condition	Description
EnableIn is true	

Revision v1.3 UI.0 Notes

Implicit message based on AZD-AEP,-CEP and -KEP.

Name	Default	Data Type	Scope
ImplicitMSGinput		INT[28]	OM_AZ_MOVE
Implicit message Input (driver->scanner)			
Usage:	InOut Parameter		
Required:	Yes		
Visible:	Yes		
Constant	No		
<i>ImplicitMSGinput - OM_AZ_MOVE/Logic - 0(CPS), 0(CPS), 0(CPS)</i>			
ImplicitMSGinput[0]	??	INT	
Remote I/O (R-OUT)			
ImplicitMSGinput[0].0	??	BOOL	
R-OUT0			
ImplicitMSGinput[0].1	??	BOOL	
R-OUT1			
ImplicitMSGinput[0].2	??	BOOL	
R-OUT2			
ImplicitMSGinput[0].3	??	BOOL	
R-OUT3			
ImplicitMSGinput[0].4	??	BOOL	
R-OUT4			
ImplicitMSGinput[0].5	??	BOOL	
R-OUT5			
ImplicitMSGinput[0].6	??	BOOL	
R-OUT6			
ImplicitMSGinput[0].7	??	BOOL	
R-OUT7			
ImplicitMSGinput[0].8	??	BOOL	
R-OUT8			
ImplicitMSGinput[0].9	??	BOOL	
R-OUT9			
ImplicitMSGinput[0].10	??	BOOL	
R-OUT10			
ImplicitMSGinput[0].11	??	BOOL	
R-OUT11			
ImplicitMSGinput[0].12	??	BOOL	
R-OUT12			
ImplicitMSGinput[0].13	??	BOOL	
R-OUT13			
ImplicitMSGinput[0].14	??	BOOL	
R-OUT14			
ImplicitMSGinput[0].15	??	BOOL	
R-OUT15			
ImplicitMSGinput[1]	??	INT	
Operation data number selection_R			
ImplicitMSGinput[1].0	??	BOOL	
M0_R			
ImplicitMSGinput[1].1	??	BOOL	
M1_R			
ImplicitMSGinput[1].2	??	BOOL	
M2_R			
ImplicitMSGinput[1].3	??	BOOL	
M3_R			
ImplicitMSGinput[1].4	??	BOOL	
M4_R			
ImplicitMSGinput[1].5	??	BOOL	
M5_R			
ImplicitMSGinput[1].6	??	BOOL	
M6_R			
ImplicitMSGinput[1].7	??	BOOL	
M7_R			
ImplicitMSGinput[1].8	??	BOOL	
Reserved			
ImplicitMSGinput[1].9	??	BOOL	

ImplicitMSGInput (Continued)		
Reserved		
ImplicitMSGInput[1].10	??	BOOL
Reserved		
ImplicitMSGInput[1].11	??	BOOL
Reserved		
ImplicitMSGInput[1].12	??	BOOL
Reserved		
ImplicitMSGInput[1].13	??	BOOL
Reserved		
ImplicitMSGInput[1].14	??	BOOL
Reserved		
ImplicitMSGInput[1].15	??	BOOL
Reserved		
ImplicitMSGInput[2]	??	INT
Fixed I/O (OUT)		
<i>ImplicitMSGInput[2] - OM_AZ_MOVE/Logic - 5(CPS)</i>		
ImplicitMSGInput[2].0	??	BOOL
SEQ-BSY		
ImplicitMSGInput[2].1	??	BOOL
MOVE		
ImplicitMSGInput[2].2	??	BOOL
IN-POS		
ImplicitMSGInput[2].3	??	BOOL
START_R		
<i>ImplicitMSGInput[2].3 - OM_AZ_MOVE/Logic - 2(XIC), 2(XIO), 5(XIC)</i>		
ImplicitMSGInput[2].4	??	BOOL
HOME-END		
ImplicitMSGInput[2].5	??	BOOL
READY		
<i>ImplicitMSGInput[2].5 - OM_AZ_MOVE/Logic - 4(XIO), 5(XIO)</i>		
ImplicitMSGInput[2].6	??	BOOL
DCMD-RDY		
<i>ImplicitMSGInput[2].6 - OM_AZ_MOVE/Logic - 4(XIO)</i>		
ImplicitMSGInput[2].7	??	BOOL
ALM-A		
<i>ImplicitMSGInput[2].7 - OM_AZ_MOVE/Logic - 4(XIC)</i>		
ImplicitMSGInput[2].8	??	BOOL
TRIG_R		
ImplicitMSGInput[2].9	??	BOOL
TRIG-MODE_R		
ImplicitMSGInput[2].10	??	BOOL
SET-ERR		
ImplicitMSGInput[2].11	??	BOOL
EXE-ERR		
ImplicitMSGInput[2].12	??	BOOL
DCMD-FULL		
ImplicitMSGInput[2].13	??	BOOL
STOP_R		
<i>ImplicitMSGInput[2].13 - OM_AZ_MOVE/Logic - 4(XIC)</i>		
ImplicitMSGInput[2].14	??	BOOL
ETO-MON		
<i>ImplicitMSGInput[2].14 - OM_AZ_MOVE/Logic - 4(XIC)</i>		
ImplicitMSGInput[2].15	??	BOOL
TLC		
ImplicitMSGInput[3]	??	INT
Present alarm		
<i>ImplicitMSGInput[3] - OM_AZ_MOVE/Logic - 0(CPS)</i>		
ImplicitMSGInput[4]	??	INT
Feedback position (lower)		
ImplicitMSGInput[5]	??	INT
Feedback position (upper)		
ImplicitMSGInput[6]	??	INT

ImplicitMSGinput (Continued)		
Feedback speed [Hz] (lower)		
ImplicitMSGinput[7]	??	INT
Feedback speed [Hz] (upper)		
ImplicitMSGinput[8]	??	INT
Command position (lower)		
ImplicitMSGinput[9]	??	INT
Command position (upper)		
ImplicitMSGinput[10]	??	INT
Torque monitor		
<i>ImplicitMSGinput[10] - OM_AZ_MOVE/Logic - 0(CPS)</i>		
ImplicitMSGinput[11]	??	INT
CST operating current		
ImplicitMSGinput[12]	??	INT
Information (lower)		
ImplicitMSGinput[13]	??	INT
Information (upper)		
ImplicitMSGinput[14]	??	INT
Reserved		
ImplicitMSGinput[15]	??	INT
Read parameter ID_R		
ImplicitMSGinput[16]	??	INT
Read/write status		
ImplicitMSGinput[16].0	??	BOOL
Reserved		
ImplicitMSGinput[16].1	??	BOOL
Reserved		
ImplicitMSGinput[16].2	??	BOOL
Reserved		
ImplicitMSGinput[16].3	??	BOOL
Reserved		
ImplicitMSGinput[16].4	??	BOOL
Reserved		
ImplicitMSGinput[16].5	??	BOOL
Reserved		
ImplicitMSGinput[16].6	??	BOOL
Reserved		
ImplicitMSGinput[16].7	??	BOOL
RD-ERR		
ImplicitMSGinput[16].8	??	BOOL
WR-END		
ImplicitMSGinput[16].9	??	BOOL
SYS-BSY		
ImplicitMSGinput[16].10	??	BOOL
Reserved		
ImplicitMSGinput[16].11	??	BOOL
WR-SET-ERR		
ImplicitMSGinput[16].12	??	BOOL
WR-IF-ERR		
ImplicitMSGinput[16].13	??	BOOL
WR-NV-ERR		
ImplicitMSGinput[16].14	??	BOOL
WR-EXE-ERR		
ImplicitMSGinput[16].15	??	BOOL
WR-ERR		
ImplicitMSGinput[17]	??	INT
Write parameter ID_R		
ImplicitMSGinput[18]	??	INT
Read data (lower)		
ImplicitMSGinput[19]	??	INT
Read data (upper)		
ImplicitMSGinput[20]	??	INT
Assignable monitor 0 (lower)		

ImplicitMSGinput (Continued)		
ImplicitMSGinput[21]	??	INT
Assignable monitor 0 (upper)		
ImplicitMSGinput[22]	??	INT
Assignable monitor 1 (lower)		
ImplicitMSGinput[23]	??	INT
Assignable monitor 1 (upper)		
ImplicitMSGinput[24]	??	INT
Assignable monitor 2 (lower)		
ImplicitMSGinput[25]	??	INT
Assignable monitor 2 (upper)		
ImplicitMSGinput[26]	??	INT
Assignable monitor 3 (lower)		
ImplicitMSGinput[27]	??	INT
Assignable monitor 3 (upper)		
ImplicitMSGoutput	INT[20]	OM_AZ_MOVE
Implicit message Output (scanner->driver)		
Usage:	InOut Parameter	
Required:	Yes	
Visible:	Yes	
Constant	No	
ImplicitMSGoutput[0]	??	INT
Remote I/O(R-IN)		
ImplicitMSGoutput[0].0	??	BOOL
R-IN0		
ImplicitMSGoutput[0].1	??	BOOL
R-IN1		
ImplicitMSGoutput[0].2	??	BOOL
R-IN2		
ImplicitMSGoutput[0].3	??	BOOL
R-IN3		
ImplicitMSGoutput[0].4	??	BOOL
R-IN4		
ImplicitMSGoutput[0].5	??	BOOL
R-IN5		
ImplicitMSGoutput[0].6	??	BOOL
R-IN6		
ImplicitMSGoutput[0].7	??	BOOL
R-IN7		
ImplicitMSGoutput[0].8	??	BOOL
R-IN8		
ImplicitMSGoutput[0].9	??	BOOL
R-IN9		
ImplicitMSGoutput[0].10	??	BOOL
R-IN10		
ImplicitMSGoutput[0].11	??	BOOL
R-IN11		
ImplicitMSGoutput[0].12	??	BOOL
R-IN12		
ImplicitMSGoutput[0].13	??	BOOL
R-IN13		
ImplicitMSGoutput[0].14	??	BOOL
R-IN14		
ImplicitMSGoutput[0].15	??	BOOL
R-IN15		
ImplicitMSGoutput[1]	??	INT
Operation data number selection		
<i>ImplicitMSGoutput[1] - OM_AZ_MOVE/Logic - *4(CPS)</i>		
ImplicitMSGoutput[1].0	??	BOOL
M0		
ImplicitMSGoutput[1].1	??	BOOL
M1		

ImplicitMSGGoutput (Continued)		
ImplicitMSGGoutput[1].2 M2	??	BOOL
ImplicitMSGGoutput[1].3 M3	??	BOOL
ImplicitMSGGoutput[1].4 M4	??	BOOL
ImplicitMSGGoutput[1].5 M5	??	BOOL
ImplicitMSGGoutput[1].6 M6	??	BOOL
ImplicitMSGGoutput[1].7 M7	??	BOOL
ImplicitMSGGoutput[1].8 Reserved	??	BOOL
ImplicitMSGGoutput[1].9 Reserved	??	BOOL
ImplicitMSGGoutput[1].10 Reserved	??	BOOL
ImplicitMSGGoutput[1].11 Reserved	??	BOOL
ImplicitMSGGoutput[1].12 Reserved	??	BOOL
ImplicitMSGGoutput[1].13 Reserved	??	BOOL
ImplicitMSGGoutput[1].14 Reserved	??	BOOL
ImplicitMSGGoutput[1].15 Reserved	??	BOOL
ImplicitMSGGoutput[2] Fixed I/O(IN)	??	INT
ImplicitMSGGoutput[2].0 FW-JOG	??	BOOL
ImplicitMSGGoutput[2].1 RV-JOG	??	BOOL
ImplicitMSGGoutput[2].2 Reserved	??	BOOL
ImplicitMSGGoutput[2].3 START	??	BOOL
<i>ImplicitMSGGoutput[2].3 - OM_AZ_MOVE/Logic - *2(OTU), *4(OTL), *5(OTU)</i>		
ImplicitMSGGoutput[2].4 ZHOME	??	BOOL
ImplicitMSGGoutput[2].5 STOP	??	BOOL
ImplicitMSGGoutput[2].6 FREE	??	BOOL
ImplicitMSGGoutput[2].7 ALM-RST	??	BOOL
ImplicitMSGGoutput[2].8 TRIG	??	BOOL
ImplicitMSGGoutput[2].9 TRIG-MODE	??	BOOL
ImplicitMSGGoutput[2].10 ETO-CLR	??	BOOL
ImplicitMSGGoutput[2].11 Reserved	??	BOOL
ImplicitMSGGoutput[2].12 FW-JOG-P	??	BOOL
ImplicitMSGGoutput[2].13 RV-JOG-P	??	BOOL
ImplicitMSGGoutput[2].14 FW-POS	??	BOOL
ImplicitMSGGoutput[2].15	??	BOOL

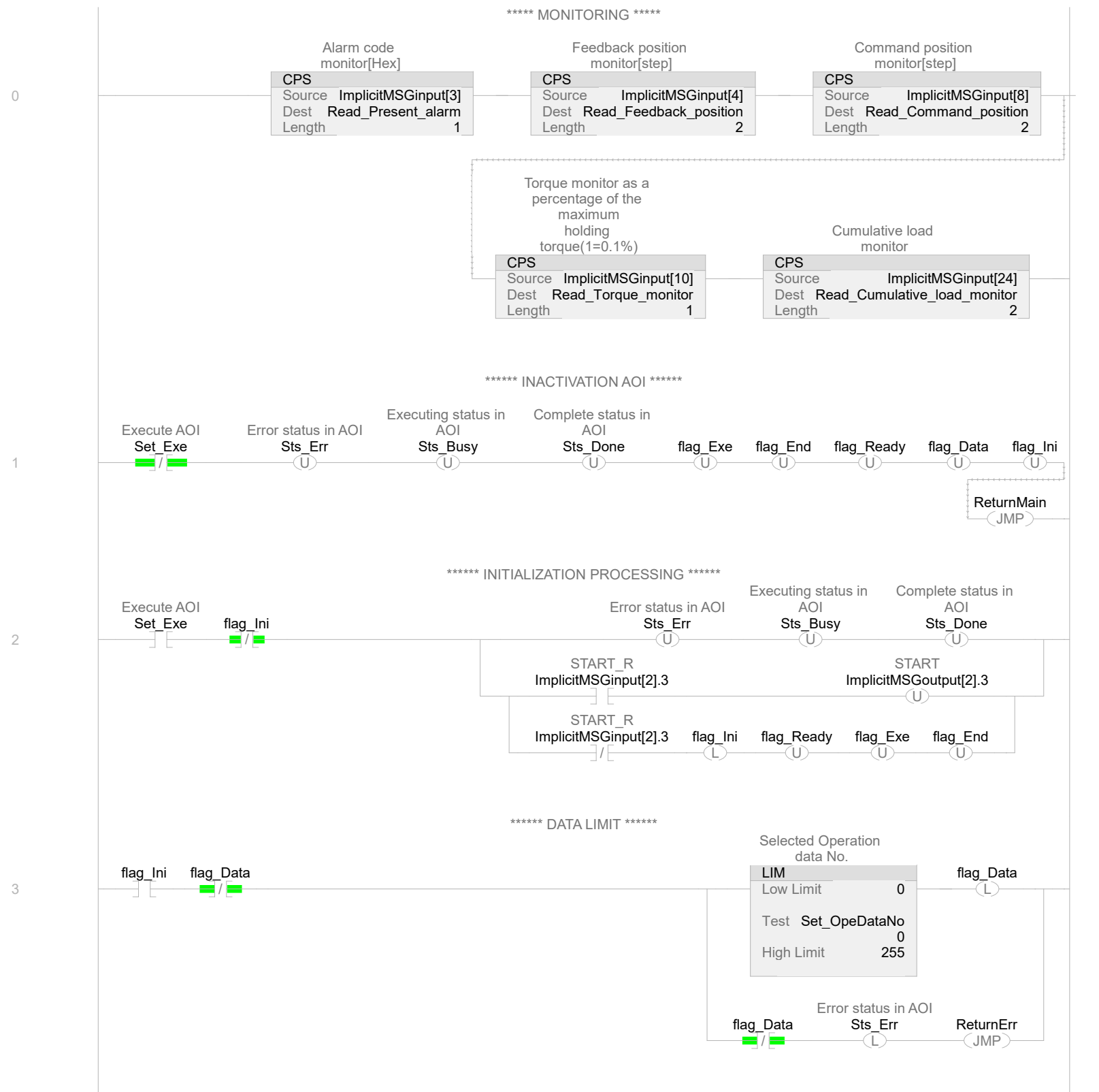
ImplicitMSGoutput (Continued)		
RV-POS		
ImplicitMSGoutput[3]	??	INT
Direct data operation operation type		
ImplicitMSGoutput[4]	??	INT
Direct data operation position (lower)		
ImplicitMSGoutput[5]	??	INT
Direct data operation position (upper)		
ImplicitMSGoutput[6]	??	INT
Direct data operation operating speed (lower)		
ImplicitMSGoutput[7]	??	INT
Direct data operation operating speed (upper)		
ImplicitMSGoutput[8]	??	INT
Direct data operation starting/changing rate (lower)		
ImplicitMSGoutput[9]	??	INT
Direct data operation starting/changing rate (upper)		
ImplicitMSGoutput[10]	??	INT
Direct data operation stopping deceleration (lower)		
ImplicitMSGoutput[11]	??	INT
Direct data operation stopping deceleration (upper)		
ImplicitMSGoutput[12]	??	INT
Direct data operation operating current		
ImplicitMSGoutput[13]	??	INT
Direct data operation forwarding destination		
ImplicitMSGoutput[14]	??	INT
Reserved		
ImplicitMSGoutput[15]	??	INT
Read parameter ID		
ImplicitMSGoutput[16]	??	INT
Write request		
ImplicitMSGoutput[16].0	??	BOOL
WR-REQ		
ImplicitMSGoutput[16].1	??	BOOL
Reserved		
ImplicitMSGoutput[16].2	??	BOOL
Reserved		
ImplicitMSGoutput[16].3	??	BOOL
Reserved		
ImplicitMSGoutput[16].4	??	BOOL
Reserved		
ImplicitMSGoutput[16].5	??	BOOL
Reserved		
ImplicitMSGoutput[16].6	??	BOOL
Reserved		
ImplicitMSGoutput[16].7	??	BOOL
Reserved		
ImplicitMSGoutput[16].8	??	BOOL
Reserved		
ImplicitMSGoutput[16].9	??	BOOL
Reserved		
ImplicitMSGoutput[16].10	??	BOOL
Reserved		
ImplicitMSGoutput[16].11	??	BOOL
Reserved		
ImplicitMSGoutput[16].12	??	BOOL
Reserved		
ImplicitMSGoutput[16].13	??	BOOL
Reserved		
ImplicitMSGoutput[16].14	??	BOOL
Reserved		
ImplicitMSGoutput[16].15	??	BOOL
Reserved		
ImplicitMSGoutput[17]	??	INT

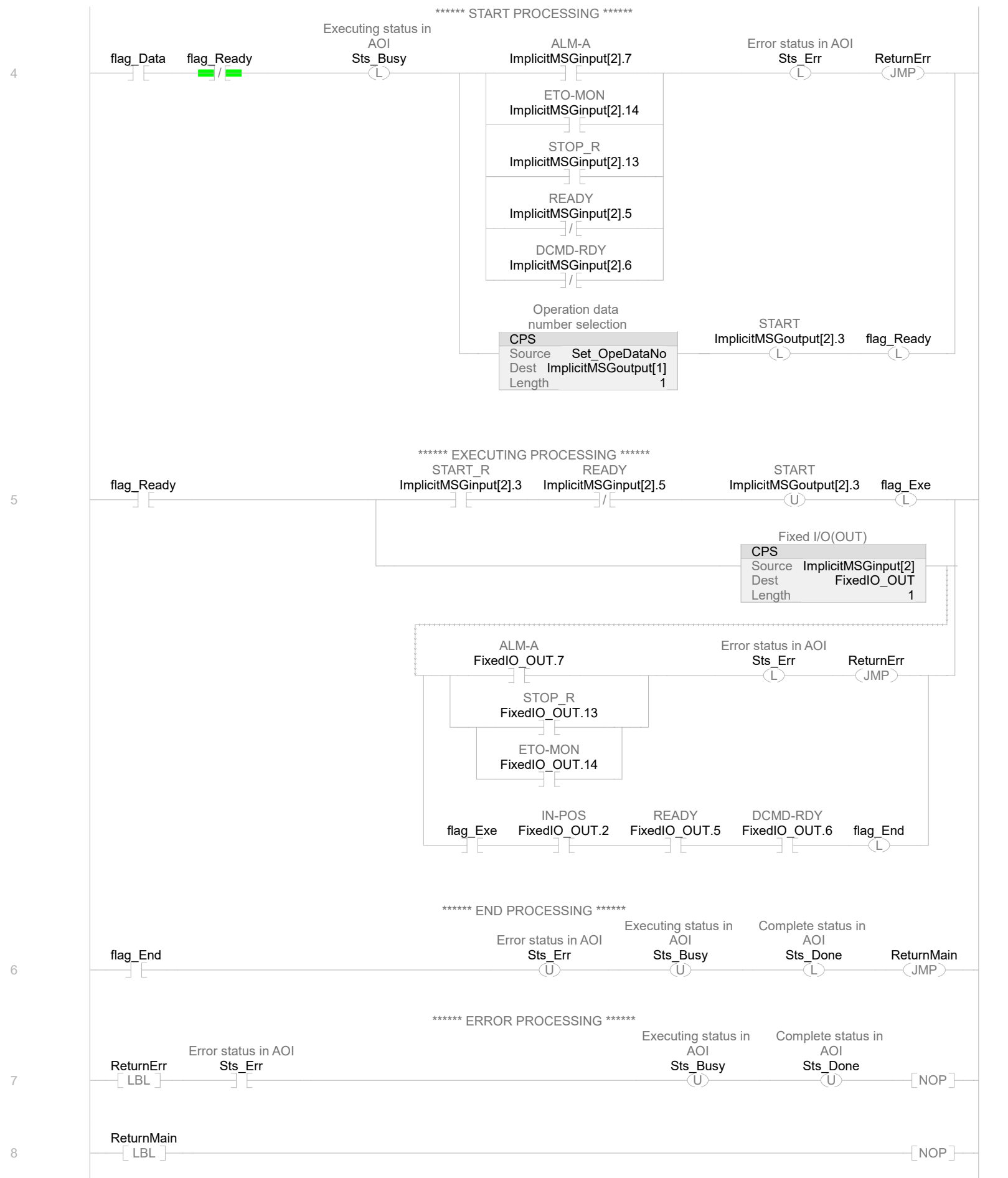
ImplicitMSGoutput (Continued)			
Write parameter ID			
ImplicitMSGoutput[18]		??	INT
Write data (lower)			
ImplicitMSGoutput[19]		??	INT
Write data (upper)			
Read_Command_position		0	DINT
Command position monitor[step]			OM_AZ_MOVE
Usage: Output Parameter			
Required: No			
Visible: Yes			
External Access: Read Only			
Read_Command_position - OM_AZ_MOVE/Logic - *0(CPS)			
Read_Cumulative_load_monitor		0	DINT
Cumulative load monitor			OM_AZ_MOVE
Usage: Output Parameter			
Required: No			
Visible: Yes			
External Access: Read Only			
Read_Cumulative_load_monitor - OM_AZ_MOVE/Logic - *0(CPS)			
Read_Feedback_position		0	DINT
Feedback position monitor[step]			OM_AZ_MOVE
Usage: Output Parameter			
Required: No			
Visible: Yes			
External Access: Read Only			
Read_Feedback_position - OM_AZ_MOVE/Logic - *0(CPS)			
Read_Present_alarm		16#0000	INT
Alarm code monitor[Hex]			OM_AZ_MOVE
Usage: Output Parameter			
Required: No			
Visible: Yes			
External Access: Read Only			
Read_Present_alarm - OM_AZ_MOVE/Logic - *0(CPS)			
Read_Torque_monitor		0	INT
Torque monitor as a percentage of the maximum holding torque(1=0.1%)			OM_AZ_MOVE
Usage: Output Parameter			
Required: No			
Visible: Yes			
External Access: Read Only			
Read_Torque_monitor - OM_AZ_MOVE/Logic - *0(CPS)			
Set_Exe		0	BOOL
Execute AOI			OM_AZ_MOVE
Usage: Input Parameter			
Required: No			
Visible: Yes			
External Access: Read/Write			
Set_Exe - OM_AZ_MOVE/Logic - 1(XIO), 2(XIC)			
Set_OpeDataNo		0	INT
Selected Operation data No.			OM_AZ_MOVE
Usage: Input Parameter			
Required: No			
Visible: Yes			
External Access: Read/Write			
Set_OpeDataNo - OM_AZ_MOVE/Logic - 3(LIM), 4(CPS)			

Sts_Busy	0	BOOL	OM_AZ_MOVE
Executing status in AOI			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read Only		
<i>Sts_Busy - OM_AZ_MOVE/Logic - *1(OTU), *2(OTU), *4(OTL), *6(OTU), *7(OTU)</i>			
Sts_Done	0	BOOL	OM_AZ_MOVE
Complete status in AOI			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read Only		
<i>Sts_Done - OM_AZ_MOVE/Logic - *1(OTU), *2(OTU), *6(OTL), *7(OTU)</i>			
Sts_Err	0	BOOL	OM_AZ_MOVE
Error status in AOI			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read Only		
<i>Sts_Err - OM_AZ_MOVE/Logic - *1(OTU), *2(OTU), *3(OTL), *4(OTL), *5(OTL), *6(OTU), 7(XIC)</i>			

Name	Default	Data Type	Scope
FixedIO_OUT Fixed I/O(OUT) Usage: External Access: <i>FixedIO_OUT - OM_AZ_MOVE/Logic - *5(CPS)</i>	0 Local Tag Read/Write	INT	OM_AZ_MOVE
FixedIO_OUT.0 SEQ-BSY	0	BOOL	
FixedIO_OUT.1 MOVE	0	BOOL	
FixedIO_OUT.2 IN-POS <i>FixedIO_OUT.2 - OM_AZ_MOVE/Logic - 5(XIC)</i>	0 	BOOL	
FixedIO_OUT.3 START_R	0	BOOL	
FixedIO_OUT.4 HOME-END	0	BOOL	
FixedIO_OUT.5 READY <i>FixedIO_OUT.5 - OM_AZ_MOVE/Logic - 5(XIC)</i>	0 	BOOL	
FixedIO_OUT.6 DCMD-RDY <i>FixedIO_OUT.6 - OM_AZ_MOVE/Logic - 5(XIC)</i>	0 	BOOL	
FixedIO_OUT.7 ALM-A <i>FixedIO_OUT.7 - OM_AZ_MOVE/Logic - 5(XIC)</i>	0 	BOOL	
FixedIO_OUT.8 TRIG_R	0	BOOL	
FixedIO_OUT.9 TRIG-MODE_R	0	BOOL	
FixedIO_OUT.10 SET-ERR	0	BOOL	
FixedIO_OUT.11 EXE-ERR	0	BOOL	
FixedIO_OUT.12 DCMD-FULL	0	BOOL	
FixedIO_OUT.13 STOP_R <i>FixedIO_OUT.13 - OM_AZ_MOVE/Logic - 5(XIC)</i>	0 	BOOL	
FixedIO_OUT.14 ETO-MON <i>FixedIO_OUT.14 - OM_AZ_MOVE/Logic - 5(XIC)</i>	0 	BOOL	
FixedIO_OUT.15 TLC	0	BOOL	
flag_Data Usage: External Access: <i>flag_Data - OM_AZ_MOVE/Logic - *1(OTU), *3(OTL), 3(XIO), 3(XIO), 4(XIC)</i>	0 Local Tag None	BOOL	OM_AZ_MOVE
flag_End Usage: External Access: <i>flag_End - OM_AZ_MOVE/Logic - *1(OTU), *2(OTU), *5(OTL), 6(XIC)</i>	0 Local Tag None	BOOL	OM_AZ_MOVE
flag_Exe Usage: External Access: <i>flag_Exe - OM_AZ_MOVE/Logic - *1(OTU), *2(OTU), *5(OTL), 5(XIC)</i>	0 Local Tag None	BOOL	OM_AZ_MOVE
flag_Ini Usage: External Access:	0 Local Tag None	BOOL	OM_AZ_MOVE

flag_Ini (Continued)			
flag_Ini - OM_AZ_MOVE/Logic - *1(OTU), *2(OTL), 2(XIO), 3(XIC)			
flag_Ready	0	BOOL	OM_AZ_MOVE
Usage:	Local Tag		
External Access:	None		
flag_Ready - OM_AZ_MOVE/Logic - *1(OTU), *2(OTU), *4(OTL), 4(XIO), 5(XIC)			





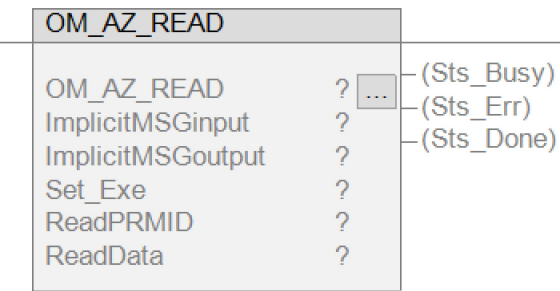
(End)	
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OM_AZ_READ v1.3
ORIENTALMOTOR CO.,LTD.

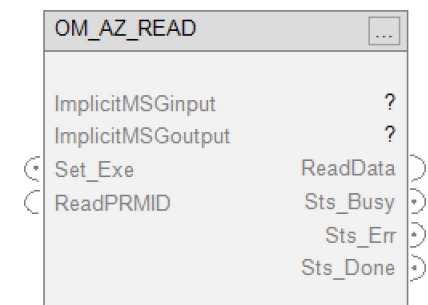
Class: Standard

Available Languages

Relay Ladder



Function Block



Structured Text

OM_AZ_READ(ImplicitMSGinput, ImplicitMSGoutput);

Parameters

Required	Name	Data Type	Usage	Description
X	OM_AZ_READ	OM_AZ_READ	InOut	
	EnableIn	BOOL	Input	
	EnableOut	BOOL	Output	
X	ImplicitMSGinput	INT	InOut	Implicit message Input (driver->scanner)
X	ImplicitMSGoutput	INT	InOut	Implicit message Output (scanner->driver)
	Set_Exe	BOOL	Input	Execute AOI
	ReadPRMID	INT	Input	Read parameter ID
	ReadData	DINT	Output	Read data
	Sts_Busy	BOOL	Output	Executing status in AOI
	Sts_Err	BOOL	Output	Error status in AOI
	Sts_Done	BOOL	Output	Complete status in AOI

Extended Description

This AOI is used to read data from the scanner to the driver via Implicit communication.

Execution

Condition	Description
EnableIn is true	

Revision v1.3 Notes

Implicit message based on AZD-AEP,-CEP and -KEP.

Name	Default	Data Type	Scope
ImplicitMSGinput		INT[28]	OM_AZ_READ
Implicit message Input (driver->scanner)			
Usage:	InOut Parameter		
Required:	Yes		
Visible:	Yes		
Constant	No		
<i>ImplicitMSGinput - OM_AZ_READ/Logic - 2(CPS)</i>			
ImplicitMSGinput[0]	??	INT	
Remote I/O (R-OUT)			
ImplicitMSGinput[0].0	??	BOOL	
R-OUT0			
ImplicitMSGinput[0].1	??	BOOL	
R-OUT1			
ImplicitMSGinput[0].2	??	BOOL	
R-OUT2			
ImplicitMSGinput[0].3	??	BOOL	
R-OUT3			
ImplicitMSGinput[0].4	??	BOOL	
R-OUT4			
ImplicitMSGinput[0].5	??	BOOL	
R-OUT5			
ImplicitMSGinput[0].6	??	BOOL	
R-OUT6			
ImplicitMSGinput[0].7	??	BOOL	
R-OUT7			
ImplicitMSGinput[0].8	??	BOOL	
R-OUT8			
ImplicitMSGinput[0].9	??	BOOL	
R-OUT9			
ImplicitMSGinput[0].10	??	BOOL	
R-OUT10			
ImplicitMSGinput[0].11	??	BOOL	
R-OUT11			
ImplicitMSGinput[0].12	??	BOOL	
R-OUT12			
ImplicitMSGinput[0].13	??	BOOL	
R-OUT13			
ImplicitMSGinput[0].14	??	BOOL	
R-OUT14			
ImplicitMSGinput[0].15	??	BOOL	
R-OUT15			
ImplicitMSGinput[1]	??	INT	
Operation data number selection_R			
ImplicitMSGinput[1].0	??	BOOL	
M0_R			
ImplicitMSGinput[1].1	??	BOOL	
M1_R			
ImplicitMSGinput[1].2	??	BOOL	
M2_R			
ImplicitMSGinput[1].3	??	BOOL	
M3_R			
ImplicitMSGinput[1].4	??	BOOL	
M4_R			
ImplicitMSGinput[1].5	??	BOOL	
M5_R			
ImplicitMSGinput[1].6	??	BOOL	
M6_R			
ImplicitMSGinput[1].7	??	BOOL	
M7_R			
ImplicitMSGinput[1].8	??	BOOL	
Reserved			
ImplicitMSGinput[1].9	??	BOOL	

ImplicitMSGinput (Continued)		
Reserved		
ImplicitMSGinput[1].10	??	BOOL
Reserved		
ImplicitMSGinput[1].11	??	BOOL
Reserved		
ImplicitMSGinput[1].12	??	BOOL
Reserved		
ImplicitMSGinput[1].13	??	BOOL
Reserved		
ImplicitMSGinput[1].14	??	BOOL
Reserved		
ImplicitMSGinput[1].15	??	BOOL
Reserved		
ImplicitMSGinput[2]	??	INT
Fixed I/O (OUT)		
ImplicitMSGinput[2].0	??	BOOL
SEQ-BSY		
ImplicitMSGinput[2].1	??	BOOL
MOVE		
ImplicitMSGinput[2].2	??	BOOL
IN-POS		
ImplicitMSGinput[2].3	??	BOOL
START_R		
ImplicitMSGinput[2].4	??	BOOL
HOME-END		
ImplicitMSGinput[2].5	??	BOOL
READY		
ImplicitMSGinput[2].6	??	BOOL
DCMD-RDY		
ImplicitMSGinput[2].7	??	BOOL
ALM-A		
ImplicitMSGinput[2].8	??	BOOL
TRIG_R		
ImplicitMSGinput[2].9	??	BOOL
TRIG-MODE_R		
ImplicitMSGinput[2].10	??	BOOL
SET-ERR		
ImplicitMSGinput[2].11	??	BOOL
EXE-ERR		
ImplicitMSGinput[2].12	??	BOOL
DCMD-FULL		
ImplicitMSGinput[2].13	??	BOOL
STOP_R		
ImplicitMSGinput[2].14	??	BOOL
ETO-MON		
ImplicitMSGinput[2].15	??	BOOL
TLC		
ImplicitMSGinput[3]	??	INT
Present alarm		
ImplicitMSGinput[4]	??	INT
Feedback position (lower)		
ImplicitMSGinput[5]	??	INT
Feedback position (upper)		
ImplicitMSGinput[6]	??	INT
Feedback speed [Hz] (lower)		
ImplicitMSGinput[7]	??	INT
Feedback speed [Hz] (upper)		
ImplicitMSGinput[8]	??	INT
Command position (lower)		
ImplicitMSGinput[9]	??	INT
Command position (upper)		
ImplicitMSGinput[10]	??	INT

ImplicitMSGinput (Continued)		
Torque monitor		
ImplicitMSGinput[11]	??	INT
CST operating current		
ImplicitMSGinput[12]	??	INT
Information (lower)		
ImplicitMSGinput[13]	??	INT
Information (upper)		
ImplicitMSGinput[14]	??	INT
Reserved		
ImplicitMSGinput[15]	??	INT
Read parameter ID_R		
ImplicitMSGinput[16]	??	INT
Read/write status		
ImplicitMSGinput[16].0	??	BOOL
Reserved		
ImplicitMSGinput[16].1	??	BOOL
Reserved		
ImplicitMSGinput[16].2	??	BOOL
Reserved		
ImplicitMSGinput[16].3	??	BOOL
Reserved		
ImplicitMSGinput[16].4	??	BOOL
Reserved		
ImplicitMSGinput[16].5	??	BOOL
Reserved		
ImplicitMSGinput[16].6	??	BOOL
Reserved		
ImplicitMSGinput[16].7	??	BOOL
RD-ERR		
<i>ImplicitMSGinput[16].7 - OM_AZ_READ/Logic - 2(XIC)</i>		
ImplicitMSGinput[16].8	??	BOOL
WR-END		
ImplicitMSGinput[16].9	??	BOOL
SYS-BSY		
ImplicitMSGinput[16].10	??	BOOL
Reserved		
ImplicitMSGinput[16].11	??	BOOL
WR-SET-ERR		
ImplicitMSGinput[16].12	??	BOOL
WR-IF-ERR		
ImplicitMSGinput[16].13	??	BOOL
WR-NV-ERR		
ImplicitMSGinput[16].14	??	BOOL
WR-EXE-ERR		
ImplicitMSGinput[16].15	??	BOOL
WR-ERR		
ImplicitMSGinput[17]	??	INT
Write parameter ID_R		
ImplicitMSGinput[18]	??	INT
Read data (lower)		
ImplicitMSGinput[19]	??	INT
Read data (upper)		
ImplicitMSGinput[20]	??	INT
Assignable monitor 0 (lower)		
ImplicitMSGinput[21]	??	INT
Assignable monitor 0 (upper)		
ImplicitMSGinput[22]	??	INT
Assignable monitor 1 (lower)		
ImplicitMSGinput[23]	??	INT
Assignable monitor 1 (upper)		
ImplicitMSGinput[24]	??	INT
Assignable monitor 2 (lower)		

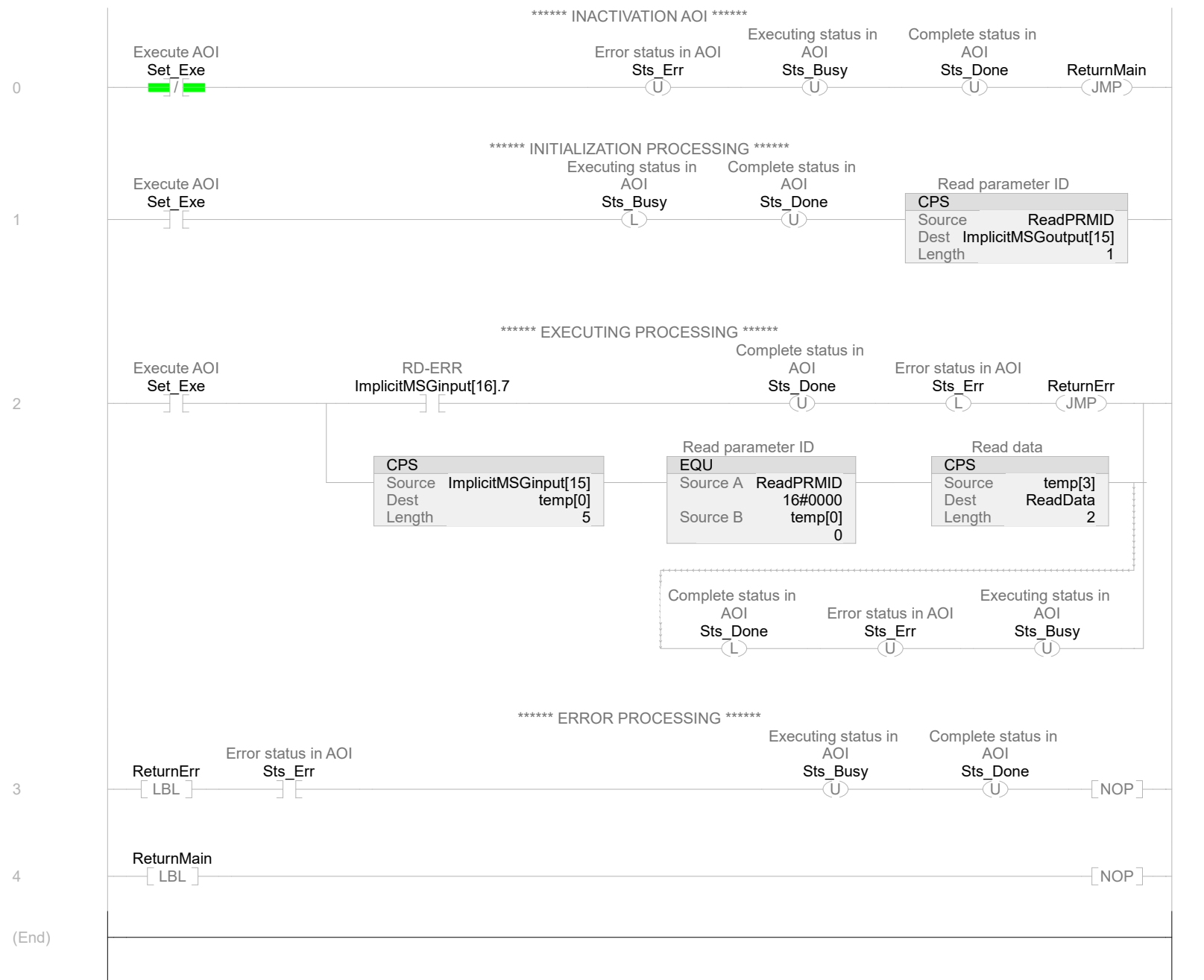
ImplicitMSGinput (Continued)		
ImplicitMSGinput[25]	??	INT
Assignable monitor 2 (upper)		
ImplicitMSGinput[26]	??	INT
Assignable monitor 3 (lower)		
ImplicitMSGinput[27]	??	INT
Assignable monitor 3 (upper)		
ImplicitMSGoutput		INT[20] OM_AZ_READ
Implicit message Output (scanner->driver)		
Usage:	InOut Parameter	
Required:	Yes	
Visible:	Yes	
Constant	No	
ImplicitMSGoutput[0]	??	INT
Remote I/O(R-IN)		
ImplicitMSGoutput[0].0	??	BOOL
R-IN0		
ImplicitMSGoutput[0].1	??	BOOL
R-IN1		
ImplicitMSGoutput[0].2	??	BOOL
R-IN2		
ImplicitMSGoutput[0].3	??	BOOL
R-IN3		
ImplicitMSGoutput[0].4	??	BOOL
R-IN4		
ImplicitMSGoutput[0].5	??	BOOL
R-IN5		
ImplicitMSGoutput[0].6	??	BOOL
R-IN6		
ImplicitMSGoutput[0].7	??	BOOL
R-IN7		
ImplicitMSGoutput[0].8	??	BOOL
R-IN8		
ImplicitMSGoutput[0].9	??	BOOL
R-IN9		
ImplicitMSGoutput[0].10	??	BOOL
R-IN10		
ImplicitMSGoutput[0].11	??	BOOL
R-IN11		
ImplicitMSGoutput[0].12	??	BOOL
R-IN12		
ImplicitMSGoutput[0].13	??	BOOL
R-IN13		
ImplicitMSGoutput[0].14	??	BOOL
R-IN14		
ImplicitMSGoutput[0].15	??	BOOL
R-IN15		
ImplicitMSGoutput[1]	??	INT
Operation data number selection		
ImplicitMSGoutput[1].0	??	BOOL
M0		
ImplicitMSGoutput[1].1	??	BOOL
M1		
ImplicitMSGoutput[1].2	??	BOOL
M2		
ImplicitMSGoutput[1].3	??	BOOL
M3		
ImplicitMSGoutput[1].4	??	BOOL
M4		
ImplicitMSGoutput[1].5	??	BOOL
M5		
ImplicitMSGoutput[1].6	??	BOOL

ImplicitMSGoutput (Continued)		
M6		
ImplicitMSGoutput[1].7	??	BOOL
M7		
ImplicitMSGoutput[1].8	??	BOOL
Reserved		
ImplicitMSGoutput[1].9	??	BOOL
Reserved		
ImplicitMSGoutput[1].10	??	BOOL
Reserved		
ImplicitMSGoutput[1].11	??	BOOL
Reserved		
ImplicitMSGoutput[1].12	??	BOOL
Reserved		
ImplicitMSGoutput[1].13	??	BOOL
Reserved		
ImplicitMSGoutput[1].14	??	BOOL
Reserved		
ImplicitMSGoutput[1].15	??	BOOL
Reserved		
ImplicitMSGoutput[2]	??	INT
Fixed I/O(IN)		
ImplicitMSGoutput[2].0	??	BOOL
FW-JOG		
ImplicitMSGoutput[2].1	??	BOOL
RV-JOG		
ImplicitMSGoutput[2].2	??	BOOL
Reserved		
ImplicitMSGoutput[2].3	??	BOOL
START		
ImplicitMSGoutput[2].4	??	BOOL
ZHOME		
ImplicitMSGoutput[2].5	??	BOOL
STOP		
ImplicitMSGoutput[2].6	??	BOOL
FREE		
ImplicitMSGoutput[2].7	??	BOOL
ALM-RST		
ImplicitMSGoutput[2].8	??	BOOL
TRIG		
ImplicitMSGoutput[2].9	??	BOOL
TRIG-MODE		
ImplicitMSGoutput[2].10	??	BOOL
ETO-CLR		
ImplicitMSGoutput[2].11	??	BOOL
Reserved		
ImplicitMSGoutput[2].12	??	BOOL
FW-JOG-P		
ImplicitMSGoutput[2].13	??	BOOL
RV-JOG-P		
ImplicitMSGoutput[2].14	??	BOOL
FW-POS		
ImplicitMSGoutput[2].15	??	BOOL
RV-POS		
ImplicitMSGoutput[3]	??	INT
Direct data operation operation type		
ImplicitMSGoutput[4]	??	INT
Direct data operation position (lower)		
ImplicitMSGoutput[5]	??	INT
Direct data operation position (upper)		
ImplicitMSGoutput[6]	??	INT
Direct data operation operating speed (lower)		
ImplicitMSGoutput[7]	??	INT

ImplicitMSGoutput (Continued)			
Direct data operation operating speed (upper)			
ImplicitMSGoutput[8]	??	INT	
Direct data operation starting/changing rate (lower)			
ImplicitMSGoutput[9]	??	INT	
Direct data operation starting/changing rate (upper)			
ImplicitMSGoutput[10]	??	INT	
Direct data operation stopping deceleration (lower)			
ImplicitMSGoutput[11]	??	INT	
Direct data operation stopping deceleration (upper)			
ImplicitMSGoutput[12]	??	INT	
Direct data operation operating current			
ImplicitMSGoutput[13]	??	INT	
Direct data operation forwarding destination			
ImplicitMSGoutput[14]	??	INT	
Reserved			
ImplicitMSGoutput[15]	??	INT	
Read parameter ID			
<i>ImplicitMSGoutput[15] - OM_AZ_READ/Logic - *I(CPS)</i>			
ImplicitMSGoutput[16]	??	INT	
Write request			
ImplicitMSGoutput[16].0	??	BOOL	
WR-REQ			
ImplicitMSGoutput[16].1	??	BOOL	
Reserved			
ImplicitMSGoutput[16].2	??	BOOL	
Reserved			
ImplicitMSGoutput[16].3	??	BOOL	
Reserved			
ImplicitMSGoutput[16].4	??	BOOL	
Reserved			
ImplicitMSGoutput[16].5	??	BOOL	
Reserved			
ImplicitMSGoutput[16].6	??	BOOL	
Reserved			
ImplicitMSGoutput[16].7	??	BOOL	
Reserved			
ImplicitMSGoutput[16].8	??	BOOL	
Reserved			
ImplicitMSGoutput[16].9	??	BOOL	
Reserved			
ImplicitMSGoutput[16].10	??	BOOL	
Reserved			
ImplicitMSGoutput[16].11	??	BOOL	
Reserved			
ImplicitMSGoutput[16].12	??	BOOL	
Reserved			
ImplicitMSGoutput[16].13	??	BOOL	
Reserved			
ImplicitMSGoutput[16].14	??	BOOL	
Reserved			
ImplicitMSGoutput[16].15	??	BOOL	
Reserved			
ImplicitMSGoutput[17]	??	INT	
Write parameter ID			
ImplicitMSGoutput[18]	??	INT	
Write data (lower)			
ImplicitMSGoutput[19]	??	INT	
Write data (upper)			
ReadData	0	DINT	OM_AZ_READ
Read data			
Usage:	Output Parameter		

ReadData (Continued)			
Required:	No		
Visible:	Yes		
External Access:	Read Only		
ReadData - OM_AZ_READ/Logic - *2(CPS)			
ReadPRMID	16#0000	INT	OM_AZ_READ
Read parameter ID			
Usage:	Input Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
ReadPRMID - OM_AZ_READ/Logic - 1(CPS), 2(EQU)			
Set_Exec	0	BOOL	OM_AZ_READ
Execute AOI			
Usage:	Input Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
Set_Exec - OM_AZ_READ/Logic - 0(XIO), 1(XIC), 2(XIC)			
Sts_Busy	0	BOOL	OM_AZ_READ
Executing status in AOI			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	None		
Sts_Busy - OM_AZ_READ/Logic - *0(OTU), *1(OTL), *2(OTU), *3(OTU)			
Sts_Done	0	BOOL	OM_AZ_READ
Complete status in AOI			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	None		
Sts_Done - OM_AZ_READ/Logic - *0(OTU), *1(OTU), *2(OTL), *2(OTU), *3(OTU)			
Sts_Err	0	BOOL	OM_AZ_READ
Error status in AOI			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	None		
Sts_Err - OM_AZ_READ/Logic - *0(OTU), *2(OTL), *2(OTU), 3(XIC)			

Name	Default	Data Type	Scope
temp		INT[5]	OM_AZ_READ
Usage:	Local Tag		
External Access:	None		
<i>temp</i> - OM_AZ_READ/Logic - *2(CPS), 2(CPS)			
temp[0]	0	INT	
<i>temp[0]</i> - OM_AZ_READ/Logic - 2(EQU)			

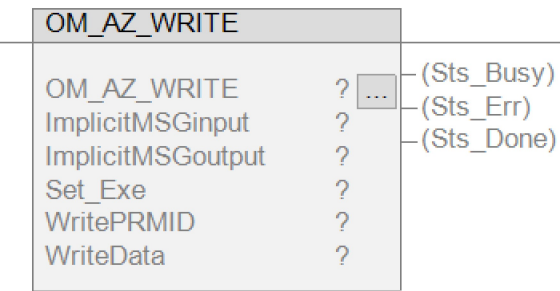


OM_AZ_WRITE v1.3
ORIENTALMOTOR CO.,LTD.

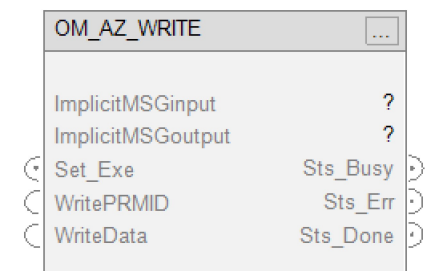
Class: Standard

Available Languages

Relay Ladder



Function Block



Structured Text

OM_AZ_WRITE(ImplicitMSGInput, ImplicitMSGOutput);

Parameters

Required	Name	Data Type	Usage	Description
X	OM_AZ_WRITE	OM_AZ_WRITE	InOut	
	EnableIn	BOOL	Input	
	EnableOut	BOOL	Output	
X	ImplicitMSGInput	INT	InOut	Implicit message Input (driver->scanner)
X	ImplicitMSGOutput	INT	InOut	Implicit message Output (scanner->driver)
	Set_Exe	BOOL	Input	Execute AOI
	WritePRMID	INT	Input	Write parameter ID
	WriteData	DINT	Input	Write data
	Sts_Busy	BOOL	Output	Executing status in AOI
	Sts_Err	BOOL	Output	Error status in AOI
	Sts_Done	BOOL	Output	Complete status in AOI

Extended Description

This AOI is used to write data from the scanner to the driver via Implicit communication.

Execution

Condition	Description
EnableIn is true	

Revision v1.3 Notes

Implicit message based on AZD-AEP,-CEP and -KEP.

Name	Default	Data Type	Scope
ImplicitMSGinput		INT[28]	OM_AZ_WRITE
Implicit message Input (driver->scanner)			
Usage:	InOut Parameter		
Required:	Yes		
Visible:	Yes		
Constant	No		
ImplicitMSGinput[0]	??	INT	
Remote I/O (R-OUT)			
ImplicitMSGinput[0].0	??	BOOL	
R-OUT0			
ImplicitMSGinput[0].1	??	BOOL	
R-OUT1			
ImplicitMSGinput[0].2	??	BOOL	
R-OUT2			
ImplicitMSGinput[0].3	??	BOOL	
R-OUT3			
ImplicitMSGinput[0].4	??	BOOL	
R-OUT4			
ImplicitMSGinput[0].5	??	BOOL	
R-OUT5			
ImplicitMSGinput[0].6	??	BOOL	
R-OUT6			
ImplicitMSGinput[0].7	??	BOOL	
R-OUT7			
ImplicitMSGinput[0].8	??	BOOL	
R-OUT8			
ImplicitMSGinput[0].9	??	BOOL	
R-OUT9			
ImplicitMSGinput[0].10	??	BOOL	
R-OUT10			
ImplicitMSGinput[0].11	??	BOOL	
R-OUT11			
ImplicitMSGinput[0].12	??	BOOL	
R-OUT12			
ImplicitMSGinput[0].13	??	BOOL	
R-OUT13			
ImplicitMSGinput[0].14	??	BOOL	
R-OUT14			
ImplicitMSGinput[0].15	??	BOOL	
R-OUT15			
ImplicitMSGinput[1]	??	INT	
Operation data number selection_R			
ImplicitMSGinput[1].0	??	BOOL	
M0_R			
ImplicitMSGinput[1].1	??	BOOL	
M1_R			
ImplicitMSGinput[1].2	??	BOOL	
M2_R			
ImplicitMSGinput[1].3	??	BOOL	
M3_R			
ImplicitMSGinput[1].4	??	BOOL	
M4_R			
ImplicitMSGinput[1].5	??	BOOL	
M5_R			
ImplicitMSGinput[1].6	??	BOOL	
M6_R			
ImplicitMSGinput[1].7	??	BOOL	
M7_R			
ImplicitMSGinput[1].8	??	BOOL	
Reserved			
ImplicitMSGinput[1].9	??	BOOL	
Reserved			

ImplicitMSGinput (Continued)		
ImplicitMSGinput[1].10	??	BOOL
Reserved		
ImplicitMSGinput[1].11	??	BOOL
Reserved		
ImplicitMSGinput[1].12	??	BOOL
Reserved		
ImplicitMSGinput[1].13	??	BOOL
Reserved		
ImplicitMSGinput[1].14	??	BOOL
Reserved		
ImplicitMSGinput[1].15	??	BOOL
Reserved		
ImplicitMSGinput[2]	??	INT
Fixed I/O (OUT)		
ImplicitMSGinput[2].0	??	BOOL
SEQ-BSY		
ImplicitMSGinput[2].1	??	BOOL
MOVE		
ImplicitMSGinput[2].2	??	BOOL
IN-POS		
ImplicitMSGinput[2].3	??	BOOL
START_R		
ImplicitMSGinput[2].4	??	BOOL
HOME-END		
ImplicitMSGinput[2].5	??	BOOL
READY		
ImplicitMSGinput[2].6	??	BOOL
DCMD-RDY		
ImplicitMSGinput[2].7	??	BOOL
ALM-A		
ImplicitMSGinput[2].8	??	BOOL
TRIG_R		
ImplicitMSGinput[2].9	??	BOOL
TRIG-MODE_R		
ImplicitMSGinput[2].10	??	BOOL
SET-ERR		
ImplicitMSGinput[2].11	??	BOOL
EXE-ERR		
ImplicitMSGinput[2].12	??	BOOL
DCMD-FULL		
ImplicitMSGinput[2].13	??	BOOL
STOP_R		
ImplicitMSGinput[2].14	??	BOOL
ETO-MON		
ImplicitMSGinput[2].15	??	BOOL
TLC		
ImplicitMSGinput[3]	??	INT
Present alarm		
ImplicitMSGinput[4]	??	INT
Feedback position (lower)		
ImplicitMSGinput[5]	??	INT
Feedback position (upper)		
ImplicitMSGinput[6]	??	INT
Feedback speed [Hz] (lower)		
ImplicitMSGinput[7]	??	INT
Feedback speed [Hz] (upper)		
ImplicitMSGinput[8]	??	INT
Command position (lower)		
ImplicitMSGinput[9]	??	INT
Command position (upper)		
ImplicitMSGinput[10]	??	INT
Torque monitor		

ImplicitMSGInput (Continued)		
ImplicitMSGInput[11]	??	INT
CST operating current		
ImplicitMSGInput[12]	??	INT
Information (lower)		
ImplicitMSGInput[13]	??	INT
Information (upper)		
ImplicitMSGInput[14]	??	INT
Reserved		
ImplicitMSGInput[15]	??	INT
Read parameter ID_R		
ImplicitMSGInput[16]	??	INT
Read/write status		
<i>ImplicitMSGInput[16] - OM_AZ_WRITE/Logic - 2(CPS), 3(CPS)</i>		
ImplicitMSGInput[16].0	??	BOOL
Reserved		
ImplicitMSGInput[16].1	??	BOOL
Reserved		
ImplicitMSGInput[16].2	??	BOOL
Reserved		
ImplicitMSGInput[16].3	??	BOOL
Reserved		
ImplicitMSGInput[16].4	??	BOOL
Reserved		
ImplicitMSGInput[16].5	??	BOOL
Reserved		
ImplicitMSGInput[16].6	??	BOOL
Reserved		
ImplicitMSGInput[16].7	??	BOOL
RD-ERR		
ImplicitMSGInput[16].8	??	BOOL
WR-END		
<i>ImplicitMSGInput[16].8 - OM_AZ_WRITE/Logic - 1(XIC), 1(XIO), 4(XIO)</i>		
ImplicitMSGInput[16].9	??	BOOL
SYS-BSY		
ImplicitMSGInput[16].10	??	BOOL
Reserved		
ImplicitMSGInput[16].11	??	BOOL
WR-SET-ERR		
ImplicitMSGInput[16].12	??	BOOL
WR-IF-ERR		
ImplicitMSGInput[16].13	??	BOOL
WR-NV-ERR		
ImplicitMSGInput[16].14	??	BOOL
WR-EXE-ERR		
ImplicitMSGInput[16].15	??	BOOL
WR-ERR		
ImplicitMSGInput[17]	??	INT
Write parameter ID_R		
ImplicitMSGInput[18]	??	INT
Read data (lower)		
ImplicitMSGInput[19]	??	INT
Read data (upper)		
ImplicitMSGInput[20]	??	INT
Assignable monitor 0 (lower)		
ImplicitMSGInput[21]	??	INT
Assignable monitor 0 (upper)		
ImplicitMSGInput[22]	??	INT
Assignable monitor 1 (lower)		
ImplicitMSGInput[23]	??	INT
Assignable monitor 1 (upper)		
ImplicitMSGInput[24]	??	INT
Assignable monitor 2 (lower)		

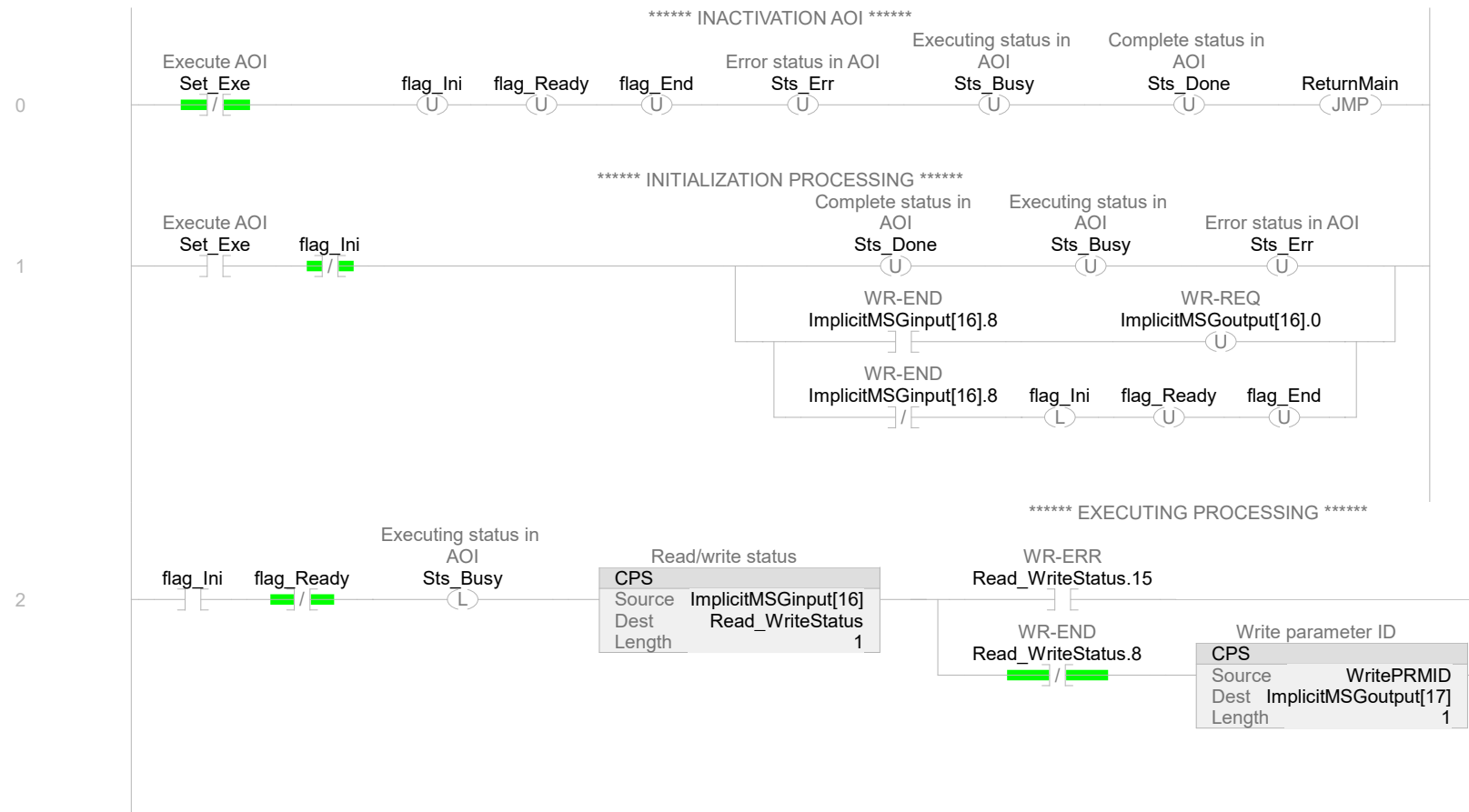
ImplicitMSGinput (Continued)		
ImplicitMSGinput[25]	??	INT
Assignable monitor 2 (upper)		
ImplicitMSGinput[26]	??	INT
Assignable monitor 3 (lower)		
ImplicitMSGinput[27]	??	INT
Assignable monitor 3 (upper)		
ImplicitMSGoutput	INT[20]	OM_AZ_WRITE
Implicit message Output (scanner->driver)		
Usage:	InOut Parameter	
Required:	Yes	
Visible:	Yes	
Constant	No	
<i>ImplicitMSGoutput - OM_AZ_WRITE/Logic - *2(CPS)</i>		
ImplicitMSGoutput[0]	??	INT
Remote I/O(R-IN)		
ImplicitMSGoutput[0].0	??	BOOL
R-IN0		
ImplicitMSGoutput[0].1	??	BOOL
R-IN1		
ImplicitMSGoutput[0].2	??	BOOL
R-IN2		
ImplicitMSGoutput[0].3	??	BOOL
R-IN3		
ImplicitMSGoutput[0].4	??	BOOL
R-IN4		
ImplicitMSGoutput[0].5	??	BOOL
R-IN5		
ImplicitMSGoutput[0].6	??	BOOL
R-IN6		
ImplicitMSGoutput[0].7	??	BOOL
R-IN7		
ImplicitMSGoutput[0].8	??	BOOL
R-IN8		
ImplicitMSGoutput[0].9	??	BOOL
R-IN9		
ImplicitMSGoutput[0].10	??	BOOL
R-IN10		
ImplicitMSGoutput[0].11	??	BOOL
R-IN11		
ImplicitMSGoutput[0].12	??	BOOL
R-IN12		
ImplicitMSGoutput[0].13	??	BOOL
R-IN13		
ImplicitMSGoutput[0].14	??	BOOL
R-IN14		
ImplicitMSGoutput[0].15	??	BOOL
R-IN15		
ImplicitMSGoutput[1]	??	INT
Operation data number selection		
ImplicitMSGoutput[1].0	??	BOOL
M0		
ImplicitMSGoutput[1].1	??	BOOL
M1		
ImplicitMSGoutput[1].2	??	BOOL
M2		
ImplicitMSGoutput[1].3	??	BOOL
M3		
ImplicitMSGoutput[1].4	??	BOOL
M4		
ImplicitMSGoutput[1].5	??	BOOL
M5		

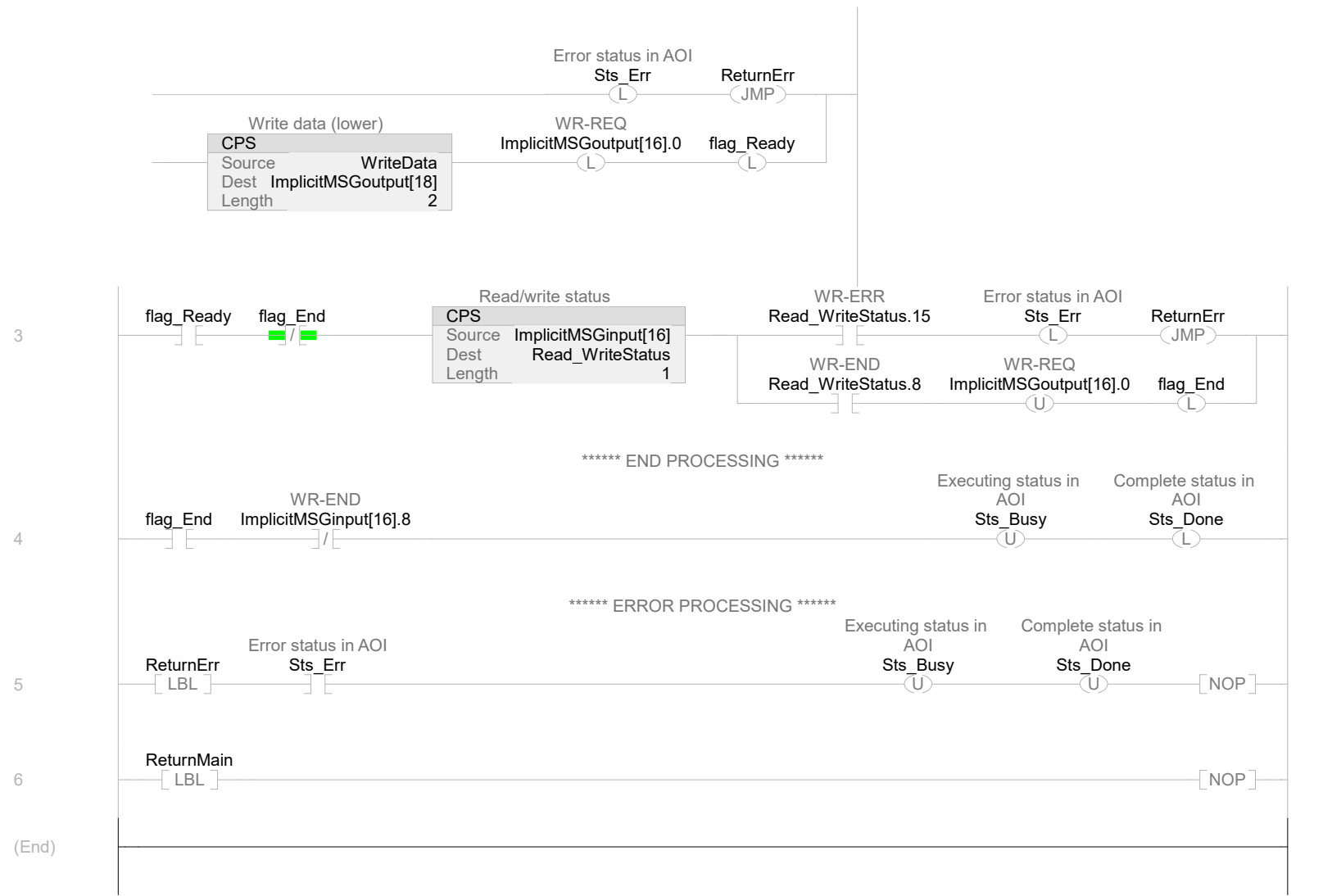
ImplicitMSGoutput (Continued)		
ImplicitMSGoutput[1].6 M6	??	BOOL
ImplicitMSGoutput[1].7 M7	??	BOOL
ImplicitMSGoutput[1].8 Reserved	??	BOOL
ImplicitMSGoutput[1].9 Reserved	??	BOOL
ImplicitMSGoutput[1].10 Reserved	??	BOOL
ImplicitMSGoutput[1].11 Reserved	??	BOOL
ImplicitMSGoutput[1].12 Reserved	??	BOOL
ImplicitMSGoutput[1].13 Reserved	??	BOOL
ImplicitMSGoutput[1].14 Reserved	??	BOOL
ImplicitMSGoutput[1].15 Reserved	??	BOOL
ImplicitMSGoutput[2] Fixed I/O(IN)	??	INT
ImplicitMSGoutput[2].0 FW-JOG	??	BOOL
ImplicitMSGoutput[2].1 RV-JOG	??	BOOL
ImplicitMSGoutput[2].2 Reserved	??	BOOL
ImplicitMSGoutput[2].3 START	??	BOOL
ImplicitMSGoutput[2].4 ZHOME	??	BOOL
ImplicitMSGoutput[2].5 STOP	??	BOOL
ImplicitMSGoutput[2].6 FREE	??	BOOL
ImplicitMSGoutput[2].7 ALM-RST	??	BOOL
ImplicitMSGoutput[2].8 TRIG	??	BOOL
ImplicitMSGoutput[2].9 TRIG-MODE	??	BOOL
ImplicitMSGoutput[2].10 ETO-CLR	??	BOOL
ImplicitMSGoutput[2].11 Reserved	??	BOOL
ImplicitMSGoutput[2].12 FW-JOG-P	??	BOOL
ImplicitMSGoutput[2].13 RV-JOG-P	??	BOOL
ImplicitMSGoutput[2].14 FW-POS	??	BOOL
ImplicitMSGoutput[2].15 RV-POS	??	BOOL
ImplicitMSGoutput[3] Direct data operation operation type	??	INT
ImplicitMSGoutput[4] Direct data operation position (lower)	??	INT
ImplicitMSGoutput[5] Direct data operation position (upper)	??	INT
ImplicitMSGoutput[6] Direct data operation operating speed (lower)	??	INT

ImplicitMSGGoutput (Continued)			
ImplicitMSGGoutput[7]		??	INT
Direct data operation operating speed (upper)			
ImplicitMSGGoutput[8]		??	INT
Direct data operation starting/changing rate (lower)			
ImplicitMSGGoutput[9]		??	INT
Direct data operation starting/changing rate (upper)			
ImplicitMSGGoutput[10]		??	INT
Direct data operation stopping deceleration (lower)			
ImplicitMSGGoutput[11]		??	INT
Direct data operation stopping deceleration (upper)			
ImplicitMSGGoutput[12]		??	INT
Direct data operation operating current			
ImplicitMSGGoutput[13]		??	INT
Direct data operation forwarding destination			
ImplicitMSGGoutput[14]		??	INT
Reserved			
ImplicitMSGGoutput[15]		??	INT
Read parameter ID			
ImplicitMSGGoutput[16]		??	INT
Write request			
ImplicitMSGGoutput[16].0		??	BOOL
WR-REQ			
ImplicitMSGGoutput[16].0 - OM_AZ_WRITE/Logic - *1(OTU), *2(OTL), *3(OTU)			
ImplicitMSGGoutput[16].1		??	BOOL
Reserved			
ImplicitMSGGoutput[16].2		??	BOOL
Reserved			
ImplicitMSGGoutput[16].3		??	BOOL
Reserved			
ImplicitMSGGoutput[16].4		??	BOOL
Reserved			
ImplicitMSGGoutput[16].5		??	BOOL
Reserved			
ImplicitMSGGoutput[16].6		??	BOOL
Reserved			
ImplicitMSGGoutput[16].7		??	BOOL
Reserved			
ImplicitMSGGoutput[16].8		??	BOOL
Reserved			
ImplicitMSGGoutput[16].9		??	BOOL
Reserved			
ImplicitMSGGoutput[16].10		??	BOOL
Reserved			
ImplicitMSGGoutput[16].11		??	BOOL
Reserved			
ImplicitMSGGoutput[16].12		??	BOOL
Reserved			
ImplicitMSGGoutput[16].13		??	BOOL
Reserved			
ImplicitMSGGoutput[16].14		??	BOOL
Reserved			
ImplicitMSGGoutput[16].15		??	BOOL
Reserved			
ImplicitMSGGoutput[17]		??	INT
Write parameter ID			
ImplicitMSGGoutput[17] - OM_AZ_WRITE/Logic - *2(CPS)			
ImplicitMSGGoutput[18]		??	INT
Write data (lower)			
ImplicitMSGGoutput[19]		??	INT
Write data (upper)			
Set_Exec		0	BOOL
			OM_AZ_WRITE

Set_Exec (Continued)			
Execute AOI			
Usage:	Input Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
Set_Exec - OM_AZ_WRITE/Logic - 0(XIO), 1(XIC)			
Sts_Busy	0	BOOL	OM_AZ_WRITE
Executing status in AOI			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	None		
Sts_Busy - OM_AZ_WRITE/Logic - *0(OTU), *1(OTU), *2(OTL), *4(OTU), *5(OTU)			
Sts_Done	0	BOOL	OM_AZ_WRITE
Complete status in AOI			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	None		
Sts_Done - OM_AZ_WRITE/Logic - *0(OTU), *1(OTU), *4(OTL), *5(OTU)			
Sts_Err	0	BOOL	OM_AZ_WRITE
Error status in AOI			
Usage:	Output Parameter		
Required:	No		
Visible:	Yes		
External Access:	None		
Sts_Err - OM_AZ_WRITE/Logic - *0(OTU), *1(OTU), *2(OTL), *3(OTL), 5(XIC)			
WriteData	0	DINT	OM_AZ_WRITE
Write data			
Usage:	Input Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
WriteData - OM_AZ_WRITE/Logic - 2(CPS)			
WritePRMID	16#0000	INT	OM_AZ_WRITE
Write parameter ID			
Usage:	Input Parameter		
Required:	No		
Visible:	Yes		
External Access:	Read/Write		
WritePRMID - OM_AZ_WRITE/Logic - 2(CPS)			

Name	Default	Data Type	Scope
flag_End	0	BOOL	OM_AZ_WRITE
Usage:	Local Tag		
External Access:	None		
<i>flag_End - OM_AZ_WRITE/Logic - *0(OTU), *1(OTU), *3(OTL), 3(XIO), 4(XIC)</i>			
flag_Ini	0	BOOL	OM_AZ_WRITE
Usage:	Local Tag		
External Access:	None		
<i>flag_Ini - OM_AZ_WRITE/Logic - *0(OTU), *1(OTL), 1(XIO), 2(XIC)</i>			
flag_Ready	0	BOOL	OM_AZ_WRITE
Usage:	Local Tag		
External Access:	None		
<i>flag_Ready - OM_AZ_WRITE/Logic - *0(OTU), *1(OTU), *2(OTL), 2(XIO), 3(XIC)</i>			
Read_WriteStatus	0	INT	OM_AZ_WRITE
Read/write status			
Usage:	Local Tag		
External Access:	None		
<i>Read_WriteStatus - OM_AZ_WRITE/Logic - *2(CPS), *3(CPS)</i>			
Read_WriteStatus.0	0	BOOL	
Reserved			
Read_WriteStatus.1	0	BOOL	
Reserved			
Read_WriteStatus.2	0	BOOL	
Reserved			
Read_WriteStatus.3	0	BOOL	
Reserved			
Read_WriteStatus.4	0	BOOL	
Reserved			
Read_WriteStatus.5	0	BOOL	
Reserved			
Read_WriteStatus.6	0	BOOL	
Reserved			
Read_WriteStatus.7	0	BOOL	
RD-ERR			
Read_WriteStatus.8	0	BOOL	
WR-END			
<i>Read_WriteStatus.8 - OM_AZ_WRITE/Logic - 2(XIO), 3(XIC)</i>			
Read_WriteStatus.9	0	BOOL	
SYS-BSY			
Read_WriteStatus.10	0	BOOL	
Reserved			
Read_WriteStatus.11	0	BOOL	
WR-SET-ERR			
Read_WriteStatus.12	0	BOOL	
WR-IF-ERR			
Read_WriteStatus.13	0	BOOL	
WR-NV-ERR			
Read_WriteStatus.14	0	BOOL	
WR-EXE-ERR			
Read_WriteStatus.15	0	BOOL	
WR-ERR			
<i>Read_WriteStatus.15 - OM_AZ_WRITE/Logic - 2(XIC), 3(XIC)</i>			



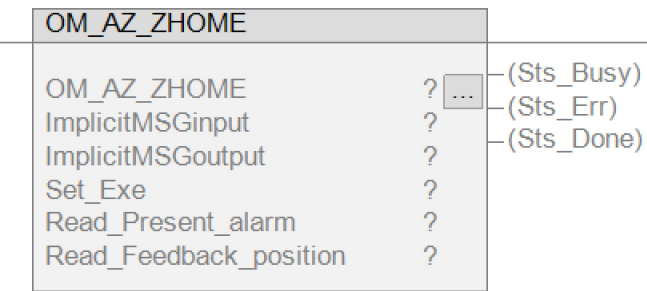


OM_AZ_ZHOME v1.3
ORIENTALMOTOR CO.,LTD.

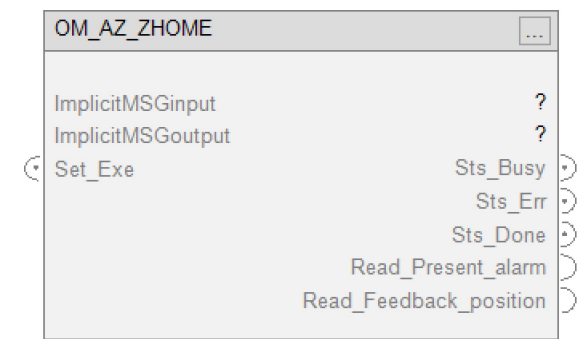
Class: Standard

Available Languages

Relay Ladder



Function Block



Structured Text

OM_AZ_ZHOME(ImplicitMSGinput, ImplicitMSGoutput);

Parameters

Required	Name	Data Type	Usage	Description
X	OM_AZ_ZHOME	OM_AZ_ZHOME	InOut	
	EnableIn	BOOL	Input	
	EnableOut	BOOL	Output	
X	ImplicitMSGinput	INT	InOut	Implicit message Input (driver->scanner)
X	ImplicitMSGoutput	INT	InOut	Implicit message Output (scanner->driver)
	Set_Exe	BOOL	Input	Execute AOI
	Sts_Busy	BOOL	Output	Executing status in AOI
	Sts_Err	BOOL	Output	Error status in AOI
	Sts_Done	BOOL	Output	Complete status in AOI
	Read_Present_alarm	INT	Output	Alarm code monitor[Hex]
	Read_Feedback_position	DINT	Output	Feedback position monitor[step]

Extended Description

This AOI is used to execute high-speed return-to-home operation.

Execution

Condition Description

EnableIn is true

Revision v1.3 Notes

Implicit message based on AZD-AEP,-CEP and -KEP.

Name	Default	Data Type	Scope
ImplicitMSGinput		INT[28]	OM_AZ_ZHOME
Implicit message Input (driver->scanner)			
Usage:	InOut Parameter		
Required:	Yes		
Visible:	Yes		
Constant	No		
<i>ImplicitMSGinput - OM_AZ_ZHOME/Logic - 0(CPS)</i>			
ImplicitMSGinput[0]	??	INT	
Remote I/O (R-OUT)			
ImplicitMSGinput[0].0	??	BOOL	
R-OUT0			
ImplicitMSGinput[0].1	??	BOOL	
R-OUT1			
ImplicitMSGinput[0].2	??	BOOL	
R-OUT2			
ImplicitMSGinput[0].3	??	BOOL	
R-OUT3			
ImplicitMSGinput[0].4	??	BOOL	
R-OUT4			
ImplicitMSGinput[0].5	??	BOOL	
R-OUT5			
ImplicitMSGinput[0].6	??	BOOL	
R-OUT6			
ImplicitMSGinput[0].7	??	BOOL	
R-OUT7			
ImplicitMSGinput[0].8	??	BOOL	
R-OUT8			
ImplicitMSGinput[0].9	??	BOOL	
R-OUT9			
ImplicitMSGinput[0].10	??	BOOL	
R-OUT10			
ImplicitMSGinput[0].11	??	BOOL	
R-OUT11			
ImplicitMSGinput[0].12	??	BOOL	
R-OUT12			
ImplicitMSGinput[0].13	??	BOOL	
R-OUT13			
ImplicitMSGinput[0].14	??	BOOL	
R-OUT14			
ImplicitMSGinput[0].15	??	BOOL	
R-OUT15			
ImplicitMSGinput[1]	??	INT	
Operation data number selection_R			
ImplicitMSGinput[1].0	??	BOOL	
M0_R			
ImplicitMSGinput[1].1	??	BOOL	
M1_R			
ImplicitMSGinput[1].2	??	BOOL	
M2_R			
ImplicitMSGinput[1].3	??	BOOL	
M3_R			
ImplicitMSGinput[1].4	??	BOOL	
M4_R			
ImplicitMSGinput[1].5	??	BOOL	
M5_R			
ImplicitMSGinput[1].6	??	BOOL	
M6_R			
ImplicitMSGinput[1].7	??	BOOL	
M7_R			
ImplicitMSGinput[1].8	??	BOOL	
Reserved			
ImplicitMSGinput[1].9	??	BOOL	

ImplicitMSGInput (Continued)		
Reserved		
ImplicitMSGInput[1].10	??	BOOL
Reserved		
ImplicitMSGInput[1].11	??	BOOL
Reserved		
ImplicitMSGInput[1].12	??	BOOL
Reserved		
ImplicitMSGInput[1].13	??	BOOL
Reserved		
ImplicitMSGInput[1].14	??	BOOL
Reserved		
ImplicitMSGInput[1].15	??	BOOL
Reserved		
ImplicitMSGInput[2]	??	INT
Fixed I/O (OUT)		
<i>ImplicitMSGInput[2] - OM_AZ_ZHOME/Logic - 4(CPS)</i>		
ImplicitMSGInput[2].0	??	BOOL
SEQ-BSY		
ImplicitMSGInput[2].1	??	BOOL
MOVE		
ImplicitMSGInput[2].2	??	BOOL
IN-POS		
ImplicitMSGInput[2].3	??	BOOL
START_R		
ImplicitMSGInput[2].4	??	BOOL
HOME-END		
<i>ImplicitMSGInput[2].4 - OM_AZ_ZHOME/Logic - 4(XIC)</i>		
ImplicitMSGInput[2].5	??	BOOL
READY		
<i>ImplicitMSGInput[2].5 - OM_AZ_ZHOME/Logic - 3(XIO), 4(XIO)</i>		
ImplicitMSGInput[2].6	??	BOOL
DCMD-RDY		
<i>ImplicitMSGInput[2].6 - OM_AZ_ZHOME/Logic - 3(XIO)</i>		
ImplicitMSGInput[2].7	??	BOOL
ALM-A		
<i>ImplicitMSGInput[2].7 - OM_AZ_ZHOME/Logic - 3(XIC)</i>		
ImplicitMSGInput[2].8	??	BOOL
TRIG_R		
ImplicitMSGInput[2].9	??	BOOL
TRIG-MODE_R		
ImplicitMSGInput[2].10	??	BOOL
SET-ERR		
ImplicitMSGInput[2].11	??	BOOL
EXE-ERR		
ImplicitMSGInput[2].12	??	BOOL
DCMD-FULL		
ImplicitMSGInput[2].13	??	BOOL
STOP_R		
<i>ImplicitMSGInput[2].13 - OM_AZ_ZHOME/Logic - 3(XIC)</i>		
ImplicitMSGInput[2].14	??	BOOL
ETO-MON		
<i>ImplicitMSGInput[2].14 - OM_AZ_ZHOME/Logic - 3(XIC)</i>		
ImplicitMSGInput[2].15	??	BOOL
TLC		
ImplicitMSGInput[3]	??	INT
Present alarm		
<i>ImplicitMSGInput[3] - OM_AZ_ZHOME/Logic - 0(CPS)</i>		
ImplicitMSGInput[4]	??	INT
Feedback position (lower)		
ImplicitMSGInput[5]	??	INT
Feedback position (upper)		
ImplicitMSGInput[6]	??	INT

ImplicitMSGinput (Continued)		
Feedback speed [Hz] (lower)		
ImplicitMSGinput[7]	??	INT
Feedback speed [Hz] (upper)		
ImplicitMSGinput[8]	??	INT
Command position (lower)		
ImplicitMSGinput[9]	??	INT
Command position (upper)		
ImplicitMSGinput[10]	??	INT
Torque monitor		
ImplicitMSGinput[11]	??	INT
CST operating current		
ImplicitMSGinput[12]	??	INT
Information (lower)		
ImplicitMSGinput[13]	??	INT
Information (upper)		
ImplicitMSGinput[14]	??	INT
Reserved		
ImplicitMSGinput[15]	??	INT
Read parameter ID_R		
ImplicitMSGinput[16]	??	INT
Read/write status		
ImplicitMSGinput[16].0	??	BOOL
Reserved		
ImplicitMSGinput[16].1	??	BOOL
Reserved		
ImplicitMSGinput[16].2	??	BOOL
Reserved		
ImplicitMSGinput[16].3	??	BOOL
Reserved		
ImplicitMSGinput[16].4	??	BOOL
Reserved		
ImplicitMSGinput[16].5	??	BOOL
Reserved		
ImplicitMSGinput[16].6	??	BOOL
Reserved		
ImplicitMSGinput[16].7	??	BOOL
RD-ERR		
ImplicitMSGinput[16].8	??	BOOL
WR-END		
ImplicitMSGinput[16].9	??	BOOL
SYS-BSY		
ImplicitMSGinput[16].10	??	BOOL
Reserved		
ImplicitMSGinput[16].11	??	BOOL
WR-SET-ERR		
ImplicitMSGinput[16].12	??	BOOL
WR-IF-ERR		
ImplicitMSGinput[16].13	??	BOOL
WR-NV-ERR		
ImplicitMSGinput[16].14	??	BOOL
WR-EXE-ERR		
ImplicitMSGinput[16].15	??	BOOL
WR-ERR		
ImplicitMSGinput[17]	??	INT
Write parameter ID_R		
ImplicitMSGinput[18]	??	INT
Read data (lower)		
ImplicitMSGinput[19]	??	INT
Read data (upper)		
ImplicitMSGinput[20]	??	INT
Assignable monitor 0 (lower)		
ImplicitMSGinput[21]	??	INT

ImplicitMSGinput (Continued)		
Assignable monitor 0 (upper)		
ImplicitMSGinput[22]	??	INT
Assignable monitor 1 (lower)		
ImplicitMSGinput[23]	??	INT
Assignable monitor 1 (upper)		
ImplicitMSGinput[24]	??	INT
Assignable monitor 2 (lower)		
ImplicitMSGinput[25]	??	INT
Assignable monitor 2 (upper)		
ImplicitMSGinput[26]	??	INT
Assignable monitor 3 (lower)		
ImplicitMSGinput[27]	??	INT
Assignable monitor 3 (upper)		
ImplicitMSGoutput	INT[20]	OM_AZ_ZHOME
Implicit message Output (scanner->driver)		
Usage:	InOut	Parameter
Required:	Yes	
Visible:	Yes	
Constant	No	
ImplicitMSGoutput[0]	??	INT
Remote I/O(R-IN)		
ImplicitMSGoutput[0].0	??	BOOL
R-IN0		
ImplicitMSGoutput[0].1	??	BOOL
R-IN1		
ImplicitMSGoutput[0].2	??	BOOL
R-IN2		
ImplicitMSGoutput[0].3	??	BOOL
R-IN3		
ImplicitMSGoutput[0].4	??	BOOL
R-IN4		
ImplicitMSGoutput[0].5	??	BOOL
R-IN5		
ImplicitMSGoutput[0].6	??	BOOL
R-IN6		
ImplicitMSGoutput[0].7	??	BOOL
R-IN7		
ImplicitMSGoutput[0].8	??	BOOL
R-IN8		
ImplicitMSGoutput[0].9	??	BOOL
R-IN9		
ImplicitMSGoutput[0].10	??	BOOL
R-IN10		
ImplicitMSGoutput[0].11	??	BOOL
R-IN11		
ImplicitMSGoutput[0].12	??	BOOL
R-IN12		
ImplicitMSGoutput[0].13	??	BOOL
R-IN13		
ImplicitMSGoutput[0].14	??	BOOL
R-IN14		
ImplicitMSGoutput[0].15	??	BOOL
R-IN15		
ImplicitMSGoutput[1]	??	INT
Operation data number selection		
ImplicitMSGoutput[1].0	??	BOOL
M0		
ImplicitMSGoutput[1].1	??	BOOL
M1		
ImplicitMSGoutput[1].2	??	BOOL
M2		

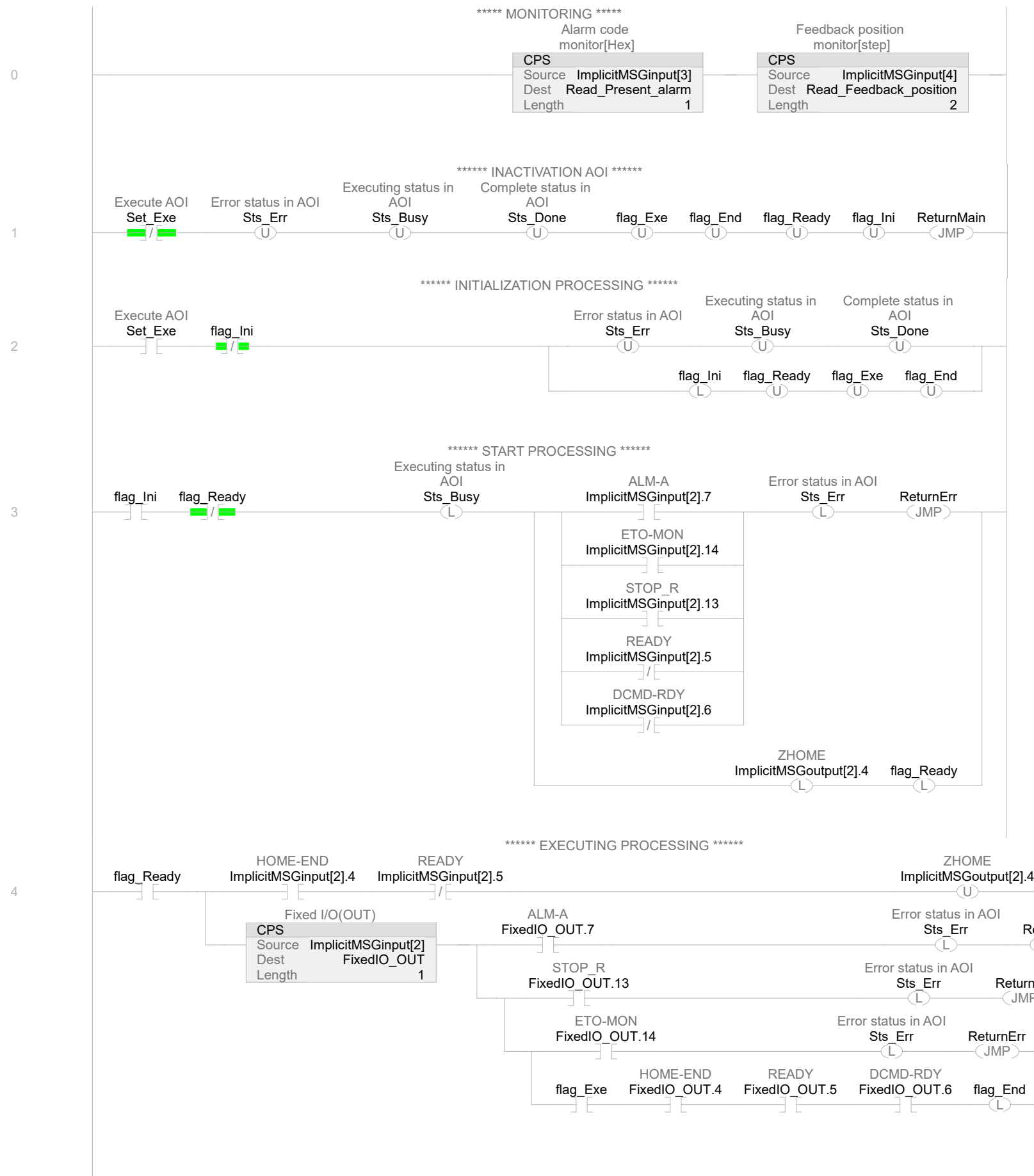
ImplicitMSGGoutput (Continued)		
ImplicitMSGGoutput[1].3	??	BOOL
M3		
ImplicitMSGGoutput[1].4	??	BOOL
M4		
ImplicitMSGGoutput[1].5	??	BOOL
M5		
ImplicitMSGGoutput[1].6	??	BOOL
M6		
ImplicitMSGGoutput[1].7	??	BOOL
M7		
ImplicitMSGGoutput[1].8	??	BOOL
Reserved		
ImplicitMSGGoutput[1].9	??	BOOL
Reserved		
ImplicitMSGGoutput[1].10	??	BOOL
Reserved		
ImplicitMSGGoutput[1].11	??	BOOL
Reserved		
ImplicitMSGGoutput[1].12	??	BOOL
Reserved		
ImplicitMSGGoutput[1].13	??	BOOL
Reserved		
ImplicitMSGGoutput[1].14	??	BOOL
Reserved		
ImplicitMSGGoutput[1].15	??	BOOL
Reserved		
ImplicitMSGGoutput[2]	??	INT
Fixed I/O(IN)		
ImplicitMSGGoutput[2].0	??	BOOL
FW-JOG		
ImplicitMSGGoutput[2].1	??	BOOL
RV-JOG		
ImplicitMSGGoutput[2].2	??	BOOL
Reserved		
ImplicitMSGGoutput[2].3	??	BOOL
START		
ImplicitMSGGoutput[2].4	??	BOOL
ZHOME		
<i>ImplicitMSGGoutput[2].4 - OM_AZ_ZHOME/Logic - *3(OTL), *4(OTU)</i>		
ImplicitMSGGoutput[2].5	??	BOOL
STOP		
ImplicitMSGGoutput[2].6	??	BOOL
FREE		
ImplicitMSGGoutput[2].7	??	BOOL
ALM-RST		
ImplicitMSGGoutput[2].8	??	BOOL
TRIG		
ImplicitMSGGoutput[2].9	??	BOOL
TRIG-MODE		
ImplicitMSGGoutput[2].10	??	BOOL
ETO-CLR		
ImplicitMSGGoutput[2].11	??	BOOL
Reserved		
ImplicitMSGGoutput[2].12	??	BOOL
FW-JOG-P		
ImplicitMSGGoutput[2].13	??	BOOL
RV-JOG-P		
ImplicitMSGGoutput[2].14	??	BOOL
FW-POS		
ImplicitMSGGoutput[2].15	??	BOOL
RV-POS		
ImplicitMSGGoutput[3]	??	INT

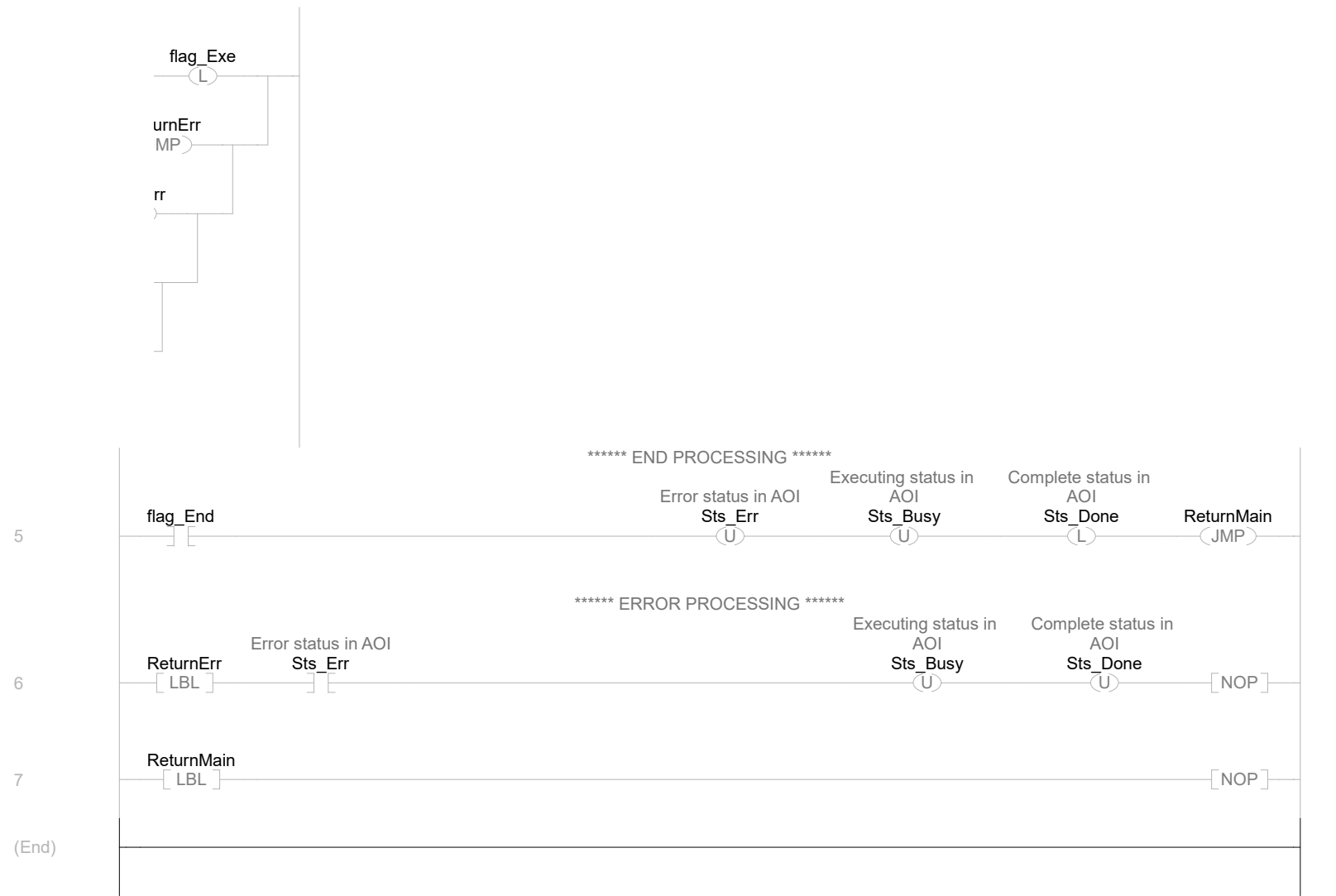
ImplicitMSGoutput (Continued)		
Direct data operation operation type		
ImplicitMSGoutput[4]	??	INT
Direct data operation position (lower)		
ImplicitMSGoutput[5]	??	INT
Direct data operation position (upper)		
ImplicitMSGoutput[6]	??	INT
Direct data operation operating speed (lower)		
ImplicitMSGoutput[7]	??	INT
Direct data operation operating speed (upper)		
ImplicitMSGoutput[8]	??	INT
Direct data operation starting/changing rate (lower)		
ImplicitMSGoutput[9]	??	INT
Direct data operation starting/changing rate (upper)		
ImplicitMSGoutput[10]	??	INT
Direct data operation stopping deceleration (lower)		
ImplicitMSGoutput[11]	??	INT
Direct data operation stopping deceleration (upper)		
ImplicitMSGoutput[12]	??	INT
Direct data operation operating current		
ImplicitMSGoutput[13]	??	INT
Direct data operation forwarding destination		
ImplicitMSGoutput[14]	??	INT
Reserved		
ImplicitMSGoutput[15]	??	INT
Read parameter ID		
ImplicitMSGoutput[16]	??	INT
Write request		
ImplicitMSGoutput[16].0	??	BOOL
WR-REQ		
ImplicitMSGoutput[16].1	??	BOOL
Reserved		
ImplicitMSGoutput[16].2	??	BOOL
Reserved		
ImplicitMSGoutput[16].3	??	BOOL
Reserved		
ImplicitMSGoutput[16].4	??	BOOL
Reserved		
ImplicitMSGoutput[16].5	??	BOOL
Reserved		
ImplicitMSGoutput[16].6	??	BOOL
Reserved		
ImplicitMSGoutput[16].7	??	BOOL
Reserved		
ImplicitMSGoutput[16].8	??	BOOL
Reserved		
ImplicitMSGoutput[16].9	??	BOOL
Reserved		
ImplicitMSGoutput[16].10	??	BOOL
Reserved		
ImplicitMSGoutput[16].11	??	BOOL
Reserved		
ImplicitMSGoutput[16].12	??	BOOL
Reserved		
ImplicitMSGoutput[16].13	??	BOOL
Reserved		
ImplicitMSGoutput[16].14	??	BOOL
Reserved		
ImplicitMSGoutput[16].15	??	BOOL
Reserved		
ImplicitMSGoutput[17]	??	INT
Write parameter ID		
ImplicitMSGoutput[18]	??	INT

ImplicitMSGoutput (Continued)			
Write data (lower)			
ImplicitMSGoutput[19]		??	INT
Write data (upper)			
Read_Feedback_position		0	DINT
Feedback position monitor[step]			OM_AZ_ZHOME
Usage:		Output Parameter	
Required:		No	
Visible:		Yes	
External Access:		Read Only	
Read_Feedback_position - OM_AZ_ZHOME/Logic - *0(CPS)			
Read_Present_alarm		16#0000	INT
Alarm code monitor[Hex]			OM_AZ_ZHOME
Usage:		Output Parameter	
Required:		No	
Visible:		Yes	
External Access:		Read Only	
Read_Present_alarm - OM_AZ_ZHOME/Logic - *0(CPS)			
Set_Exec		0	BOOL
Execute AOI			OM_AZ_ZHOME
Usage:		Input Parameter	
Required:		No	
Visible:		Yes	
External Access:		Read/Write	
Set_Exec - OM_AZ_ZHOME/Logic - 1(XIO), 2(XIC)			
Sts_Busy		0	BOOL
Executing status in AOI			OM_AZ_ZHOME
Usage:		Output Parameter	
Required:		No	
Visible:		Yes	
External Access:		Read Only	
Sts_Busy - OM_AZ_ZHOME/Logic - *1(OTU), *2(OTU), *3(OTL), *5(OTU), *6(OTU)			
Sts_Done		0	BOOL
Complete status in AOI			OM_AZ_ZHOME
Usage:		Output Parameter	
Required:		No	
Visible:		Yes	
External Access:		Read Only	
Sts_Done - OM_AZ_ZHOME/Logic - *1(OTU), *2(OTU), *5(OTL), *6(OTU)			
Sts_Err		0	BOOL
Error status in AOI			OM_AZ_ZHOME
Usage:		Output Parameter	
Required:		No	
Visible:		Yes	
External Access:		Read Only	
Sts_Err - OM_AZ_ZHOME/Logic - *1(OTU), *2(OTU), *3(OTL), *4(OTL), *4(OTL), *4(OTL), *5(OTU), 6(XIC)			

Name	Default	Data Type	Scope
FixedIO_OUT Fixed I/O(OUT) Usage: External Access: <i>FixedIO_OUT - OM_AZ_ZHOME/Logic - *4(CPS)</i>	0 Local Tag Read/Write	INT	OM_AZ_ZHOME
FixedIO_OUT.0 SEQ-BSY	0	BOOL	
FixedIO_OUT.1 MOVE	0	BOOL	
FixedIO_OUT.2 IN-POS	0	BOOL	
FixedIO_OUT.3 START_R	0	BOOL	
FixedIO_OUT.4 HOME-END <i>FixedIO_OUT.4 - OM_AZ_ZHOME/Logic - 4(XIC)</i>	0	BOOL	
FixedIO_OUT.5 READY <i>FixedIO_OUT.5 - OM_AZ_ZHOME/Logic - 4(XIC)</i>	0	BOOL	
FixedIO_OUT.6 DCMD-RDY <i>FixedIO_OUT.6 - OM_AZ_ZHOME/Logic - 4(XIC)</i>	0	BOOL	
FixedIO_OUT.7 ALM-A <i>FixedIO_OUT.7 - OM_AZ_ZHOME/Logic - 4(XIC)</i>	0	BOOL	
FixedIO_OUT.8 TRIG_R	0	BOOL	
FixedIO_OUT.9 TRIG-MODE_R	0	BOOL	
FixedIO_OUT.10 SET-ERR	0	BOOL	
FixedIO_OUT.11 EXE-ERR	0	BOOL	
FixedIO_OUT.12 DCMD-FULL	0	BOOL	
FixedIO_OUT.13 STOP_R <i>FixedIO_OUT.13 - OM_AZ_ZHOME/Logic - 4(XIC)</i>	0	BOOL	
FixedIO_OUT.14 ETO-MON <i>FixedIO_OUT.14 - OM_AZ_ZHOME/Logic - 4(XIC)</i>	0	BOOL	
FixedIO_OUT.15 TLC	0	BOOL	
flag_End Usage: External Access: <i>flag_End - OM_AZ_ZHOME/Logic - *1(OTU), *2(OTU), *4(OTL), 5(XIC)</i>	0 Local Tag None	BOOL	OM_AZ_ZHOME
flag_Exe Usage: External Access: <i>flag_Exe - OM_AZ_ZHOME/Logic - *1(OTU), *2(OTU), *4(OTL), 4(XIC)</i>	0 Local Tag None	BOOL	OM_AZ_ZHOME
flag_Ini Usage: External Access: <i>flag_Ini - OM_AZ_ZHOME/Logic - *1(OTU), *2(OTL), 2(XIO), 3(XIC)</i>	0 Local Tag None	BOOL	OM_AZ_ZHOME
flag_Ready Usage: External Access:	0 Local Tag None	BOOL	OM_AZ_ZHOME

flag_Ready (Continued)
*flag_Ready - OM_AZ_ZHOME/Logic - *1(OTU), *2(OTU), *3(OTL), 3(XIO), 4(XIC)*





Data type Name: AOI_Motor

Description:

Size: 272 byte(s)

Name	Value	Data Type	Style
DDO External Access:	Read/Write	OM_AZ_DDO	NullType
Move External Access:	Read/Write	OM_AZ_MOVE	NullType
Monitor External Access:	Read/Write	OM_AZ_MONITOR	NullType
ReadPosition External Access:	Read/Write	OM_AZ_READ	NullType
WritePosition External Access:	Read/Write	OM_AZ_WRITE	NullType
Home External Access:	Read/Write	OM_AZ_ZHOME	NullType
WriteSpeed External Access:	Read/Write	OM_AZ_WRITE	NullType
ReadSpeed External Access:	Read/Write	OM_AZ_READ	NullType

Data type Name: Coil

Description:

Size: 36 byte(s)

Name	Value	Data Type	Style
Inductance External Access:	Read/Write	REAL	Float
Turns External Access:	Read/Write	INT	Decimal
Mass External Access:	Read/Write	REAL	Float
Diameter External Access:	Read/Write	REAL	Float
Turns_X External Access:	Read/Write	INT	Decimal
Turns_Z External Access:	Read/Write	INT	Decimal
Current External Access:	Read/Write	REAL	Float
Length External Access:	Read/Write	REAL	Float
WireRequested External Access:	Read/Write	BOOL	Decimal
Gauge External Access:	Read/Write	DINT	Decimal

Data type Name: DCMotorDirectionControl

Description:

Size: 4 byte(s)

Name	Value	Data Type	Style
FWD1		BOOL	Decimal
External Access:	Read/Write		
FWD2		BOOL	Decimal
External Access:	Read/Write		
FWD3		BOOL	Decimal
External Access:	Read/Write		
REV1		BOOL	Decimal
External Access:	Read/Write		
REV2		BOOL	Decimal
External Access:	Read/Write		
REV3		BOOL	Decimal
External Access:	Read/Write		

Data type Name: HMIControl

Description:

Size: 28 byte(s)

Name	Value	Data Type	Style
Manual		HMIManual	NullType
External Access:	Read/Write		
StartPB		BOOL	Decimal
External Access:	Read/Write		
ResetPB		BOOL	Decimal
External Access:	Read/Write		
Temp		REAL	Float
External Access:	Read/Write		
StopPB		BOOL	Decimal
External Access:	Read/Write		
RobotFaultReset		BOOL	Decimal
External Access:	Read/Write		

Data type Name: HMIManual

Description:

Size: 16 byte(s)

Name	Value	Data Type	Style
Wind External Access:	Read/Write	DCMotorDirectionControl	NullType
RED External Access:	Read/Write	BOOL	Decimal
YELLOW External Access:	Read/Write	BOOL	Decimal
GREEN External Access:	Read/Write	BOOL	Decimal
Spool External Access:	Read/Write	DCMotorDirectionControl	NullType
Glue External Access:	Read/Write	BOOL	Decimal
Cut External Access:	Read/Write	BOOL	Decimal
CameraLight External Access:	Read/Write	BOOL	Decimal
ExtrudeHeater External Access:	Read/Write	BOOL	Decimal
ExtrudeMotor External Access:	Read/Write	BOOL	Decimal
WindingSolenoid External Access:	Read/Write	BOOL	Decimal
ClampMotorFWD External Access:	Read/Write	BOOL	Decimal
ClampMotorREV External Access:	Read/Write	BOOL	Decimal
HeightMotorUp External Access:	Read/Write	BOOL	Decimal
HeightMotorDown External Access:	Read/Write	BOOL	Decimal

Data type Name: MicroScan

Description:

Size: 8 byte(s)

Name		Value	Data Type	Style
In	External Access:	Read/Write	MicroScanIn	NullType
Out	External Access:	Read/Write	MicroScanOut	NullType

Data type Name: MicroScanIn

Description:

Size: 4 byte(s)

Name	Value	Data Type	Style
Mode		SINT	Decimal
External Access:	Read/Write		
Monitor_Case		SINT	Decimal
External Access:	Read/Write		
Field_Status		SINT	Decimal
External Access:	Read/Write		
Error		SINT	Decimal
External Access:	Read/Write		

Data type Name: MicroScanOut

Description:

Size: 4 byte(s)

Name	Value	Data Type	Style
Set_Mode		SINT	Decimal
External Access:	Read/Write		
Control_Input		SINT	Decimal
External Access:	Read/Write		

Data type Name: ServoMotor

Description:

Size: 20 byte(s)

Name	Value	Data Type	Style
POS request angle relative to current pos External Access:	Read/Write	REAL	Float
SPEED deg/s External Access:	Read/Write	DINT	Decimal
ACCEL deg/s/s External Access:	Read/Write	DINT	Decimal
DECCEL deg/s/s External Access:	Read/Write	DINT	Decimal
MotorBusy External Access:	Read/Write	BOOL	Decimal

Data type Name: Spool

Description:

Size: 20 byte(s)

Name	Value	Data Type	Style
REV		BOOL	Decimal
out			
External Access:	Read/Write		
FWD		BOOL	Decimal
out			
External Access:	Read/Write		
Mass		REAL	Float
current mass reading			
External Access:	Read/Write		
Length		REAL	Float
approximated legth based on Mass			
External Access:	Read/Write		
Loaded		BOOL	Decimal
if is in loaded state			
External Access:	Read/Write		
Gauge		DINT	Decimal
Wire Gauge			
External Access:	Read/Write		

Data type Name: STRING

Description:

Size: 88 byte(s)

Name	Value	Data Type	Style
LEN		DINT	Decimal
External Access:	Read/Write		
DATA		SINT[82]	ASCII
External Access:	Read/Write		

Data type Name: STRING_16

Description:

Size: 20 byte(s)

Name	Value	Data Type	Style
LEN		DINT	Decimal
External Access:	Read/Write		
DATA		SINT[16]	ASCII
External Access:	Read/Write		

Data type Name: STRING_32

Description:

Size: 36 byte(s)

Name	Value	Data Type	Style
LEN		DINT	Decimal
External Access:	Read/Write		
DATA		SINT[32]	ASCII
External Access:	Read/Write		

Data type Name: AB:5000_String16_Struct:I:0

Description:

Size: 20 byte(s)

Name	Value	Data Type	Style
LEN		DINT	Decimal
External Access:	Read/Write		
DATA		SINT[16]	ASCII
External Access:	Read/Write		

Data type Name: AB:5000_String32_Struct:I:0

Description:

Size: 36 byte(s)

Name	Value	Data Type	Style
LEN		DINT	Decimal
External Access:	Read/Write		
DATA		SINT[32]	ASCII
External Access:	Read/Write		

5069 Backplane : Local Modules

Local: [0] 5069-L330ERMS3 Capstone_Test			
Type:	5069-L330ERMS3 Compact GuardLogix® 5380 Safety Controller	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	0	Safety Network Number:	4C2C_03D9_C3CA
Electronic Keying:	Disabled	Revision:	33.11
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off		
Local: [1] 5069-IB8S/A Safety_Input			
Type:	5069-IB8S/A 8 Point 24V DC Safety Input, Sink	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	1	Safety Network Number:	4C2C_03D9_C3CA
Electronic Keying:	Compatible Module	Revision:	2.1
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off	Use Unicast:	n/a
Local: [2] 5069-OBV8S/A Safety_Output			
Type:	5069-OBV8S/A 8 Point 24V DC Bipolar/Sourcing Safety Output	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	2	Safety Network Number:	4C2C_03D9_C3CA
Electronic Keying:	Compatible Module	Revision:	2.11
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off	Use Unicast:	n/a
Local: [3] 5069-IB16/A DC_Input			
Type:	5069-IB16/A 16 Point 24V DC Input, Sink	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	3	Electronic Keying:	Compatible Module
Revision:	2.1	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
Use Unicast:	n/a	Enable Automatic Diagnostics:	Yes

Module Defined Configuration Tag	Value	Data Type
Local:3:C	AB:5000_DI16:C:0	
.Pt00	AB:5000_DI_Channel:C:0	
.Pt01	AB:5000_DI_Channel:C:0	
.Pt02	AB:5000_DI_Channel:C:0	
.Pt03	AB:5000_DI_Channel:C:0	
.Pt04	AB:5000_DI_Channel:C:0	
.Pt05	AB:5000_DI_Channel:C:0	
.Pt06	AB:5000_DI_Channel:C:0	
.Pt07	AB:5000_DI_Channel:C:0	
.Pt08	AB:5000_DI_Channel:C:0	
.Pt09	AB:5000_DI_Channel:C:0	
.Pt10	AB:5000_DI_Channel:C:0	

.Pt11	AB:5000_DI_Channel:C:0
.Pt12	AB:5000_DI_Channel:C:0
.Pt13	AB:5000_DI_Channel:C:0
.Pt14	AB:5000_DI_Channel:C:0
.Pt15	AB:5000_DI_Channel:C:0

Local: [4] 5069-OW16/A DC_Relay_Outut			
Type:	5069-OW16/A 16 Point AC/DC Relay Output, N.O.	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	4	Electronic Keying:	Compatible Module
Revision:	3.1	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
Use Unicast:	n/a		

Module Defined Configuration Tag	Value	Data Type
Local:4:C	AB:5000_DO16:C:0	
.Pt00	AB:5000_DO_Channel:C:0	
.Pt01	AB:5000_DO_Channel:C:0	
.Pt02	AB:5000_DO_Channel:C:0	
.Pt03	AB:5000_DO_Channel:C:0	
.Pt04	AB:5000_DO_Channel:C:0	
.Pt05	AB:5000_DO_Channel:C:0	
.Pt06	AB:5000_DO_Channel:C:0	
.Pt07	AB:5000_DO_Channel:C:0	
.Pt08	AB:5000_DO_Channel:C:0	
.Pt09	AB:5000_DO_Channel:C:0	
.Pt10	AB:5000_DO_Channel:C:0	
.Pt11	AB:5000_DO_Channel:C:0	
.Pt12	AB:5000_DO_Channel:C:0	
.Pt13	AB:5000_DO_Channel:C:0	
.Pt14	AB:5000_DO_Channel:C:0	
.Pt15	AB:5000_DO_Channel:C:0	

Local: [5] 5069-IF8/A Analog_Input			
Type:	5069-IF8/A 8 Channel Voltage/Current Analog Input	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	5	Electronic Keying:	Compatible Module
Revision:	2.1	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
Use Unicast:	n/a	Enable Automatic Diagnostics:	Yes

Module Defined Configuration Tag	Value	Data Type
Local:5:C	AB:5000_AI8:C:0	
.Ch00	AB:5000_AI_Channel:C:0	
.Ch01	AB:5000_AI_Channel:C:0	
.Ch02	AB:5000_AI_Channel:C:0	
.Ch03	AB:5000_AI_Channel:C:0	
.Ch04	AB:5000_AI_Channel:C:0	
.Ch05	AB:5000_AI_Channel:C:0	
.Ch06	AB:5000_AI_Channel:C:0	
.Ch07	AB:5000_AI_Channel:C:0	

A2, Ethernet : Local Modules

ETHERNET-MODULE Fanuc_Robot

Type:	ETHERNET-MODULE Generic Ethernet Module	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Electronic Keying:	Disabled	Revision:	1.1
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off	RPI:	10 ms
Use Unicast:	Yes		

Module Defined Configuration Tag

ValueData Type

Fanuc_Robot:C

.Data

AB:ETHERNET_MODULE:C:0

SINT[400]

1732E-IB16M12R/A Input_RemoteIO

Type:	1732E-IB16M12R/A 16 Point 24V DC Input, 2-Port	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Electronic Keying:	Compatible Module	Revision:	1.1
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off	RPI:	20 ms
Use Unicast:	Yes		

Module Defined Configuration Tag

ValueData Type

Input_RemoteIO:C

.FilterOffOn

.FilterOnOff

1000

1000

AB:1732_DI16:C:0

INT

INT

ETHERNET-MODULE Cognex_5000

Type:	ETHERNET-MODULE Generic Ethernet Module	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Electronic Keying:	Disabled	Revision:	1.1
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off	RPI:	10 ms
Use Unicast:	Yes		

Module Defined Configuration Tag

ValueData Type

Cognex_5000:C

.Data

AB:ETHERNET_MODULE:C:0

SINT[400]

<EDS not registered> Winding_Servo

Type:		Parent:	Local
Vendor:		Vendor ID:	187
Electronic Keying:	Compatible Module	Revision:	1.1
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off		
RPI:	10 ms	Input Type:	Unicast
Input Trigger:	Cyclic	Use Unicast:	n/a

? <EDS not registered> microScan

Type:		Parent:	Local
Vendor:		Vendor ID:	808
Safety Network Number:	4C33_0505_D74C	Electronic Keying:	Compatible Module
Revision:	1.2	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
RPI:	20 ms	Input Type:	Unicast
Input Trigger:	Application		
RPI:	20 ms	Input Type:	Unicast
Input Trigger:	Application		
RPI:	20 ms	Input Type:	Unicast
Input Trigger:	Cyclic	Use Unicast:	n/a

🔌 1732E-OB16M12R/A Output_RemoteIO

Type:	1732E-OB16M12R/A 16 Point 24V DC Output, 2-Port	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Electronic Keying:	Compatible Module	Revision:	1.1
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off	RPI:	20 ms
Use Unicast:	Yes		

Module Defined Configuration Tag	Value	Data Type
Output_RemoteIO:C		AB:1732_DO16:C:0
.FaultMode	0	BOOL
.FaultValue	0	BOOL
.ProgMode	0	BOOL
.ProgValue	0	BOOL

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